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Appendix B: Technical Report

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Summary

The goal of this project is to develop model-free, real-time adaptive control frameworks for long-range detection and localization of targets in uncertain dynamic environments where measured signals are distorted, diverted, or diffused by environmental factors. The control strategies will steer multiple surface and underwater vehicles to the location of static or moving signal sources, without prior knowledge of their position or the position of the sources. The proposed approach utilizes extremum seeking control (ESC) to localize precise global signal extrema, i.e. the signal source, within spatially distributed nonconvex fields. The work entails both theoretical and operational frameworks to enable robust open-world application of the seekers and will provide mathematical proof of stability for the designed control to ensure desired behavior in the presence of uncertainties. Instead of relying on mere static processing of sensory data, we will develop robust control frameworks that empower the agents to adaptively construct their intelligence and autonomously address ambiguities in the sensed signals. The methods portray the natural process of perception in humans and animals for whom the perception task is not simply about analysis of passively sampled data, but rather active exploration and search to thrive in an uncertain world. The outcome is a perception system that guarantees robust performance in dynamic and unpredictable settings, offering the following advantages: (1) power efficiency by developing a systematic computationally low-cost search methodology and providing the versatility to use a single sensor rather than multi-sensor systems; (2) integrated strategies for addressing signal interference and degradation within the control scheme; (3) novel safety mechanisms enabled through control log barrier functions; (4) position-free operation, thus eliminating the need for data association; (5) multi-source localization and multi-vehicle operation; (6) provision of coordination-free and limited coordination operation options for multi-vehicle cases; and (7) convergence to sources in user-prescribed time.

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1. Introduction

The domain of source seeking is incredibly rich with applications for tasks such as search-and-rescue, target pursuit, and system parameter optimization. In these tasks, one would construct the problem in such a way that is similar to optimization problems, e.g. optimizing a spatial dependent sensor signal in the target pursuit task where the sensor signal is maximized at the target. The naive first consideration would be to use the existing optimizers found in numerical optimization i.e. discrete-time optimizers. However, discrete methods are not favored for applications onto real vehicles as most vehicles have physics-driven dynamics. Therefore, it is advantageous to consider the continuous-time variations of these optimizers and how they might be used on vehicles. In particular, one would wish to use methods in continuous-time where the method did not have direct information of the spatial derivatives of the signal and thus the focus of this work on extremum seeking control (ESC).

ESC is a family of continuous-time optimization algorithms. They were first introduced in 1922 [2] but remained largely uninvestigated until 2000 when the local practical stability was proven for general nonlinear systems [3]. This work views ESC in general sense as shown in Fig. 1. Here the vehicle states are encoded into a state vector x with the dynamics governed by an ordinary differential equation. The differential equation is a function f with arguments of time t, states x, and a control input u. Often u is a feedback controller that stabilizes the vehicle to some position as parameterized by some parameters θ. For the source seeking task, the vehicle senses a spatially varying cost function J (e.g., a sensor output y) which may also be slowly varying in time. Thus, the goal of ESC is to update the parameters θ so that the vehicle moves towards extrema of J.

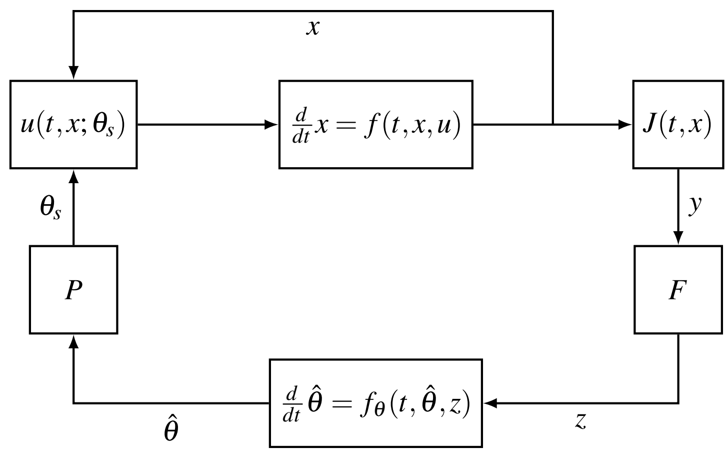


Figure 1: Generalized Extremum Seeker for Adaptive and Accelerated Methods

To accomplish this without *a priori* knowledge of J, it is necessary to know or estimate the spatial derivatives of J to adjust estimates of the optimal parameters $\hat{\theta}$. In numerical optimization, one might estimate these derivatives by using evaluations of J around $\hat{\theta}$ before updating $\hat{\theta}$. This does not work for vehicles with single sensors operating in continuous time. Therefore, ESCs will continuously perturb $\hat{\theta}$ through the block P while simultaneously estimating derivatives z from the perturbed sensor output through block F. Thus, u does not take the estimate of the optimal parameter but rather a perturbed version for searching θ_s. Historically, the F is a dynamical system to high pass, convolve, and then low pass the signal y while P is simply an additive perturbation. Additionally, since ESCs are always locally exploring around $\hat{\theta}$, their stability is talked about in terms of practical stability. In practice, a vehicle performing source seeking would start far from the source before progressively moving closer until the vehicle was sufficiently close to the source. At this point, the vehicle would verify the source or remain sufficiently close until the task ended using a different controller and sensor. However, we still need to prove practical stability since this stability is how to rigorously show that the vehicle will be able to get sufficiently close to the source.

With estimates of the derivatives, the last unexplained part of the ESC is the strategy of how to update $\hat{\theta}$ via a differential equation governed by f_θ. The choice of f_θ depends on the type of ESC. Classically, this would be a Gradient-based ESC (GESC) with $f_\theta(t,\hat{\theta},z) = -k\hat{g}$ where k is the gain, either as a matrix or a scalar value, and $\hat{g}$ is the point-wise in time estimate of the cost function gradient, ∇J. Note that $\hat{g}$ is not necessarily ∇J whenever the estimates use continuous perturbations, instead the analysis for ESC stability is done on the average. What this means is that if $\hat{g}$ is averaged over time (with $\hat{\theta}$ held constant) to $\bar{\hat{\theta}}$, ESC desires that $\bar{\hat{\theta}}$ is ∇J (or at least sufficiently close).

Currently, most ESC literature is based on either gradient descents [4, 5] or Newton methods [6–10]. However, there are a few theoretical and operational shortcomings associated with these methods that prevent their application to source seeking using real vehicles in cluttered dynamic environments with complex signal propagation paths. **(1)** In traditional ESC, the estimation of the spatial signal derivatives is done by perturbing the estimate through the vehicle motions. The perturbation of vehicle position would result in significant time and energy consumption, making it impractical for most operations. **(2)** GESC and Newton-based ESC (NESC) lack a regularized convergence rate. The GESC exhibits a high convergence rate when the gradient is strong (far from the extremum), but this rate significantly drops as it approaches the source because of a diminishing gradient. While exponential convergence may be a desirable local property for NESC, it becomes problematic when the seeker is far from the extremum. In such cases, the exponential convergence rate will force the vehicle to move too fast, violating the speed constraints and saturating the control input. **(3)** The ESC literature has only proved the stability to equilibrium when the sensor field is strongly or strictly convex [6, 10–12]. In practice, the sensor fields are almost never strongly/strictly convex as required by ESCs. This can cause two issues. **(3-A)** The seeker becomes trapped in a local minimum (i.e., false sources) or an undesired global minimum (i.e., undesired sources). **(3-B)** In scenarios with multiple desired and undesired sources, the seeker's inability to isolate the signal of interest within the measured mixture signal can considerably prolong the convergence time.

In **Task 1**, we address issue **(1)** by formulating a method for determining estimation signals for m-th order derivatives from arbitrary sensor perturbations. When integrated into a source seeking controller for unmanned vehicles in **Task 10-2**, the method enables significant time and energy savings. It does so by distributing the local map exploration to a rotating onboard sensor, eliminating the need to perturb the vehicle position for estimation.

In **Tasks 2 and 3**, we tackle the issue **(2)**. In **Task 3**, we address the issue **(2)** by expanding ESCs to other first-order methods, particularly adaptive gradient methods (AdaGM). These methods attempt to normalize the descent speed based on historical information of the gradient. **Task 3** integrates these optimizers with the ESC and **Task 2** develops fundamental stability theorems to allow showing the stability for model-free ESC systems which use adaptive or accelerated gradient optimizers.

Tasks 4, 5, and **6** further build upon this foundation to formulate techniques that enable escaping undesired extrema and isolating the signals of interest in highly non-convex and multi-source fields and thus resolving issues **(3-B)** and **(3-C)**, respectively. **Task 4** integrates another class of first-order optimizers known as accelerated gradient methods (AccGM) with the ESC. The formulation for the AccGM includes a momentum term that accelerates convergence and under specific circumstances allows the seeker to operationally escape undesired extrema. **Task 5** merges the properties of AdaGM and AccGM and leverage their respective strengths to regularize the convergence rate and improve convergence along flat regions and non-convex regions. Accelerated gradient-based ESC can escape some extrema, but not all extrema, and not always. The success depends on the depth and geometry of the minimum relative to the optimizer's current momentum. **Task 6** builds upon the blend ESC (BESC) developed under **Task 5** by creating additional strategies designed to ensure escape from undesired extrema particularly in cases where accelerated gradient-based ESC fails.

Tasks 7 and **8** will address additional challenges involved in complex real-life environments, namely the need to guarantee safety during ESC explorations in cluttered environments and tracking moving sources, respectively.

Task 9 will further improve the convergence speed and overall optimization outcome by enabling cooperation between multiple vehicles to seek single or multiple signal sources and studying the feasibility of a prescribed-time (PT) controller.

Tasks 1-9 form the theoretical framework required to enable robust and successful operations in real-life environments, while **Task (10)** will establish the operational framework by implementing these methods onboard various types of unmanned vehicles. The implementation is performed in parallel with the theoretical studies to inform the theoretical design and minimize modifications.
Derivative Work — Publications that are a result of this project are:

- A study on the stability of NESC on strictly convex scalar maps [13];
- A study on integrating the RMSprop optimizer with ESC and its stability [14];
- A necessary and sufficient condition for determining perturbation-based derivative estimates for arbitrary perturbation signals [15];
- The relationship between model-based system stability and the model-free system stability [16];
- The stability of interconnected systems [17].

2. Project Tasks, Relations, and Progress

Section 1 described the project tasks. Charts shown in Figs. 2 and 3 illustrate the relation between the tasks and their progress for the theoretical and operational frameworks, respectively.

Theoretical Framework

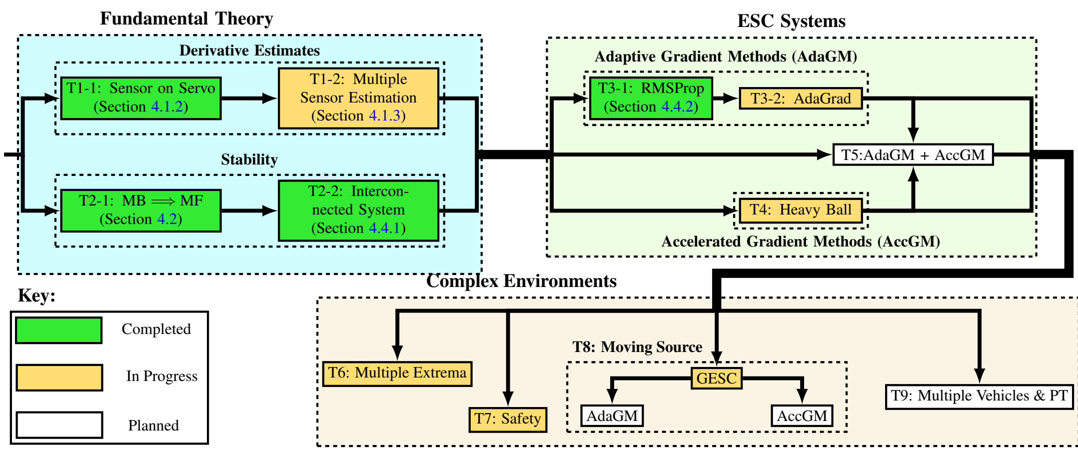


Figure 2: The theoretical framework tasks and their progress (**T** denotes Task).

Our first priority is to determine fundamental theory of the ESCs, namely how does one estimate derivatives and prove the practical stability of ESCs. We have derived conditions for perturbation-based derivative estimation in ESCs under Section 4.1. This allows to have an equivalency between the limit of an integral and differentiation for calculating the associated model-based system from our model-free ESC system, and also frees us from using traditional sinusoidal signals for the perturbations. Derivative estimates for 2D signal maps can be obtained using a single source of information (or sensor), while 3D maps require multiple sources (a minimum of two). **Task 1-1** (discussed in [15] and under Section 4.1.2) establishes the theory for estimating the derivatives using a single rotating sensor while **Task 1-2** (under Section 4.1.3) develops a new approach that integrates information from multiple sensors to generate a gradient estimate of the cost function J.

The model-based systems are the result of averaging the vector field before taking a limit and these model-based systems can be then used to prove the practical stability of the model-free systems. This is discussed in [16] (**Task 2-1**, Section 4.2), where we show that if the model-based ESC is globally asymptotically stable (GAS), then the average system is semiglobally practically asymptotically stable (sGPAS), and the model-free ESC is semiglobally practically uniformly asymptotically stable (sGPUAS). While this research is fundamental, it allows us to investigate the specific stability of certain extremum seekers such as the adaptive gradient-based ESCs and the Heavy Ball ESC.

Additionally, we have found that most ESCs, and in particular interest the adaptive gradient-based ESC, are interconnected systems, with auxiliary parameters attempting to estimate derivative information about the cost function J. Interconnected systems have been widely studied in the literature, with a focus on interconnected systems whose subsystems are solely input-to-state stable (ISS) or passive systems. However, interconnected systems with subsystems which are not ISS or passive are rarely studied but frequently arise in the context of adaptive gradient optimization. In adaptive gradient optimization, one subsystem seeks to move parameters of a cost function towards a minimizer while the other subsystem works to estimate some derivative information about the cost function to help determine the direction of the parameter update. In **Task 2-2** (discussed in [17] and under Section 4.4.1), we investigate the stability of these systems as a means to prove the stability of ESCs when we introduce auxiliary parameters. We study instances of such interconnected systems and give various Lyapunov functions, using different techniques, for adaptive gradient optimization systems. In doing so, adaptive gradient optimizers are proven to be globally asymptotically stable (GAS).

The five systems we are interested in investigating are: the gradient-based ESC (GESC), the RMSprop-based ESC (RMSpESC), the AdaGrad ESC (AGESC), Heavy Ball ESC (HBESC), and a blended AdaGM-ESC and HBESC. The Root Mean Squared Propagation (RMSProp) optimizer [18] and the Adaptive Gradient (AdaGrad) optimizer [19] are the two prominent optimizers in the machine learning community. Heavy Ball is an AccGM method selected here due to showing greater convergence properties compared to the other AccGM methods (such as Nestrov's) in our preliminary studies. We have shown the practical stability of the GESC on scalar and multivariable strictly convex maps (in [13] and under Section 4.3). Furthermore, using our studies [16, 17], we showed that the model-free RMSpESC system is sGPUAS on multivariable strictly convex maps (in **Task 3-1**, Section 4.4.2) [14].

From the lessons learned, we are investigating the stability of the AGESC and HBESC, hoping to utilize our work on interconnected systems [17] and/or stability of model-free systems [16] to prove their practical stability on maps which meet second-order sufficient conditions of strict convexity. Additionally, we have made significant progress in establishing the foundation for methods required in complex environments, including approaches to escape multiple extrema, showing stability and safety of Log Barrier Functions (LBFs) for unicycle vehicle models, and showing the stability for tracking moving sources using a unicycle vehicle with GESC.

Operational Framework

The operational framework is being established in parallel to the theoretical framework. We have implemented the GESC using both static and rotating sensor frames onboard unmanned ground vehicles (UGV) to seek both static and moving sources (see Sections 5.5.2 and 5.5.4). The experiments included both light and acoustic sources. The RMSpESC and AGESC implementations with rotating sensors onboard UGVs have been as well completed and examined for both static and moving signal sources (see Section 5.5.5). All tests were conducted in indoor environments, which constrained the maximum distance between the source and the seeker. We have made significant progress in instrumenting an unmanned surface vessel (USV) and an unmanned underwater vehicle

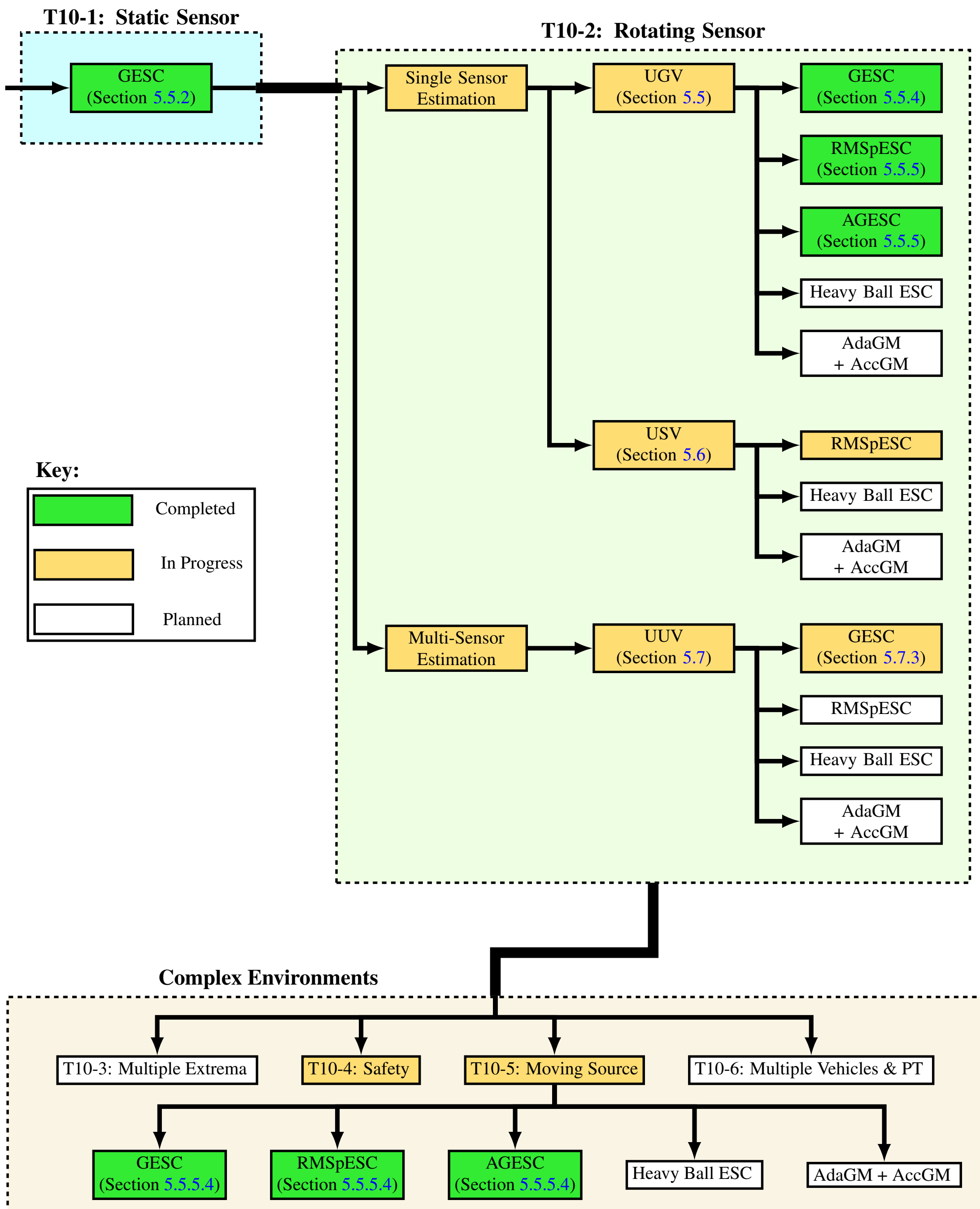


Figure 3: The operational framework tasks and their progress (**T** denotes Task).

(UUV) to perform experiments on the surface of water (a 2D signal map) and in underwater environments (a 3D signal maps), respectively. These experiments allow expansion of test environments to more complex environments and effectively remove the distance constraint between the seeker and the source. The UUV experiments aim to validate the multiple sensor estimation theory, developed under **Task 1-2**, for derivative estimation in 3D maps. Lastly, we are working to implement methods to overcome operational challenges such as multiple extrema and safety.

3. Definitions and Nomenclature

3.1 Nomenclature

Mathematics in this work uses lower case letters x for either real scalar numbers ($x \in \mathbb{R}$) or column real vectors ($x \in \mathbb{R}^n$) and upper case letters X for matrices $X \in \mathbb{R}^{n\times m}$. Row vectors are denoted by x^T where $\cdot^T$ is the general matrix transpose. Whenever subscripts are used, each subscript indicates the index in the specified direction, e.g., x_i is the ith element of x and X_{ij} is the element in the ith row and jth column of X. The gradient of a vector function $f : \mathbb{R}^n \mapsto \mathbb{R}^m$ is represented as $\nabla f_{ij} = \frac{\partial f_i}{\partial x_j}$ while the gradient of a scalar is a column vector $\nabla f_i = \frac{\partial f}{\partial x_i}$. The Hessian of the scalar functions is $\nabla^2 f$ where $\nabla^2 f_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

3.2 Definitions

3.2.1 Matrices

The main matrices of interest are the Hessians of scalar functions. By definition, Hessians are symmetric as $\frac{\partial f}{\partial x_i \partial x_j} = \frac{\partial f}{\partial x_j \partial x_i}$ so they are an element of the symmetric real matrices $\mathbb{S}^{n\times n}$. Often we would like the Hessian to be a positive definite matrix.

Definition 3.1 (Positive Definite Matrices)**.** *A matrix A is a positive definite matrix (PDM) if*

$$x^T A x > 0 \,, \quad \forall\, x \in \mathbb{R} \setminus \{0\} \tag{3.1}$$

If the inequality is relaxed from a strict inequality to a general inequality, then the matrix is said to be a positive semidefinite matrix (PSDM).

The special sets for matrices which are both symmetric and a PDM or symmetric and a PSDM are denoted as $\mathbb{S}^{n\times n}_{\succ 0}$ and $\mathbb{S}^{n\times n}_{\succeq 0}$, respectively.

3.2.2 Functions

Definition 3.2 (Positive and Negative Definite Functions)**.** *A function $f : \mathbb{R}^n \mapsto \mathbb{R}$ is a positive (semi-)definite function if $f(0) = 0$ and $f(x) > 0$ (≥ 0) if $x \neq 0$. A function f is a negative (semi-)definite function if $-f$ is a positive (semi-)definite function.*

Other special functions used for establishing stability of dynamical systems are the class $\mathscr{K}$ and class $\mathscr{K}\mathscr{L}$ functions.

Definition 3.3 (Class $\mathscr{K}$ functions)**.** *A continuous function $\alpha : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a class $\mathscr{K}$ function if*

- $\alpha(0) = 0$ *and*
- $\alpha(r)$ *is monotonically increasing with respect to the argument r.*

Furthermore, if $\alpha(r) \to \infty$ *as* $r \to \infty$*, then* α *is a class* $\mathscr{K}_\infty$ *function.*

Definition 3.4 (Class $\mathscr{K}\mathscr{L}$ functions)**.** *A continuous function* $\beta : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ *is a class* $\mathscr{K}\mathscr{L}$ *function if*

- $\beta(r,s)$ *is a class* $\mathscr{K}$ *function with respect to r for every fixed s;*
- $\beta(r,s)$ *is monotonically decreasing with respect to the argument s for every fixed r;*
- *and* $\beta(r,s) \to 0$ *as* $s \to \infty$*.*

3.2.3 Stability

We denote trajectories $x(t) \in \mathbb{R}^n$ as a solution to the differential equation

$$\frac{d}{dt}x = f(t,x) \tag{3.2}$$

for a given initial condition $x(t_0) = x_0$. Often we consider a family of differential equations

$$\frac{d}{dt}x = f^{\varepsilon}(t,x) \tag{3.3}$$

where $f^{\varepsilon} : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ is parameterized by a vector of small parameters $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_m) \in \mathbb{R}^m_{>0}$. Stability of trajectories in the parameterized differential equations are given in terms of practical stability as given in the following Definitions 3.5-3.8. The reader should note that the phrase "there exists $\varepsilon^* > 0$ such that for all $\varepsilon \in (0, \varepsilon^*)$" is a short hand for "there exists $\varepsilon_1^* > 0$ such that for all $\varepsilon_1 \in (0, \varepsilon_1^*)$, there exists ε_2^* such that for all $\varepsilon_2 \in (0, \varepsilon_2^*)$, ..." when dealing with multiple small parameters.

Definition 3.5 (Practically Stable)**.** *The origin is practically stable for* (3.3) *if for every* $c_2 > 0$*, there exists* $c_1 > 0$ *and* $\varepsilon^* > 0$ *such that for all* $\varepsilon \in (0, \varepsilon^*)$*,*

$$\|x(t_0)\| < c_1 \implies \|x(t)\| < c_2 \;\forall\, t \in [t_0, \infty) \tag{3.4}$$

Definition 3.6 (Practically Bounded)**.** *The origin is practically bounded of* (3.3) *if for every* $c_1 > 0$*, there exists* $c_2 > 0$ *and* $\varepsilon^* > 0$ *such that for all* $\varepsilon \in (0, \varepsilon^*)$*,*

$$\|x(t_0)\| \leq c_1 \implies \|x(t)\| \leq c_2 \;\forall\, t \in [t_0, \infty) \tag{3.5}$$

Definition 3.7 (δ-Practically Uniformly Attractiveness)**.** *The origin is* δ*-practically uniformly attractive of* (3.3) *if for* $c_1 = \delta > 0$*, for all* $c_2 > 0$*, there exists* $T = T(c_1, c_2) > 0$ *and* $\varepsilon^* > 0$ *such that for all* $\varepsilon \in (0, \varepsilon^*)$*,*

$$\|x(t_0)\| \leq c_1 \implies \|x(t)\| \leq c_2 \;\forall\, t \in [t_0 + T, \infty) \tag{3.6}$$

If a stricter bounds can be placed on the trajectories

$$\|x(t)\| \leq c_1 M e^{-\lambda(t-t_0)} + c_2 \;\forall\, t \in [t_0, \infty) \tag{3.7}$$

where $M,\lambda > 0$ are constants independent of δ, then the origin is locally practically exponentially stable (PES).

If δ can be made arbitrarily large, then the origin is said to be semiglobally practically uniformly attractive of (3.3).

Definition 3.8 (sGPUAS)**.** *The origin is semiglobally practically uniformly asymptotically stable (sGPUAS) of (3.3) if it is practically stable, practically bounded, and semiglobally practically uniformly attractive of (3.3).*

If the system of (3.3) is independent of the small parameters as in (3.2), then the notion of "practically" can be dropped and the stability definitions follow those presented in [20, Ch 4]. For proving the stability of these differential equations, we often use the following theorems.

Theorem 3.1 ([20, Th 4.8] and [20, Th 4.9])**.** *Let $x = 0$ be an equilibrium point for (3.2) and $\mathscr{D} \subset \mathbb{R}^n$ be a domain containing $x = 0$. Let $V : [0,\infty) \times \mathscr{D} \to \mathbb{R}$ be a continuous differentiable function ($V \in \mathscr{C}^1$) such that*

$$W_1(x) \leq V(t,x) \leq W_2(x) \tag{3.8}$$

$$\frac{\partial}{\partial t}V + \nabla V(t,x) \cdot f(t,x) \leq 0 \tag{3.9}$$

$\forall\, t \geq 0$ and $\forall\, x \in \mathscr{D}$, where W_1 and W_2 are continuous positive definite functions on $\mathscr{D}$. Then, $x = 0$ is uniformly stable. Furthermore, if (3.9) is strengthened to

$$\frac{\partial}{\partial t}V(t,x) + \nabla V(t,x) \cdot f(t,x) \leq -W_3(x) \tag{3.10}$$

where W_3 is a continuous positive definite function on $\mathscr{D}$. Then, $x = 0$ is uniformly asymptotically stable. Finally, if $\mathscr{D} = \mathbb{R}^n$ and W_1 is radially unbounded, then $x = 0$ is a globally uniformly asymptotically stable.

Theorem 3.2 (GUAS for a autonomous average system implies sGPUAS for the original system [21, Th 2])**.** *Consider a singular perturbed system*

$$\frac{d}{dt}x = f\left(\frac{t}{\varepsilon},x\right), \quad \varepsilon > 0 \tag{3.11}$$

where f is locally Lipschitz in x uniformly in t and $f(t,0)$ is uniformly bounded. Suppose $f(t,x)$ has an average $\overline{f}(x)$ and that the origin is of $\frac{d}{dt}x = \overline{f}(x)$ is semiglobally uniformly asymptotically stable. Then, the origin of (3.11) is sGPUAS.

Corollary 3.1. *Consider the perturbed system of*

$$\frac{d}{dt}x = f_{\varepsilon_{m-}}\left(\frac{t}{\varepsilon_m},x\right), \quad \varepsilon_m > 0 \tag{3.12}$$

where $f_{\varepsilon_{m-}}$ is a function depending on the small parameters $\varepsilon_1,\ldots,\varepsilon_{m-1}$ and $f_{\varepsilon_{m-}}$ is locally Lipschitz in x uniformly in t and $f_{\varepsilon_{m-}}(t,0)$ is uniformly bounded. Suppose $f_{\varepsilon_{m-}}(t,x)$ has an average $\overline{f}_{\varepsilon_{m-}}(x)$ and that the origin is semiglobally practically uniformly asymptotically stable for $\frac{d}{dt}x = \overline{f}_{\varepsilon_{m-}}(x)$. Then, the origin of (3.11) is sGPUAS.

Proof. Select a $c_1, c_2 > 0$ for the practical stability Definitions 3.5-3.7. Find ε_{m-1}^* such that the origin for $\overline{f}_{\varepsilon_{m-}}$ is uniformly attracted to a ball $\mathscr{B}_{c_2'} \subset \mathscr{B}_{c_2}$ for all trajectories starting in the ball $\mathscr{B}_{c_1}$. Then there is a ε_m^* such that for all $\varepsilon_m \in (0, \varepsilon_m^*)$, the trajectories of the original system will stay sufficiently close to the trajectories of the average system as to not leaving a ball $\mathscr{B}_{c_3}$ where $c_3 > c_2$ and be in $\mathscr{B}_{c_2}$ for all $t \in [t_0 + T(c_1, c_2'), \infty)$ by the usual arguments presented in [21, 22]. □

4. Theoretical Framework for ESC

The general problem statement of extremum seeking control acting directly on maps is to practically minimize a sensor output y using model-free methods. The sensor output is generally considered as $y = J(\theta)$ where the cost function $J : \mathbb{R}^n \to \mathbb{R}$ depends on the parameter vector $\theta \in \mathbb{R}^n$ and is minimized by some optimal parameter θ_*. This work will use the standard additive dither/perturbation signals, such as those in [6], to decompose the parameter into $\theta = \hat{\theta} + ap(\omega t)$ where $\hat{\theta}$ is the estimate of θ_* and p is the baseline periodic (or almost-periodic) signal. We use the small parameter $a \ll 1$ to control the size of the additive dither signal and the large parameter $\omega \gg 1$ to control the relative speed of the dither signal. For regularization, we will assume that $\sup_{t\in\mathbb{R}} \|p(t)\| = 1$. The sinusoidal dither signal s is defined as a vector whose elements are $s_i(t) = r_i \sin(\omega_i' t)$ where $r_i \neq 0$ for any element. If it can be written that $\omega_i' \in \mathbb{Z}_{\neq 0}$ for all i, then s is a periodic signal. Otherwise the signal is almost-periodic since there does not exist a common period share between all the signal elements. See Reference [23] for a more formal and thorough treatement of almost-periodic functions.

The following assumptions are made about the cost function for this work:

Assumption 4.1. *The cost function $J : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable ($J \in \mathscr{C}^1$) on the domain $\theta \in \mathbb{R}^n$.*

Assumption 4.2. *The cost function J has a unique minimum θ_* such that $\forall\ \theta \in \mathbb{R}^n$, $J(\theta_*) < J(\theta)$ if $\theta \neq \theta_*$.*

Assumption 4.3. *The gradient vector $\nabla J(\theta) = 0$ if and only if $\theta = \theta_*$.*

Assumption 4.4. *The cost function J is a radially unbounded function, i.e. $\lim_{|\theta|\to\infty} J(\theta) = \infty$.*

The following sections are arranged to cover perturbation-based derivative estimation, practical stability of the model-free system, and then some select ESC systems. Perturbation-based derivative estimates are used to estimate the derivatives which are required for the ESC systems. However, these derivative estimates are generally not correct at any give time but rather convergent in the limits of $\omega \to \infty$ and $a \to 0$. Thus, we need to show the conditions for when the model-free system of ESCs is practically stable with respect to the parameters (ω, a), namely that their model-based systems are classically stable. Afterwards, we examine the ESC systems to prove when they are stable.

4.1 Perturbation-Based Derivative Estimation

Derivatives are crucial for optimization but since ESCs are model-free, meaning they cannot directly obtain derivatives of the cost function, it is necessary to estimate them based on the local changes in the sensor value. The results for obtaining perturbation-based derivative estimates rely on a

Taylor series of the cost function, which requires multi-index notation for multivariable J. A multi-index α is an n-tuple of nonnegative integer entries ($\alpha \in \mathbb{Z}^n_{\geq 0}$). The sum of α is $|\alpha| = \sum_{i=1}^n \alpha_i$ and the factorial is defined similarly as $\alpha! = \prod_{i=1}^n \alpha_i!$. A vector $x \in \mathbb{R}^n$ can be given as a power $x^\alpha = \prod_{i=1}^n {x_i}^{\alpha_i}$. For Taylor series, multi-index notation can be used to compactly express the m-th order Taylor series expansion of a function $f : \mathbb{R}^n \to \mathbb{R}$ at the point $x_0 \in \mathbb{R}^n$ as

$$f(x) = \sum_{|\alpha|=0}^{m} \frac{1}{\alpha!} D^\alpha f(x_0)(x - x_0)^\alpha \tag{4.1}$$

where $D^\alpha = \left(\frac{\partial}{\partial x_1}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n}$. We assume that there is a set $\mathscr{A}$ that contains all α that we will consider for the ESCs. For brevity, we will assume that $\mathscr{A}$ has some implicit ordering so that, from now on, α_i is the i-th element of $\mathscr{A}$ rather than the i-th element of the n-tuple.

For the dynamics of the ESC, we let the estimate state vector be represented as $\hat{x} \in \mathbb{R}^{n+n'}$, which is an augmented state vector containing the n parameters and n' auxiliary states ($\hat{\phi} \in \mathbb{R}^{n'}$) which often estimates the derivative information. Typically, these systems are affine with respect to the derivative estimates, so we can use the affine dynamics

$$\frac{d}{dt}\hat{x} = f_0(\hat{x}) + f_1(\hat{x})\,\hat{\xi}\left(\omega t, \hat{\theta}, a\right) \tag{4.2}$$

where $\hat{\xi} \in \mathbb{R}^p$ is a vector of the estimates of the derivatives $\xi_i(\hat{\theta}) = D^{\alpha_i} J\left(\hat{\theta}\right)$. To better illustrate how often ESCs with affine dynamics appear in the literature, we give some examples of ESCs with differential equations in this form.

4.1.1 Examples of Affine Extremum Seekers

4.1.1.1 Gradient Descent

The Gradient-based Extremum Seeking Control [24] is the baseline ESC design. It relies on estimating the gradient of a scalar objective function with respect to tunable parameters through periodic perturbations and demodulation. The parameter estimate update dynamics are given by

$$\frac{d}{dt}\hat{\theta} = -k\hat{g}(\omega t, \hat{\theta}, a), \tag{4.3}$$

where $\hat{x} = \hat{\theta}$ as there are no auxiliary states and $\hat{g} = \hat{\xi}$ is the perturbation-based estimate of the gradient ∇J. This system can be expressed in the form of (4.2) with $f_0(\hat{\theta}) = 0$ and $f_1(\hat{\theta}) = -k$.

4.1.1.2 Heavy-ball

The Heavy-ball ESC [25] has the dynamics of

$$\frac{d}{dt}\hat{\theta} = \phi \tag{4.4}$$

$$\frac{d}{dt}\phi = -\beta\phi - k\hat{g}(\omega t, \hat{\theta}, a) \tag{4.5}$$

where the state vector $\hat{x} = [\theta^T, \phi^T]^T$ is augmented with the auxiliary state $\phi \in \mathbb{R}^n$ and $\beta \in \mathbb{R}$ is the dampening term. This system can be rewritten with affine dynamics using $f_0(\hat{x}) = [\phi^T,\ -\beta\phi^T]^T$ and $f_1(\hat{x}) = [0,\ -kI_{n\times n}]^T$ where $I_{n\times n}$ is the $n \times n$ identity matrix.

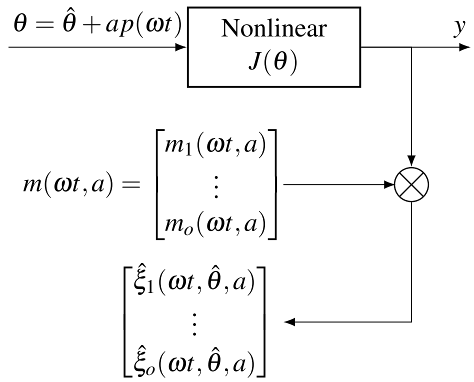


Figure 4: Block diagram of derivative estimator

4.1.1.3 Newton Method

The Newton-based ESC (NESC) [6, 13] corrects for second-order derivatives $\nabla^2 J$ by estimating the inverse of the Hessian. The dynamical system is given in [6] as

$$\frac{d}{dt}\hat{\theta} = -k\hat{\Gamma}\hat{g}\left(\omega t, \hat{\theta}, a\right) \tag{4.6}$$

$$\frac{d}{dt}\hat{\Gamma} = \omega_\ell\left(\hat{\Gamma} - \hat{\Gamma}\hat{H}(\omega t, \hat{\theta}, a)\hat{\Gamma}\right) \tag{4.7}$$

where $\hat{\Gamma}$ is the estimate of the inverse of the Hessian and $\hat{H}$ is the perturbation-based estimate of the Hessian. As it stands, the NESC system is a combination of a vector differential equation and a matrix differential equation, so its initially unclear how this can be expressed as an affine vector differential equation. However, the system can be expressed as

$$\frac{d}{dt}\begin{bmatrix} \hat{\theta} \\ \mathrm{vech}(\hat{\Gamma}) \end{bmatrix} = \begin{bmatrix} 0 \\ \omega_\ell \,\mathrm{vech}(\hat{\Gamma}) \end{bmatrix} + \begin{bmatrix} -k\hat{\Gamma} \\ \omega_\ell L_n(\hat{\Gamma}\otimes\hat{\Gamma})D_n \end{bmatrix}\begin{bmatrix} \hat{g}(\omega t, \hat{\theta}, a) \\ \mathrm{vech}(\hat{H}(\omega t, \hat{\theta}, a)) \end{bmatrix}, \tag{4.8}$$

where $\otimes$ is the Kronecker product, vech is the half-vectorization operator, and the matrices L_n and D_n are elimination and duplication matrices to account for the symmetry of the symmetric matrices. In this form, the state vector of the NESC exists in $\hat{x} \in \mathbb{R}^{n+n(n+1)/2}$.

4.1.2 Perturbation-based Derivative Estimates

While affine systems are prevalent for ESCs, one still needs to average the dynamics to obtain the average system for stability analysis. The following assumptions are necessary for an m-th order ESC.

Assumption 4.5. *The cost function $J : \mathbb{R}^n \to \mathbb{R}$ is m-times differentiable ($J \in \mathscr{C}^m$) for the m-th order algorithm.*

Assumption 4.6. *The dither signal $p : \mathbb{R} \to \mathbb{R}^n$ is a continuous function, periodic or almost periodic, and*

$$\sup_{t\in\mathbb{R}} \|p(t)\|_2 = 1 \tag{4.9}$$

Under these assumptions, one can formulate the derivative estimates $\hat{\xi}$. Although $\hat{\xi}$ may result from differential equations [26, Eq 14], the simplest formulation of $\hat{\xi}$ is element-wise by

$$\hat{\xi}_i\left(\omega t, \hat{\theta}, a\right) = h_i(\omega t, a) J\left(\hat{\theta} + a p(\omega t)\right) \tag{4.10}$$

where the signal h_i is used to "demodulate" the derivative information from the cost function output variation. As expected, h_i depends on the system designer's choice of p. In [6], the dither signal was a sinusoidal signal s. The sinusoidal dither signal has corresponding h_i for each α_i as: $h_i(t,a) = \frac{2}{ar_k}\sin(\omega_k' t)$ if α_i has k-th element 1 and the rest zero; $h_i(t,a) = \frac{16}{a^2 r_k^2}\left(\sin^2(\omega_k' t) - \frac{1}{2}\right)$ if α_i has k-th element 2 and the rest zero; and $h_i(t,a) = \frac{4}{a^2 r_j r_k}\sin(\omega_j' t)\sin(\omega_k' t)$ if α_i has j-th and k-th elements 1 and the rest zero. One can continue to find h_i for higher than second-order derivatives using [26] but these results are only for sinusoidal p and scalar J. Unlike [6], we also consider almost periodic p. Well-known texts on almost periodic functions [23, 27] should be consulted for a thorough treatment of the subject, but for illustrative purposes, an example of an almost periodic p is the sinusoidal dither signal with $\omega_1' = 1$ and $\omega_2' = \sqrt{2}$ since $\sin(t)$ and $\sin(\sqrt{2}t)$ do not share a common period between them.

The output dimension of $\hat{\xi}$ depends on the necessary m-th order derivatives required for the method in question. Methods using just the gradient ($m = 1$) will have $\hat{\xi}(\omega t, \hat{\theta}, a) \in \mathbb{R}^n$ whereas Newton-based methods ($m = 2$) need both the gradient and the Hessian of J and thus $\hat{\xi}(\omega t, \hat{\theta}, a) \in \mathbb{R}^{\frac{1}{2}n^2 + \frac{3}{2}n}$. To generalize, we define $c(k)$ as

$$c(k) = \left|\{\alpha \mid |\alpha| = k\}\right| = \binom{n+k-1}{k} \tag{4.11}$$

to keep count of the total number of unique k-th order partial derivatives. Now we can state that $\hat{\xi}(\omega t, \hat{\theta}, a) \in \mathbb{R}^o$ where $o = \sum_{i=0 \text{ or } 1}^{m} c(i)$ with the initial i depending on if the algorithm requires $D^0 J = J$.

The analysis of ESCs as mentioned previously, does not happen in the original system, but rather in some nominal system, which is the average system for this work. The average system is defined as

$$\frac{d}{dt}\hat{\bar{x}} = f_0\left(\hat{\bar{x}}\right) + f_1\left(\hat{\bar{x}}\right)\hat{\bar{\xi}}\left(\hat{\bar{\theta}}, a\right) \tag{4.12}$$

where $\hat{\bar{x}}$ and $\hat{\bar{\theta}}$ are the average state and parameter estimates with the average derivative estimate $\hat{\bar{\xi}}$ defined as

$$\hat{\bar{\xi}}_i\left(\hat{\bar{\theta}}, a\right) = \lim_{T\to\infty}\frac{1}{T}\int_0^T h_i(\tau, a) J\left(\hat{\bar{\theta}} + a p(\tau)\right) d\tau \tag{4.13}$$

One would hope that $\lim_{a\to 0}\hat{\bar{\xi}}_i(\hat{\bar{\theta}}, a) = D^{\alpha_i} J(\hat{\bar{\theta}})$ for all i so that the average system resembles some model-based system

$$\frac{d}{dt}z = f_0(z) + f_1(z)\,\xi(\vartheta) \tag{4.14}$$

for sufficiently small a. Here, the adjective "model-based" indicates that the system has direct access to the derivative information of J and we use z and ϑ to represent the state vector and parameter estimates in this model-based system. One would like to study the model-based system to glean stability insights into the original model-free system. However, we first need to find an h if even one exists so that one can relate the model-free systems with their respective model-based systems.

In order to show the existence of an m, we first consider the problem of finding various derivatives ξ for a perturbed polynomial $P_m\left(ap(t),\hat{\hat{\theta}}\right)$ of order m which is centered at $\hat{\hat{\theta}}$. This polynomial can be expressed in multi-index notation as

$$P_m(ap(t),\hat{\hat{\theta}}) = \sum_{|\alpha|\leq m} \frac{1}{\alpha!} a^{|\alpha|} p(t)^{\alpha} \xi_i\left(\hat{\hat{\theta}}\right) \tag{4.15}$$

$$= \rho(t)^T A \xi\left(\hat{\hat{\theta}}\right) \tag{4.16}$$

where $\xi_i\left(\hat{\hat{\theta}}\right) = D^{\alpha_i} J(\hat{\hat{\theta}})$ are the derivative values of interest, the extended dither signal vector includes the signals $\rho_i(t) = p(t)^{\alpha_i}$ associated with the ξ_i, and A is a diagonal matrix with $a^{|\alpha_i|}/\alpha_i!$ as the i-th element on the main diagonal and zero everywhere else. Note that ρ is a continuous (almost) periodic function as a result of p being a continuous (almost) periodic function in Assumption 4.6. Also, since $\sup_{t\in\mathbb{R}} \|p(t)\|_2 = 1$ by Assumption 4.6, the dither signal components being upper-bounded by $|p(t)| \leq 1$ implies that $|\rho_i(t)| \leq 1$ and $\|\rho(t)\|_2 \leq \sqrt{o}$ for all $t \in \mathbb{R}$.

The average derivative estimate for this polynomial is

$$\hat{\hat{\xi}}\left(\hat{\hat{\theta}},a\right) = \lim_{T\to\infty} \frac{1}{T} \int_0^T m(\tau,a) P_m(ap(\tau),\hat{\hat{\theta}}) d\tau \tag{4.17}$$

$$= \left(\lim_{T\to\infty} \frac{1}{T} \int_0^T m(\tau,a)\rho(\tau)^T d\tau\right) A\xi\left(\hat{\hat{\theta}}\right) \tag{4.18}$$

and the desired outcome is that m results in $\hat{\hat{\xi}}\left(\hat{\hat{\theta}},a\right) = \xi\left(\hat{\hat{\theta}}\right)$ for any nonzero a. Thus, m must satisfy the condition

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T m(\tau,a)\rho(\tau)^T d\tau = A^{-1} \tag{4.19}$$

so that m achieves the desired result $\hat{\hat{\xi}}\left(\hat{\hat{\theta}},a\right) = \xi\left(\hat{\hat{\theta}}\right)$. Here we can tell that a necessary and sufficient condition for the existence of m is that ρ is a vector of linearly independent signals, which is to say that

$$\nexists \beta \neq 0 \text{ s.t. } \rho(t)^T\beta = 0 \; \forall t \in \mathbb{R} \tag{4.20}$$

The condition is necessary since, if there existed a $\beta \neq 0$, then right multiplying both sides of (4.19) by β would result in the contradiction of the left side being a zero vector but the right side a non-zero vector. The condition is also sufficient since

$$m(t,a) = A^{-1}Q^{-1}\rho(t) \tag{4.21}$$

with Q being the covariance matrix of ρ as defined by

$$Q = \lim_{T\to\infty} \frac{1}{T} \int_0^T \rho(\tau)\rho(\tau)^T d\tau \tag{4.22}$$

would satisfy (4.19). The demodulation signal m is well defined since Q is a positive definite matrix, and therefore invertible, since

$$\beta^T Q\beta = \lim_{T\to\infty} \frac{1}{T} \int_0^T \left(\beta^T\rho(\tau)\right)^2 d\tau \tag{4.23}$$

$$\leq \lambda_{\max}(Q)\,\|\beta\|_2^2 \leq \operatorname{tr}(Q)\,\|\beta\|_2^2 \leq o\,\|\beta\|_2^2 \tag{4.24}$$

is finite and strictly positive [23, Th 3.8] for any nonzero β. Note that to meet the condition in (4.20), ρ only needs to be comprised of linearly independent signals, not necessarily orthogonal signals as stated in [26]. This implies, particularly for the $m = 1$ case, that it is not necessary for p to be comprised of orthogonal signals when estimating the derivative as required by [28].

The extended dither function is linearly independent if and only if Q is invertible. In practice, it is easier to calculate Q and check its determinate to determine if Q is invertible than to check the linear independence of ρ by calculating an orthogonal basis using the Gram-Schmidt reduction process. This is because: 1) we already need Q^{-1} for (4.21) and 2) the computational overhead to determine and simplify the signals in the orthogonal basis can be quite complex.

Additionally, we do not claim that (4.21) results in the only possible solution to (4.19) since it may be possible to find another m' with

$$m'(t,a) = A^{-1}R^{-1}r(t) \tag{4.25}$$

where $r(t) \in \mathbb{R}^o$ is vector of continuous, bounded, linear independent, (almost) periodic signals and R is the cross-variance between the signals

$$R = \lim_{T\to\infty} \frac{1}{T}\int_0^T r(\tau)\rho(\tau)^T d\tau \tag{4.26}$$

This requires finding r and ensuring that the associated R is invertible, a task which may not be trivial. Hence, determining m is easier with the covariance version (4.21) rather than the cross-variance version (4.25).

Having determined the necessary and sufficient conditions for the existence of m, we present the main theorem of this work which states that, for a non-polynomial J, the limit of $\hat{\hat{\xi}}(\vartheta,a) \to \xi((\vartheta)$ is valid as a approaches zero.

Theorem 4.1. *Let $J : \mathbb{R}^n \to \mathbb{R}$ satisfies Assumption 4.5 for an m-th order ESC and p satisfies Assumption 4.6. Possible demodulation signal vectors m exist if and only if the associated ρ is a vector of linearly independent signals. The use of h in (4.10) will result in the limit $\hat{\hat{\xi}}(\hat{\hat{\theta}},a) \to \xi(\hat{\hat{\theta}})$ as $a \to 0$ for all $\hat{\hat{\theta}} \in \mathbb{R}^n$.*

The condition for the existence of m has already been shown and so we prove the convergence portion of Theorem 4.1.

Proof. In order to prove convergence of $\hat{\hat{\xi}}$, we need to use an $\varepsilon - \delta$ definition of convergence

$$\forall \hat{\hat{\theta}} \in \mathbb{R}^n, \forall \varepsilon > 0,\ \exists \delta = \delta\left(\hat{\hat{\theta}},\varepsilon\right) > 0 \text{ s.t. } \forall a\ \in (0,\delta),\ \left\|\hat{\hat{\xi}}(\hat{\hat{\theta}},a) - \xi(\hat{\hat{\theta}})\right\|_2 < \varepsilon \tag{4.27}$$

where the estimate error $\left\|\hat{\hat{\xi}}(\hat{\hat{\theta}},a) - \xi(\hat{\hat{\theta}})\right\|_2$ is the Euclidean norm

$$\left\|\hat{\hat{\xi}}(\hat{\hat{\theta}},a) - \xi(\hat{\hat{\theta}})\right\|_2 = \left(\sum_{i=1}^{o}\left|\hat{\hat{\xi}}_i(\hat{\hat{\theta}},a) - \xi_i(\hat{\hat{\theta}})\right|^2\right)^{1/2} \tag{4.28}$$

Furthermore, we can focus on the individual average derivative estimates by finding every $\delta_i = \delta_i\left(\hat{\theta}, \varepsilon/\sqrt{o}\right) > 0$ such that

$$\left|\hat{\bar{\xi}}_i\left(\hat{\theta}, a\right) - \xi_i\left(\hat{\theta}\right)\right| < \frac{\varepsilon}{\sqrt{o}} \tag{4.29}$$

for all $a \in (0, \delta_i)$ as $\delta = \min_{i=1,\ldots,o} \delta_i$ would be sufficient for (4.27).

For the individual derivative estimates, we calculate the difference between the average estimate and the true derivative value with

$$\hat{\bar{\xi}}_i\left(\hat{\theta}, a\right) - \xi_i\left(\hat{\theta}\right) = \lim_{T\to\infty} \frac{1}{T}\int_0^T m_i(\tau, a) \cdot \left[J\left(\hat{\theta} + ap(\tau)\right) - P_m\left(ap(\tau), \hat{\theta}\right)\right] d\tau \tag{4.30}$$

by utilizing the fact that $\hat{\bar{\xi}}_i\left(\hat{\theta}, a\right) = \xi_i\left(\hat{\theta}\right)$ for the polynomial P_m by our construction of m. Utilizing Taylor's Theorem [29, Th 1], we can further simplify (4.30) to

$$\begin{aligned}\hat{\bar{\xi}}_i\left(\hat{\theta}, a\right) - \xi_i\left(\hat{\theta}\right) = \lim_{TT\to\infty} \frac{1}{T}\int_0^T & m_i(\tau, a) \\ & \cdot \left[\sum_{|\alpha_j|=m} \frac{a^m}{\alpha_j!}\left(D^{\alpha_j} J\left(\hat{\theta} + \eta_j(\tau)\right) - D^{\alpha_j} J\left(\hat{\theta}\right)\right)\rho_j(\tau)\right] d\tau\end{aligned} \tag{4.31}$$

where $\eta_j(\tau) \in \mathscr{B}_a$ for all $\tau \in (0, T)$. We now bound the magnitude of the error by

$$\begin{aligned}\left|\hat{\bar{\xi}}_i\left(\hat{\theta}, a\right) - \xi_i\left(\hat{\theta}\right)\right| \le \sum_{|\alpha_j|=m} & \sup_{\tau\in\mathbb{R}} |m_i(\tau, a)| \\ & \cdot \left[\frac{a^m}{\alpha_j!}\left|D^{\alpha_j} J\left(\hat{\theta} + \eta_j(\tau)\right) - D^{\alpha_j} J\left(\hat{\theta}\right)\right| \left|\rho_j(\tau)\right|\right]\end{aligned} \tag{4.32}$$

The demodulation signal component, when formulated with (4.25), can be bounded by

$$\begin{aligned}|m_i(\tau)|^2 \le \|m(\tau)\|_2^2 &= r(\tau)^T \left(R^{-1}\right)^T A^{-2} R^{-1} r(\tau) && (4.33)\\ &\le \sigma_o(A)^{-2} r(\tau)^T \left(R^{-1}\right)^T R^{-1} r(\tau) && (4.34)\\ &\le \sigma_o(A)^{-2} \sigma_o(R)^{-2} \|r(\tau)\|_2^2 && (4.35)\end{aligned}$$

where $\sigma_o(\cdot)$ gives the o-th, i.e., the smallest, singular value of a matrix. For A, the smallest singular value is known to be $a^m/m!$ when $a < 1$ while $\sigma_o(R)$ is some constant dependent on the selection of r. Since r is a boundable (almost) periodic function and $\exists\ r_{\max} > 0$ such that $\sup_{t\in\mathbb{R}} \|r(t)\| \le r_{\max} < \infty$, then (4.32) has the upper-bound

$$\left|\hat{\bar{\xi}}_i\left(\hat{\theta}, a\right) - \xi_i\left(\hat{\theta}\right)\right| \le \frac{m! r_{\max}}{\sigma_o(R)} \cdot \left(\sum_{|\alpha_j|=m} \sup_{\tau\in\mathbb{R}} \left|D^{\alpha_j} J\left(\hat{\theta} + \eta_j(\tau)\right) - D^{\alpha_j} J\left(\hat{\theta}\right)\right|\right) \tag{4.36}$$

This gives bound with h formulated by (4.25). If m had been formulated by (4.21) instead, then one should replace $r_{\max}$ and R with $\sqrt{o}$ and Q, respectively.

Since $J \in \mathscr{C}^m$ by Assumption 4.5, all derivatives $D^{\alpha_j} J$ are continuous functions. Therefore we can make the components of the summation in the right hand side of (4.36) to be as small as possible. Let δ_j be the δ function in the $\varepsilon - \delta$ continuity definition for the j-th derivative $D^{\alpha_j}J$ where $|\alpha_j| = m$. Then

$$\delta_i\left(\hat{\hat{\theta}}, \frac{\varepsilon}{\sqrt{o}}\right) = \min_j \delta_j\left(\hat{\hat{\theta}}, \frac{\varepsilon\sigma_o(R)}{c(m)\cdot r_{\max}\cdot m!\cdot\sqrt{o}}\right) \tag{4.37}$$

ensures that the right hand side of (4.36) meets the condition in (4.29). Thus the δ function for the definition in (4.27) is

$$\delta\left(\hat{\hat{\theta}}, \varepsilon\right) = \min_{j=1,\dots,c(m)} \delta_j\left(\hat{\hat{\theta}}, \frac{\varepsilon\sigma_o(R)}{c(m)\cdot r_{\max}\cdot m!\cdot\sqrt{o}}\right) \tag{4.38}$$

which completes the proof. □

Remark 4.1. *It is important to show the convergence in the derivative estimates because there are cases where the residual terms cause instabilities in the average system as shown in [11].*

Remark 4.2. *While the convergence proof of Theorem 4.1 seems more complicated than necessary with the use of the $\varepsilon - \delta$ proof, this is a necessary level of complexity given Assumption 4.5. Let $\kappa \in \mathscr{K}$ be the slowest local convergence rate for the derivatives $D^{\alpha_i}J$ so that*

$$\left|D^{\alpha_i}J(\theta) - D^{\alpha_i}J\left(\hat{\hat{\theta}}\right)\right| \in \mathscr{O}\left(\kappa\left(\left\|\theta - \hat{\hat{\theta}}\right\|\right)\right) \tag{4.39}$$

for all $|\alpha_i| = m$ and θ in a neighborhood of $\hat{\hat{\theta}}$. By an examination of (4.36), it is clear that the derivative estimates will share the same local convergence rate of the worst m-th order derivative local convergence rate. If $J \in \mathscr{C}^{m+1}$, like in [30] and [28], then the local convergence rate is linear with $\kappa(r) = r$ but the local convergence is not necessarily linear. Consider the scalar cost function

$$J(\theta) = \frac{4}{15}|\theta|^{5/2} \tag{4.40}$$

whose second-order derivative (i.e. the Hessian) is $\frac{d^2J}{d\theta^2} = \sqrt{|\theta|}$. The second-order derivative satisfies the constraint

$$\left|\frac{d^2J}{d\theta^2}(\theta) - \frac{d^2J}{d\theta^2}(0)\right| \leq \sqrt{|\theta|} \tag{4.41}$$

around $\theta = 0$ which means that the second-order derivative estimate should have a convergence rate of $\kappa(r) = \sqrt{r}$, which is slower than linear. We show in Fig. 5 the convergence of the average Hessian estimate [6]

$$\hat{\hat{H}}(0,a) = \frac{1}{2\pi}\int_0^{2\pi} \frac{-8}{a^2}\cos(2\tau)J(a\sin(\tau))d\tau \tag{4.42}$$

as a function of a in a log scale along with the functions a and $\sqrt{a}$. The convergence rate can be gleaned by the slope in the log scale and $\hat{\hat{H}}(0,a)$ matches the shallower slope of $\sqrt{a}$ rather than the steeper a. Thus, $\hat{\hat{H}}$ locally converges with respect to a at a rate slower *than linear.*

While Theorem 4.1 gives the conditions for the existence of m whose signal components may be non-zero mean, it is possible to estimate the other derivatives without necessarily estimating $J(\hat{\hat{\theta}})$.

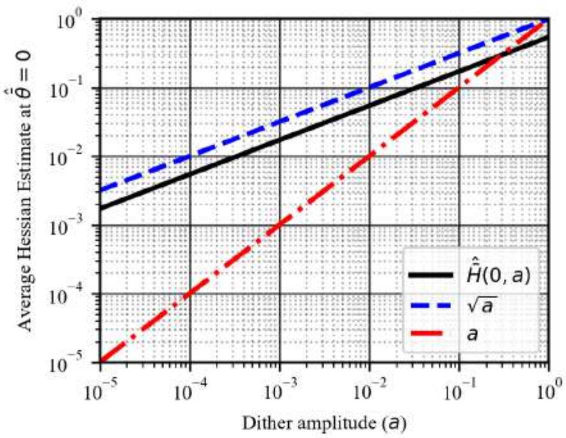


Figure 5: 1D example of the convergence explained by the proof in more detail.

Proposition 4.1. *Let $J : \mathbb{R}^n \to \mathbb{R}$ satisfies Assumption 4.5 for an m-th order ESC and p satisfies Assumption 4.6. If one wants to use strictly zero mean signals to estimate derivatives of order* 1 *through m without estimating $J(\hat{\theta})$ ($m = 0$), then possible demodulation signal vectors m exist if and only if the associated $\widetilde{\rho}$ is a vector of linearly independent signals where $\widetilde{\rho}(t) = \rho(t) - \overline{\rho}$ with $\overline{\rho} = \lim_{T\to\infty} \frac{1}{T}\int_0^T \rho(\tau)d\tau$ is linearly independent. The use of h in* (4.10) *will result in the limit $\hat{\hat{\xi}}(\hat{\hat{\theta}}, a) \to \xi(\hat{\hat{\theta}})$ as $a \to 0$ for all $\hat{\hat{\theta}} \in \mathbb{R}^n$.*

Proof. Consider again the average derivative estimate

$$\hat{\hat{\xi}}\left(\hat{\hat{\theta}}, a\right) = \lim_{T\to\infty} \frac{1}{T}\int_0^T m(\tau, a) P_m(ap(\tau), \hat{\hat{\theta}}) d\tau \tag{4.43}$$

$$= \left(\lim_{T\to\infty} \frac{1}{T}\int_0^T m(\tau, a) J\left(\hat{\hat{\theta}}\right) d\tau\right) + \left(\lim_{T\to\infty} \frac{1}{T}\int_0^T m(\tau, a)\rho(\tau)^T d\tau\right) A\xi\left(\hat{\hat{\theta}}\right) \tag{4.44}$$

where $\hat{\hat{\xi}}$ are estimates of the derivatives ξ which now range from order 1 to m. By m being a zero mean signal, this average estimate is simplified to

$$\hat{\hat{\xi}}\left(\hat{\hat{\theta}}, a\right) = \left(\lim_{T\to\infty} \frac{1}{T}\int_0^T m(\tau, a)\widetilde{\rho}(\tau)^T d\tau\right) A\xi\left(\hat{\hat{\theta}}\right) \tag{4.45}$$

since $\rho(\tau) = \widetilde{\rho}(\tau) - \overline{\rho}$. Since (4.45) is now the same as (4.18) with $\widetilde{\rho}$ instead of ρ, it follows that the necessary and sufficient condition for the existence of m is the linear independence of $\widetilde{\rho}$. The proof of the limit $\hat{\hat{\xi}}(\hat{\hat{\theta}}, a) \to \xi(\hat{\hat{\theta}})$ as $a \to 0$ is then identical to that in Theorem 4.1. □

4.1.3 Multi-Sensor Derivative Estimation

Based on the theory presented in 4.1.2, it was demonstrated through the averaging technique that derivative estimates $\hat{\xi}$ of the nonlinear field J can be obtained from a single rotating sensor. Expanding on this idea, it is possible to use additional sensors to provide more than a single source of information about J. This enables a new approach that integrates the information from multiple sensors to generate a single fitted gradient estimate $\hat{g}$ of J. This theory will serve as the foundation for implementing rotating sensor extremum seeking in three dimensions on the BlueROV2 platform, discussed in Section 5.7.

Assuming that the nonlinear field J is polynomial and m-times differentiable, it becomes possible—using the Taylor series expansion and polynomial regression techniques—to estimate the gradient from a determined or overdetermined set of derivative information sources. Moreover, this approach allows for the estimation of higher-order derivatives beyond those directly provided by the sensors.

The multi-derivative estimation method was developed starting from the multi-indexed Taylor series expansion from [29] which can be presented in the following form

$$J(\theta+d) = \sum_{|\alpha|\geq m} \frac{1}{\alpha!} d^{\alpha} D^{\alpha} J(\theta) \tag{4.46}$$

where α is a multi-index that defines the order of partial derivatives in each estimation. We assume that there is a set $\mathscr{A}$ that contains all α that we will consider. For brevity, we will assume that $\mathscr{A}$ has some implicit ordering so that, from now on, α_ℓ is the ℓ-th element of $\mathscr{A}$ rather than the ℓ-th element of the n-tuple.

Although the multi-index Taylor series expansion approximates the nonlinear field as a polynomial, it can also be differentiated to approximate the field's higher-order behavior. If sensors are capable of gathering higher-order information about J, that data can be incorporated into the approximation. To account for this, the sensor multi-index set μ_i, the index ℓ for elements of multi-index α, the sensor displacement vector d_i, and the derivative estimate term $\hat{\xi}_\ell(\theta) = D^{\alpha_\ell} J(\theta)$ are added to (4.46).

$$D^{\mu_{i,j}} J(\theta+d_i) = \sum_{|\alpha|\geq m} \frac{1}{(\alpha_\ell - \mu_{i,j})!} d_i^{\alpha_\ell - \mu_{i,j}} \hat{\xi}_\ell(\theta) \tag{4.47}$$

The first term introduced to (4.47), μ_i, is a multi-index used to represent the order of derivative information gathered by the ith sensor s_i. For example, if two sensors are gathering derivative information at displacement d_i where s_1 can gather 0th and 1st-order information and s_2 can only gather 0th-order information, the resulting sensor mutli-index sets would be

$$\mathscr{A}_{s_1} = \{\mu_{1,1}, \mu_{1,2}, \mu_{1,3}\} = \{(0,0),(1,0),(0,1)\}, \qquad \mathscr{A}_{s_2} = \{\mu_{2,1}\} = \{(0,0)\} \tag{4.48}$$

Each sensor i has a maximum measurable derivative order which determines the elements of $\hat{\xi}$ that the sensor's measurement contributes to estimating. To illustrate, the sensor 1 in the example above contributes to all three elements of $\mathscr{A}$ (shown in (4.49)) while sensor 2 would only contribute to its first element (α_1).

$$\mathscr{A} = \{\alpha_1, \alpha_2, \alpha_3\} = \{(0,0),(1,0),(0,1)\} \tag{4.49}$$

The other addition to (4.47) is the sensor displacement vector d_i used to show the distance of each sensor from the estimation's origin θ.

In addition to the terms added in equation (4.47), a boundary condition on the summation, $\alpha : \alpha - \mu \geq 0$, is introduced. This condition prevents the inclusion of terms that would correspond to indefinite components and finalizes the equation to

$$D^{\mu_{i,j}}J(\theta+d_i)=\sum_{|\alpha|\geq m\,\&\,\alpha:\alpha-\mu\geq 0}\frac{1}{(\alpha_\ell-\mu_{i,j})!}d_i^{\alpha_\ell-\mu_{i,j}}\hat{\xi}_\ell(\theta) \tag{4.50}$$

As a result of these modifications, the multi-indexed Taylor series has been reformulated to accommodate higher orders. The next step is to apply the polynomial regression method from [31] fitted by least squares. This process combines each sensor's information to estimate the gradient. The matrix form of the polynomial regression is as follows

$$\bar{y}=X\hat{\beta}+\hat{\varepsilon} \tag{4.51}$$

The polynomial regression can then be solved using the least squares solution in (4.52) excluding random errors ε.

$$\hat{\hat{\beta}}=(X^TX)^{-1}X^T\bar{y} \tag{4.52}$$

Combining (4.50) with the polynomial regression equation from (4.51) results in

$$D^{\mu_{i,j}}J(\theta+d_i)=(X_{i,j})\,\hat{\xi}(\theta),\quad X_{i,j}\in\mathbb{R}^{1\text{x}K} \tag{4.53}$$

where K is the total number of distinct estimated derivatives and,

$$(X_{i,j})_{1,\ell}=\begin{cases}\frac{(d_i)^{\alpha_\ell-\mu_{i,j}}}{(\alpha_\ell-\mu_{i,j})!} & \text{if}\quad \alpha_\ell\geq\mu_{i,j}\\ 0 & \text{otherwise}\end{cases} \tag{4.54}$$

The expanded form of Eq.(4.53) is given by

$$\begin{bmatrix} D^{\mu_{1,1}}J(\theta+d_1)\\ D^{\mu_{1,2}}J(\theta+d_1)\\ \vdots\\ D^{\mu_{1,K_1}}J(\theta+d_1)\\ D^{\mu_{2,1}}J(\theta+d_2)\\ \vdots\\ D^{\mu_{2,K_2}}J(\theta+d_2)\\ \vdots\\ D^{\mu_{I,K_I}}J(\theta+d_I)\end{bmatrix}=\begin{bmatrix} \frac{(d_1)^{\alpha_1-\mu_{1,1}}}{(\alpha_1-\mu_{1,1})!} & \frac{(d_1)^{\alpha_2-\mu_{1,1}}}{(\alpha_2-\mu_{1,1})!} & \frac{(d_1)^{\alpha_3-\mu_{1,1}}}{(\alpha_3-\mu_{1,1})!} & \cdots & \frac{(d_1)^{\alpha_K-\mu_{1,1}}}{(\alpha_K-\mu_{1,1})!}\\ \frac{(d_1)^{\alpha_1-\mu_{1,2}}}{(\alpha_1-\mu_{1,2})!} & \cdots & \cdots & \cdots & \vdots\\ \vdots & & & & \\ \frac{(d_1)^{\alpha_1-\mu_{1,K_1}}}{(\alpha_1-\mu_{1,K_1})!} & \cdots & \cdots & \cdots & \vdots\\ \frac{(d_2)^{\alpha_1-\mu_{2,1}}}{(\alpha_1-\mu_{2,1})!} & \cdots & \cdots & \cdots & \vdots\\ \vdots & & & & \\ \frac{(d_2)^{\alpha_1-\mu_{2,K_2}}}{(\alpha_1-\mu_{2,K_2})!} & \cdots & \cdots & \cdots & \vdots\\ \vdots & & & & \\ \frac{(d_I)^{\alpha_1-\mu_{I,K_I}}}{(\alpha_1-\mu_{I,K_I})!} & \cdots & \cdots & \cdots & \frac{(d_I)^{\alpha_K-\mu_{I,K_I}}}{(\alpha_K-\mu_{I,K_I})!}\end{bmatrix}\begin{bmatrix}\hat{\xi}_1(\theta)\\ \hat{\xi}_2(\theta)\\ \hat{\xi}_3(\theta)\\ \vdots\\ \hat{\xi}_K(\theta)\end{bmatrix} \tag{4.55}$$

where I is the total number of sensors and K_i is the total number of distinct derivatives measured by sensor i.

To illustrate the method, consider a simple 1D example without rotations. Let the nonlinear field be the cost function $J=x^2$, with the center point of the sensors located at $\theta=1$. Two sensors are positioned symmetrically from the center point represented by the vector $d=[-0.3,0.3]$. An illustration of the problem can be seen below in Fig. 6.

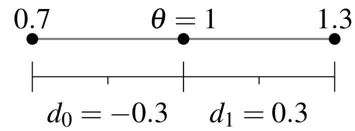


Figure 6: 1D multi-derivative estimation example

Each sensor provides only 0th-order information, resulting in sensor multi-index sets $\mathscr{A}_{s_1} = \{\mu_1\} = \{(0)\}$ and $\mathscr{A}_{s_2} = \{\mu_2\} = \{(0)\}$. However, since the estimation contains overdetermined information, it is possible to estimate the derivative values up to the maximum order of 1. The resulting multi-index for the prediction becomes $\mathscr{A} = \{\alpha_1, \alpha_2\} = \{(0), (0)\}$.

Substituting the problem's indexing into equation (4.53) provides

$$\begin{bmatrix} D^{\mu_{1,1}} J(\theta + d_1) \\ D^{\mu_{2,1}} J(\theta + d_2) \end{bmatrix} = \begin{bmatrix} \frac{(d_1)^{\alpha_1 - \mu_{1,1}}}{(\alpha_1 - \mu_{1,1})!} & \frac{(d_1)^{\alpha_2 - \mu_{1,1}}}{(\alpha_2 - \mu_{1,1})!} \\ \frac{(d_2)^{\alpha_1 - \mu_{2,1}}}{(\alpha_1 - \mu_{2,1})!} & \frac{(d_2)^{\alpha_2 - \mu_{2,1}}}{(\alpha_2 - \mu_{2,1})!} \end{bmatrix} \begin{bmatrix} \hat{\xi}_1(\theta) \\ \hat{\xi}_2(\theta) \end{bmatrix} \tag{4.56}$$

Then, inserting the problem's parameters gives

$$\begin{bmatrix} (1 - 0.3)^2 \\ (1 + 0.3)^2 \end{bmatrix} = \begin{bmatrix} \frac{(-0.3)^{0-0}}{(0-0)!} & \frac{(-0.3)^{1-0}}{(1-0)!} \\ \frac{(0.3)^{0-0}}{(0-0)!} & \frac{(0.3)^{1-0}}{(1-0)!} \end{bmatrix} \begin{bmatrix} \hat{\xi}_1(\theta) \\ \hat{\xi}_2(\theta) \end{bmatrix} \tag{4.57}$$

Which simplifies to

$$\begin{bmatrix} 0.49 \\ 1.64 \end{bmatrix} = \begin{bmatrix} 1 & -0.3 \\ 1 & 0.3 \end{bmatrix} \begin{bmatrix} \hat{\xi}_1(\theta) \\ \hat{\xi}_2(\theta) \end{bmatrix} \tag{4.58}$$

Finally, applying the least squares solution from (4.53) to solve for the vector of estimated derivatives results in the estimated gradient $\hat{g}$.

$$\hat{g} = \left(\begin{bmatrix} 1 & 1 \\ -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 1 & -0.3 \\ 1 & 0.3 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 \\ -0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.49 \\ 1.64 \end{bmatrix} \tag{4.59}$$

$$\tag{4.60}$$

$$\hat{g} = \begin{bmatrix} 1.065 \\ 1.916 \end{bmatrix} \tag{4.61}$$

When compared to the true gradient g at the center of the estimate, $\theta = 1$ (as shown in (4.62)), the multi-sensor derivative estimation method accurately estimates the cost function's solution and derivatives.

$$g = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \tag{4.62}$$

In future work, this multi-derivative estimation method will be extended to incorporate rotating sensors, as described in Section 4.1.2. This enhancement will enable multi-dimensional ESC applications, including 3D ESC experiments on the BlueROV2 platform.

4.2 Stability of the Model-Based ESC Implies Semiglobal Practical Stability of the Model-Free ESC

This section covers the results presented in [16]. We now consider how we can prove practical stability of the model-free system. As we generally only consider seekers with two parameters, we write the model-free ESC as

$$\frac{d}{dt}\hat{x} = f(\omega t, \hat{x}, a) \tag{4.63}$$

where $\hat{x} \in \mathbb{R}^{n+n'}$ is the augmented space with both the parameter estimates $\hat{\theta}$ as well as any auxiliary states, e.g., $\hat{\gamma}$ in the NESC algorithm. The model-free ESC has an average system

$$\frac{d}{dt}\hat{\bar{x}} = \bar{f}(\hat{\bar{x}}, a) := \lim_{T\to\infty} \frac{1}{T}\int_0^T f(\tau, \hat{\bar{x}}, a)d\tau\,. \tag{4.64}$$

We define the model-based ESC as

$$\frac{d}{dt}z = h(z) := \lim_{a\to 0^+} \bar{f}(z, a) \tag{4.65}$$

because h evolves the augmented state vector z based on the true derivative values of J with respect to z, such as the gradient $\nabla J(z)$ or the Hessian $\nabla^2 J(z)$. We now give the main stability results, which are that for determining practical stability of the model-free ESC, we only need to prove the model-based ESC is stable.

Theorem 4.2. *Consider the model-free, perturbation-based ESC in (4.63) and its associated model-based, perturbation-free ESC in (4.65). If the model-based system exists and is GAS to the origin, then*

1. *the average system defined in (4.64) is semiglobally practically asymptotically stable (sGPAS) to the origin with respect to the small parameter a and*
2. *the model-free ESC system is sGPUAS to the origin with respect to the small parameter vector (a, ω^{-1}).*

The proof of Theorem 4.2 is broken into two parts, the proof that the average system is sGPAS with respect to the small parameter a and then the proof that the original model-free system is sGPUAS with respect to the small parameter vector (a, ω^{-1}).

4.2.1 The Average System is sGPAS

We will use a Lyapunov function in order to prove that the average system meets the definitions of stability, boundedness, and attractivity. Since the model-based system is GAS, then by a converse Lyapunov theorem [32, Th 23], there exists a continuously differentiable $V : \mathbb{R}^m \to \mathbb{R}$, two class $\mathcal{K}_\infty$ functions (α_1 and α_2) [20, Ch 4], and a positive definite function W which satisfies the conditions

$$\alpha_1(\|z\|) \le V(z) \le \alpha_2(\|z\|) \tag{4.66}$$

$$\frac{\partial V}{\partial z}(z) \cdot h(z) \le -W(\|z\|) \tag{4.67}$$

If we rewrite the vector field $\bar{f}$ as

$$\bar{f}(\hat{\tilde{x}},a) = h(\hat{\tilde{x}}) + R_f(\hat{\tilde{x}},a) \tag{4.68}$$

where R_f is defined as

$$R_f(\hat{\tilde{x}},a) = \bar{f}(\hat{\tilde{x}},a) - h(\hat{\tilde{x}}), \tag{4.69}$$

then we can bound the Lie derivative of V along the trajectories of the average system with

$$\frac{\partial V}{\partial \hat{\tilde{x}}}(\hat{\tilde{x}}) \cdot f(\hat{\tilde{x}},a) = \frac{\partial V}{\partial \hat{\tilde{x}}}(\hat{\tilde{x}}) \cdot h(\hat{\tilde{x}}) + \frac{\partial V}{\partial \hat{\tilde{x}}}(\hat{\tilde{x}}) \cdot R_f(\hat{\tilde{x}},a) \tag{4.70}$$

$$\leq -W(\|\hat{\tilde{x}}\|) + \|R_f(\hat{\tilde{x}},a)\| \left\| \frac{\partial V}{\partial \hat{\tilde{x}}}(\hat{\tilde{x}}) \right\| \tag{4.71}$$

Since $\bar{f}(\hat{\tilde{x}},a) \to h(\hat{\tilde{x}})$ as $a \to 0$, $\|R_f(\hat{\tilde{x}},a)\| \to 0$ as $a \to 0$ for all $\hat{\tilde{x}} \in \mathbb{R}^m$. Thus, for any ρ_1, ρ_2 where $0 < \rho_1 < \rho_2 < \infty$, we can always find $a^* = a^*(\rho_1, \rho_2) > 0$ such that the Lie derivative bound shown in (4.71) is strictly negative when $\|\hat{\tilde{x}}\| \in [\rho_1, \rho_2]$ for all $a \in (0, a^*)$. We use V and selection of ρ_1, ρ_2 to prove that the average system meets the three required definitions of sGPAS; that the average system is practically stable, practically bounded, and semiglobally practically attractive.

4.2.1.1 The Average System is Practically Stable

Given some final radius $c_2 > 0$, we select c_1, ρ_1, and ρ_2 such that $\alpha_2(c_1) < \alpha_1(c_2)$, $0 < \rho_1 < c_1$, and $c_2 < \rho_2$. The boundary of the level set $V_{\alpha_2(c_1)} = \{\hat{\tilde{x}} \mid V(\hat{\tilde{x}}) < \alpha_2(c_1)\}$ is within the difference of balls $\mathscr{B}_{\rho_2} \setminus \mathscr{B}_{\rho_1}$ and thus $\frac{dV}{dt} < 0$ everywhere on the boundary whenever $a \in (0, a^*(\rho_1, \rho_2))$. Therefore, this level set is forward invariant. From the above selection of the constants c_1, ρ_1, and ρ_2, we can see

$$\left\|\hat{\tilde{x}}(t_0)\right\| < c_1 \implies \hat{\tilde{x}}(t_0) \in V_{\alpha_2(c_1)} \implies \hat{\tilde{x}}(t) \in V_{\alpha_2(c_1)} \subseteq V_{\alpha_1(c_2)} \subseteq \mathscr{B}_{c_2} \; \forall t \in [t_0, \infty) \tag{4.72}$$

when $a \in (0, a^*(\rho_1, \rho_2))$ which is the definition of practical stability that we were trying to achieve. We conclude that the average system is practically stable with respect to the small parameter a. All sets and relations can be seen in Fig. 7.

4.2.1.2 The Average System is Practically Bounded

The proof of practical boundedness follows in a way similar to the proof of practical stability. Given some initial radius $c_1 > 0$, we select c_1, ρ_1, and ρ_2 to meet once again the requirements of $\alpha_2(c_1) < \alpha_1(c_2)$, $0 < \rho_1 < c_1$, and $c_2 < \rho_2$. With $a \in (0, a^*(\rho_1, \rho_2))$, the level set $V_{\alpha_2(c_1)}$ is again forward invariant and thus (4.72) is once again true. We therefore conclude that the average system is practically bounded with respect to the small parameter a.

4.2.1.3 The Average System is Semiglobally Practically Attractive

Given an initial and final radius of $c_1 > 0$ and $c_2 > 0$, we select ρ_1 so that $0 < \rho_1 < \alpha_2^{-1} \circ \alpha_1(\min(c_1, c_2))$ and ρ_2 so that $\alpha_1^{-1} \circ \alpha_2(\max(c_1, c_2)) < \rho_2$. Note that the previous proofs of practical stability and boundedness explicitly had the ordering $c_1 < c_2$ but the definition of practical attractiveness does not have this ordering. Hence, our selection of ρ_1 and ρ_2 must be based on the minimum

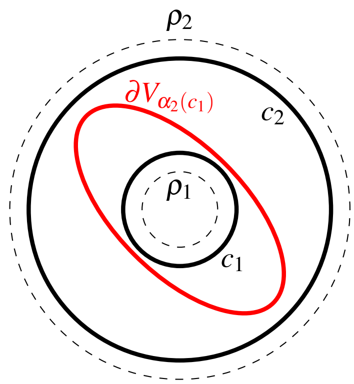


Figure 7: Graphical representation of the proof for the average system being practical stability and bounded. Balls with radii ρ_1 and ρ_2 are shown as dashed while balls with radii c_1 and c_2 are solid. $\partial V_{\alpha_2(c_1)}$ is the boundary of the forward invariant levels set.

and maximum c's, respectively, but our selection means that the level sets $V_{\alpha_1(c_2)} = \{\hat{\bar{x}} \mid V(\hat{\bar{x}}) < \alpha_1(c_2)\}$ and $V_{\alpha_2(c_1)} = \{\hat{\bar{x}} \mid V(\hat{\bar{x}}) < \alpha_2(c_1)\}$ are forward invariant sets when $a \in (0, a^*(\rho_1, \rho_2))$. Thus,

$$\hat{\bar{x}}(t_0) \in V_{\alpha_1(c_2)} \implies \|\hat{\bar{x}}(t)\| < c_2 \;\forall t \in [t_0, \infty) \tag{4.73}$$

If $\mathscr{B}_{c_1} \subseteq V_{\alpha_1(c_2)}$, then all the starting trajectories are within $V_{\alpha_1(c_2)}$ and our proof is done with $T(c_1, c_2) \equiv 0$. Otherwise, we must consider the trajectories when $\hat{\bar{x}}(t_0) \in \mathscr{B}_{c_1} \setminus V_{\alpha_1(c_2)}$. These trajectories are still in the forward invariant level set $V_{\alpha_2(c_1)}$ by choice of ρ_1 and ρ_2 so it is now a question of whether there exists a trajectory $\hat{\bar{x}}(t) \in V_{\alpha_2(c_1)} \setminus V_{\alpha_1(c_2)}$ for all $t \in [t_0, \infty)$? The existence of such a trajectory would serve as a counter-example to the average system being practically attractive.

Note that V would decrease along such a trajectory at least as fast as c_3 where

$$c_3 = \min_{\rho_1 \leq \|\hat{\bar{x}}\| \leq \rho_2} \inf_{b \in (0, a^*)} W(\hat{\bar{x}}) - \|h(\hat{\bar{x}}, b)\| \left\| \frac{\partial V}{\partial x}(\hat{\bar{x}}) \right\| \tag{4.74}$$

and $c_3 > 0$ by $a \in (0, a^*(\rho_1, \rho_2))$. After a time $T(c_1, c_2)$ where

$$T(c_1, c_2) = \max\left(\frac{\alpha_2(c_1) - \alpha_1(c_2)}{c_3}, 0\right) \tag{4.75}$$

has elapsed, we can form the upper bound of $V(\hat{\bar{x}}(t_0 + T))$ for trajectories with $\hat{\bar{x}}(t_0) \in V_{\alpha_2(c_1)} \setminus V_{\alpha_1(c_2)}$ by

$$V(\hat{\bar{x}}(t_0 + T)) = V(\hat{\bar{x}}(t_0)) + \int_{t_0}^{t_0+T} \underbrace{\frac{\partial V}{\partial \hat{\bar{x}}}(\hat{\bar{x}}(\tau))^T \bar{f}(\hat{\bar{x}}(\tau), a)}_{\leq -c_3 \;\forall \hat{\bar{x}}(t) \in V_{\alpha_2(c_1)} \setminus V_{\alpha_1(c_2)}} d\tau < \alpha_2(c_1) - c_3 T \leq \alpha_1(c_2) \tag{4.76}$$

This means that $\hat{\bar{x}}(t_0 + T)$ is inside the forward invariant set $V_{\alpha_1(c_2)}$ and thus

$$\|\hat{\bar{x}}(t_0)\| < c_1 \implies \hat{\bar{x}}(t) \in V_{\alpha_1(c_2)} \subseteq \mathscr{B}_{c_2} \;\forall t \in [t_0 + T, \infty) \tag{4.77}$$

Since c_1 can be arbitrarily large, then we conclude that we meet the definition of δ-practically attractive (δ-PA) with δ being able to be made arbitrarily large by choice of a.

4.2.2 The Model-Free ESC is sGPUAS

The analysis of the model-free ESC system will prove that the system is sGPUAS to the origin by proving the system meets the required definitions. This analysis is aided by the fact that the average system is sGPAS and that we can make trajectories of the model-free and average systems arbitrarily close from the same initial condition, for any given a, over a finite time interval by taking ω arbitrarily large [22, Th 4.3.6], [20, Th 10.5]. However, these are singular perturbation proofs so one must consider how values of a would effect choice of ω. As a likely influences the local Lipschitz constant of f with respect to $\hat{x}$ in (4.63), a impacts the growth of the bounds between trajectories of the two systems, as seen in the details of the proof of [22, Lem 4.2.7]. Thus we summarize the ability to make the trajectories arbitrarily close in the context of this work with the following lemma.

Lemma 4.1 (Adapted from [20, Th 10.5])**.** *Consider trajectories for systems* (4.63) *and* (4.64) *and compact sets* $\mathscr{D}_0 \subset \mathscr{D} \subset \mathbb{R}^m$ *such that for the initial conditions* $\hat{\bar{x}}(t_0) \in \mathscr{D}_0$*, the trajectories satisfy* $\hat{\bar{x}}(t) \in \mathscr{D}$ *for all* $t \in [t_0, t_0+T]$ *and some* $T \in (0,\infty)$*. Then for a given* $a > 0$ *and* $\Delta > 0$*,* $\exists \omega^* = \omega^*(T,\Delta,a) > 0$ *such that if* $\hat{x}(t_0) = \hat{\bar{x}}(t_0) \in \mathscr{D}_0$*, then* $\forall \omega \in (\omega^*,\infty)$*, the trajectories of the original and average system satisfy*

$$\left\|\hat{x}(t) - \hat{\bar{x}}(t)\right\| < \Delta, \ \forall t \in [t_0, t+T] \tag{4.78}$$

The remainder of this section merely verifies that the original model-free ESC system meets the required definitions of sGPUAS; that the model-free ESC system is practically stable, practically bounded, and semiglobally practically uniformly attractive. The proofs will follow similarly to the proof of [33, Th 1].

4.2.2.1 The Model-Free ESC is Practically Stable

Given a final radius of $c_2 > 0$, we select parameters c_1, d, ρ_1, and ρ_2 such that $\alpha_2(c_1) < \alpha_1(c_2)$, $\alpha_2(d) < \alpha_1(c_1)$, $0 < \rho_1 < \alpha_2^{-1}(d)$, and $\alpha_1(\rho_2) > \alpha_2(c_2)$. By the average system being sGPUAS, $\exists\, T(c_1,d) = T,\ a^* > 0$ such that $\forall a \in (0,a^*)$

$$\|\hat{\bar{x}}(t_0)\| < c_1 \implies \|\hat{\bar{x}}(t_0+T)\| < d \tag{4.79}$$

$$\|\hat{\bar{x}}(t_0)\| < c_1 \implies \|\hat{\bar{x}}(t)\| < c_2,\ \forall\, t \in [t_0, t_0+T] \tag{4.80}$$

where (4.80) is a consequence of the level set $V_{\alpha_2(c_1)} = \{\hat{\bar{x}} \mid V(\hat{\bar{x}}) < \alpha_2(c_1)\} \subseteq \mathscr{B}_{c_2}$ being forward invariant by choices of ρ_1 and ρ_2. If $\Delta = \min\{c_2 - \alpha_1^{-1} \circ \alpha_2(c_1), c_1 - d\}$, then $\omega^* = \omega^*(T,\Delta,a)$ and $\forall\ \omega \in (\omega^*,\infty)$ the statements

$$\|\hat{x}(t_0)\| < c_1 \implies \|\hat{x}(t_0+T)\| < c_1 \tag{4.81}$$

$$\|\hat{x}(t_0)\| < c_1 \implies \|\hat{x}(t)\| < c_2,\ \forall\, t \in [t_0, t_0+T] \tag{4.82}$$

hold. Since the average system is autonomous, we can use the conditions (4.81) and (4.82) in an inductive manner by considering the following time intervals $[t_n, t_{n+1}]$ where $t_n = t_0 + nT$, $\|\hat{x}(t_n)\| < c_1$, and $n \in \{1,2,\ldots\}$. In these time intervals, we have the statements

$$\|\hat{x}(t_n)\| < c_1 \implies \|\hat{x}(t_{n+1})\| < c_1 \tag{4.83}$$

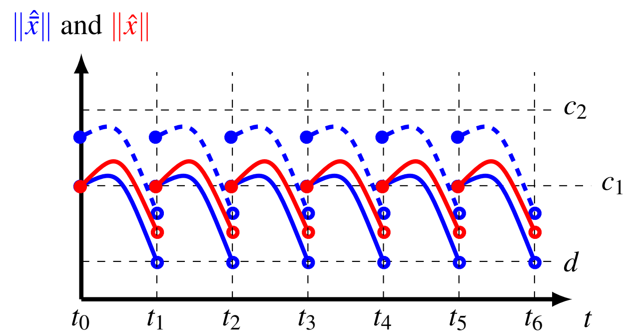


Figure 8: Graphical representation of the proof of the model-free ESC being practically stable and bounded. Bounds for $\|\hat{\hat{x}}\|$ are shown with a blue line —, bounds for $\|\hat{x}\|$ are shown with a red line —, and $\|\hat{\hat{x}}\| + \Delta$ is shown with a dashed blue line - - -. All curves are inclusive on the left endpoint and exclusive on the right endpoint. These discontinuities is a result of the use of induction in the proof. The dashed blue line forms an upper bound for $\|\hat{x}(t)\|$ to show that the model-free ESC system meets the definitions of practical stability and boundedness.

$$\|\hat{x}(t_n)\| < c_1 \implies \|\hat{x}(t)\| < c_2,\ \forall\, t \in [t_n, t_{n+1}] \tag{4.84}$$

due to the average system being autonomous. Thus, we can extend the upper bound of the trajectories over finite time intervals to

$$\|\hat{x}(t_0)\| < c_1 \implies \|\hat{x}(t)\| < c_2,\ \forall\, t \in [t_0, \infty) \tag{4.85}$$

and conclude that the model-free ESC is practically stable with respect to the small parameter vector (a, ω^{-1}). See Fig. 8 for a graphical representation of this proof.

4.2.2.2 The Model-Free ESC is Practically Bounded

Given some initial radius c_1, we choose the strictly positive constants c_2, d, ρ_1, and ρ_2 such that $c_2 > \alpha_1^{-1} \circ \alpha_2(c_1)$, $d < \alpha_2^{-1} \circ \alpha_1(c_1)$, $\rho_1 < \alpha_2^{-1}(d)$, and $\rho_2 > \alpha_1^{-1} \circ \alpha_2(c_2)$. Then the arguments are identical to the previous practical stability arguments. Thus, the model-free ESC system is practically bounded with respect to the small parameter vector (a, ω^{-1}).

4.2.2.3 The Model-Free ESC is Semiglobally Practically Uniformly Attractive

The proof of the model-free ESC being δ-PUA with δ being arbitrarily large is broken up into two parts. The initial part is to show that for the initial conditions $\|\hat{x}(t_0)\| < c_1$, there are critical parameter values a^* and $\omega^*(a)$ such that all those trajectories are within a ball of radius d_1 after time T has elapsed when $a \in (0, a^*)$ and $\omega \in (\omega^*(a), \infty)$. The inductive part is to show that since all trajectories are in $\mathscr{B}_{d_1}$ after T seconds elapsed, they will never leave $\mathscr{B}_{c_2}$.

Initial Part:

Given an initial and final radius of c_1 and c_2, we choose $d_1, d_2 > 0$ such that $d_1 < \alpha_2^{-1} \circ \alpha_1(\min(c_1, c_2))$ and $d_2 < \alpha_2^{-1} \circ \alpha_1(d_1)$. In addition, we must choose ρ_1 and ρ_2 such that $0 < \rho_1 < \alpha_2^{-1} \circ \alpha_1(d_2)$ and $\rho_2 > \alpha_1^{-1} \circ \alpha_2(\max(c_1, c_2))$. From the average system being sGPUAS,

$\exists\, T(c_1, d_2) = T, a^*(\rho_1, \rho_2) = a^*$ such that $\forall\, a \in (0, a^*)$

$$\|\hat{\bar{x}}(t_0)\| < c_1 \implies \|\hat{\bar{x}}(t_0 + T)\| < d_2 < c_2 \tag{4.86}$$

Next we make the trajectories between the average and model-free systems to be $\Delta = \min\{c_2 - \alpha_1^{-1} \circ \alpha_2(d_1),\ d_1 - d_2\}$ close over a time interval of T length from the same initial condition using Lemma 4.1. With $\omega^* = \omega^*(T, \Delta, a)$, $\forall\, \omega \in (\omega^*, \infty)$

$$\|\hat{x}(t_0)\| < c_1 \implies \|\hat{x}(t_0 + T)\| < d_1 < c_2 \tag{4.87}$$

Inductive Part:

For this inductive part, the analysis will take place over time intervals of $[t_n, t_{n+1}]$ where $t_n = t_0 + nT$ with $n \in \{1, 2, \ldots\}$. From the initial part, and by the average system being autonomous, we know that the same a^* allows for

$$\|\hat{\bar{x}}(t_n)\| < d_1 < c_1 \implies \|\hat{\bar{x}}(t_{n+1})\| < d_2 \tag{4.88}$$

and

$$\|\hat{\bar{x}}(t_n)\| < d_1 \implies \|\hat{\bar{x}}(t)\| < \alpha_1^{-1} \circ \alpha_2(d_1),\ \forall\, t \in [t_n, t_{n+1}] \tag{4.89}$$

whenever $a \in (0, a^*)$. With the ω^* from the initial part granting the Δ closeness between trajectories, we can conclude that

$$\|\hat{x}(t_n)\| < d_1 \implies \|\hat{x}(t_{n+1})\| < d_1 \tag{4.90}$$

$$\|\hat{x}(t_n)\| < d_1 \implies \|\hat{x}(t)\| < c_2,\ \forall\, t \in [t_n, t_{n+1}] \tag{4.91}$$

$\forall\, \omega \in (\omega^*, \infty)$. With this we can say that $\forall\, a \in (0, a^*)$ and $\forall\, \omega \in (\omega^*, \infty)$, the trajectories satisfy $\|\hat{x}(t)\| < c_2\ \forall\, t \in [t_0 + T, \infty)$, and thus complete the induction part of the proof. Now we can combine both the initial and inductive parts to state

$$\|\hat{x}(t_0)\| < c_1 \implies \|\hat{x}(t)\| < c_2 \forall\, t \in [t_0 + T, \infty) \tag{4.92}$$

for all $a \in (0, a^*)$ and for all $\omega \in (\omega^*(T, \Delta, a), \infty)$. Hence, the model-free ESC system is δ-PUA with respect to the small parameter vector (a, ω^{-1}) and δ being able to be made arbitrarily large. A graphical representation of this proof can be found in Fig. 9.

4.3 Gradient-Based Extremum Seeking

The simplest and baseline method for optimization in the context of ESC is the gradient descent. For the baseline model of gradient-based extremum seeking control (GESC), the differential equation is

$$\frac{d}{dt}\hat{\theta} = -k\hat{g}\left(\omega t, \hat{\theta}, a\right) \tag{4.93}$$

where the gradient estimate $\hat{g}$ is defined as

$$\hat{g}\left(t, \hat{\theta}, a\right) = m(\omega t, a) J\left(\hat{\theta} + a p(t)\right) \tag{4.94}$$

and obtained by multiplying the sensor value J by a time-varying gradient estimation filter $m : \mathbb{R} \times \mathbb{R} \to \mathbb{R}^n$. For sinusoidal dither signals, the filter is defined as $m_i(\omega t, a) = \frac{2}{a r_i} \sin(\omega \omega' t)$. One

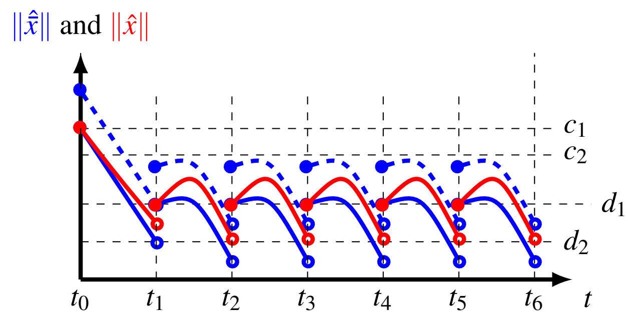


Figure 9: Graphical representation of the practical uniform attractivity of the model-free ESC systems. Bounds for $\|\hat{\hat{x}}\|$ are shown with a blue line, bounds for $\|\hat{x}\|$ are shown with a red line, and $\|\hat{\hat{x}}\| + \Delta$ is shown with a dashed blue line. Again, all curves are inclusive on the left endpoint and exclusive on the right endpoint with discontinuities a result of the use of induction in the proof. The dashed blue line forms an upper bound for $\|\hat{x}(t)\|$ to show that the model-free ESC system meets the definitions of practical uniform attractiveness.

does not care about $\hat{g}$ being accurate to ∇J at any particular instance of t but rather in the average (as $\omega \to \infty$). This will result in the average system, which is defined as

$$\frac{d}{dt}\hat{\hat{\theta}} = -k\hat{\hat{g}}\left(t, \hat{\theta}\right), \quad \hat{\hat{g}}\left(\hat{\hat{\theta}}, a\right) = \lim_{T\to\infty} \frac{1}{T}\int_0^T \hat{g}\left(\tau, \hat{\hat{\theta}}, a\right) d\tau \tag{4.95}$$

While we still do not expect $\hat{\hat{g}}$ to be exactly ∇J, we expect that $\hat{\hat{g}} \to \nabla J$ as $a \to 0$ by Theorem 4.1. Thus we obtain the model-based system of

$$\frac{d}{dt}\vartheta = -k\nabla J(\vartheta) \tag{4.96}$$

For proving the practical stability of the GESC, we consider a special case where J is a strictly convex scalar map and the more general case when J is a multivariable map. While we are interested in the general case, we specifically investigate strictly convex scalar maps to show a practical stability with respect to a single parameter.

4.3.1 Practical Stability on Scalar Strictly Convex Maps

This section covers the stability results for the GESC as presented in [13, 11]. In addition to assumptions 4.1-4.4, we require the following two assumptions.

Assumption 4.7. *The map $J : \mathbb{R} \to \mathbb{R}$ is a twice differentiable function ($J \in \mathscr{C}^2$).*

Assumption 4.8. *The map J is a strictly convex function, that is, for any distinct pairs of $\theta_1,\ \theta_2 \in \mathbb{R}^n$ and $\forall \lambda \in (0,1)$ the following inequality holds over the convex line segment [34].*

$$J(\lambda\theta_1 + (1-\lambda)\theta_2) < \lambda J(\theta_1) + (1-\lambda)J(\theta_2) \tag{4.97}$$

For the later analysis, we state two useful lemmas for strictly convex functions and give a proposition.

Lemma 4.2 ([34])**.** *A differentiable map is strictly convex if and only if for* $\theta_1,\ \theta_2 \in \mathbb{R}^n$ *where* $\theta_1 \neq \theta_2$ *the following inequality holds.*

$$J(\theta_2) > J(\theta_1) + \nabla J(\theta_1)^T(\theta_2 - \theta_1) \tag{4.98}$$

Lemma 4.3 ([34])**.** *A twice differentiable map is convex if and only if*

$$\nabla^2 J \succeq 0, \quad \forall\ \theta \in \mathbb{R}^n \tag{4.99}$$

Proposition 4.2. *A scalar, strictly convex map* J *implies that* $\forall\ \theta_1, \theta_2 \in \mathbb{R}$ *where* $\theta_1 < \theta_2$*, the following proper integral of the second derivative of the map is strictly positive.*

$$\int_{\theta_1}^{\theta_2} \frac{d^2 J}{d\phi^2}(\phi) d\phi > 0 \tag{4.100}$$

Proof. Suppose that the proposition was not true, then J is strictly convex and there is a finite interval where $\int_{\theta_1}^{\theta_2} \frac{d^2J}{d\phi^2}(\phi)d\phi \leq 0$. From Lemma 4.3, the integrand must be strictly non-negative so $\frac{d^2J}{d\phi^2}(\phi) = 0$ for all $\phi \in (\theta_1, \theta_2)$, i.e., the function is a linear function in the interval. Thus, J cannot be strictly convex as (4.97) would be an equality rather than a strict inequality, leading to a contradiction that J was strictly convex. Therefore, (4.100) must hold for any $\theta_1, \theta_2 \in \mathbb{R}$ where $\theta_1 < \theta_2$. □

Now, for the assumptions presented, we can prove two propositions for the average gradient estimate using sinusoidal perturbations.

$$\hat{\hat{g}}(\hat{\hat{\theta}}, a) = \frac{1}{2\pi} \int_0^{2\pi} \frac{2}{a} \sin(\tau) J(\hat{\hat{\theta}} + a\sin(\tau)) d\tau \tag{4.101}$$

Proposition 4.3. *Under Assumptions 4.7 and 4.8, the average system gradient estimate* $\hat{\hat{g}}$ *defined in* (4.101) *is a strictly monotonically increasing function with respect to* $\hat{\hat{\theta}}$.

Proof. Rearranging (4.101) gives in a more suitable form for differentiation

$$\hat{\hat{g}}(\hat{\hat{\theta}}) = \int_0^{\pi} \frac{\sin(\tau)}{a\pi} \left(J(\hat{\hat{\theta}} + as(\tau)) - J(\hat{\hat{\theta}} - as(\tau)) \right) d\tau$$

which when differentiating and then replacing the difference with an integration yields

$$\hat{\hat{g}}'(\hat{\hat{\theta}}) = \frac{1}{a\pi} \int_0^{\pi} \sin(\tau) \int_{\hat{\hat{\theta}} - a\sin(\tau)}^{\hat{\hat{\theta}} + a\sin(\tau)} J''(\phi) d\phi d\tau > 0.$$

The strict inequality is a consequence of Proposition 4.2 and $\sin\tau > 0$ for all $\tau \in (0,\ \pi)$. With the derivative of $\hat{\hat{g}}$ being strictly positive, $\hat{\hat{g}}$ must be a strictly monotonically increasing function with respect to $\hat{\hat{\theta}}$. □

In addition to showing the monotonicity of $\hat{\hat{g}}$, we establish the existence and uniqueness of θ_* when the scalar map has a global minimizer.

Proposition 4.4. *Under Assumptions 4.7 and 4.2 the gradient estimate in the average system* $\hat{\hat{g}}$ *as defined by* (4.101) *has a unique zero* $\theta_* \in (\theta_* - a, \theta_* + a)$.

Proof. The proof follows by providing an upper and lower bound to $\hat{\hat{g}}$ and applying Assumption 4.2 to the found bounds. First, (4.101) is simplified by periodicity to

$$\hat{\hat{g}}(\hat{\hat{\theta}}) = \frac{2}{a\pi}\int_{-\pi/2}^{\pi/2} \sin(\tau) J(\hat{\hat{\theta}} + a\sin(\tau))d\tau \tag{4.102}$$

Next, we introduce an auxiliary function f_ℓ using a convex line segment and support line

$$f_\ell(\phi) = \begin{cases} \lambda J\left(\hat{\hat{\theta}} - a\right) + (1-\lambda)J\left(\hat{\hat{\theta}}\right) & \text{for } \phi < \hat{\hat{\theta}} \\ J\left((\hat{\hat{\theta}}\right) + J'\left(\hat{\hat{\theta}}\right)\cdot\left(\phi - \hat{\hat{\theta}}\right) & \text{for } \phi \geq \hat{\hat{\theta}} \end{cases}$$

where $\lambda = (\bar{\theta} - \phi)/a$ to form a strict lower bound on the integral in (4.102) by replacing J with f_ℓ in the integrand. The lower bound is justified by the convex line segment and the support line regions. The inequality

$$\sin(\tau) f_\ell\left(\hat{\hat{\theta}} + a\sin(\tau)\right) \leq \sin(\tau) J\left(\hat{\hat{\theta}} + a\sin(\tau)\right)$$

may then be formed with equality met only when $\sin\tau = 0$. Hence, replacing f_ℓ for J in (4.102) forms the lower bounds of

$$\begin{aligned} \hat{\hat{g}}(\hat{\hat{\theta}}) &> \frac{2}{a\pi}\int_{-\pi/2}^{\pi/2} \sin(\tau) f_\ell\left(\hat{\hat{\theta}} + a\sin(\tau)\right)d\tau \\ &= \frac{1}{2}\left(J'\left(\hat{\hat{\theta}}\right) + \frac{1}{a}\left(J\left(\hat{\hat{\theta}}\right) - J\left(\hat{\hat{\theta}} - a\right)\right)\right) \\ &= \frac{1}{2}\left(J'(\hat{\hat{\theta}}) + \frac{1}{a}\int_{\hat{\hat{\theta}}-a}^{\hat{\hat{\theta}}} J'(\phi)d\phi\right) \end{aligned}$$

While we have the lower bound in terms of J', it would be a more applicable bound if we use a single value instead of having an integral calculating the average value of J' over an interval. As J' is a strictly monotonically increasing function by Proposition 4.2, the extrema of J' over finite intervals occur at the interval endpoints. Thus, both $J'(\hat{\hat{\theta}}) > J'(\hat{\hat{\theta}} - a)$ and $\frac{1}{a}\int_{\hat{\hat{\theta}}-a}^{\hat{\hat{\theta}}} J'(\phi)d\phi > J'(\hat{\hat{\theta}} - a)$. The lower bound for $\hat{\hat{g}}$ is now given as

$$\hat{\hat{g}}\left(\hat{\hat{\theta}}\right) > \frac{1}{2}\left(J'(\hat{\hat{\theta}}) + \frac{1}{a}\int_{\hat{\hat{\theta}}-a}^{\hat{\hat{\theta}}} J'(\phi)d\phi\right) > J'(\hat{\hat{\theta}} - a) \tag{4.103}$$

The upper bound for $\hat{\hat{g}}$ follows similar logic but uses the auxiliary function

$$f_u(\phi) = \begin{cases} J\left(\hat{\hat{\theta}}\right) + J'\left(\hat{\hat{\theta}}\right)\cdot\left(\phi - \hat{\hat{\theta}}\right) & \text{for } \phi < \hat{\hat{\theta}} \\ \lambda J\left(\hat{\hat{\theta}} + a\right) + (1-\lambda)J\left(\hat{\hat{\theta}}\right) & \text{for } \phi \geq \hat{\hat{\theta}} \end{cases}$$

where $\lambda = (\phi - \bar{\theta})/a$ for the inequality

$$\sin(\tau) f_u\left(\hat{\hat{\theta}} + a\sin(\tau)\right) \geq \sin(\tau) J\left(\hat{\hat{\theta}} + a\sin(\tau)\right) \tag{4.104}$$

This gives the upper bound of $\hat{\bar{g}}$ by

$$\begin{aligned}\hat{\bar{g}}\left(\hat{\bar{\theta}}\right) &< \frac{2}{a\pi}\int_{-\pi/2}^{\pi/2}\sin(\tau)f_u\left(\hat{\bar{\theta}}+a\sin(\tau)\right)d\tau \\ &= \frac{1}{2}\left(J'\left(\hat{\bar{\theta}}\right)+\frac{1}{a}\int_{\hat{\bar{\theta}}}^{\hat{\bar{\theta}}+a}J'(\phi)d\phi\right) \\ &< J'\left(\hat{\bar{\theta}}+a\right)\end{aligned} \tag{4.105}$$

With the upper and lower bounds established on $\hat{\bar{g}}$, we combine both (4.103) and (4.105) to show that

$$\hat{\bar{g}}(\theta_* - a) < J'(\theta_*) < \hat{\bar{g}}(\theta_* + a)$$

and $J'(\theta_*) = 0$ from Assumption 4.2. By the Intermediate Value Theorem, there is a zero $\theta_* \in (\theta_* - a,\ \theta_* + a)$ and this zero is unique as $\hat{\bar{g}}$ is a strictly monotonically increasing function by Proposition 4.3. □

With propositions about $\hat{\bar{g}}$ established, we can give the stability of the GESC for strictly convex scalar maps.

Theorem 4.3 (GESC is sGPUAS for Strictly Convex Scalar Maps)**.** *Under Assumptions 4.2, 4.3, 4.7, and 4.7, the GESC system defined in* (4.93) *using sinusoidal perturbations is sGPUAS to the* $\theta_* \in (\theta_* - a, \theta_* + a)$ *with respect to the large parameter* ω*.*

Proof. We define the error coordinates $\widetilde{\theta} = \hat{\theta} - \theta_*$ with the dynamics of

$$\frac{d}{dt}\widetilde{\theta} = -k\hat{g}(\omega t, \widetilde{\theta}+\theta_*, a) = -\frac{2k}{a}\sin(\omega t)J\left(\widetilde{\theta}+\theta_*+a\sin(\omega t)\right) \tag{4.106}$$

It has the associated average system of

$$\frac{d}{dt}\bar{\widetilde{\theta}} = -k\hat{\bar{g}}\left(\bar{\widetilde{\theta}}+\theta_*\right) \tag{4.107}$$

where $\hat{\bar{g}}$ was defined in (4.101). The Lyapunov function

$$V\left(\bar{\widetilde{\theta}}\right) = \frac{1}{2}\bar{\widetilde{\theta}}^2 \tag{4.108}$$

is trivially radially unbounded and has the Lie derivative of

$$\frac{d}{dt}V\left(\bar{\widetilde{\theta}}\right) = -k\bar{\widetilde{\theta}}\hat{\bar{g}}\left(\bar{\widetilde{\theta}}+\theta_*\right) \tag{4.109}$$

which is a negative definite function by Propositions 4.4. Thus the average system (4.107) is GAS to the origin and the original model-free system in (4.93) is sGPUAS to θ_* with respect to the large parameter ω [21]. □

4.3.2 A Multivariable Example

Having establishing practical stability of the GESC for scalar maps with only a single parameter for practical stability, it is natural to question if this can be accomplished for multivariable maps as well. To do this, we will need to establish that the average system is GAS. For local stability and consequently global stability of (4.95), $-k\nabla\hat{\hat{g}}(\theta_*, a)$ must be Hurwitz, i.e., all the eigenvalues of $\nabla\hat{\hat{g}}(\theta_*, a)$ must have positive real components. The necessary condition for stability then is that the trace of $\nabla\hat{\hat{g}}(\theta_*, a)$ must be strictly positive, i.e., a sufficient condition for instability is that the trace of $\nabla\hat{\hat{g}}(\theta_*, a)$ is negative.

Now consider the cost function

$$J(\theta) = \theta_1^4 + (\theta_1 + \theta_2)^4 \tag{4.110}$$

$$= 2\theta_1^4 + 4\theta_1^3\theta_2 + 6\theta_1^2\theta_2^2 + 4\theta_1\theta_2^3 + \theta_2^4 \tag{4.111}$$

being acted upon by a GESC using sinusoidal perturbations. The average gradient estimate for the sinusoidal perturbations is

$$\hat{\hat{g}}\left(\hat{\hat{\theta}}, a\right) = \lim_{T\to\infty} \frac{1}{T} \int_0^T m(\tau, a) J\left(\hat{\hat{\theta}} + as(\tau)\right) d\tau = \nabla J\left(\hat{\hat{\theta}}\right) + R_g\left(\hat{\hat{\theta}}, a\right) \tag{4.112}$$

where $m_i(\tau, a) = \frac{2}{a}\sin(\omega'\tau, a)$ for $i = 1, 2$. With the choice of $\omega_2'/\omega_1' = 3$, this cost function has an averaged gradient estimation vector, defined in (4.112), of

$$\hat{\hat{g}}(\hat{\hat{\theta}}) = 4\begin{bmatrix} \hat{\hat{\theta}}_1^3 + (\hat{\hat{\theta}}_1 + \hat{\hat{\theta}}_2)^3 \\ (\hat{\hat{\theta}}_1 + \hat{\hat{\theta}}_2)^3 \end{bmatrix} + 6a\begin{bmatrix} r_1^2 - \frac{r_1 r_2}{2} + r_2^2 & \frac{r_1^2}{2} - \frac{r_1 r_2}{2} + r_2^2 \\ \frac{r_1^3}{3r_2} + r_1^2 + \frac{r_2^2}{2} & -\frac{r_1^3}{6r_2} + r_1^2 + \frac{r_2^2}{2} \end{bmatrix} \hat{\hat{\theta}} \tag{4.113}$$

The equilibrium point of the GESC system is the origin $\theta_* = 0$. Trivially, the Jacobian of the GESC system at the origin is simply the linear coefficients in (4.113).

Using the sufficient condition for instability, we arrive at the inequality

$$2r_1^2 - \frac{r_1 r_2}{2} + \frac{3r_2^2}{2} - \frac{r_1^3}{6r_2} \le 0 \tag{4.114}$$

Substituting the ratio $r_a = r_1/r_2$ and simplifying leads to the new inequality

$$r_a^3 - 12r_a^2 + 3r_a - 9 \ge 0$$

that is satisfied when $r_a \ge 15c^{-1/3} + 4 + c^{1/3}$ where

$$c = \frac{5}{2}\sqrt{85} + \frac{125}{2}$$

The lower bound for r_a to generate this instability is approximately 11.81, which means the GESC is unstable on this map when using highly asymmetric dither amplitudes. Thus, there are system designer choices that may destabilize the GESC for this particular cost function.

The GESC acting on (4.110) is unstable if the dither amplitude associated with the faster perturbation signal (r_2 here) is relatively too small. It is important to note that this is a relative magnitude between r_1 and r_2 and not their absolute magnitude. This is a crucial detail to consider when faced with the claim that the model-free GESC system with arbitrarily dither amplitudes is nearly identical to the model-based GESC systems. After all, $\hat{\hat{g}} \to \nabla J$ in (4.113) as $a \to 0$ which means the average system is practically stable, but not stable by traditional definitions. However, while the model-based GESC can be shown to be GUAS using a Lyapunov stability analysis on the Lyapunov function $V = J$, the model-free GESC can always be unstable no matter how small r_1 and r_2.

The specific cause of the instability studied in this example is the $\theta_1^3\theta_2$ product term in (4.111), which contributes to the r_1^3/r_2 components in (4.112). Therefore, we expect that terms composed of products of different parameters could potentially lead to instability in the GESC algorithm. Note that most other choices of ω_1' and ω_2' give the average gradient estimate in (4.113) with the r_1/r_2 terms. The reason that these terms exist is due to the choice of $\omega_2'/\omega_1' = 3$. For other relative dither rates, these destabilizing terms do not appear in (4.112) and thus the average system often traditionally stable. However, in the worst case, we cannot say that the average system is traditionally stable but merely practically stable by Section 4.2.

4.3.3 Practical Stability on Multivariable Maps

Since GESC is the baseline ESC algorithm, it is rather straightforward to derive the practical stability with two parameters.

Theorem 4.4 (GESC is sGPUAS). *Let $J : \mathbb{R}^n \to \mathbb{R}$ satisfy Assumptions 4.1-4.4. Then the GESC is sGPUAS to θ_* with the small parameter vector (a_0, ω^{-1}).*

Proof. We define the error coordinate system as $\widetilde{\vartheta} = \vartheta - \vartheta_*$ with the model-based dynamics of

$$\frac{d}{dt}\widetilde{\vartheta} = -k\nabla J\left(\vartheta_* + \widetilde{\vartheta}\right) \tag{4.115}$$

Let the Lyapunov function for the model-based system be

$$V\left(\widetilde{\vartheta}\right) = J\left(\widetilde{\vartheta} + \vartheta_*\right) - J\left(\vartheta_*\right) \tag{4.116}$$

Its Lie derivative is

$$\frac{d}{dt}V\left(\widetilde{\vartheta}\right) = -k\nabla J(\vartheta)^2 \leq 0 \tag{4.117}$$

which is a negative definite function by Assumption 4.3. Thus, the model-based system is GAS to the origin [20, Th 4.9] and consequently, the model-free system is sGPUAS to the origin with respect to the small parameter vector (a, ω) by Theorem 4.2. □

4.4 Adaptive Methods

It is a common observation that the baseline GESC has local convergence rates towards extremums that are dependent on the eigenvalues of an unknown Hessian of the cost function at the extrema. These eigenvalues could be arbitrarily small or large which may lead for the GESC to be impractical to implement in real world scenarios. Too small of an eigenvalue would have a system designer deem the GESC too slow while too large will drive the partial derivatives of system to increase. The increased stiffness of the system due to large eigenvalues leads to dither rates having to increase to potentially mechanically unobtainable levels in order to achieve the same practical convergence criteria. This nuance is never overtly stated in the previous literature but can be gleaned by looking at the trajectory distance bounds between original and average systems such as in [22, Th 2.8.1] where the bounds between the original and average systems depend on the Lipschitz constant of the differential equation. Thus system designers may want to choose ESC based on optimization algorithms which seek to regularize the convergence rates.

Another ESC is the Newton-based ESC (NESC) [6] which is based on Newton algorithms and attempts to correct for the unknown Hessian by estimating its inverse. This can lead to, in specific circumstances, assignable convergence rates. However, the algorithms require that the Hessian be invertible for convergence. This additional requirement and the need to estimate more derivatives of the cost function tend to complicate their implementation into real-world scenarios. Instead of the Newton algorithms, one can use some existing adaptive algorithms such as the RMSprop [18], AdaDelta, or AdaGrad which use only the gradient information to adaptively alter the convergence rate in different directions.

The first method considered in the following sections is RMSprop, which inversely scales the convergence rate in each parameter direction by a low pass estimate of the root mean squared (RMS) value of the gradient component. This scaling seeks to achieve a speed of update for any parameter to be an assignable k. From an implementation perspective, the RMSprop is simple to implement as there is only n more scalar variables to account for. The later methods of AdaDelta and AdaGrad are more complicated, as they seek to alter the gain with an estimate of the gradient outer product. This complicates the differential equations as now there are matrix differential equations but the benefit is that AdaDelta and AdaGrad seek to achieve a seeker with a total velocity of the assignable k, rather than just a speed in a parameter update.

It should be noted that all methods examined used a low pass estimations for their direction. This means that the time history of the gradient along the trajectories plays a significant role in the current parameter direction. When looking at the gradient, the seeker will be largely insensitive to directions with previously large gradient estimates but highly sensitive to new directions which have not previously appeared in past. This is also a reason why the speed in any parameter update direction may not achieve k but instead be approximately k. This is still an improvement over the baseline GESC in which one does not have any control on the speed of the seeker.
Adaptive systems can be seen as an interconnected system with

$$\dot{\vartheta} = -K(\varphi)\nabla J(\vartheta) \tag{4.118}$$

$$\dot{\varphi} = \omega_\ell(q(\vartheta) - \varphi) \tag{4.119}$$

Here, $J : \mathbb{R}^n \to \mathbb{R}$ is a cost function that depends on the state vector $\vartheta \in \mathbb{R}^n$, $\varphi \in \mathbb{R}^m$ is a low-pass filter estimate of some properties of J, $K(\cdot)$ denotes an adaptive gain function whose output is positive definite ($K(\varphi) \succ 0$) for all $\varphi \in \mathbb{R}^m$, and $\omega_\ell > 0$ is the low-pass filter parameter. The term $q(\vartheta)$ represents an adaptation law and represents a mapping of ϑ to some property of J, potentially involving elements of the Hessian $\nabla^2 J(\vartheta)$ or elements of the gradient $\frac{\partial J}{\partial \vartheta_i}(\vartheta)$. The cost function J is assumed to satisfy Assumptions 4.1 - 4.4 and the following assumption.

Assumption 4.9 (Maximum Growth Rate)**.** *J is sufficiently differentiable so that $\exists \rho_q \in \mathscr{K}$ such that*

$$|q(\vartheta) - q(0)| \leq \frac{1}{2}\rho_q(|\vartheta|) \tag{4.120}$$

where $q(\vartheta^*) = 0$ wlog.

Under Assumptions 4.1 - 4.4 and 4.9, we note the following properties of the two subsystems.

Remark 4.3. *The Lie derivative of J is*

$$\dot{J}(\vartheta,\varphi) = -\nabla J(\vartheta)^T\ K(\varphi)\ \nabla J(\vartheta) \le -\lambda_{\min}\left(K(\varphi)\right)\ |\nabla J(\vartheta)|^2 = -W(\vartheta,\varphi) \tag{4.121}$$

where W is a positive definite function with respect to ϑ. Thus, J is always strictly decreasing along the trajectories of the system given in (4.118)-(4.119) *if the initial condition $\vartheta(t_0) \neq 0$.*

Remark 4.4. *The subsystem* (4.119) *is ISS to $\varphi = 0$. The corresponding ISS-Lyapunov function is $V_{\mathrm{ISS}}(\varphi) = \alpha_\varphi(|\varphi|) = |\varphi|^2$ with*

$$\dot{V}_{\mathrm{ISS}}(\varphi,\vartheta) \le -\omega_\ell|\varphi|^2 \quad \forall\, |\varphi| \ge \rho_q(|\vartheta|) \tag{4.122}$$

The gain of the system is ρ_q.

4.4.1 Stability of Interconnected Systems

The two properties mentioned in Remarks 4.3 and 4.4 allow us to give the following main theoretical results of this work.

Theorem 4.5. *Consider the system described by* (4.118)-(4.119)*. If Assumptions Assumptions 4.1 - 4.4 and 4.9 hold, then the system* (4.118)-(4.119) *is GAS at the origin.*

We give three methodologies for constructing Lyapunov functions for systems of the form (4.118)-(4.119). The first methodology is the way to prove Theorem 4.5 with only Assumptions 4.1 - 4.4 and 4.9. We also give two additional methodologies that exploit additional structure in the interconnected system.

4.4.1.1 Lyapunov Function with a Maximum

The main idea behind this construction of Lyapunov functions is that we relate the gain of (4.122) to the value of $J(\vartheta)$ rather than $|\vartheta|$. Since Remark 4.3 shows that J is strictly decreasing along the trajectories of the interconnected system, we naturally expect the argument in the ISS gain for (4.119) to go to zero. It is this realization that motivates the following Lyapunov construction for proving Theorem 4.5.

Proof of Theorem 4.5. We relax the condition in (4.122) using the radially unboundedness of J so that

$$\dot{V}_{\mathrm{ISS}}(\varphi) \le -|\varphi|^2 \quad \forall\, |\varphi| \ge \rho_q \circ \alpha_{\vartheta,1}^{-1} \circ J(\varphi) \ge 0 \tag{4.123}$$

and further relax the condition to get

$$\dot{V}_{\mathrm{ISS}}(\varphi) \le -|\varphi|^2 \quad \forall \varphi\, \mathrm{s.t.}\, |\varphi|^2 \ge \alpha_{\vartheta,\varphi} \circ J(\vartheta) \ge 0 \tag{4.124}$$

where $\alpha_{\vartheta,\varphi} = \alpha_\varphi \circ \rho_q \circ \alpha_{\vartheta,1}^{-1}$. The function $\alpha_{\vartheta,\varphi} \in \mathscr{K}$ since it is a composition of class $\mathscr{K}$ functions. This function is used to determine a lower bound, using the cost function $J(\vartheta)$, for which we can use to guarantee that $|\varphi|^2$ decreases when $|\varphi|^2$ exceeds the lower bound. Additionally, note that the bound $\alpha_{\vartheta,\varphi} \circ J(\vartheta)$ is always decreasing along the trajectories by Remark 4.3, so the upper right-handed Dini derivative along the trajectories is

$$D^+\alpha_{\vartheta,\varphi} \circ J(\vartheta) = -W(\vartheta,\varphi) \cdot D_-\alpha_{\vartheta,\varphi}(J(\vartheta)) \le 0 \tag{4.125}$$

To utilize the bound, we construct the candidate Lyapunov function for the overall system (4.118)-(4.119)

$$V_{\text{Max}}(\vartheta,\varphi) = J(\vartheta) + \max\left\{\alpha_{\vartheta,\varphi}\circ J(\vartheta)\,,\,|\varphi|^2\right\} \tag{4.126}$$
$$= J(\vartheta) + \alpha_{\vartheta,\varphi}\circ J(\vartheta) + \max\left\{|\varphi|^2 - \alpha_{\vartheta,\varphi}\circ J(\vartheta)\,,\,0\right\} \tag{4.127}$$

This candidate Lyapunov function is meant to penalize three things:

1. the excess "value" in the cost function $J(\vartheta)$;
2. the magnitude of the lower bound of $|\varphi|^2$ given by $\alpha_{\vartheta,\varphi}\circ J(\vartheta)$;
3. and the amount that $|\varphi|^2$ exceeds the bound.

While the max function is an unusual addition to the Lyapunov function, there is precedent for using max-type Lyapunov functions in the context of interconnected systems [35–37]. Note that the candidate Lyapunov function is radially unbounded as

$$V_{\text{Max}}(\vartheta,\varphi) \geq \alpha_{\vartheta,1}(|\vartheta|) + |\varphi|^2 \tag{4.128}$$

since $\max\{a,b\} \geq a$ and $\max\{a,b\} \geq b$ for any $a,b \in \mathbb{R}$. Additionally, we form the upper bound

$$V_{\text{Max}}(\vartheta,\varphi) \leq \alpha_{\vartheta,2}(|\vartheta|) + \alpha_{\vartheta,\varphi}\circ\alpha_{\vartheta,2}(|\vartheta|) + |\varphi|^2 \tag{4.129}$$

as $\max\{a,b\} \leq a+b$ if $a,b \geq 0$. Lastly, since the max function is nonsmooth at the switching surface

$$S(t) = \left\{(\vartheta,\varphi)\mid|\varphi|^2 = \alpha_{\vartheta,\varphi}\circ J(\vartheta)\right\} \tag{4.130}$$

we do not expect that the time derivative of V_{Max} to be continuous and instead have to rely on the upper right Dini derivative of V_{Max} along trajectories of the system (4.118)-(4.119). To this end, we must consider three cases: when (ϑ,φ) is "above" the switching surface with $|\varphi|^2 > \alpha_{\vartheta,\varphi}\circ J(\vartheta)$; when (ϑ,φ) is "below" the switching surface with $|\varphi|^2 < \alpha_{\vartheta,\varphi}\circ J(\vartheta)$; and when (ϑ,φ) is on the switching surface with $|\varphi|^2 = \alpha_{\vartheta,\varphi}\circ J(\vartheta)$.

Case 1: Above the switching surface. In this region, $\varphi \neq 0$ and the Lyapunov function is

$$V_{\text{Max}}(\vartheta,\varphi) = J(\vartheta) + |\varphi|^2 \tag{4.131}$$

Since $J(\vartheta)$ is continuously differentiable by Assumption 4.1, the Dini derivative is the normal Lie derivative. This Lie derivative is

$$D^+V_{\text{Max}}(\vartheta,\varphi) = \dot{J}(\vartheta,\varphi) + \frac{d}{dt}|\varphi|^2 \tag{4.132}$$
$$\leq -W(\vartheta,\varphi) - \omega_\ell|\varphi|^2 \tag{4.133}$$

which is a negative definite function by virtue of being the sum of two negative definite functions whose arguments are different variables.

Case 2: Below the switching surface. When the trajectories are only in this region, which can only be achieved when $\vartheta \neq 0$, the Lyapunov function is

$$V_{\text{Max}}(\vartheta,\varphi) = J(\vartheta) + \alpha_{\vartheta,\varphi}\circ J(\vartheta) \tag{4.134}$$

While $J(\vartheta)$ is still differentiable, we do not know if the composite function $\alpha_{\vartheta,\varphi}$ is differentiable, so one must use Dini derivatives when analyzing V_{Max}. The upper right-hand Dini derivative of V_{Max} is

$$\begin{aligned} D^+V_{\text{Max}}(\vartheta,\varphi) &= \dot{J}(\vartheta,\varphi) + D^+\left(\alpha_{\vartheta,\varphi}\circ J(\vartheta)\right) && (4.135)\\ &\le -W(\vartheta,\varphi)\cdot\left(1+(D_-\alpha_{\vartheta,\varphi})\circ J(\vartheta)\right) && (4.136)\\ &\le -W(\vartheta,\varphi) && (4.137)\end{aligned}$$

which is a strictly negative function since $\vartheta \neq 0$.

Case 3: On the switching surface. Whenever the trajectories are on the switching surface, the Dini derivative D^+V_{Max} satisfies

$$D^+V_{\text{Max}}(\vartheta,\varphi) = \dot{J}(\vartheta,\varphi) + \max\left\{\frac{d}{dt}|\varphi|^2, D^+\alpha_{\vartheta,\varphi}\circ J(\vartheta)\right\} \tag{4.138}$$

which is the maximum Lie derivative between the two previously discussed cases. Thus,

$$\begin{aligned} D^+V_{\text{Max}}(\vartheta,\varphi) &\le -W(\vartheta,\varphi) - \min\left\{|\varphi|^2, (\vartheta,\varphi)\cdot D_-\alpha_{\vartheta,\varphi}\circ J(\vartheta)\right\} && (4.139)\\ &\le -W(\vartheta,\varphi) && (4.140)\end{aligned}$$

The derivative above is zero only if the function $W(\vartheta,\varphi) = 0$, which occurs only if $\vartheta = 0$. If $\vartheta = 0$ then $\varphi(t)$ satisfies $|\varphi|^2 = \alpha_{\vartheta,\varphi}(0) = 0$ to be on the switching surface, which is only possible if $\varphi = 0$. Thus, $D^+V_{\text{Max}}(\vartheta,\varphi)$ is a negative definite function on the switching surface.

Combining the cases: Since all three cases show that D^+V_{Max} was a negative definite function with respect to its arguments, we conclude that V_{Max} is a proper, radially unbounded function whose derivative is a negative definite function. Thus, the system described by (4.118)-(4.119) is GAS to the origin by [20, Thm. 4.9]. □

4.4.1.2 Lyapunov Function for Scalar Cost Functions

This section demonstrates that a different Lyapunov function can be constructed when the cost function is a scalar function. The approach for the construction of this Lyapunov function is inspired by the Lyapunov function used in [11, 13], and give the following proposition.

Proposition 4.5. *Consider a scalar cost function $J:\mathbb{R}\to\mathbb{R}$ and the system described by* (4.118)-(4.119). *If Assumptions 4.1 - 4.4 are satisfied in addition to $q\in\mathscr{C}^1$, then the system* (4.118)-(4.119) *is GAS at the origin.*

Proof. First, we introduce an error coordinate system of $\widetilde{\varphi} = \varphi - q(\vartheta)$. The new dynamics are

$$\begin{aligned} \dot{\vartheta} &= -K(\widetilde{\varphi}+q(\vartheta))J'(\vartheta) && (4.141)\\ \dot{\widetilde{\varphi}} &= -\omega_\ell\widetilde{\varphi} - q'(\vartheta)\cdot\dot{\vartheta} && (4.142)\end{aligned}$$

Now consider the candidate Lyapunov function

$$V_{\text{Scalar}}(\vartheta,\widetilde{\varphi}) = \vartheta^2 + \text{sgn}(\vartheta)\int_0^\vartheta \left|q'(x)\right| dx + \ln(1+\widetilde{\varphi}^2) \tag{4.143}$$

where $\mathrm{sgn}(\cdot)$ is the usual sign function and the integral term in (4.143) is used to account for the interconnected term in (4.142). We note that V_{Scalar} is a radially unbounded function, as the first and third terms in (4.143) are radially unbounded functions with respect to their arguments. The Lie derivative of V_{Scalar} is

$$\dot{V}_{\text{Scalar}}(\vartheta,\widetilde{\varphi}) = -\frac{2\omega_\ell\widetilde{\varphi}^2}{1+\widetilde{\varphi}^2} - 2K(\widetilde{\varphi}+q(\vartheta))\,\vartheta J'(\vartheta) - K(\widetilde{\varphi}+q(\vartheta))J'(\vartheta) \cdot \left(\mathrm{sgn}(\vartheta)\cdot\left|q'(\vartheta)\right| - \frac{2\widetilde{\varphi}\cdot q'(\vartheta)}{1+\widetilde{\varphi}^2}\right) \quad (4.144)$$

By Assumptions 4.1 - 4.4, $\mathrm{sgn}(J'(\vartheta)) = \mathrm{sgn}(\vartheta)$ which allows us to bound (4.144) by

$$\dot{V}_{\text{Scalar}}(\vartheta,\widetilde{\varphi}) \le -\frac{2\omega_\ell\widetilde{\varphi}^2}{1+\widetilde{\varphi}^2} - 2K(\widetilde{\varphi}+q(\vartheta))\vartheta J'(\vartheta) - K(\widetilde{\varphi}+q(\vartheta))\left|J'(\vartheta)\right|\cdot\left(\left|q'(\vartheta)\right| - \frac{2|\widetilde{\varphi}|\cdot|q'(\vartheta)|}{1+\widetilde{\varphi}^2}\right) \quad (4.145)$$

By exploiting the fact that

$$\max_{\widetilde{\varphi}\in\mathbb{R}}\left|\frac{2\widetilde{\varphi}}{1+\widetilde{\varphi}^2}\right| = 1 \quad (4.146)$$

the bound on $\dot{V}_{\text{Scalar}}$ further simplifies to

$$\dot{V}_{\text{Scalar}}(\vartheta,\widetilde{\varphi}) \le -\frac{2\omega_\ell\widetilde{\varphi}^2}{1+\widetilde{\varphi}^2} - 2K(\widetilde{\varphi}+q(\vartheta))\vartheta J'(\vartheta) \quad (4.147)$$

which is a negative definite function. Thus, V_{Scalar} proves that the system is GAS to the origin using [20, Thm. 4.9]. □

4.4.1.3 Lyapunov Function with Integral Transforms

In this section, we construct a Lyapunov function via an integral transform, inspired by [38, 39, 35, 40, 41], to establish GAS property under additional conditions. We then state the following proposition.

Proposition 4.6. *Consider the system described by (4.118)-(4.119). If Assumptions 4.2-4.9 hold and there exists $\gamma_2 : \mathbb{R}\to\mathbb{R}$ such that $\lambda_{\min}(K(\varphi)) \ge \gamma_2(|\varphi|^2) > 0$ and $\lim_{c\to\infty}\int_0^c \gamma_2(r)dr = \infty$, then the system (4.118)-(4.119) is GAS at the origin.*

Proof. Consider the candidate Lyapunov function

$$V_{\text{Integral}}(\vartheta,\varphi) = J(\vartheta) + \int_0^{J(\vartheta)} \frac{\left(\rho_q\circ\alpha_{\vartheta,1}^{-1}(r)\right)^2}{4b_g(r)}dr + \frac{1}{2\omega_\ell}\int_0^{|\varphi|^2}\gamma_2(r)dr \quad (4.148)$$

where

$$b_g(c) := \min_{\vartheta'\in\mathbb{R}^n \text{ s.t. } J(\vartheta')=c}|\nabla J(\vartheta')|^2 \quad (4.149)$$

Then the Lie derivative of V_{Integral} is

$$\dot{V}_{\text{Integral}}(\vartheta,\varphi) \leq -\left(\frac{1}{8}\left(\rho_q \circ \alpha_{\vartheta,1}^{-1} \circ J(\vartheta)\right)^2 + |\nabla J(\vartheta)|^2 + \frac{1}{2}|\varphi|^2\right) \cdot \gamma_2(|\varphi|^2) \tag{4.150}$$

which is a negative definite function. Thus, V_{Integral} proves that the system is GAS to the origin using [20, Thm. 4.9].

□

The main step of how we obtain the Lyapunov function (4.148) is inspired by [38, 39, 35, 40, 41]. We know that J is a Lyapunov function for proving (4.118) is strictly decreasing and that $|\varphi|^2$ worked as a Lyapunov function for proving (4.119) was ISS to the origin. Generally, we do not expect a naive sum of $J(\vartheta) + |\varphi|^2$ to have a negative definite Lie derivative. We seek a radially unbounded V_{Integral} in the form of

$$V_{\text{Integral}}(\vartheta,\varphi) = \int_0^{J(\vartheta)} \gamma_1(r)dr + \frac{1}{2\omega_\ell}\int_0^{|\varphi|^2} \gamma_2(r)dr, \tag{4.151}$$

which does have a negative definite Lie derivative. We require that functions γ_1 and γ_2 satisfying the following properties:

P1: $\lim_{c\to\infty}\int_0^c \gamma_1(r)dr = \lim_{c\to\infty}\int_0^c \gamma_2(r)dr = \infty$.

P2: $\gamma_1(r) > 0$ and $\gamma_2(r) > 0$ for all $r \in (0,\infty)$;

We require P1 so that V_{Integral} is a radially unbounded and P2 so that $\dot{V}_{\text{Integral}}$ is still a negative definite function, even if $\vartheta = 0$ or $\varphi = 0$. In the subsequent analysis, we identify additional functional conditions that ensure the validity of this Lyapunov candidate and demonstrate how to choose γ_1 and γ_2 so that V_{Integral} is radially unbounded and $\dot{V}_{\text{Integral}}$ is a negative definite function.

To determine the functional conditions on γ_1 and γ_2, we take the derivative of the Lyapunov function along the trajectories

$$\dot{V}_{\text{Integral}}(\vartheta,\varphi) = -\gamma_1 \circ J(\vartheta) \cdot \nabla J(\vartheta)^T K(\varphi) \nabla J(\vartheta) + \gamma_2\left(|\varphi|^2\right) \cdot \varphi^T\left(q(\vartheta) - \varphi\right) \tag{4.152}$$

$$\dot{V}_{\text{Integral}}(\vartheta,\varphi) \leq -\gamma_1 \circ J(\vartheta) \cdot \lambda_{\min}(K(\varphi)) \cdot |\nabla J(\vartheta)|^2 - \gamma_2(|\varphi|^2) \cdot \left[|\varphi|^2 - \varphi^T q(\vartheta)\right] \tag{4.153}$$

By using Young's inequality, we obtain

$$\begin{aligned}\dot{V}_{\text{Integral}}(\vartheta,\varphi) \leq -\Big[&\gamma_1 \circ J(\vartheta) \cdot \lambda_{\min}(K(\varphi)) \cdot |\nabla J(\vartheta)|^2 \\ &-\frac{1}{2}|q(\vartheta)|^2 \cdot \gamma_2\left(|\varphi|^2\right)\Big] - \frac{1}{2}|\varphi|^2 \cdot \gamma_2\left(|\varphi|^2\right)\end{aligned} \tag{4.154}$$

The first condition is that

$$\gamma_1 \circ J(\vartheta) \cdot \lambda_{\min}(K(\varphi)) \cdot |\nabla J(\vartheta)|^2 > \frac{1}{2}|q(\vartheta)|^2 \cdot \gamma_2(|\varphi|^2) \tag{4.155}$$

whenever $\vartheta \neq 0$ for $\dot{V}_{\text{Integral}}$ to be a negative definite function. To achieve this, we choose γ_1 and γ_2 such that

$$|\nabla J(\vartheta)|^2 \cdot \gamma_1 \circ J(\vartheta) > \frac{1}{2}|q(\vartheta)|_2^2 \quad \forall \vartheta \neq 0 \tag{4.156}$$

and

$$\lambda_{\min}(K(\varphi)) \geq \gamma_2(|\varphi|^2) > 0 \tag{4.157}$$

Using Assumption 4.9, the RHS of (4.156) can be relaxed to

$$\frac{1}{2}|q(\vartheta)|_2^2 \leq \frac{1}{8}\rho_q(|\vartheta|)^2 \leq \frac{1}{8}\left(\alpha_q \circ \alpha_{\vartheta,1}^{-1} \circ J(\vartheta)\right)^2 \tag{4.158}$$

According to (4.149), we relax the LHS of (4.156) to

$$|\nabla J(\vartheta)|^2 \cdot \gamma_1 \circ J(\vartheta) \geq b_g \circ J(\vartheta) \cdot \gamma_1 \circ J(\vartheta) \tag{4.159}$$

With the two simplifications of (4.156), the condition of (4.156) is satisfied when

$$b_g \circ J(\vartheta) \cdot \gamma_1 \circ J(\vartheta) > \frac{1}{8}\left(\rho_q \circ \alpha_{\vartheta,1}^{-1} \circ J(\vartheta)\right)^2 \quad \forall \vartheta \neq 0 \tag{4.160}$$

Now we have a clear choice of γ_1 with

$$\gamma_1(r) = \frac{\left(\rho_q \circ \alpha_{\vartheta,1}^{-1}(r)\right)^2}{4b_g(r)} + 1 \tag{4.161}$$

where the addition of one in (4.161) ensures P1 is satisfied for γ_1. P2 is also satisfied for γ_1 as γ_1 is constructed with positive definite functions.

While we have given a γ_1 defined based on the bounds given in Assumptions 4.4 and 4.9, the function γ_2 still depends on the undefined behavior of $K(\varphi)$. This means that it may not be possible to use V_{Integral} to prove that the system is GAS for any given adaptive gradient method but it was possible for the RMSprop and AdaHessian method.

4.4.2 RMSprop Extremum Seeking Control

The RMSprop method seeks to have a uniform convergence speed in all parameter directions by scaling each gradient component by an estimate of the inverse gradient component magnitude. This uniform convergence speed is an attractive property to try and achieve with ESCs, thus we propose a novel RMSprop ESC (RMSpESC). The continuous time differential equations defining RMSprop can be found in [18] but the ESC version of the RMSprop (RMSpESC) simply substitutes the perturbation-based estimates in place of the true values. For a parameter space of dimension n, the RMSpESC is defined variable-wise by the $2n+1$ equations

$$\frac{d}{dt}\hat{\theta}_i = -k\frac{\hat{g}_i\left(t,\hat{\theta},\xi\right)}{\sqrt{\hat{v}_i}+\varepsilon} \tag{4.162}$$

$$\frac{d}{dt}\hat{v}_i = \omega_{l,i}\left(\hat{g}_i(t,\hat{\theta},\xi)^2 - \hat{v}_i\right) \tag{4.163}$$

$$\frac{d}{dt}\xi = = \omega_\xi\left(y-\xi\right) \tag{4.164}$$

for every $i \in \{1,\ 2,\ \ldots,n\}$. Here $\hat{g}_i$ is the ith element of the gradient estimate $\hat{g}$, $\hat{v}_i$ is the low pass filter estimate of $\hat{g}_i^2$, and ξ is a washout filter state. The $\hat{g}_i$ is formed using the output of the washout filter and is defined as

$$\hat{g}_i\left(t,\hat{\theta},\xi\right) = m_i(t)\left(J\left(\hat{\theta}+p(t)\right) - \xi\right) \tag{4.165}$$

where m_i is a sinusoidal signal defined as $m_i(t) = \frac{2}{a_i}\sin(\omega r_i t)$.

We need to know the corresponding average system for the RMSpESC in order to prove the semiglobal practical uniform asymptotic stability (sGPUAS) of the RMSpESC, where practical stability is defined in other previous works [13, 42]. We later prove that this average system is autonomous and globally uniformly asymptotically stable (GUAS). The corresponding average system is

$$\frac{d}{dt}\hat{\bar{\theta}}_i = -k\frac{\hat{\bar{g}}_i\left(\hat{\bar{\theta}}\right)}{\sqrt{\hat{\bar{v}}_i+\varepsilon}} \tag{4.166}$$

$$\frac{d}{dt}\hat{\bar{v}}_i = \omega_{l,i}\left(\overline{\hat{g}_i^2}\left(\hat{\bar{\theta}},\bar{\xi}\right)-\hat{\bar{v}}_i\right) \tag{4.167}$$

$$\frac{d}{dt}\bar{\xi} = \omega_{\xi}\left(\bar{J}\left(\hat{\bar{\theta}}\right)-\bar{\xi}\right) \tag{4.168}$$

where

$$\bar{J}\left(\hat{\bar{\theta}}\right) = \frac{1}{T}\int_t^{t+T} J\left(\hat{\bar{\theta}}+p(t+\tau)\right)d\tau \tag{4.169}$$

$$\hat{\bar{g}}_i\left(\hat{\bar{\theta}}\right) = \frac{1}{T}\int_t^{t+T} \hat{g}_i\left(t+\tau,\hat{\bar{\theta}},\cdot\right)d\tau \tag{4.170}$$

$$\overline{\hat{g}_i^2}\left(\hat{\bar{\theta}},\bar{\xi}\right) = \frac{1}{T}\int_t^{t+T} \hat{g}_i\left(t+\tau,\hat{\bar{\theta}},\bar{\xi}\right)^2 d\tau \tag{4.171}$$

An important feature of the average system is that $\hat{\bar{g}}$ is independent of $\bar{\xi}$ since $\int_t^{t+T} m(t+\tau)\bar{\xi}d\tau = 0$. Other notable properties of the average system is the convergence of $\hat{\bar{g}} \to \nabla J$ as $a_0 \to 0$ and the functions $\bar{J}, \overline{\hat{g}_i^2} \in \mathscr{C}^1$, as proven in Section 4.4.2.1.

The equilibrium of (4.166) is θ_* which then implies that the equilibrium for the system defined by (4.166)-(4.168) is $(\theta_*, v_*, \xi_*) \in \mathbb{R}^n \times \mathbb{R}^n_{\geq 0} \times \mathbb{R}$ where the equilibrium components ξ_* and $v_{i,*}$ are $\xi_* = \bar{J}(\theta_*)$ and $v_{i,*} = \overline{\hat{g}_i^2}(\theta_*, \xi_*)$. While the domain as stated is not unbounded, one could modify (4.162) to take the absolute value of $\hat{v}_i$ inside the square root and there would be no issues using theorems requiring unbounded domains. This is not done in this work to stay consistent with previous literature of the RMSprop such as [18]. In the limit as $a_0 \to 0$, $\bar{J} \to J$ by Assumption 4.1 (and hence $\xi_* \to J(\theta_*)$) and $v_{i,*} \to 0$ by Corollary 4.1. The equilibrium is then $(\theta_*, 0, J(\theta_*))$ when considering both a_0 and ω^{-1} as small parameters.

4.4.2.1 Properties

We give some general properties for some the estimates in RMSpESC.

Corollary 4.1. *The equilibrium of the average gradient magnitude squared estimate $v_{i,*} \to 0$ as $a_0 \to 0$.*

Proof. It is already known that $\bar{\xi}$ will remove the constant terms in $\bar{J}$ at the equilibrium. Thus one can use (4.101) and Assumption 4.1 to get

$$\left|\overline{\hat{g}_i^2}(\theta_*)\right| \leq \frac{4a_0^2}{a_i^2}\varepsilon^2 \tag{4.172}$$

for any $\varepsilon > 0$ where $a_0 < \delta$, where ε, δ are from the $\varepsilon - \delta$ continuity of ∇J. □

Remark 4.5. *Given a cost function $J : \mathbb{R}^n \to \mathbb{R}$ that satisfies 4.1 and the sinusoidal signals p and m, the functions $\overline{J}$, $\overline{g}_i$, and $\overline{\hat{g}_i{}^2}$ for $i = 1, \ldots, n$ are all continuously differentiable with respect to $\hat{\overline{\theta}}$. Their derivatives are simply differentiation of* (4.169)-(4.171).

4.4.2.2 Local Practical Stability

To illustrate the usefulness of the RMSpESC, we first look at the local stability properties of the system. Consider the scalar cost function

$$J(\theta) = J_* + \frac{1}{2}H\theta^2, \quad H > 0. \tag{4.173}$$

When we perturb θ and "freeze" the variables $\hat{\theta}$, $\hat{v}$, and ξ for the averaging process, we find that J asymptotically approaches a periodic signal

$$J(t,\hat{\theta}) = J_0 + \frac{1}{2}H\left(\hat{\theta}^2 + 2a\hat{\theta}\sin(\omega t) + a^2\sin^2(\omega t)\right) \tag{4.174}$$

This is better written in the form of a Fourier series

$$J(t,\hat{\theta}) = b_0(\hat{\theta}) + b_1(\hat{\theta})\sin(\omega t) + b_2\cos(2\omega t) \tag{4.175}$$

where the coefficients are

$$b_0(\hat{\theta}) = \left(J_* + \frac{1}{2}H\hat{\theta}^2 + \frac{a^2}{4}H\right) \tag{4.176}$$

$$b_1(\hat{\theta}) = aH\hat{\theta} \tag{4.177}$$

$$b_2 = -\frac{a^2}{4}H \tag{4.178}$$

This conveniently simplifies the calculation of the average gradient and average squared magnitude of the gradient.

$$\overline{g}\left(\hat{\overline{\theta}}\right) = \frac{1}{a}b_1\left(\hat{\overline{\theta}}\right) = H\hat{\overline{\theta}} \tag{4.179}$$

$$\overline{\hat{g}_i{}^2}\left(\hat{\overline{\theta}},\overline{\xi}\right) = \frac{4}{a^2}\left[\frac{1}{2}\left(b_0\left(\hat{\overline{\theta}}\right) - \frac{1}{2}b_2 - \xi\right)^2 + \frac{3}{2}\left(\frac{1}{2}b_1\left(\hat{\overline{\theta}}\right)\right)^2 + \frac{1}{2}\left(\frac{1}{2}b_2\right)^2\right] \tag{4.180}$$

Note that the washout filter does not influence the average gradient but only has influence on the average squared gradient magnitude. However, it is revealed the importance of the washout filter for the RMSpESC, as the filter prevents the local convergence rate from depending on the extremum of J. Without the filter, larger magnitudes of the extremum would slow down the convergence rate.

The equilibrium point for $\hat{\overline{\theta}}$ can be easily seen to be $\theta_* = 0$ by examination of $\hat{\overline{\theta}}$ in (4.179). Therefore the equilibrium for $\overline{\xi}$ and $\hat{\overline{v}}$ are $\xi_* = b_0(0)$ and $v_* = b_2^2/a^2 = a^2H^2/16$. Changing to the error variables $\tilde{\hat{\overline{\theta}}} = \hat{\overline{\theta}} - \theta_*$, $\tilde{\overline{\xi}} = \overline{\xi} - \xi_*$, and $\tilde{\hat{\overline{v}}} = \hat{\overline{v}} - v_*$, the new dynamics in this error coordinate system is

$$\frac{d}{dt}\tilde{\hat{\overline{\theta}}}_i = -k\frac{\hat{\overline{g}}_i\left(\theta_* + \tilde{\hat{\overline{\theta}}}\right)}{\sqrt{v_{i,*} + \tilde{\hat{\overline{v}}}_i + \varepsilon}} \tag{4.181}$$

$$\frac{d}{dt}\tilde{\bar{v}}_i = \omega_{l,i}\left(\widetilde{\overline{\hat{g}_i^2}}\left(\tilde{\bar{\theta}},\tilde{\bar{\xi}}\right) - \tilde{\bar{v}}_i\right) \tag{4.182}$$

$$\frac{d}{dt}\tilde{\bar{\xi}} = \omega_\xi\left(\tilde{\bar{J}}\left(\tilde{\bar{\theta}}\right) - \tilde{\bar{\xi}}\right) \tag{4.183}$$

where

$$\hat{\bar{g}}\left(\theta_* + \tilde{\bar{\theta}}\right) = H\tilde{\bar{\theta}} \tag{4.184}$$

$$\tilde{\bar{J}}\left(\tilde{\bar{\theta}}\right) = \bar{J}\left(\theta_* + \tilde{\bar{\theta}}\right) - \bar{J}\left(\theta_*\right) \tag{4.185}$$

$$\widetilde{\overline{\hat{g}_i^2}}\left(\tilde{\bar{\theta}},\tilde{\bar{\xi}}\right) = \overline{\hat{g}_i^2}\left(\theta_* + \tilde{\bar{\theta}}, \xi_* + \tilde{\bar{\xi}}\right) - \overline{\hat{g}_i^2}\left(\theta_*, \xi_*\right) \tag{4.186}$$

Dropping the unnecessary i subscript and calculating the linearization of the system about the origin gives

$$\frac{d}{dt}\begin{bmatrix}\tilde{\bar{\theta}}\\ \tilde{\bar{\xi}}\\ \tilde{\bar{v}}\end{bmatrix} = \begin{bmatrix}\frac{-kH}{\frac{1}{4}|a|H+\varepsilon} & 0 & 0\\ 0 & -\omega_\xi & 0\\ 0 & -\frac{1}{2}\omega_l H & -\omega_l\end{bmatrix}\begin{bmatrix}\tilde{\bar{\theta}}\\ \tilde{\bar{\xi}}\\ \tilde{\bar{v}}\end{bmatrix} \tag{4.187}$$

This indicates that the Jacobian is Hurwitz as the matrix in (4.187) is a lower triangular matrix with strictly negative terms on the main diagonal. Thus the RMSpESC for the quadratic case is exponentially practically stable by [20, Th 10.4]. Due to the dependence of $\overline{\hat{g}_i^2}$ on $\bar{\xi}$, the matrix is not strictly diagonal.

The eigenvalues of the Jacobian in (4.187) are the diagonal elements. Of the three, two are directly assignable by means of the filter gains. The last eigenvalue, which corresponds to the first element of the Jacobian, depends on an unknown H. Although the system designer does have a choice of ε and a which influence the behavior of the eigenvalue depending on H. When the curvature of J becomes steep ($H \to \infty$), unknown eigenvalue approaches a lower limit of $-4k/|a|$. In the other extreme when the curvature of J becomes shallow ($H \to 0$), the eigenvalue asymptotically approaches $-(k/\varepsilon)H$. This allows for some correction to convergence rate over a traditional GESC since ε is chosen as a small factor.

Given the behavior of first eigenvalue in the limits of H, one can see that the RMSpESC is somewhat between the GESC and NESC algorithms. It still shows a convergence rate dependent on H when $H \ll 1$ like the GESC but a convergence rate that is almost independent of H like the NESC when $H \gg 1$. Things are more uncertain in multivariable case but we can still prove a general local exponential practical stability.

Theorem 4.6 (RMSpropESC Local Practical Stability)**.** *For a cost function $J : \mathbb{R}^n \to \mathbb{R}$ satisfying $J \in \mathscr{C}^2$, the Assumptions 4.2-4.4, and $\nabla^2 J(\theta_*)$ being strictly positive definite, the RMSpESC defined by Eqs (4.162)-(4.164) is locally practically exponential stable to $(\theta_*, 0, J(\theta_*))$ with the small parameter vector $\left(a_0, \omega^{-1}\right)$.*

Proof of Theorem 4.6. We can consider the general local quadratic Taylor series about θ_*

$$J(\theta) = J_* + \frac{1}{2}(\theta - \theta_*)^T H (\theta - \theta_*), \quad H \succ 0 \tag{4.188}$$

which is valid as $a_0 \to 0$. Linearizing about the equilibrium point $(\theta_*, 0, J(\theta_*))$ results in

$$\frac{d}{dt}\begin{bmatrix}\tilde{\tilde{\theta}} \\ \tilde{\tilde{\xi}} \\ \tilde{\tilde{v}}\end{bmatrix} = \begin{bmatrix} -\left(\frac{k}{\varepsilon}\right)H & 0 & 0 \\ \star & -\omega_\xi & 0 \\ \star & \star & -\operatorname{diag}(\omega_l)\end{bmatrix}\begin{bmatrix}\tilde{\tilde{\theta}} \\ \tilde{\tilde{\xi}} \\ \tilde{\tilde{v}}\end{bmatrix} \tag{4.189}$$

where the Jacobian is a block lower triangular matrix with $\star$ denoting potentially non-zero off-diagonal elements and $\operatorname{diag}(\omega_l)$ is strictly diagonal matrix with $\omega_{l,i}$ on the ith diagonal entry. Since eigenvalues of a block lower triangular matrix are the eigenvalues of its diagonal blocks, the Jacobian is Hurwitz as all of its diagonal blocks are Hurwitz. Thus the average system is local uniformly asymptotically stable [20, Th 4.7] and consequently the original system is locally practically exponentially stable [20, Th 10.4].

If the cost function is a quadratic, not just in the local sense but globally, then for any $a_0 > 0$ and a potentially nonzero vector v_*, the Jacobian of the system is

$$\begin{bmatrix} -k\left(\operatorname{diag}(v_*)^{1/2} + \varepsilon I\right)^{-1} H & 0 & 0 \\ \star & -\omega_\xi & 0 \\ \star & \star & -\operatorname{diag}(\omega_l)\end{bmatrix} \tag{4.190}$$

Since left multiplying by a strictly diagonal matrix does not change the stability of a linear system, the first triangular block $-\left(\operatorname{diag}(v_*)^{1/2} + \varepsilon I\right)^{-1} H$ is still Hurwitz with different eigenvalues than H. Subsequently the Jacobian is also Hurwitz so the average system is GUAS [20, Th 4.5] and thus the original system is globally practically exponentially stable [20, Th 10.4]. □

4.4.2.3 Semiglobal Practical Stability

For semiglobal practical stability, we analyze the average system since an autonomous average system which is GUAS to the origin implies that the original system is sGPUAS [21]. However, this still leads us with the task of showing that the average system is GUAS. We will use Lyapunov stability but still need to construct a suitable Lyapunov function which is not in general an easy task, although we wish to use the following properties outlined in Lemma 4.4.

Lemma 4.4. *For RMSpESC acting on a cost function J that satisfies Assumptions 4.1-4.4, the level sets of*

$$V_\theta\left(\tilde{\tilde{\theta}}\right) = J\left(\tilde{\tilde{\theta}} + \theta_*\right) - J(\theta_*) \tag{4.191}$$

are compact and connected. Additionally any non-empty level set of V_θ which is not a singleton set can be made forward invariant using a sufficiently small a_0.

Proof. The compact property of the level sets of V_θ comes from Assumptions 4.1 and 4.4 since Assumption 4.1 ensures J has no singularities and Assumption 4.4 ensures that there is no finite upper limit for any sequence $\left\{V_\theta\left(\tilde{\tilde{\theta}}_k\right)\right\}_{k=1}^{\infty}$ where $\left\|\tilde{\tilde{\theta}}_k\right\| \to \infty$ as $k \to \infty$. The forward invariance property is similarly easy to show by taking the derivative of (4.191) to get

$$\frac{d}{dt}V_\theta\left(\tilde{\tilde{\theta}}\right) \le \sum_{i=1}^{n} -k\frac{\left(\frac{\partial J}{\partial \theta_i}\left(\theta_* + \tilde{\tilde{\theta}}\right)^2 - \eta_i\left|\frac{\partial J}{\partial \theta_i}\left(\theta_* + \tilde{\tilde{\theta}}\right)\right|\right)}{\sqrt{\hat{\tilde{v}}_i} + \varepsilon} \tag{4.192}$$

where $\eta_i > 0$ can all be made arbitrarily small by the convergence of $\hat{\tilde{g}} \to \nabla J$ from Section 4.1. Thus for any level set of V_θ and a set of considered $\hat{\tilde{v}}$, there is a sufficiently small a_0 such that the level set is forward invariant for all considered $\hat{\tilde{v}}$ by ensuring $\frac{d}{dt}V_\theta < 0$.

The last property of connectedness is the only property that requires a more in-depth proof which is done by contradiction. We start by assuming the negative that the level sets are not always connected. Then there exists $c_\theta > 0$ such that the level set is comprised of at least two disjoint, connected, compact sets. Specifically, $\exists c_\theta > 0$ where one of these sets is a singleton set, containing only a point $\tilde{\bar{\theta}} \neq 0$. The gradient at this point must be zero by Assumption 4.1 since it is a local minimum but that contradicts Assumption 4.2. Therefore the level sets of V_θ must always be connected sets. □

In light of Lemma 4.4, we can also Remark 4.6.

Remark 4.6. *Given the conditions of Lemma 4.4, the error states $\tilde{\bar{\xi}}$ and $\tilde{\bar{v}}_i$ are bounded.*

Proof. Since the dynamics of $\tilde{\bar{\xi}}$ is linear, we can write its solution explicitly as

$$\tilde{\bar{\xi}}(t) = e^{-\omega_\xi t}\tilde{\bar{\xi}}_0 + \int_0^t \omega_\xi e^{\omega_\xi(\tau - t)}\tilde{\bar{J}}\left(\tilde{\bar{\theta}}(\tau)\right)d\tau \tag{4.193}$$

where $\tilde{\bar{\xi}}_0 = \tilde{\bar{\xi}}(0)$. Since $\tilde{\bar{J}}$ is a continuous function by Remark 4.5 and level sets of V_θ are compact and forward invariant by Lemma 4.4, then the images of the level sets of V_θ through $\tilde{\bar{J}}$ are also compact [43, Prop 1.5.2] and forward invariant. Hence we can bound $\left|\tilde{\bar{\xi}}(t)\right|$ by

$$\left|\tilde{\bar{\xi}}(t)\right| \leq e^{-\omega_\xi t}\left|\tilde{\bar{\xi}}_0\right| + \left(1 - e^{-\omega_\xi t}\right) r_\xi \leq \max\left\{\left|\tilde{\bar{\xi}}_0\right|, r_\xi\right\} \tag{4.194}$$

where r_ξ is the radius of a bounding ball for the level sets of V_θ through $\tilde{\bar{J}}$. Equation (4.194) shows that $\tilde{\bar{\xi}}$ is bounded. The steps to show that $\tilde{\bar{v}}_i$ is bounded are the same as those for $\tilde{\bar{\xi}}$ except we use the fact that the image of the level sets of V_θ and a ball $\mathscr{B}_{\max\left\{\left|\tilde{\bar{\xi}}_0\right|, r_\xi\right\}}$ through $\overline{\hat{g}_i{}^2}$ is compact and forward invariant. □

One insight that we can glean from Remark 4.6 is that V_θ is sGPUAS to 0 with respect to the small parameter a_0 since $\hat{\tilde{v}}_i$ is bounded. However, as this work deals with practical stability so the nominal average system arrived in the limit as $a_0 \to 0$ treats $\frac{d}{dt}V_\theta$ as a negative definite function and consequently θ_* as GUAS [20, Th 4.9]. Another is that as (4.163) and (4.164) are low pass filters and therefore attracted to the images of $\tilde{\bar{J}}$ and $\overline{\hat{g}_i{}^2}$. Combining these two insights gives the central idea for constructing a Lyapunov function to prove Theorem 4.7, that *the error variables of the low pass filters are being attracted to bounding balls which are shrinking to zero as $t \to \infty$.*

Theorem 4.7 (RMSpropESC is sGPUAS)**.** *For a cost function $J : \mathbb{R}^n \to \mathbb{R}$ satisfying the Assumptions 4.1-4.4, the RMSpESC defined by Eqs (4.162)-(4.164) is sGPUAS to the point $(\theta_*, 0, J(\theta_*))$ with the small parameter vector $\left(a_0, \omega^{-1}\right)$.*

Proof of Theorem 4.7. The Lyapunov function in the error coordinate system is

$$V\left(\tilde{\bar{\theta}}, \tilde{\bar{\xi}}, \tilde{\bar{v}}\right) = V_\theta\left(\tilde{\bar{\theta}}\right) + V_\xi\left(\tilde{\bar{\xi}}; \tilde{\bar{\theta}}\right) + \sum_{i=1}^{n} V_{v_i}\left(\tilde{\bar{v}}_i; \tilde{\bar{\theta}}, \tilde{\bar{\xi}}\right) \tag{4.195}$$

where V_θ was defined earlier in (4.191) and

$$V_\xi\left(\tilde{\tilde{\xi}};\tilde{\tilde{\theta}}\right) = \max\left\{r_\xi\left(V_\theta\left(\tilde{\tilde{\theta}}\right)\right), \left|\tilde{\tilde{\xi}}\right|\right\} \tag{4.196}$$

$$V_{v_i}\left(\tilde{\tilde{v}}_i;\tilde{\tilde{\theta}},\tilde{\tilde{\xi}}\right) = \max\left\{r_{v_i}\left(V_\theta\left(\tilde{\tilde{\theta}}\right), V_\xi\left(\tilde{\tilde{\xi}};\tilde{\tilde{\theta}}\right)\right), \left|\tilde{\tilde{v}}_i\right|\right\} \tag{4.197}$$

The variables r_ξ and r_v are radii of bounding balls for the images of the functions $\tilde{\tilde{J}}$ and $\widetilde{\widetilde{\hat{g}_i{}^2}}$ along the forward trajectories. The formulation behind (4.195) is to penalize: the suboptimality of $J\left(\hat{\hat{\theta}}\right)$; the minimum radii required for the bounding balls of the images of $\tilde{\tilde{J}}$ and $\widetilde{\widetilde{\hat{g}_i{}^2}}$; and the distance that variables $\tilde{\tilde{\xi}}$ and $\tilde{\tilde{v}}$ are from the bounding balls. Note that (4.196) and (4.197) each include both the penalties for the size of the bounding balls and the error variables away from these balls using the max function as, for example, $\left|\tilde{\tilde{\xi}}\right| = \left\|\tilde{\tilde{\xi}}\right\|_{\mathscr{B}_{r_\xi}} + r_\xi$ whenever $\left|\tilde{\tilde{\xi}}\right| > r_\xi$.

The minimum radii of the bounding balls can be expressed as solutions to the optimization problems of

$$r_\xi\left(c_\theta\right) = \max_{\phi\in\mathbb{R}^n}\left|\tilde{\tilde{J}}_0(\phi)\right| \tag{4.198}$$

$$\text{s.t. } V_\theta\left(\phi\right) \le c_\theta \tag{4.199}$$

and

$$r_{v_i}\left(c_\theta, c_\xi\right) = \max_{\phi\in\mathbb{R}^n,\eta\in\mathbb{R}}\left|\widetilde{\widetilde{\hat{g}_i{}^2}}(\phi,\eta)\right| \tag{4.200}$$

$$\text{s.t. } V_\theta\left(\phi\right) \le c_\theta \tag{4.201}$$

$$V_\xi\left(\eta;\phi\right) \le c_\xi \tag{4.202}$$

The connected, compact sublevel sets of V_θ and the differentiability of $\tilde{\tilde{J}}$ and $\widetilde{\widetilde{\hat{g}_i{}^2}}$ with respect to θ means that r_ξ and r_{v_i} are all continuous, monotonically increasing functions. However, r_ξ and r_{v_i} are not expected to be continuously differentiable everywhere. Hence, a more general notion of derivatives are the Dini derivatives, particularly the upper right Dini derivative D^+ and the lower left Dini derivative D_- [44], are used to differentiate the radii. The Dini deriviates are found by using first order sensitivity analysis of the Karush-Kuhn-Tucker (KKT) optimality conditions [34, Ch 5.6]. Starting with r_ξ, the Lagrangian of the standard minimization problem for (4.198) and (4.199) is

$$\mathscr{L}_\xi\left(\tilde{\tilde{\theta}},\lambda\right) = -\left|\tilde{\tilde{J}}\left(\tilde{\tilde{\theta}}\right)\right| + \lambda\left(V_\theta\left(\tilde{\tilde{\theta}}\right) - c_\theta\right) \tag{4.203}$$

where $\lambda \ge 0$ is a Lagrange multiplier for the problem. The constraint sensitivity of the Lagrangian at optimal points of this function is $\frac{\partial\mathscr{L}}{\partial c_\theta} = -\frac{\partial r_\xi}{\partial c_\theta} = -\lambda$. Thus we can write the upper right Dini derivatives of r_ξ as

$$\begin{aligned} D^+ r_\xi\left(c_\theta\right) = \max\Big\{\lambda \;\Big|\; &\tilde{\tilde{\theta}} \in \mathbb{R}^n,\ \lambda \ge 0,\ \left|\tilde{\tilde{J}}\left(\tilde{\tilde{\theta}}\right)\right| = r\left(c_\theta\right), \\ &V_\theta\left(\tilde{\tilde{\theta}}\right) \le c_\theta, -\nabla\left|\tilde{\tilde{J}}\left(\theta\right)\right| + \lambda\nabla V_\theta\left(\tilde{\tilde{\theta}}\right) = 0\Big\} \end{aligned} \tag{4.204}$$

for $c_\theta > 0$. The lower left Dini derivative is the same as in (4.204) except it is a minimization rather than a maximization. The procedure to find the various Dini derivatives for r_{v_i} follows in nearly the same way. One of the upper right Dini derivatives is

$$D^+ r_{v_i}\left((c_\theta, c_\xi), e_{c_\theta}\right) = \max\Big\{\lambda_\theta \;\Big|\; \phi \in \mathbb{R}^n,\ \eta \in \mathbb{R},\ \lambda_\theta, \lambda_\xi \in \mathbb{R}_{\geq 0},\ V_\theta(\phi) \leq c_\theta,\ V_\xi\left(\eta; V_\theta(\phi)\right) \leq c_\xi,$$
$$\left|\widetilde{\hat{g}_i{}^2}(\phi, \eta)\right| = r_{v_i}(c_\theta, c_\xi),\ -\nabla\left|\widetilde{\hat{g}_i{}^2}(\phi, \eta)\right| + \lambda_\theta\left(V_\theta(\phi) - c_\theta\right) + \lambda_\xi\left(V_\xi\left(\eta; V_\theta(\phi)\right) - c_\xi\right) = 0\Big\} \tag{4.205}$$

while the other is similar, maximizing the other Lagrange multiplier at the extremum points. The lower left Dini derivatives in those directions can be found by replacing the maximization with a minimization. As the direction in the given Dini derivatives does not appear in the variable of the set in (4.205), the Dini derivatives in any other direction $d = (d_\theta, d_\xi)$ satisfying $d_\theta, d_\xi \geq 0$ is simply a linear combination of the Dini derivatives given, e.g.,

$$D^+ r_{v_i}\left((c_\theta, c_\xi), d\right) = d_\theta D^+ r_{v_i}\left((c_\theta, c_\xi), e_{c_\theta}\right) + d_\xi D^+ r_{v_i}\left((c_\theta, c_\xi), e_{c_\xi}\right) \tag{4.206}$$

Having established Dini derivatives for bounding ball radii, we now find the Lie derivative of V in (4.195). We have already established that the component $\frac{d}{dt}V_\theta$ is a negative definite function with respect to its argument in the nominal average system where $a_0 \to 0$. The other components, V_ξ and V_{v_i}, have Lie derivatives of

$$\frac{dV_\xi}{dt} = \begin{cases} \operatorname{sgn}\left(\tilde{\tilde{\xi}}\right) \cdot \dfrac{d\tilde{\tilde{\xi}}}{dt} = -\left|\dfrac{d\tilde{\tilde{\xi}}}{dt}\right| & \left|\tilde{\tilde{\xi}}\right| > r_\xi \quad (4.207a) \\ \dfrac{dr_\xi}{dt} & \left|\tilde{\tilde{\xi}}\right| < r_\xi \quad (4.207b) \\ \max\left\{\dfrac{dr_\xi}{dt}, -\left|\dfrac{d\tilde{\tilde{\xi}}}{dt}\right|\right\} & \left|\tilde{\tilde{\xi}}\right| = r_\xi \quad (4.207c) \end{cases}$$

and

$$\frac{dV_{v_i}}{dt} = \begin{cases} -\left|\dfrac{d\tilde{\tilde{v}}_i}{dt}\right| & \left|\tilde{\tilde{v}}_i\right| > r_{v_i} \quad (4.208a) \\ \dfrac{dr_{v_i}}{dt} & \left|\tilde{\tilde{v}}_i\right| < r_{v_i} \quad (4.208b) \\ \max\left\{\dfrac{dr_{v_i}}{dt}, -\left|\dfrac{d\tilde{\tilde{v}}_i}{dt}\right|\right\} & \left|\tilde{\tilde{v}}_i\right| = r_{v_i} \quad (4.208c) \end{cases}$$

We need to establish that these Lie derivatives are negative semi-definite functions with respect to their arguments. We shall individually examine the cases of the Lie derivatives, namely whenever the error variable is outside, in the interior, or on the boundary of the bounding ball. Whenever the error variable is outside of the bounding ball as in (4.207a) and (4.208a), the Lie derivative of the magnitude of the error is negative, for example in $\tilde{\tilde{\xi}}$ where

$$\frac{d}{dt}\left|\tilde{\tilde{\xi}}\right| = -\omega_\xi \operatorname{sgn}\left(\tilde{\tilde{\xi}}\right)\left(\tilde{\tilde{\xi}} - \tilde{\tilde{J}}\left(\tilde{\tilde{\theta}}\right)\right) \tag{4.209}$$

$$\leq -\omega_l \left(\left| \tilde{\tilde{\xi}} \right| - r_\xi \left(V_\theta \left(\tilde{\tilde{\theta}} \right) \right) \right) \leq 0, \text{ if } \left| \tilde{\tilde{\xi}} \right| \geq r_\xi \tag{4.210}$$

with equality only being possible whenever the error variable is on the boundary of the bounding ball ($\left| \tilde{\tilde{\xi}} \right| = r_\xi$). The other error filter states $\tilde{\tilde{v}}_i$ have similar arguments for $\frac{d}{dt} \left| \tilde{\tilde{v}}_i \right| \leq 0$ if $\left| \tilde{\tilde{v}}_i \right| \geq r_{v_i}$.

The other consideration is whenever the error variable is contained within the bounding ball. In such an instance, it is radii of the bounding balls which are the active element in the maximization in (4.196) and (4.197). The Lie derivative of r_ξ can be upper bounded by zero since

$$\frac{d}{dt} r_\xi \left(\tilde{\tilde{\theta}} \right) \leq -D_- r_\xi \left(V_\theta \left(\tilde{\tilde{\theta}} \right) \right) \cdot \frac{d}{dt} V_\theta \left(\tilde{\tilde{\theta}} \right) \leq 0 \tag{4.211}$$

and therefore $\frac{d}{dt} r_\xi$ is a negative semidefinite function with respect to its arguments. We also know that the condition (4.207c) is a maximization of two negative semidefinite functions and thus a negative semidefinite function itself. All components of $\frac{d}{dt} V_\xi$ are negative semidefinite functions which means $\frac{d}{dt} V_\xi$ must be as well. By a similar argument, $\frac{d}{dt} V_{v_i}$ is a negative semidefinite function as seen in

$$\begin{aligned} \frac{d}{dt} r_{v_i} \left(V_\theta \left(\tilde{\tilde{\theta}} \right), V_\xi \left(\tilde{\tilde{\xi}}; \tilde{\tilde{\theta}} \right) \right) \leq D_- r_{v_i} \left(\left(V_\theta \left(\tilde{\tilde{\theta}} \right), V_\xi \left(\tilde{\tilde{\xi}}; \tilde{\tilde{\theta}} \right) \right), e_{c_\theta} \right) \cdot \frac{d}{dt} V_\theta \left(\tilde{\tilde{\theta}} \right) \\ + D_- r_{v_i} \left(\left(V_\theta \left(\tilde{\tilde{\theta}} \right), V_\xi \left(\tilde{\tilde{\xi}}; \tilde{\tilde{\theta}} \right) \right), e_{c_\xi} \right) \cdot \frac{d}{dt} V_\xi \left(\tilde{\tilde{\xi}}; \tilde{\tilde{\theta}} \right) \leq 0 \end{aligned} \tag{4.212}$$

We have proven that $\frac{d}{dt} V_\theta$, $\frac{d}{dt} V_\xi$, and $\frac{d}{dt} V_{v_i}$ are all negative semidefinite functions, what remains to be shown is that the sum of all these negative semidefinite functions is a negative definite function. If $\frac{d}{dt} V$ is zero, then all of its component terms must also be zero. However, $\frac{d}{dt} V_\theta = 0$ implies that both $\tilde{\tilde{\theta}} = 0$ and $r_\xi(0) = 0$ which in turn implies $\tilde{\tilde{\xi}} = 0$ and $r_{v_i}(0,0) = 0$ meaning $\tilde{\tilde{v_i}} = 0$ for all $i = 1, \ldots, n$. Consequently, the only zero of $\frac{d}{dt} V$ is at the origin of the error coordinate system. Thus we can conclude that the origin of the error coordinate system is GUAS for the average RMSpESC system [20, Th 4.10] and sGPUAS for the oringal RMSpESC system [21]. $\square$

4.4.3 AdaGrad and AdaDelta

While RMSprop is an optimizer that is adaptive in the element-wise directions, i.e., in each of the n $\hat{\theta}_i$ directions, one might seek an optimizer that is adaptive in a multivariable sense so that it independent of the orthogonal coordinate systems selected. This is to say if the inertial coordinate system is rotated, then the trajectories of the seeker are merely rotated in space for AdaGrad and not warped as in the case of RMSprop. One of the first options presented is the AdaGrad optimizer

$$G_t = \sum_{\tau=1}^{t} g_\tau g_\tau^T \tag{4.213}$$

$$\hat{\theta}_{t+1} = \hat{\theta}_t - k \left(G + \eta I \right)^{-1/2} g_t \tag{4.214}$$

where g_t is the gradient obtained numerically. AdaGrad to seeks to update the gradient in a way that is responsive to new directions in the gradient. However, the current structure of these cumulative outer product G_t defined in (4.213) is not ideal for continuous time. It was designed so that the learning rate in the parameters would eventually decay to zero, which is not ideal for the proving uniform convergence of G to any specific point.

An alternative variation is AdaDelta

$$G_t = \rho_1 G_{t-1} + (1-\rho_1) g_\tau g_\tau^T \tag{4.215}$$

$$\hat{\theta}_{t+1} = \hat{\theta}_t - k\sqrt{\mathbb{E}\left[\Delta\hat{\theta}^2\right]_{t-1} + \eta} \cdot (G_t + \eta I)^{-1/2} g_t \tag{4.216}$$

$$\mathbb{E}\left[\Delta\hat{\theta}^2\right]_t = \rho_2 \mathbb{E}\left[\Delta\hat{\theta}^2\right]_{t-1} + (1-\rho_2)\left\|\hat{\theta}_t - \hat{\theta}_{t-1}\right\|^2 \tag{4.217}$$

which seeks to have the gradient outer product be low pass filtered but changes the step size in parameters adaptively based on the previous magnitude of parameter estimate updates. Not only is this responsive to changes in the gradient's direction but also accounts for the previous step size of the parameter estimate updates.

For the continuous time implementation, the proposed differential for a AdaGrad ESC (AGESC) equations are

$$\frac{d}{dt}\hat{\theta} = -k\hat{\Gamma}\hat{g} \tag{4.218}$$

$$\hat{\Gamma}\left(\frac{d}{dt}\hat{\Gamma}\right) + \left(\frac{d}{dt}\hat{\Gamma}\right)\hat{\Gamma} = \omega_l\left(\hat{\Gamma}^2 - \hat{\Gamma}^2\left(\hat{G} + \eta I\right)\hat{\Gamma}^2\right) \tag{4.219}$$

where $\hat{\Gamma}$ is the estimate of $\left(\hat{G} + \eta I\right)^{-1/2}$. While (4.219) has a similar structure to the low pass Ricatti inverse filter from NESC [6], as it is derived the same way by this study, it is a more complicated matrix differential equation. Future work on this method will analyze the stability of the differential equations and attempt to prove the practical stability of the system.

4.5 Accelerated Methods

While the baseline gradient descent algorithm tends to preliminary results for convergence, one would wish to improve convergence to optimal points of a function. One of the natural ways to potentially improve the convergence is to change (4.93) from a first order method in time to a second order or higher in time. The fundamental idea is that previous information of the gradient in the past could give insight into the large scale directions of improvement, so averaging out the small scale information would remove small scale noise.

4.5.1 Heavy Ball

The Heavy Ball ESC (HBESC) is defined by the differential equation

$$\frac{d^2}{dt^2}\hat{\theta} + \beta\frac{d}{dt}\hat{\theta} + k\hat{g}\left(t, \hat{\theta}\right) = 0 \tag{4.220}$$

which is asymptotic in the average system to

$$\frac{d^2}{dt^2}\hat{\bar{\theta}} + \beta\frac{d}{dt}\hat{\bar{\theta}} + k\nabla J\left(\hat{\bar{\theta}}\right) = 0 \tag{4.221}$$

as $a_0 \to 0$. Traditionally (4.221) has been treated as an integratable equation

$$\frac{d}{dt}\left(\frac{1}{2}\left\|\frac{d}{dt}\hat{\bar{\theta}}\right\|^2 + k\left(J\left(\hat{\bar{\theta}}\right) - J_*\right)\right) = -\beta\left\|\frac{d}{dt}\hat{\theta}\right\|^2 \tag{4.222}$$

to arrive at the conclusion that the average system was GAS by Theorem 3.1 and LaSalle's Invariance Theorem. Regretfully, this stability is not strong enough to use Theorem 3.2 to prove anything about the average system.

The Heavy Ball system is still worthy of study though as often previous literature will attempt to low pass filter the gradient estimate to remove the high frequency variations. This would give a set of differential equations

$$\frac{d}{dt}\hat{\theta} = -k\xi \tag{4.223}$$

$$\frac{d}{dt}\xi = \omega_l\left(\hat{g}\left(t,\hat{\theta}\right) - \xi\right) \tag{4.224}$$

which is actually equivalent to the singular equation

$$\frac{d^2}{dt^2}\hat{\theta} + \frac{\omega}{k}\frac{d}{dt}\hat{\theta} + \omega_l\hat{g}\left(t,\hat{\theta}\right) = 0 \tag{4.225}$$

which has the same form as (4.220).

The future work on this method involves attempting to prove stability of the Heavy Ball for ESC. While current literature does have some results for stability on very restricted J [25], this study seeks to extend the stability proofs of the Heavy Ball method to larger classes of J by removing the strict convexity requirement needed in [25].

4.5.2 Nesterov's Method

An alternative to Heavy Ball is Nesterov ESC (NesESC), which seeks to minimize a regret function [45]. This leads to a new differential equation

$$\frac{d^2}{dt^2}\hat{\theta} + \frac{3}{t+t_0}\frac{d}{dt}\hat{\theta} + \hat{g}\left(t,\hat{\theta}\right) = 0 \tag{4.226}$$

with a time-varying dampening coefficient. Initially this dampening coefficient is high as $0 < t_0 \ll 1$ but diminishes over time. This trend means that the initially overdamped system behaves like a gradient descent but the long term solution behaves like an underdamped system. This behavior allows for Nesterov's method to have sometimes better convergence rates to the standard gradient descent, particularly when $\nabla J \approx 0$. However, Nesterov's method is not an autonomous one so while the average system may be GUAS [46] , one cannot use Theorem 3.2 to prove that the original system is sGPUAS. From a physical perspective, the diminishing dampening means that the system is unable to properly dissipate energy added into the system by the perturbation-based exploration which leads to the potential loss of the practical convergence property. Thus for practical convergence, a modified differential equation can be proposed

$$\frac{d^2}{dt^2}\hat{\theta} + \min\left\{\frac{3}{t+t_0},\beta\right\}\frac{d}{dt}\hat{\theta} + \hat{g}\left(t,\hat{\theta}\right) = 0 \tag{4.227}$$

to ensure a minimum dampening coefficient.

Initial simulation studies have been performed for this method in Section 5.1, and while the performance of NesESC is better than the GESC around flat regions of the map, it often has lower convergence rate compared to HBESC. Unlike HBESC, NesESC cannot as well escape the local minimum if it starts near it. Therefore we decided to only focus on HBESC proof and implementation for now as NesESC does not seem to be competitive.

5. Operational Framework for ESC

Implementation of source seeking control on real-life unmanned vehicles (seeking real signal sources) even in convex signal maps, remains mostly unexplored [47–49, 24, 50]. This study is one of the first, to the authors' knowledge, to implement ESC on robots to seek real acoustic signal sources. The rest of the section is organized to: first show theoretical simulations of the various optimization algorithms introduced before illustrating how ESC is implemented onto real-life vehicles, specifically a UGV.

5.1 Theoretical Simulations

The simulation results presented in this section illustrate various apsects of the proposed methods. In this way we hope to illuminate some characteristics mentioned about the methods in the earlier Sections 4.4 and 4.5.

5.1.1 Adaptive Methods

Having proven the theoretical aspects of the RMSpESC, we will demonstrate some qualitative aspects. Consider the scalar quartic $J(\theta) = \frac{1}{24}\theta^4$, a function more difficult than the quadratic discussed earlier since the Hessian $\nabla^2 J(\theta_*) = 0$. We therefore cannot claim the results of Theorem 4.6 although the previous work [13] does prove the local PES for the GESC (and subsequently the RMSpESC) albeit with some unknown rate.

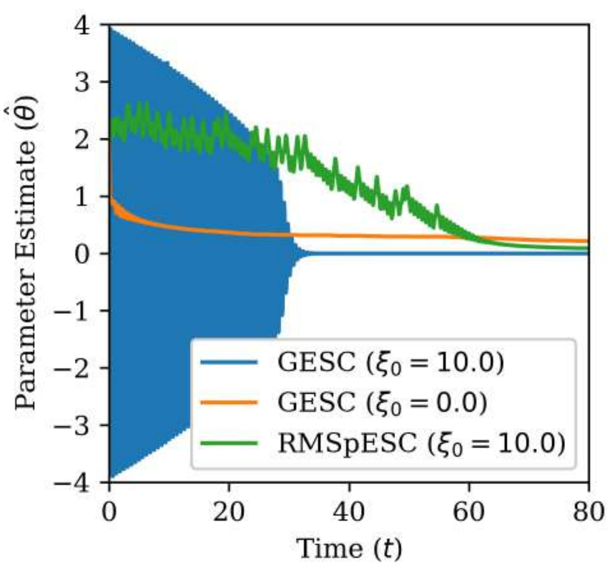


Figure 10: Parameter estimates for the example quartic scalar cost function with different initial washout filter states ξ_0. Simulation parameters were $a = 0.02$, $\omega = 10$, $\omega_\xi = 1$, $\omega_l = \frac{1}{4}$, $\varepsilon = 0.05$, and $k = 1$. The initial conditions for the other states were $\hat{\theta}_0 = 2$ and $\hat{v}_0 = 0.81$.

While the effects of the washout filter on gradient estimate were nonexistent in the average system, there is a noticeable, adverse impact on the trajectories in the original system. Observing the trajectories in Fig 10, it becomes apparent that the trajectories of the GESC can be rapidly changing with fast oscillations if the initial washout filter state is chosen poorly as would frequently occur when guessing for J which is *a priori* unknown. In contrast, the RMSpESC does not experience oscillations as severe and instead a more moderate convergence rate. In particular, the convergence is somewhat linear for the time between roughly 30 and 60 seconds which is what was desired of the RMSpESC. In addition, when all trajectories in Fig 10 reach a region of little variation in J, such as after 60 seconds, the RMSpESC will have a faster convergence rate than the GESC for appropriately selected simulation parameters. It is this combination of the reduction of fast oscillations observed in the GESC as well as the increased convergence rate around shallow minima that makes the RMSpESC so attractive for future work and implementation onto vehicles. Future simulations studies will examine further behavior of the RMSpESC, and eventually AGESC, on multivariate and non-convex cost functions.

5.1.2 Accelerated Methods

To study the HBESC and NesESC, we examine the cost function

$$J(\theta) = 1 - \frac{1}{\sqrt{\pi}} e^{-\theta^2} \tag{5.1}$$

which is similar to a standard Gaussian function. However, unlike the Gaussian, (5.1) has a minimum rather than a maximum. This is intentional as in real-world operations, one might try to maximize a sensor output but all of the theory developed for optimization is generally about minimization. Thus the simple practical solution for real sensors, as illustrated here, is simply to minimize the negative of the sensor output. Additionally, this cost function shares a similar behavior to acoustic signals as there are large variations around the extremum, where the source presumably is, but these variations quickly diminish as parameters move away from the extremum. Thus this cost function is a remarkably applicable test function for the case of acoustic source seeking.

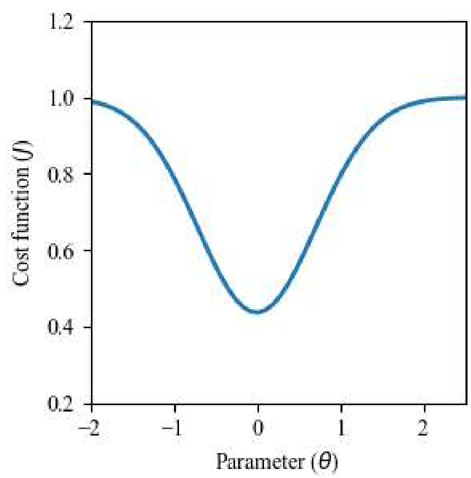


Figure 11: Gaussian-like cost function

The intended advantage of the accelerated methods is that they should be move in directions that were previously beneficial. In the case of the trajectories shown in Fig. 12, this takes the form of the HBESC and the NesESC picking up parameter update speed as initially $\hat{\theta}$ moves towards the extrema of $\theta = 0$. However, this still leads to the expected oscillations as both methods will eventually pass the extrema. In real-world scenarios, this may be inconsequential as perhaps the vehicle is carrying a second sensor, e.g. a camera, to verify whenever the vehicle is at an extrema and thus the vehicle would stop and avoid overshooting. While both HBESC and NesESC have overshoots and develop oscillatory motion, the characteristics of the oscillatory motion are different. HBESC has decaying oscillations whereas NesESC has oscillations which appear to be noticably growing after $t = 60$. In stark contrast, GESC has no oscillations since it is a first order method.

The speed at which each seeker reaches the extremum is also different. HBESC is the first, followed by GESC and then NesESC. The reason why GESC is able to reach the extremum faster is

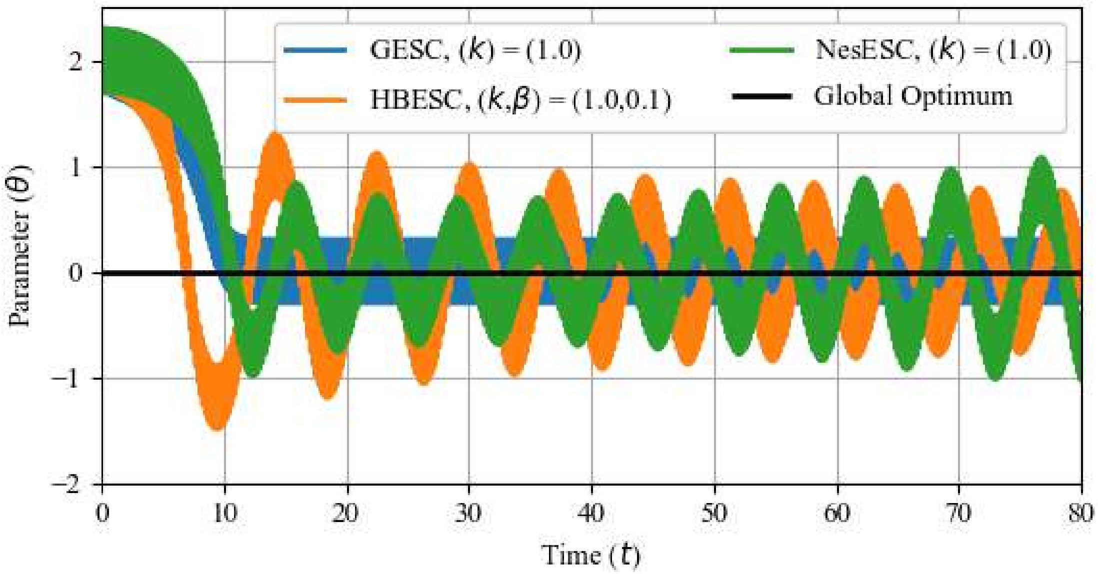


Figure 12: Extremum Seekers acting on a Gaussian-like cost function. Starting conditions where $(\hat{\theta}, \dot{\hat{\theta}}) = (2, 0)$ and the various gains and dampening ratios are shown in the legend. Sinusoidal perturbations were used with $a = 0.3$ and $\omega = 30$.

due to the initial conditions. As NesESC starts with a drastically high dampening ratio, NesESC is slow to alter its initial search direction. GESC does not suffer this setback as its search is always in the direction of the gradient. Starting farther way from the extrema would have resulted in NesESC reaching the optimal parameters faster, meaning that although GESC may have instances of faster convergence, NesESC (and HBESC) will still outperform GESC wherever there is extremely little variations of the cost function.

Additional simulations with non-convex test functions showed that HBESC can frequently escape local extremum, even relatively deep ones, if the starting position is closer to the local extremum. GESC may occasionally escape shallow local extremum when starting sufficiently close, but at a much slower rate than the HBESC.

5.2 ESC Implementation in ROS2

We utilize ROS2 in order to implement extremum seeking control schemes onto robotic vehicles. ROS2 is an open source message passing framework with easily accessible development tools and a plethora of pre-existing applications for robotic projects. ROS2 works by having processes, called nodes in the framework, which publish and subscribe to topics. When a node publishes data to a topic, all other nodes subscribed receive that data through internet packet protocols. This allows for easy communication between different processes (which may not be running on the same machine) and quick prototyping of software for robots.

Using ROS2 architecture, we have been developing our own software packages for use in ESC design and implementation. The extremum seeking control problem is broken down into subproblems: 1) calculating the cost function from sensor data; 2) updating internal states by numerically propagating differential equations; and 3) sending motor commands based on the internal states.

Each of these subproblem is a task managed by a different ROS2 node. By breaking the ESC problem into smaller subproblems, we can create a modular framework where the ROS2 nodes can take on different configurations to implement arbitrary ESC scheme designs. This breakdown can be seen in the ROS2 node diagrams of Figs. 13 and 14 for Gazebo simulations and real-world experiments, respectively. If a user needs to change the functionality of a particular item, it simply involves reconfiguring one particular node in the system. With this modular design, the goal is to allow for rapid testing and deployment of ESC schemes, first in Gazebo simulation, then on real-world robotic vehicles.

Our methodology for tuning ESC parameters is to first test our ESC designs in numerical simulations to determine initial gains before implementation of the designs in Gazebo simulations. After verifying the parameter selections in Gazebo simulations, configuration files used for the Gazebo simulations can then be uploaded to the robotic vehicle to conduct real-life experiments.

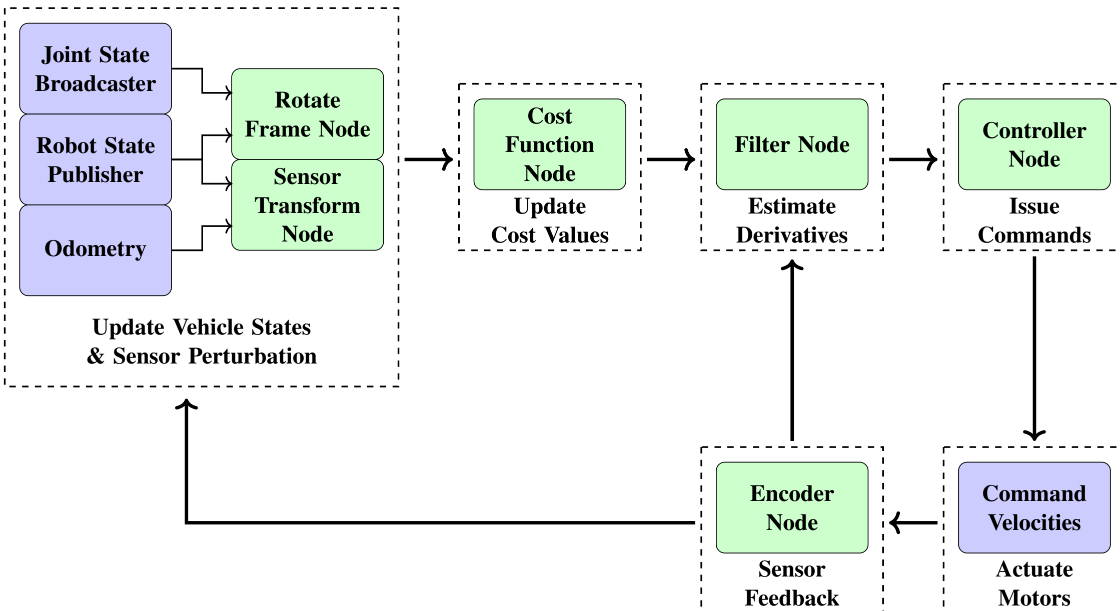


Figure 13: Block diagram of ROS2 nodes used in a Gazebo simulation. Note that nodes in green come from our custom software package, while nodes in blue indicate built-in ROS2 software.

5.2.1 Sensor Feedback Information

The Sensor Feedback Information nodes are used to collect and publish data streams for sensors present on the vehicle. Common sensors may include Lidar data, IMU data, rotary encoder data, microphone data, and sensor readings. Each individual sensor has its own associated ROS2 node to publish its data using ROS2 communication. In particular, we use a ROS2 node for communicating sensor readings from a microphone and/or photoresistor. Either data stream can be used to publish cost function values depending on whether the ESC algorithm is seeking an acoustic or light source. In addition, for rotating sensor applications later discussed in Section 5.5, we use an encoder node to broadcast the angular position of the rotating frame. This information is essential for calculating derivative estimation signals in our filters.

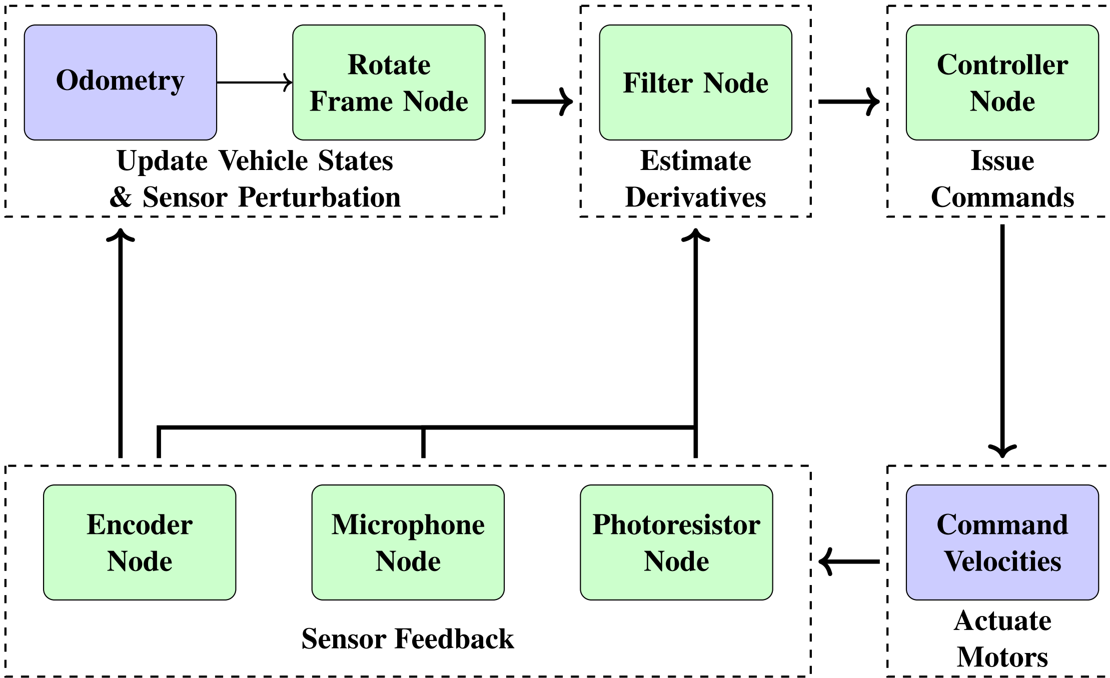


Figure 14: Block diagram of ROS2 nodes used in a real life experiment. Note that nodes in green come from our custom software package, while nodes in blue indicate built in ROS2 software.

5.2.2 Perturbation Signal with Rotating Sensors

Physically, the perturbation signal on robots describes how the sensor's position changes as a function of time with respect to the robot's center of mass. By varying the perturbation signal, one can decide how the rotating sensor explores the vehicle's surrounding environment. Knowing the desired sensor position as a function of time, one can differentiate to yield a velocity. This velocity is used to command the speed of a servo motor actuating the sensor to achieve a desired sensor trajectory.

The Rotate Frame node is used to command the angular velocity of the rotating sensor frame as it spins on the vehicle. Our work so far has used perturbation signals corresponding to a full rotation of the sensor frame, and back and forth rotation within a set angular position envelope. These rotations are done at relatively low speeds, less than 20 revolutions per minute (RPM), to avoid damaging the rotating sensor frame attachment.

The Sensor Transform node is used to gather transformation matrices corresponding to each sensor (i.e. microphone or photoresistor) on the vehicle. These transformation matrices describe the position and orientation of the sensors in 3D space. This information is used to compute virtual cost values in Gazebo simulations in lieu of a physical sensor.

5.2.3 Cost Function Design

Our package uses a Cost Function node to allow for custom cost functions to be used in experimentation. This node is given a list of transformation matrices as an input, then uses its custom cost function to operate on the matrices and produce cost values. This node is used only in Gazebo simulations (as seen in Fig. 13), its sole goal is to generate a cost value based on the sensor's position and orientation relative to the source. We have used this node to conduct simulations with quadratic cost functions and the virtual acoustic cost function seen in Section 5.3.2. These are strictly position-based cost functions, meaning only a small portion of the transformation matrix describing the sensor's frame is used to compute a theoretical cost value. We have also used this node to simulate with a virtual light-based cost function seen in Section 5.3.4.1. With this light source model, both the sensor's position and orientation data are used to compute a cost value. Virtual cost functions are described in further detail in Section 5.3.

5.2.4 Filter Design

We have developed the Filter node to handle any filtering needed for an ESC scheme. The user can pass a configuration file where a custom filter's architecture is specified. A custom filter is an object built up of several baseline filters (for example, low pass, washout, or convolution filters) in a unique configuration. Within this file, one can easily specify parameters such as gains and cutoff frequencies. The Filter node will parse this file, create a custom filter object, and use it to updated filter state values based on the current cost function values. Currently, we have been developing filters for various gradient-based and adaptive extremum seeking methods. These filters provide the derivative estimations needed for the user's choice of extremum seeking scheme.

5.2.5 Controller Design

Finally, we have a Controller node to implement various control laws into our ESC scheme. This node operates on the derivative estimation data stream published by the Filter node. The controller operates on this input with the custom control law corresponding to the user's choice of extremum seeking method. This results in a set of velocity commands which are published to move the robotic vehicle in the environment. Within the custom controller, the user has the ability to tune gains, and also impose constraints on the commanded velocities. We have been using this controller node to test our gradient-based and adaptive ESC schemes in both Gazebo simulation and real-life experiments.

5.3 Source Models

5.3.1 Virtual Quadratic Sources

The ultimate goal of this study is to enable the ESC algorithms to track real acoustic sources. However, to enable integration with real acoustic sources, preliminary tests are often conducted using a simpler virtual quadratic source to verify controller performance. The 2D virtual quadratic source is located at a distance (x, y) as shown in Fig. 15b. The cost function for the virtual source

$$J(x,y) = q_x(x-x^*)^2 + q_y(y-y^*)^2 + J_* \tag{5.2}$$

is based on the positional error where (x^*, y^*) represents the source's coordinates, J_* is the extremum of the cost function, and q_x and q_y are positive constants that govern the growth of the cost function in the testing environment. The ESC uses gradient estimations of the cost function to guide the robot towards the source, at least in a practical stability sense.

For preliminary tests of 3D extremum seeking (e.g., for underwater vehicle), a third z component should be added to the virtual quadratic source. This results in

$$J(x,y) = q_x(x-x^*)^2 + q_y(y-y^*)^2 + q_z(z-z^*)^2 + J_* \tag{5.3}$$

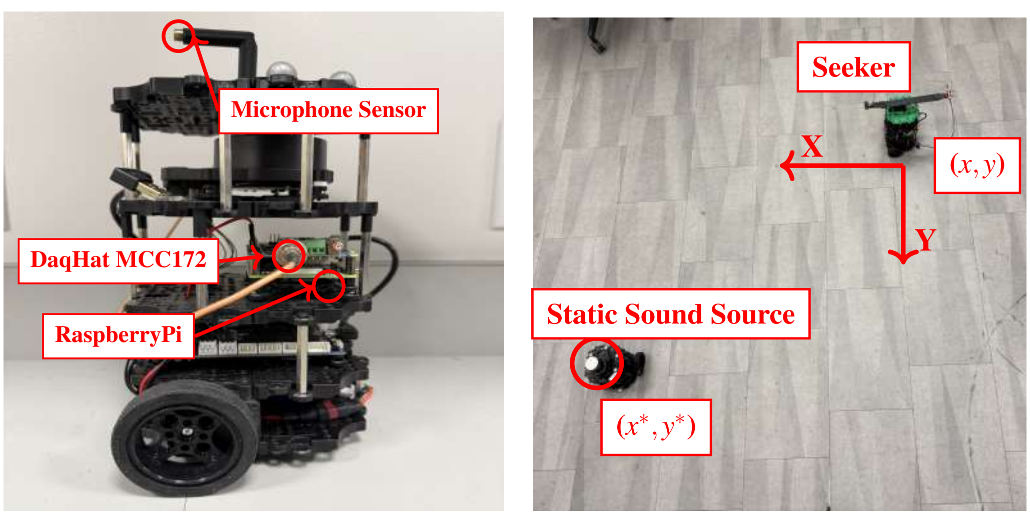
where q_z is the sensitivity of J in the z-direction.

5.3.2 Virtual Acoustic Monopole Source

The quadratic cost function is primarily utilized to ensure that our ESC design functions perform as intended. To achieve a more accurate representation, we developed an alternative cost function that serves as a virtual acoustic source, providing a more realistic basis for our simulations. Specifically, we model the sound intensity of a monopole source at a given sensor position r_s as

$$L_p(r_\mathrm{s}) = -20\log_{10}\left(\max\left\{\|r_\mathrm{s} - r^*\|, r_0\right\}\right) + 20\log_{10}\left(S/p_\mathrm{ref}\right) \tag{5.4}$$

where $p_\mathrm{ref} = 20\ \mu\mathrm{Pa}$ is the reference pressure, r^* is the source position, S is an arbitrary chosen value representing the source intensity, and r_0 is a small positive constant to avoid singularities when r_s goes to r^*. The sound pressure level (SPL) $L_p(r_\mathrm{s})$ represents the magnitude of a complex pressure amplitude [51, Eq. B.5]. We set S to a value so that the SPL of the source is 80 decibels (dB) when the sensor is located one meter from the source. This enables us to test out a more realistic cost function that can demonstrate a more comparable experiment with what we expect in our laboratory experiments as opposed to only using a quadratic from simulations. In later sections, we go into more detail on the comparison between Gazebo simulations and real-world experiments with the virtual acoustic cost and real acoustic cost value.



(a) (b)

Figure 15: (a) TurtleBot3 with horizontal static directional microphone, (b) DSIM lab test environment with the seeker robot and the static sound source.

5.3.3 Real Acoustic Source

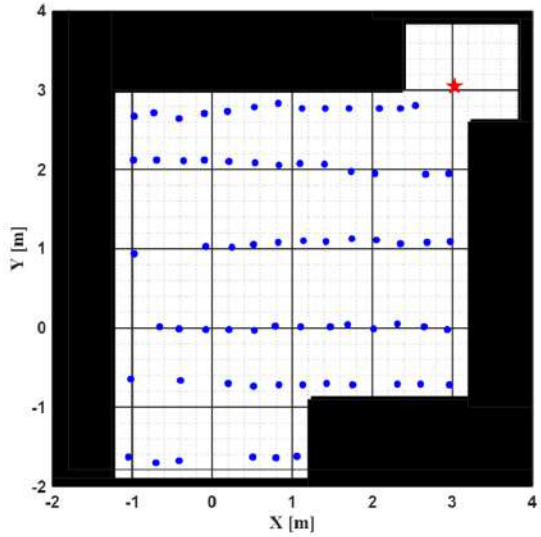
To incorporate real acoustic sources, the seeker robot has been equipped with a Hottinger Brüel & Kjær (HBK) Type 4958 array microphone. This $\frac{1}{4}$" pre-polarized directional microphone is mounted horizontally on top of the robot, as shown in Fig. 15a. This microphone is connected to a DAQ-HAT MCC-172 data acquisition system, which enables the Raspberry Pi to process the voltage readings from it. These readings are then converted to sound pressure levels (SPL) L_p using

$$L_p = 10 \log_{10} \frac{\overline{p^2}}{p_{\text{ref}}^2} \tag{5.5}$$

where the average squared pressure reading $\overline{p^2}$ is obtained by calculating the Root Mean Squared (RMS) of the acoustic data in Pascals and $p_{\text{ref}} = 20\ \mu$Pa is the reference sound pressure for air. The ESC uses the spatial variation of SPL data measured as recorded over its trajectory to estimate the gradient of the acoustic signal. For acoustic extremum seeking experiments, two different types of test tones were played from the sound source. In the acoustic experiments in the laboratory featured in Sections 5.5.2.2 ((A) and (B)) and Section 5.5.4, a linear chirp test tone was used which sweeps frequencies from 0 up to 7,500 Hz. However, this chirp's volume level is not constant, therefore for the additional acoustic experiments with the rotating sensor in Section 5.5.5, a 2000 Hz test tone at a constant volume was used. In future acoustic source seeking experiments with a USV and UUV, constant test tones will be utilized.

Existing ESC methods are capable of converging only on convex maps. Therefore, to ensure their successful implementation, it is important to verify that the seeker does not encounter any local minima within the DSIM lab environment. Sound reflections can give rise to local extrema, potentially causing the vehicle using the current ESC to settle at a suboptimal local solution. To test if there are local minimums, we wish to create an acoustic profile of the lab environment via experiment.

(a)



(b)

Figure 16: DSIM Laboratory acoustic source profile experiment: (a) laboratory setup for the experiment with the TurtleBot3 in the center of the room and the speaker placed atop a water bottle near the door in the top left, and (b) positions where data was collected.

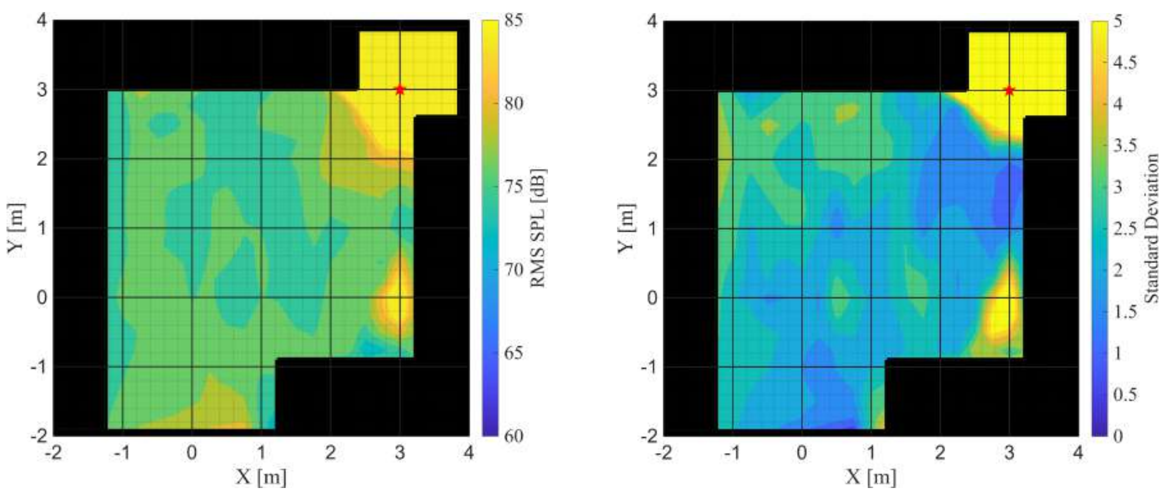
The goal of this experiment is to characterize the acoustic profile of the DSIM lab testing area. We wish to determine if and where local minimums in the sound pressure level (SPL) field may appear. To do so, we place an EWA bluetooth portable speaker playing a 2000 Hz test tone at max volume placed in a corner with the speaker emitting sound towards the center of the room. We collect SPL data with our microphone sensor at various positions around the room. At each data collection position, we set up the microphone at several incidence angles β, where β is the angle between the vector normal to the sensor's surface and the vector pointing from the sensor to the signal source (as shown later in Fig. 24). The angle β is varied from $0°$ to $330°$ in increments of 30 degrees to study how the sensor incidence angle may impact the SPL readings. Our DAQ-HAT MCC-172 data acquisition board samples at 51.2 kHz, we utilize this to collect ten seconds of data at each position and incidence angle. From the ten seconds of audio data, we compute a SPL value via Eq. (5.5) in batches of 512 samples, this yields 1000 readings per recording. In addition to computing the SPL value, we are also interested in the frequency contents of our samples. We perform a Discrete Fourier Transform on each bach of 512 samples, and organize the results into the standard octave band frequency bins. We then select the amplitude of the 2000 Hz octave band frequency bin as another quantity that we are interested in studying.

Figures 17 - 19 show the sound pressure level results for choice values of $\beta = 0°,\ 180°,\ 210°$. Note that the position of the sound source is shown with a red star. The typical initial position of the TurtleBot3 vehicle is at (0 m, 0 m) to provide largest possible initial distance between the source and seeker. Black areas indicate walls or desks in the DSIM lab environment, we draw a heatmap over the interior area where acoustic experiments can be conducted. Note, we do not pay much attention to the results close to the source's position in the corner since an acoustic extremum seeking vehicle would have no issue locating the source from short distances. For this experiment, we are more interested in the results in the middle and close to the borders of the testing area.

Subfigure (a) in figures 17 - 19 show the distribution of the mean SPL value measured in decibels over the DSIM lab test area. These figures show local extrema appearing both in the interior and along the borders of the test area. From these figures and further experiments (not shown here), we have noticed that varying the incidence angle β changes both the relative size and location of these local extrema. In addition to computing the mean SPL value, we also computed the standard deviation of our datasets and plotted its distribution over the test area in Subfigure (b) of figures 17 - 19. These provide information about the areas with high variation in the collected SPL data. The results indicate areas of high variation along the borders of the test area, likely caused by sound reflections.

Figures 20 through 22 show the magnitude results of the 2000 Hz octave band for the same incidence angles β shown previously. In these figures, we see large local extrema distributed in pockets in the center of the room. In addition, we can see high variance in our measurements, particularly concentrated along one border of the test area. The blank areas (near the position of the source) in this figure are extrapolations using Matlab's scattered-interpolant function. These blank areas can be disregarded, we did not sample in this region because we are more interested in the acoustic profile of the center of the lab space.

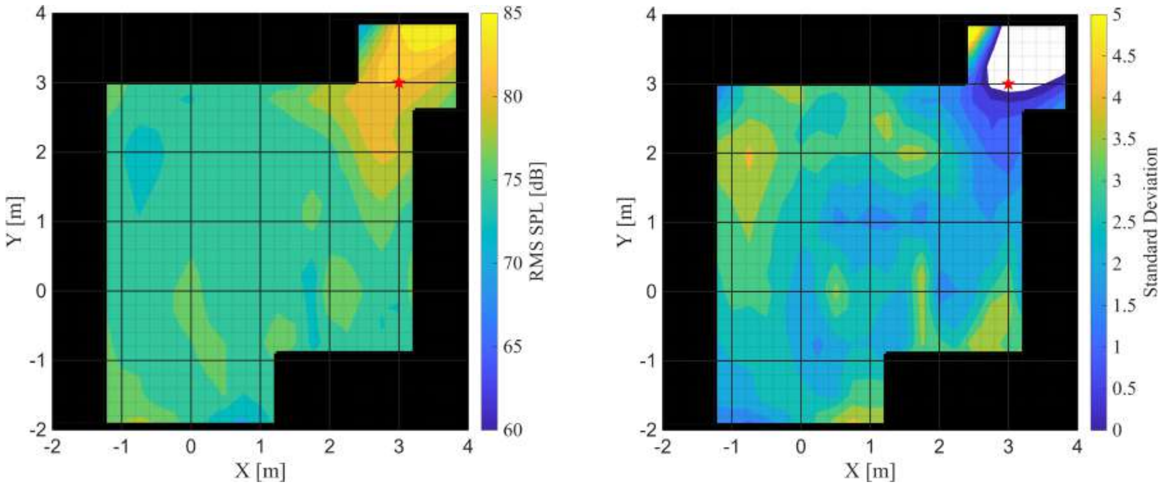
These experiments show clear issues with the acoustic field in our lab environment. For an ideal acoustic source seeking experiment, we would only have one local extremum at the position of the source, and the noise affecting our acquisition of the acoustic signal would be minimal. These results show that local extrema appear in various locations in the environment and that they change with the sensor incidence β. We can see that sound reflections and background noise cause the high variance we see in the plots, especially in parts of the room near walls or desks. These are issues to keep in mind when conducting tests in the indoor lab environment and motivate us to search for an alternative testing space for acoustic ESC experiments.



(a)

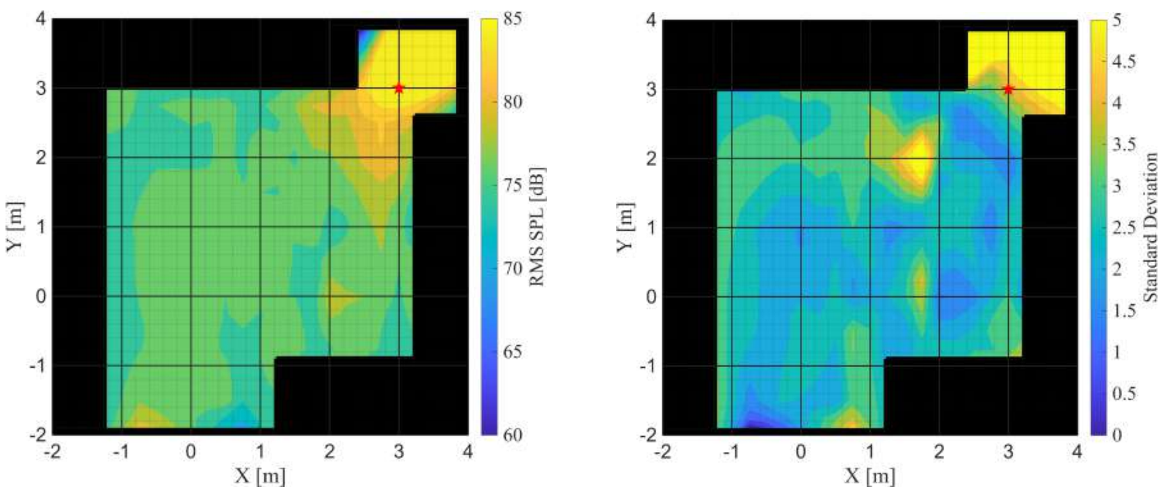
(b)

Figure 17: Results for incidence angle $\beta = 0°$: (a) mean dB value, and (b) the standard deviation of the dataset. **KEY:** Source Position = ★.



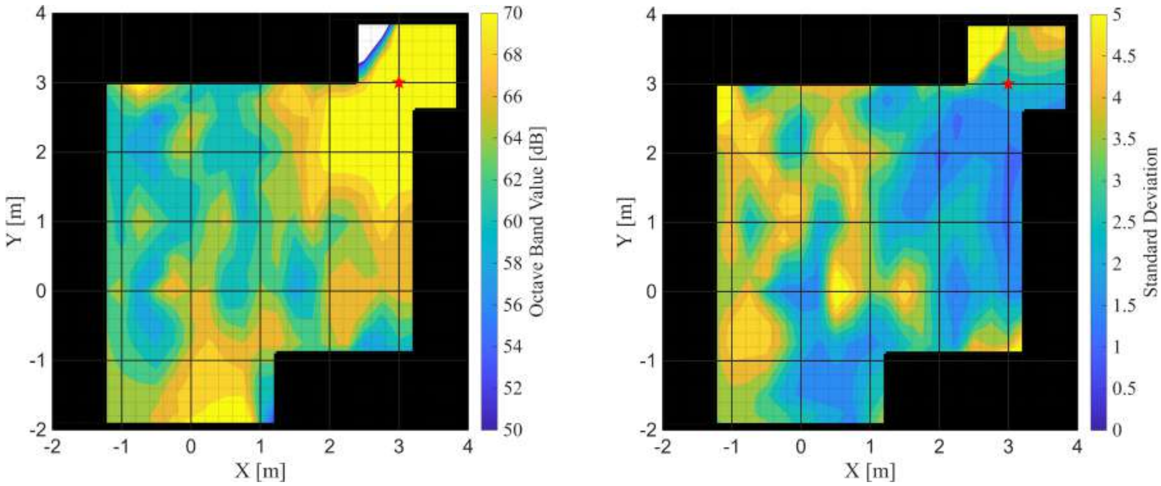
(a) (b)

Figure 18: Results for incidence angle $\beta = 180°$: (a) mean dB value, and (b) the standard deviation of the dataset. **KEY:** Source Position = ★.



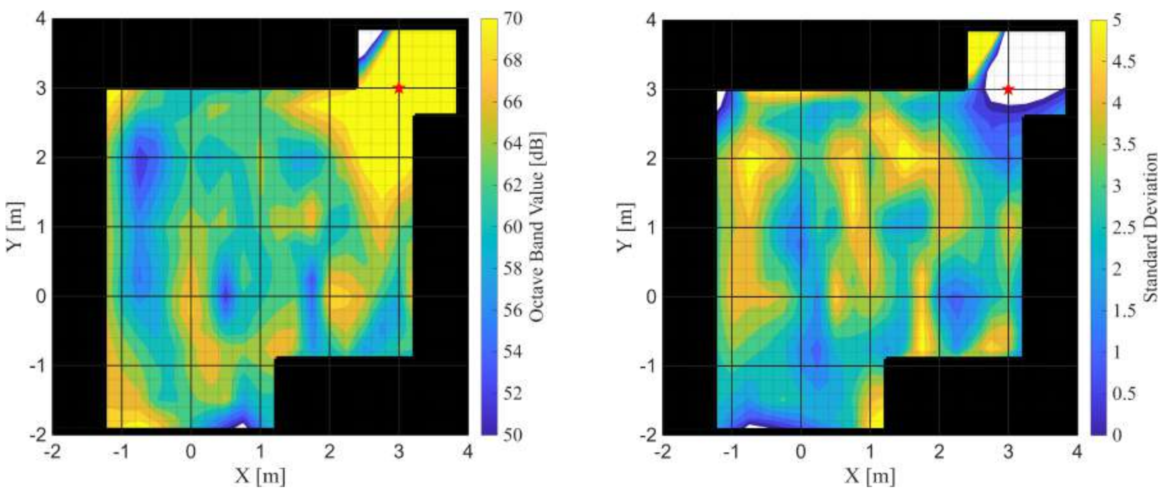
(a) (b)

Figure 19: Results for incidence angle $\beta = 210°$: (a) mean dB value, and (b) the standard deviation of the dataset. **KEY:** Source Position = ★.



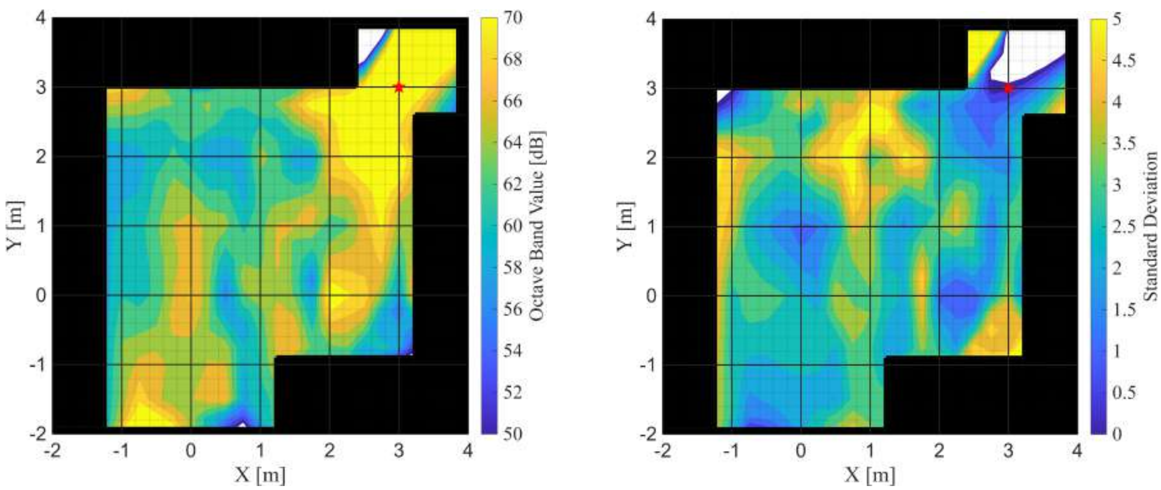
(a) (b)

Figure 20: Results for incidence angle $\beta = 0°$: (a) octave band value, and (b) the standard deviation of the dataset. **KEY:** Source Position = ★.



(a) (b)

Figure 21: Results for incidence angle $\beta = 180°$: (a) octave band value, and (b) the standard deviation of the dataset. **KEY:** Source Position = ★.



(a) (b)

Figure 22: Results for incidence angle $\beta = 210°$: (a) octave band value, and (b) the standard deviation of the dataset. **KEY:** Source Position = ★.

5.3.4 Light Source

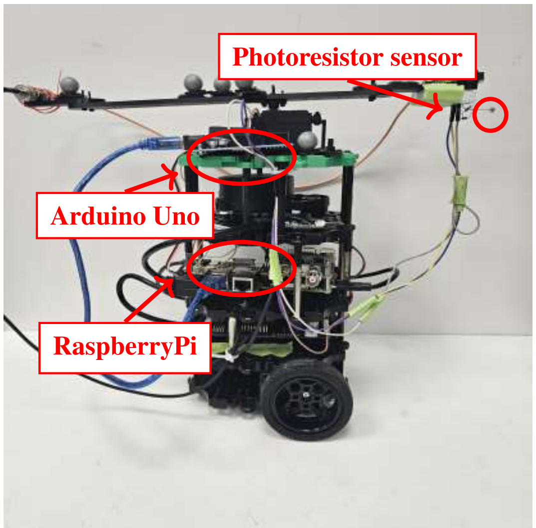
Along with conducting acoustic-based ESC experiments, we wish to validate our ESC schemes on light sources. Taking some measurement of light intensity as the cost value and optimizing the vehicle's position on a real-life light-based cost function offers several key advantages over a real life acoustic cost function. Light-based experiments are easy to setup and maintain, and outside effects like background light and reflections can be easily managed. Since it is easy to control these environmental factors and outside effects, we have found light-based ESC experiments easier to replicate in practice. Light-based experiments give our ESC methods the most "ideal conditions" for extremum seeking based navigation that we can easily achieve in our laboratory environment. We would like to use results from light-based experiments to inform our choices with ESC methods and tuning parameters for acoustic-based experiments. The light intensity of a monopole source in a 2D plane can be given by the following equation

$$I_\ell(r_s) = \frac{P}{4\pi} \cdot \frac{1}{(\max\{\|r_s - r^*\|, r_0\})^2} \tag{5.6}$$

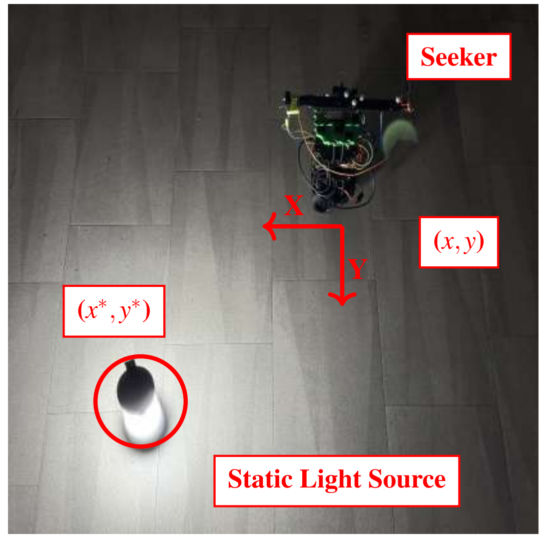
where again r^* is the source (or extremum) position, r_0 is a small number to avoid singularities, and P is an arbitrary number describing the total power radiated from the light source. The $I_\ell(r_s)$ here describes the light intensity as a function of the radius or distance from the light source. The equation is an implementation of the inverse square law, where the light intensity is inversely proportional to the radius squared [52].

5.3.4.1 Real Light Source

To incorporate real light sources, the seeker robot uses a CdS photoresistor sensor. The resistance of a photoresistor is inversely proportional to the light intensity hitting the surface of the sensor. Since the light intensity is inversely proportional to the square of the radius to the source, the resistance



(a)



(b)

Figure 23: Photoresistor experiment setup: (a) mounted photoresistor on TurtleBot3, (b) light sensor in lab environment.

value of this sensor scales proportionally with the distance to the light source. The photoresistor is wired in a voltage divider circuit seen in Fig. 25. An Arduino Uno board is used to sample the voltage drop across a 330 Ω resistor. This measured voltage, V_{meas}, is then used to calculate the photoresistor resistance using

$$I = \frac{V_{\text{meas}}}{330\ \Omega} \tag{5.7}$$

$$R_{\text{pr}} = \frac{5 - V_{\text{meas}}}{I} \tag{5.8}$$

where I denotes electrical current and R_{pr} denotes the photoresistor resistance. In this implementation, the photoresistor sensor's resistance value R_{pr} becomes the cost value in our light-based source seeking experiments. The ESC algorithm will use the variation of the sensor's resistance value to estimate the gradient of the light signal. In all experiments with the light source, the same camping lantern, seen in Fig. 23b, was used at its maximum brightness setting.

Similar to the microphone, the photoresistor is mounted in a horizontal configuration seen in Fig. 23a. This sensor is strongly directional, meaning it measures light intensity best when the source is positioned somewhere in front of it. We define an angle β, as seen in Fig. 24, to denote the angle between the vector normal to the photoresistor sensor's surface and the vector pointing from the sensor to the light source. At a $\beta = 0$, the entire surface of the photoresistor is facing the light source, the sensor sees peak intensity at this particular position. As β increases, the photoresistor turns away from the source, thus less light hits the sensor's surface, and the measured intensity falls off. The sensor's resistance value, R_{pr}, becomes a function of the distance to the light source and the angle β as described by:

$$J(r_s, \beta) = R_{\text{pr}}(r_s, \beta) \tag{5.9}$$

To visualize this R_{pr} cost function, we sampled the photoresistor's resistance values over multiple distances and β angles to make the visualizations seen in Fig. 26. Here, the source is assumed

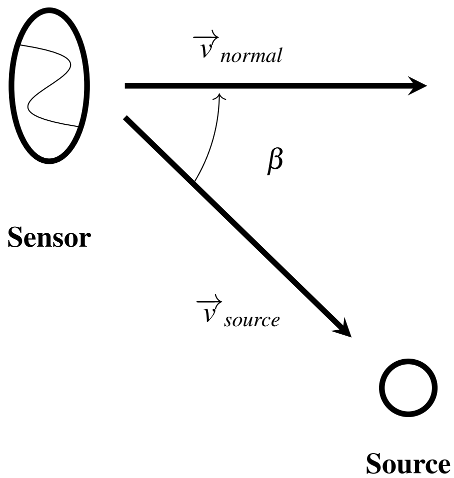


Figure 24: Definition of β.

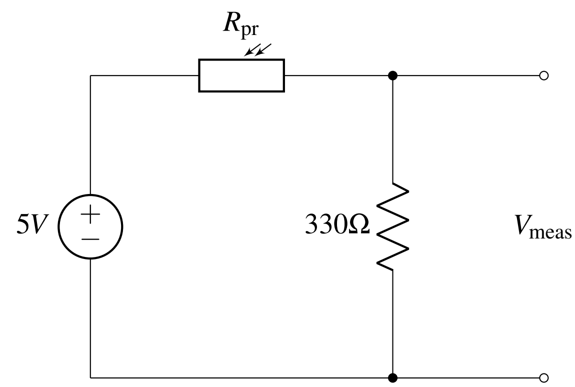
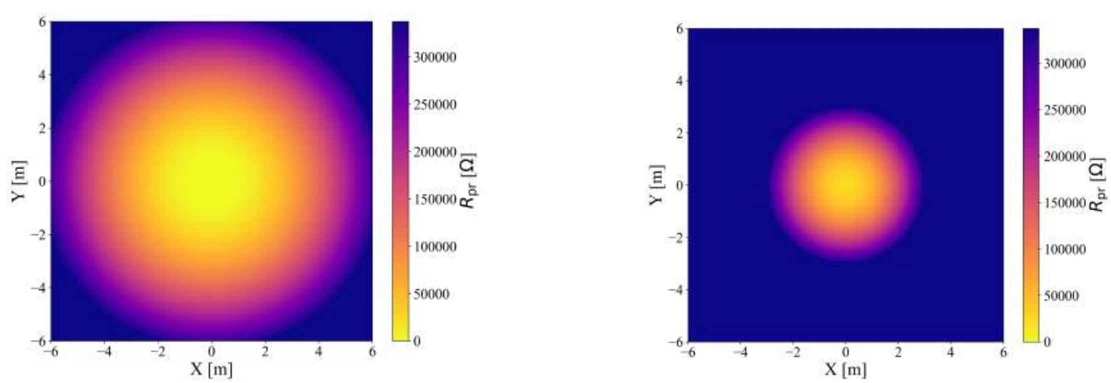


Figure 25: Photoresistor wiring diagram.



(a)

(b)

Figure 26: Photoresistor resistance comparison with different values of β mapped over the (x, y) plane with the source position at (0 m, 0 m): (a) $\beta = 0$ and (b) $\beta = 90$ degrees.

to be at (0, 0), and the function is mapped over the (x, y) plane for some choice values of β. Our results show that for small values of β, the resistance value increases proportional to the radius to the source like one would expect from the inverse square law. However for larger values of β, this law no longer holds. Since the photoresistor's surface is angled away from the light, the sensor cannot capture the full magnitude of the light intensity at a given position. To the sensor, this configuration appears darker and the resistance value increases. We note that the photoresistor has a maximum resistance value of about 337 $k\Omega$. Our sample data shows that for a bound of $r_s \lessapprox 5m$ and small values of $\beta \lessapprox 45^o$, we see spatial variations in the R_{pr} cost value which enable us to operate with our ESC controllers. However outside of these bounds, the photoresistor reading is often at its maximum resistance, and it becomes difficult for an ESC algorithm to operate. The sensor value is maxed out and unchanging at long distances and large values of β, and the ESC algorithm fails under these conditions.

5.3.4.2 Virtual Light Source Model

We used our experimental measurements of the photoresistor resistance with the real light source to develop a quadratic curve fit to model the variation of R_{pr} with respect to independent variables r_s and β. This curve fit is seen in (5.10) and corresponds to a minimum at (0, 0) and R-squared of 0.7845. We can then use this fit to write the cost function for the virtual light source in Eq. (5.11) which imposes minimum and maximum constraints on the cost value. Typical values for these parameters are $R_{min} = 100\ \Omega$ and $R_{max} = 337,260\ \Omega$. Note that β must be entered as an angle in degrees. The interpolated map in Fig. 27 can be visualized by a contour plot in Fig. 28a.

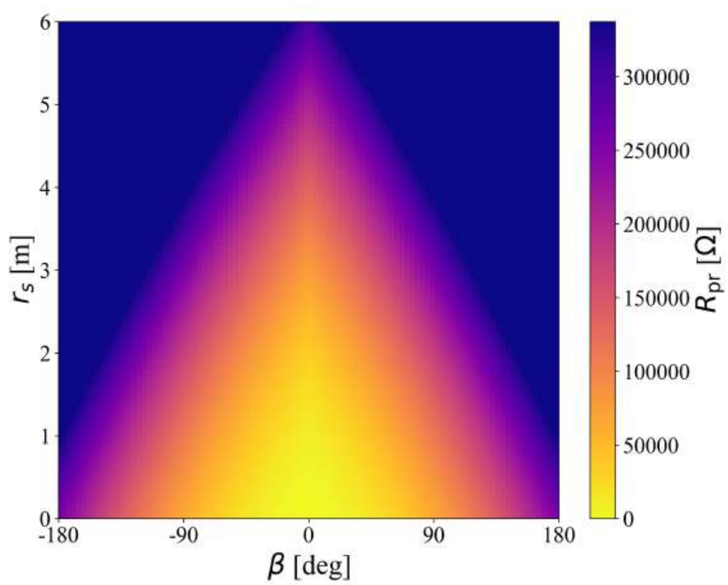


Figure 27: The quadratic interpolated map for R_{pr} seen in Eq. (5.11).

$$R_{\mathrm{pr}}(r_{\mathrm{s}},\beta) = 828.082r_{\mathrm{s}}^2 + 7.904\beta^2 + 442.130r_{\mathrm{s}} * \beta + 1.986\times10^{-13}r_{\mathrm{s}} + 5.309\times10^{-17}\beta + 8.829\times10^{-19} \quad (5.10)$$

$$J(r_{\mathrm{s}},\beta) = \begin{cases} R_{\max} & if\ \ R_{\max} \leq R_{\mathrm{pr}} \\ R_{\mathrm{pr}} & if\ \ R_{\min} \leq R_{\mathrm{pr}} \leq R_{\max} \\ R_{\min} & if\ \ R_{\mathrm{pr}} \leq R_{\min} \end{cases} \quad (5.11)$$

Figure 28 provides 3D plots of the virtual light source cost function. Figure 28a plots (5.11) and Fig. 28b shows an alternate cost function where a transformation mimicking an analog to digital conversion (ADC) was applied. We use an Arduino Uno board in order to collect readings from our photoresistor sensor. This board has a 10 bit resolution on its analog pins which we use to collect voltage measurements from our circuit pictured in Fig. 25. To more closely match the readings one would expect to see from a laboratory experiment, the cost value $J(r_{\mathrm{s}},\beta) = R_{\mathrm{pr}}$ can be correlated with one of 1024 distinct resistance levels given by the following sequence in (5.12). Note that $R_a \approx 330\Omega$ is the approximate resistance of the components wired in series with the photoresistor, and $\delta V = 0.0049\,V$ is the change in voltage from one distinct level to another. One can first compute a cost value with (5.11), then match that cost value with the closest entry that arises from the sequence in (5.12), this creates the alternate cost function visualized in Fig. 28b.

$$R_a\left(\frac{5}{n*\delta V} - 1\right), \quad n = 1,2,\ldots,1024 \quad (5.12)$$

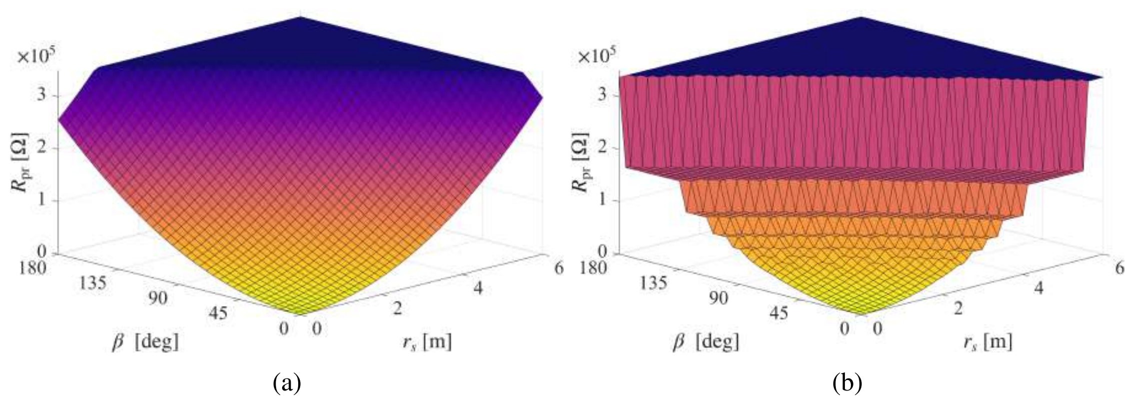
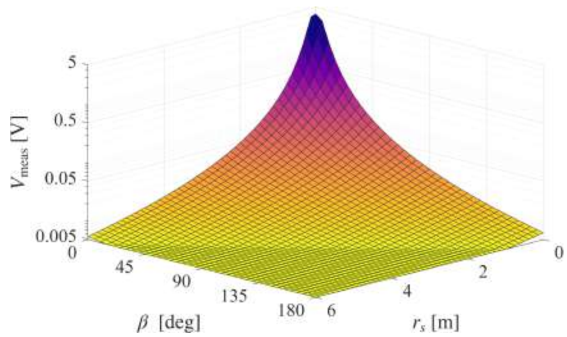


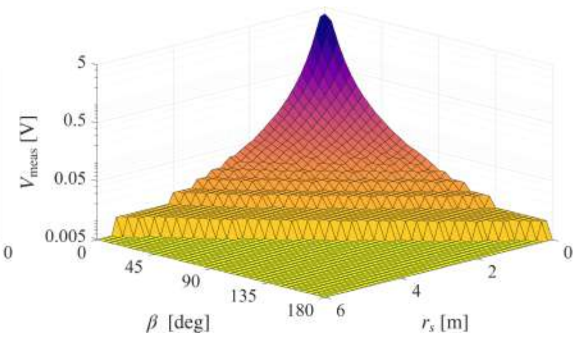
Figure 28: A 3D visualization of the resistance-based cost function with: (a) visualizing the cost value in Eq. (5.11), and (b) representing the distinct resistance levels mimicking an analog to digital conversion performed by an Arduino board utilizing Eq. (5.12).

With a resistance value computed from this cost function model, one can calculate the corresponding voltage value using (5.13). One can utilize this equation to change the resistance-based cost function in Fig. 28 into a voltage-based cost function in Fig. 29. With the resistance-based cost function, the magnitude of the gradient increases with the distance from the source's position, but diminishes as the seeker closes in on the source. The voltage-based cost function has the reciprocal relationship.

$$V_{\text{meas}} = \frac{5}{(R_{\text{pr}}/330+1)} \tag{5.13}$$



(a)



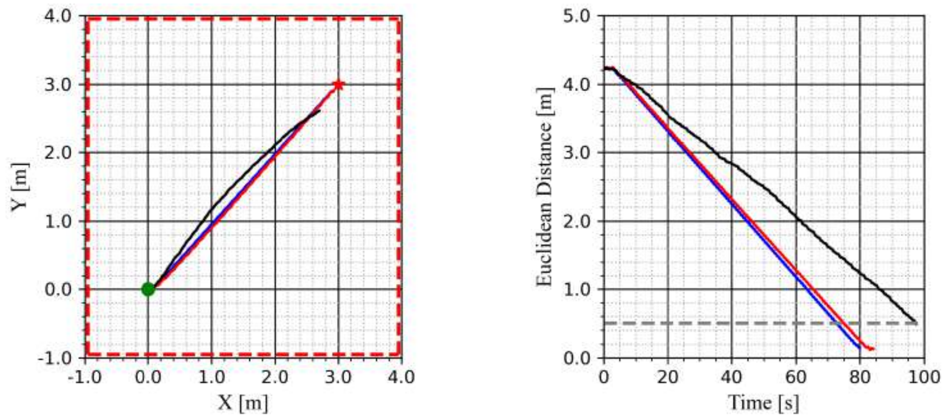
(b)

Figure 29: A 3D visualization of the voltage-based cost function: (a) shows an implementation of Eq. (5.13), and (b) mimics an analog to digital conversion applied on the previous voltage values.

5.3.4.3 Light Model Validation

In this section, we perform experiments to validate our simulated light source model against results of laboratory experiments. The corresponding simulations and laboratory experiments are performed using the TurtleBot3 vehicle and the RMSpESC method (theory and detailed implementation of RMSpESC discussed in Sections 4.4 and Section 5.5.5, respectively). Numerical and Gazebo simulations utilize the virtual cost function described in Eq. (5.11), while the laboratory experiments utilize the photoresistor sensor mounted on the TurtleBot3 vehicle seen in Fig. 23. For laboratory experiments, the light source used was a camping lantern set on its maximum brightness setting. In the following experiments, the resistance reading of the photoresistor sensor was used as the cost value measured.

The convergence time suggested by simulations with the virtual light source model will be compared to convergence times with the real light source to validate the effectiveness of the simulation. For this set of experiments, we utilize a rotating sensor frame (which will be introduced later in Section 5.5.4). The results for the simulation and laboratory experiments are shown in Fig. 30. The same ESC parameters were used for both simulations and the laboratory experiment, as given in Fig. 30). The vehicle trajectory in Fig. 30a shows the seeker taking a rather direct route towards the source. There is no significant difference between the predicted trajectory in simulation, and the one achieved experimentally. The time history of the seeker distance from the source in Fig. 30b is shown to give a sense of the convergence rates for these experiments. The simulations predict a convergence rate of about 20 seconds faster than the one obtained in experiments. In other validation experiments (not shown here), we observed that simulations predict faster convergence rates than seen in experimental tests; however, the difference in time is rather consistent at around 10 to 20 seconds. It seems that simulations over the virtual light-based cost function can provide a decent estimate of the convergence time of a laboratory experiment. The simulation results using the virtual light source model provide good estimates of the trajectory and convergence rate when compared with laboratory extremum seeking experiments. These demonstrate the reliability of the virtual light source model as we utilize it to simulate with and tune other ESC schemes.



(a) (b)

Figure 30: RMSpESC light source seeking with resistance-based cost values: (a) the seeker trajectory, and (b) the Euclidean distance to the source. The parameters used for simulations and laboratory experiment are given below.

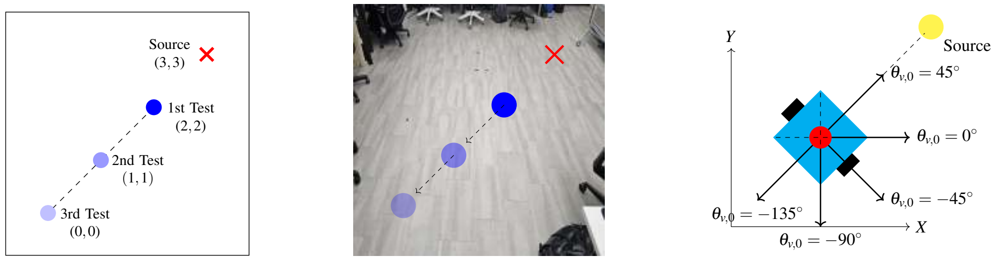
Parameters: $\phi_0 = -90°$, $a = 0.1524$ m, $\omega_s = 2.094$ rad/s, $\omega_\xi = 1.0$, $\omega_\ell = 1.0$, $\varepsilon = 0.1$, $k_v = 0.1$, $k_\omega = 0.5$, $v_{\max} = 0.1$ m/s, $\omega_{\max} = 0.5$ rad/s

KEY: Seeker Initial Position = ●, Source Position = ★. Room Walls = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Numerical Simulation = ▬▬, Gazebo Simulation = ▬▬, Laboratory Experiment = ▬▬.

5.4 Testing Environments

We conduct tests using non-holonomic robot to test the ESCs in three different environments: (1) in Python3 by numerically propagating differential equations; (2) simulations in the Gazebo toolbox; and (3) the physical laboratory. We prioritize the numerical simulation as our initial test of the theory. It is the simplest to implement among all three environments as it omits real-world complexities such as friction or sound reflections and refractions. This allows us to quickly verify the method's functionality and convergence before moving to more realistic environments like Gazebo or the real-world. In numerical simulations, we test three positions of the source relative to the starting robot position: (1m, 1m), (2m, 2m), and (3m, 3m). These represent the permissible testing area within our laboratory environment, whose walls are lined with cabinets. In each test, we record the functioning parameters for different cost functions including quadratic and virtual acoustic functions.

Once the system has been tested and verified to converge in numerical simulations, we proceed to the Gazebo simulator, where we evaluate a simulation of the robot within a realistic environment that incorporates real-world factors such as gravity and speed constraints. While there are other similar options available, Gazebo integrates seamlessly with our ROS2 implementation, ensuring smooth communication between the node framework. For example, our simulation features an open-world design shown in Fig. 31, we do not model the cabinets or walls in the simulation, as our primary focus is to test whether the robot will converge to the source. To verify the performance



(a) (b) (c)

Figure 32: (a) The outlined rectangle represents the extent of the testable laboratory environment used for the three primary experiments conducted to test the various sources used (acoustic and light), (b) The laboratory area representing the limited testing space, (c) The initial vehicle orientations ($\theta_{v,0}$) tested based on the source's location (which is always located at (1m,1m)). The robot distance from the source is ranged from (1m, 1m) to (3m, 3m) units. Position data is captured with a Vicon motion tracking system, the initial position of the vehicle is always represented with the coordinate (0m, 0m) in plots of the vehicle's path travelled.

of safety filters in obstacle avoidance, we would consider modifying the simulation to include obstacles for the robot to avoid, but this falls outside the scope of the current experiment. The Gazebo simulation is tested to simulate how the algorithm would work on the robot as well as to determine proper parameters that can be used on the robot and not cause damage. The numerical simulation does not include these constraints, so the parameters that were effective for the numerical model are unlikely to be suitable for the Gazebo simulation. This step is essential before testing the control design on the physical robot to ensure that we do not cause any damage to the robot's actuators by violating the physical constraints. Additionally, Gazebo enables more efficient testing to identify a range of parameters that are both physically viable and lead to convergence. It is much easier to test in the Gazebo as we can adjust parameters more easily and speed up the simulation time, something which cannot be done in real-world experiments.

Figure 31: Gazebo simulation of TurtleBot3 in an empty world.

After identifying these parameters in Gazebo, we implement them on the real robot and test the initial source position of (1m, 1m) seen in Fig. 32. Some additional parameter adjustments may be necessary as the simulator is a representation of the real world and cannot capture all its aspects. This will be discussed in more detail in Section 5.5, where discrepancies between laboratory experiment and Gazebo simulations are observed in the robot's trajectory. We are not necessarily optimizing the parameters; instead we are determining which parameters work for the robot in order to achieve convergence. Assuming there are good results with the first test, we complete the other two distances mentioned in Fig. 32a.

Additionally, we often examine the robustness of our design by changing the initial orientation of the robot. We start out at an initial orientation of $\theta_{v,0} = 0^\circ$ and then test the select orientations shown in Fig. 32c which range from 0° to 180°. This ensures that our system can effectively converge in dynamic settings, as it may not initially be aligned with the source. Other tests that were not conducted across all methods include assessing whether the robot can successfully converge when dealing with a time-varying source or when relocating the source to a different area within the room.

The experiments using the light source are performed in the Dynamic Systems and Intelligent Machines (DSIM) Lab. However, for acoustic extremum seeking experiments, we introduce an alternative testing environment. From the discussion in Section 5.3.3, we note that the DSIM Lab Environment pictured in Fig. 33 gives us issues with sound reflections negatively impacting our source seeking experiments. As an alternative, we also conduct some acoustic extremum seeking experiments at the San Diego State University Backdoor Recording Studio. This recording studio seen in Fig. 34 provides a space to test our ESC schemes in much more acoustically controlled environment than the DSIM lab. Figure 35 shows a diagram of the recording studio environment, the borders of the test area are marked with ArUco markers, and special coordinates are denoted on the floor by masking tape. Unfortunately, we lose the ability to track the vehicle's position in this testing environment due to the lack of motion capture system in this room. To address this challenge, all acoustic extremum seeking experiments will be recorded with a camera so that one can watch exactly how the vehicle navigates through its environment. Although the recording studio is quite limited in terms of floor space (with only about 9 square meters available as seen in Fig. 35), the heavy curtains around the borders of the room reduce the effect of the sound reflections in the room by absorbing the sound.

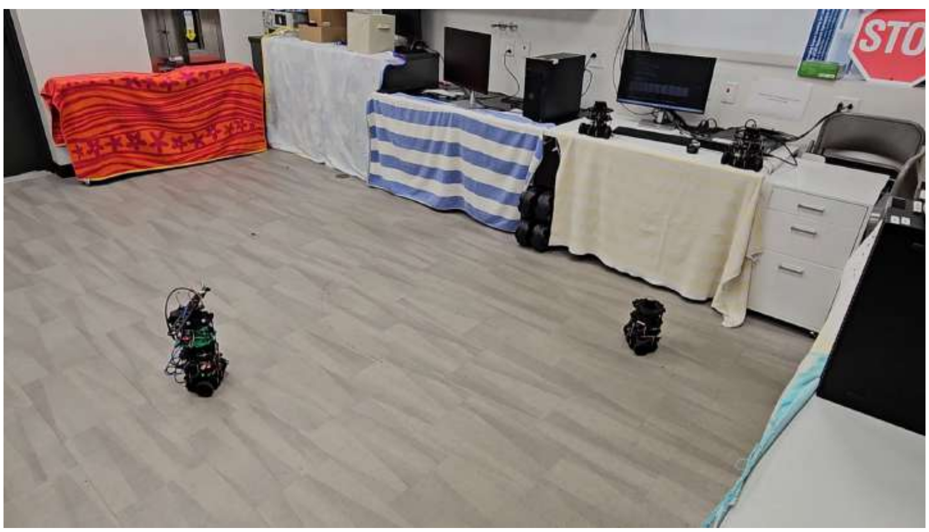


Figure 33: DSIM Lab environment setup for acoustic extremum seeking experiments. Towels were used as temporary sound dampeners over parts of the room with desks.

We began our acoustic source seeking experiments in the DSIM laboratory environment. Here we introduce results obtained with the Lie Bracket extremum seeking method implemented on the TurtleBot3. The Lie Bracket scheme is described by the block diagram in figure 40 and was initialized with the parameters in table 1. Note that the Lie Bracket controller commands a constant forward velocity, only the angular velocity of the vehicle is tuned by the ESC scheme. To compare with the Lie Bracket extremum seeker, we utilize the RMSpESC method shown in the block

Figure 34: SDSU Backdoor Recording Studio environment setup for acoustic source seeking experiments.

diagram in figure 73. We used the rotating sensor frame with the full rotation spin profile and the parameters shown in table 2.

Table 1: Parameters tuned for the Lie Bracket extremum seeking vehicle in the DSIM lab environment.

Symbol	Value
ω	0.35 rad/s
k	1.0
ε	0.1

Some successful experimental results obtained with the Lie Bracket and RMSpESC methods are shown in figure 36. Figure 36a shows a path comparison between the two methods, while figure 36b shows the time history of the cost value in the RMSpESC experiment. The Lie Bracket seeker took over 20 minutes to converge to the acoustic source's position, while the RMSpESC completes the same objective in about 10 minutes. These results show that the rotating sensor

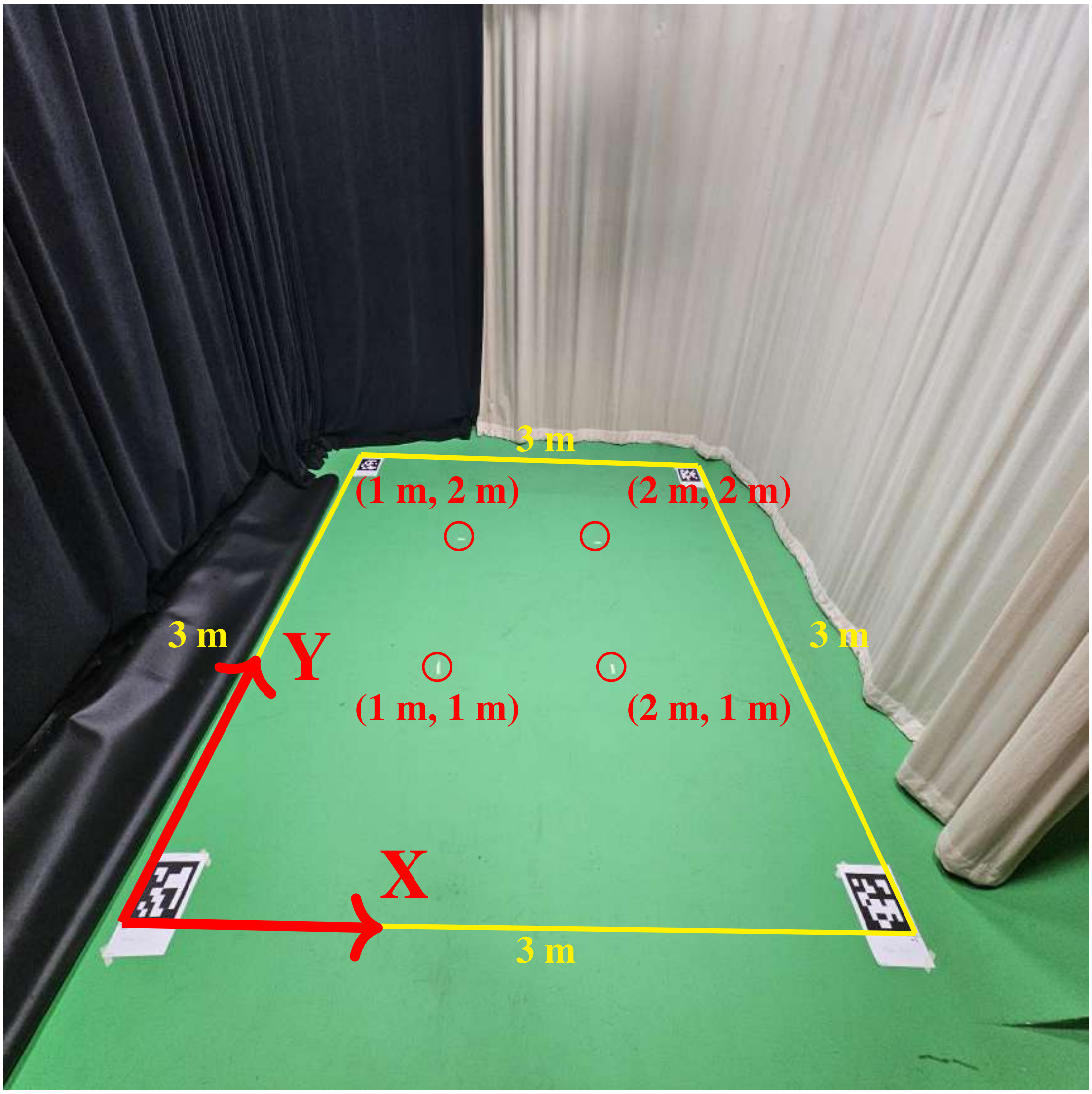


Figure 35: Diagram of test environment in the recording studio with some coordinates of interest marked on the ground with strips of tape.

provides a convergence time advantage over the previously tested static sensor Lie Bracket method.

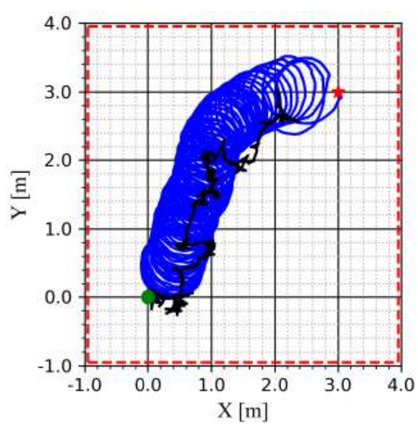
Compared to experiments with a light source, this RMSpESC has a significantly longer convergence time. If we look closely at the video of the successful result in this YouTube video, we can see that the seeker appears stuck in place for long periods of time during the experiment. The vehicle does not appear to have a clear sense of the direction it should be heading in, and it may even turn and drive the wrong way sometimes. In addition to the poor convergence rates, the RMSpESC proved to be unreliable at actually converging to the source's position. We conducted extensive testing (not pictured here) and exhaustive parameter tuning with the RMSpESC, but we could not get the seeker to reliably converge to the sound source's position in the DSIM lab environment. The result pictured in figure 36 is the only success out of dozens of failed experiments. Figure 37

Table 2: Parameters tuned for the RMSProp acoustic extremum seeking vehicle in the DSIM lab environment.

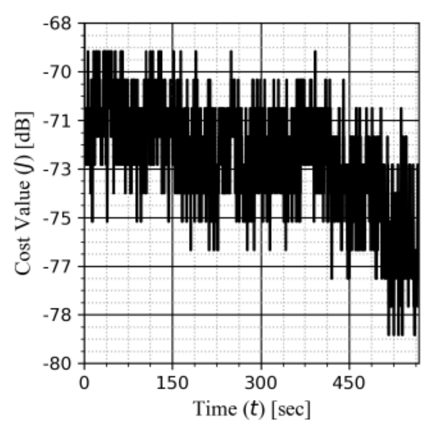
Symbol	Value
a	18 cm
ω_s	1.047 rad/s
ω_j	2.0
ω_ξ	1.0
ω_ℓ	1.0
ε	0.1
k_v	0.1
k_ω	0.5
$v_{\max}$	0.1 m/s
$\omega_{\max}$	0.5 rad/s

provides an example of a failed acoustic source seeking experiment with the same RMSpESC. We can see much of the same poor behavior in the failed experiment in this YouTube video, the seeker does not know which direction to go and it is stuck in place for 10 entire minutes.

From the work in section 5.3.3 it is clear that sound reflections in the DSIM lab environment have a harmful affect on all of acoustic source seeking experiments. Reflections from the walls and other objects in the room could create local extrema via constructive or destructive interference that rival the extrema created by the source. It is clear that we are running into some limitations for acoustic source seeking in an indoor environment. In order to get consistent acoustic source seeking results indoors, it is clear that a change of testing environment is needed.



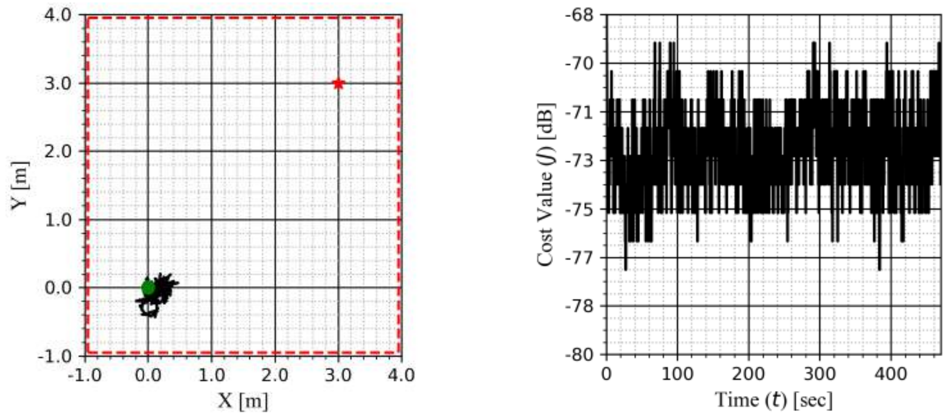
(a)



(b)

Figure 36: Two successful acoustic source seeking experiments conducted in the DSIM laboratory environment with: (a) a path comparison of the RMSpESC and Lie Bracket seekers, and (b) the cost value time history of the RMSpESC seeker.

KEY: Seeker Initial Position : ●, Source Position : ★, Room Boundaries = ▪▪▪▪, RMSProp ESC Seeker = ▬▬, Lie Bracket Seeker = ▬▬.



(a) (b)

Figure 37: Failed acoustic source seeking experiment conducted in the DSIM laboratory environment with: (a) the path of the RMSpESC seeker, and (b) the cost value time history.

KEY: Seeker Initial Position : ●, Source Position : ★, Room Boundaries = ▪▪▪▪, RMSProp ESC Seeker = ▬

5.5 Unmanned Ground Vehicle

This section describes implementation of ESC algorithms on an unmanned ground vehicle (UGV), which are easily accessible for experimentation in readily available low-cost platforms. The first algorithm to implement is the GESC since ESC is the most simplest and baseline method for the ESC. The GESC implementation will be conducted for two different sensor types on the UGV. The first is a statically mounted sensor and the other is a sensor mounted to move relative to the UGV. The existing ESC designs use a static sensor and perturb the vehicle's position to achieve spatial derivatives estimates of the signal field. Section 5.5.2 enables a UGV with the GESC algorithm to use a single-unit static microphone and steer toward an acoustic signal source. This is considered a novel contribution in the realm of ESC physical implementation since the state-of-the-art have not implemented ESC on real robots for acoustic source seeking.

While the statically mounted sensor does allow unmanned vehicles to perform extremum seeking, the required perturbation or local exploration of the vehicle's environment by the vehicle itself leads to significant time and energy consumption, making it impractical for most real-world operations. In Section 5.5.4, we formulate a novel approach to achieve the necessary perturbations (for estimation of the derivatives) by moving the sensor attached to the vehicle, i.e., by using a rotating sensor mount.

Insights gained from UGV implementation stages, with both static and rotating sensors, will support the implementation of more complex accelerated and adaptive ESC algorithms while reducing the number of experiments required for underwater and surface vehicles.

5.5.1 Ground Robot Model

We consider a mobile agent modeled as a unicycle with a sensor mounted at the center. The TurtleBot3 robot, manufactured by ROBOTIS, is a widely used platform for robotics research and exper-

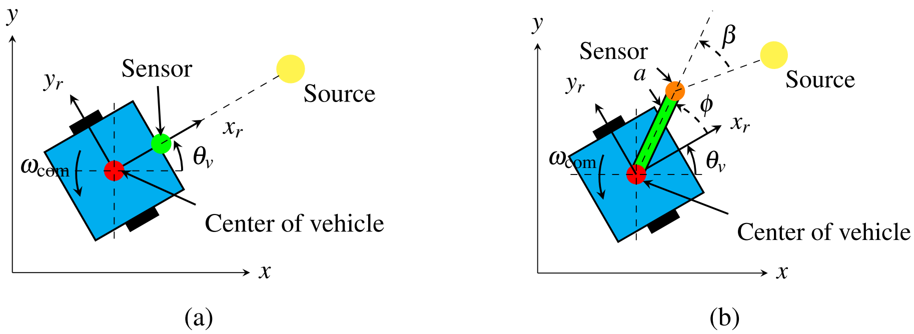


Figure 38: Unicycle model with: (a) a statically mounted sensor and (b) a rotating mounted sensor.

imentation. The dynamics of the TurtleBot3 follow a nonholonomic unicycle model

$$\frac{d}{dt}x_c = v_{\text{com}} \cos(\theta_v) \tag{5.14}$$

$$\frac{d}{dt}y_c = v_{\text{com}} \sin(\theta_v) \tag{5.15}$$

$$\frac{d}{dt}\theta_v = \omega_{\text{com}} \tag{5.16}$$

where x_c, y_c are the position of the vehicle's center in the global frame, v_{com} is the commanded forward velocity, and ω_{com} is the commanded angular velocity. Figs. 38a and 38b show the unicycle model with a statically and a rotating mounted sensor, respectively. The TurtleBot3 will be used through all the experiments to evaluate various ESC algorithms and their effectiveness in indoor environments. The maximum speed for both angular and forward velocities is 1 rad/s and 0.5 m/s respectively. If the output from the ESC exceeds this speed, the TurtleBot3 will not recognize it as an acceptable speed command. This means that if the control output is 1 m/s for forward velocity, the robot will not travel at that speed and will instead operate at the maximum allowable speed.

5.5.2 GESC with a Static Sensor

For the statically mounted sensor setup, we implemented three GESC designs for the UGV; a Lie Bracket system with angular velocity tuning and Direct Averaging with forward and angular velocity tuning. The experiments with these two designs incorporate a virtual source as well as a real-life acoustic source as the signals to be optimized. The results will compare the theoretical simulations, the Gazebo simulations, and laboratory experiments. The following subsections present the GESC designs and their performance results.

5.5.2.1 Experiment Setup

In our real-world testing, we adhere to the format detailed in the test types section (Section 5.4). Prior to conducting the tests with the acoustic source (Section 5.3.3), we needed to establish the optimal height for the source speaker, whether it should be positioned at the same height as the robot sensor or at a greater height. We experimented with two height variations: one aligned with the sensor height, as shown in Fig. 15b, and another set at a height of 0.76 meters above the sensor, illustrated in Fig. 39. Additionally, we aimed to assess the sensor orientation to determine whether a vertical or horizontal configuration would yield better results. Our findings indicate that mounting the microphone in a horizontal position is preferable to a vertical one, as it improves the speed of convergence.

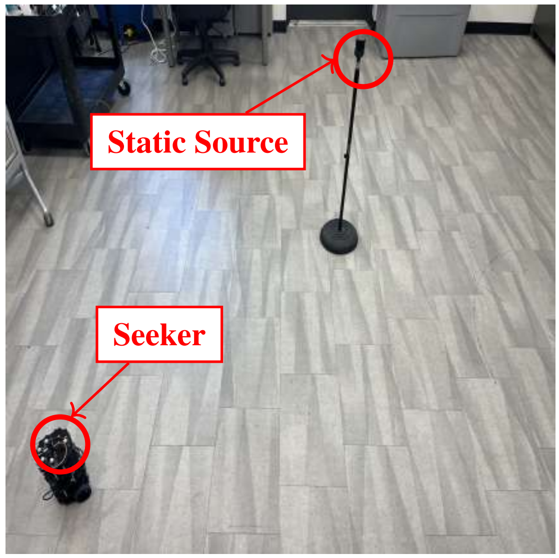


Figure 39: Taller setup of source speaker, determining the best height of the source.

Since the microphone is directional, a horizontal configuration was considered to be ideal since the seeker should perform the best when the sensor is angled toward the source speaker and at the same height. This was proven with comparative tests between vertical and horizontal configurations using a speaker as an acoustic source. The experiments revealed consistently faster convergence with the horizontal microphone. The vertical microphone configuration appears to pick up more sound reflections, particularly from the room's ceiling, which appears to generate shallow local minima that mislead the robot and prolong the convergence time. When the microphone and speaker were positioned at the same height with a horizontal microphone configuration, the system achieved better performance. Therefore, to ensure comparable test results, the source and the robot's microphone were kept at the same height throughout the experiments. Future work should explore strategies that enhance convergence in scenarios where the source and microphone are positioned at different heights.

5.5.2.2 GESC for UGVs in Previous Literature

We implemented three GESC designs: two using a Direct Averaging technique and another based on the Lie Bracket method. The Direct Averaging method was tested under two configurations: one tuning linear velocity [24], and one tuning angular velocity [53]. In [24], the UGV's forward commanded linear velocity had an additive sinusoidal perturbation while in [53], the commanded angular velocity had the additive sinusoidal perturbation. Both [24, 53] assumed a quadratic cost function with a local maximum and proved that the point mass model successfully converges this local maximum. The Direct Averaging method obtains the average system locally exponentially stable to the source and thus the model-free system is practically locally stable to the source by

[20, Th 10.4]. In contrast, the Lie Bracket method uses theories of fast, oscillatory inputs for affine systems to analyze trajectories using the Lie Bracket system, which is obtained from a sequence of higher and higher orders of Lie Bracket operators. The Lie Bracket method, as seen in [9], ultimately converges to the neighborhood of the source. According to [42, Th 3], if the Lie Bracket is GUAS it indicates that the original system is sGPUAS. While there are similarities between the Lie Bracket and average systems and their ability to prove practical stability of the orininal model-free system, the analysis techniques used to arrive at these systems differs. Nevertheless, we provide all three controllers as a comparison between our work and previous literature.

(A) Lie Bracket

Figure 40 shows the block diagram of the controller from [9]. The dynamical system is primarily governed by the heading angle dynamics

$$\frac{d}{dt}\theta_v = -\frac{k}{\omega_\xi}\left(J\left(r_c, r^*(t)\right) - \xi\right) + \omega \tag{5.17}$$

where r_c is the position of the vehicle in the global frame, r^* is the optimal position of the source, ξ is the washout filter state, ω is the dither frequency, and t is time, and the commanded forward velocity is kept constant at $v_{\text{com}} = \sqrt{\omega}$. The gain k and the washout filter parameter ω_ξ are parameters which are meant to assist in tuning the controller. Note that the unknown cost function J depends on the relative position between the seeker and the source, such that $J(r_c, r^*) = J(r_c - r^*)$. As $\omega_\xi \to \infty$ and $\omega \to \infty$, the robot's trajectory resembling the Lie Bracket system

$$\frac{d}{dt}\begin{bmatrix} x_c \\ y_c \end{bmatrix} = -k\nabla J(r_c - r^*) \tag{5.18}$$

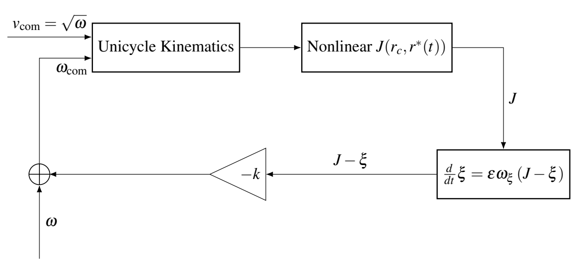


Figure 40: Lie Bracket for unicycle model tuning the angular velocity

Having given the controller, we question how close can we match simulations of the Lie Bracket ESC with a virtual acoustic signal to real-world experiments of a real acoustic signal in the laboratory. Initial testing with a numerical and Gazebo simulation was conducted where the vehicle's trajectory successfully converges to the virtual source. Confirming the preliminary results, we integrated the acoustic cost with the robot and successfully implemented this approach into the GESC design. The results comparing the laboratory result obtained with a real signal with the the numerical and Gazebo simulations results using a virtual acoustic signal are shown in Fig. 41. Note the discrepancy between the Gazebo simulations and laboratory experiment results, where the Gazebo simulation converges significantly faster than the real-world test. We anticipate some differences between the simulation and real-world outcomes, but the extent of the variation is unexpectedly large. Several factors could contribute to this discrepancy, including real-world influences such as friction and gravity, which introduce physical variations that may not be accurately represented in the simulation. Another factor is software communication issues which can lead to delays or latency between the ROS2 nodes and may cause the system to slow down during real-world testing. The problem lies in the message timing between nodes within our software; for example, a node in Gazebo may be perfectly synchronized with the callbacks, but the real-world simulation may not communicate with the same timing as other nodes, such as velocity commands. This can result in latency when nodes transmit data over wireless networks, potentially causing the wheel motors to ignore velocity commands which slows the convergence rate. These discrepancies may also arise from estimating the gradient of the cost function, indicating that small fluctuations in the cost value aren't detected through sensor noise until the seeker approaches the source. Incorporating random or Gaussian noise could help simulate real-world uncertainties not currently represented in the simulation. We will continue to discuss the challenges associated with the cost function in the subsequent sections.

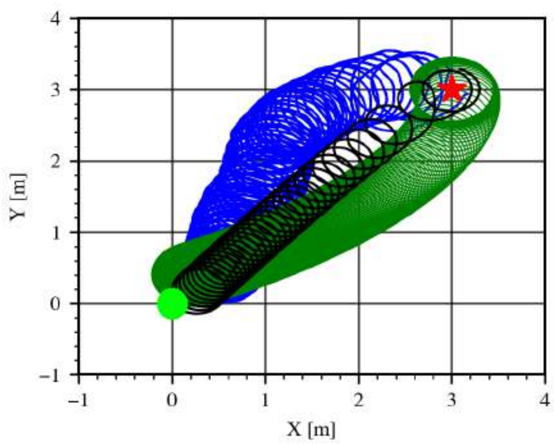


Figure 41: Lie bracket comparison between the numerical simulation, Gazebo simulation, and laboratory experiment. Numerical and Gazebo simulations seek a virtual acoustic source and the laboratory experiment converges to a real acoustic source. Parameters for the numerical simulation were $\omega_\xi = 0.1$, $\omega = 0.5$, $k = 0.5$. Parameters for the Gazebo simulation and laboratory experiment were $\varepsilon = 0.1$, $\omega = 0.35$, and $k = 1.0$.
KEY: Gazebo simulation = ▬, Laboratory experiment = ▬, Numerical simulation = ▬, Seeker initial position = ●, Source = ★.

The results presented in Fig. 41 indicate successful convergence. Due to space constraints in our laboratory, the distance used was the maximum achievable with our allowable test area. As mentioned in Section 5.4, additional tests were conducted but we only show the test with maximum distance of (3m, 3m). The robot is able to converge at various distances, which suggests that the seeker using a Lie Bracket ESC can effectively converge to the source for all possible distances tested in the laboratory. Further testing will be conducted to address the issues observed with the cost function to determine an appropriate resolution in the DAC. We will also evaluate various initial robot orientations, as these tests currently demonstrate convergence only when the initial orientation is angled 45° away from the source. Later, in Section 5.5.4, we provide an overview of the results from the different initial orientation angles. Tests shown in Fig. 42 were done to determine the impact of the gain k on the dither amplitude of the vehicle trajectory. These plots show that by adjusting to a larger k value, the seeker is able to converge faster towards the source. This is encouraging for a system with a time constraint or a time-varying source as it is able to converge faster by properly tuning the learning rate.

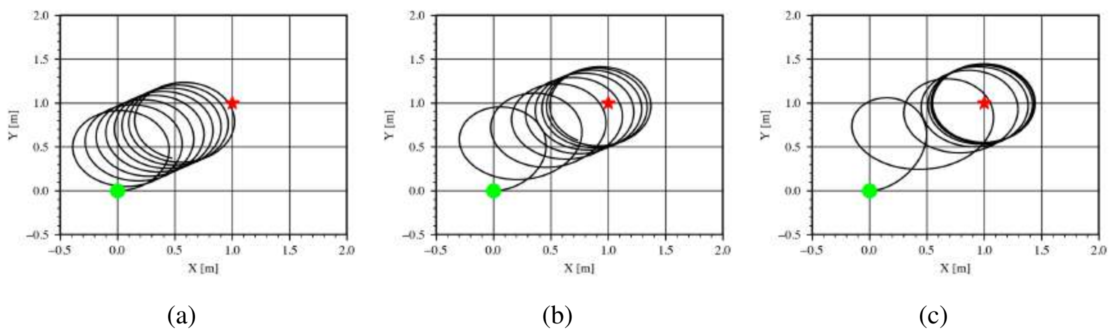


Figure 42: Effect of the gain k on the seeker's trajectory using a numerical simulation and a virtual quadratic source: (a) $k = 1$, (b) $k = 3$, and (c) $k = 7$. Common parameters used for all the tests were $\omega = 5$, $\omega_\xi = 0.1$, and $v_{\text{com}} = \sqrt{|\omega|}$.
KEY: Seeker trajectory = ▬ , Seeker initial position = ● , Source = ★ .

(B) Direct Averaging

The experiments conducted in this part aimed to determine how well the classic averaging design of GESC works in a practical setting. We examined two existing designs for the direct averaging method: the first design tunes linear velocity [24], and the second tunes the angular velocity [53] to achieve accurate estimate of the field's gradient.

(B-1) Forward Tuning

Forward tuning uses perturbs the robot's forward velocity and keeps the angular velocity as a constant. The commanded forward velocity is

$$v_{\text{com}} = a\omega \cos(\omega t) + k_p \xi \sin(\omega t) \tag{5.19}$$

where ξ is the washout filter state, k_p is a proportional gain, a is the dither amplitude, and ω is the dither frequency. Figure 43 shows the block diagram of the ESC with forward velocity being perturbed with a sinusoidal signal so that the UGV can estimate the gradient of the cost function along the UGV's forward axis. Figures 44a and 44b compare the numerical simulation and Gazebo virtual acoustic source seeking results. The theory presented in [24] demonstrates that the system can converge to the source based on estimating the projected gradient of the J. We were able to replicate a similar result to [24] in numerical and Gazebo simulations using a virtual acoustic source, as seen in Figs. 44a and 44b. Similar results were expected when implementing the algorithm for the laboratory experiments on the real robot. However, the algorithm resulted in a failed test since, as shown in Fig. 44c, the seeker remains at the initial position and does not converge to the real sound source during the testing period.

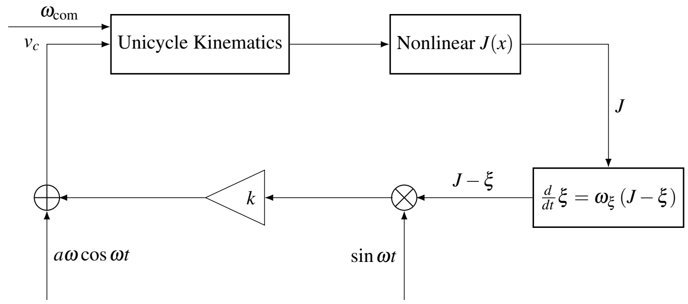


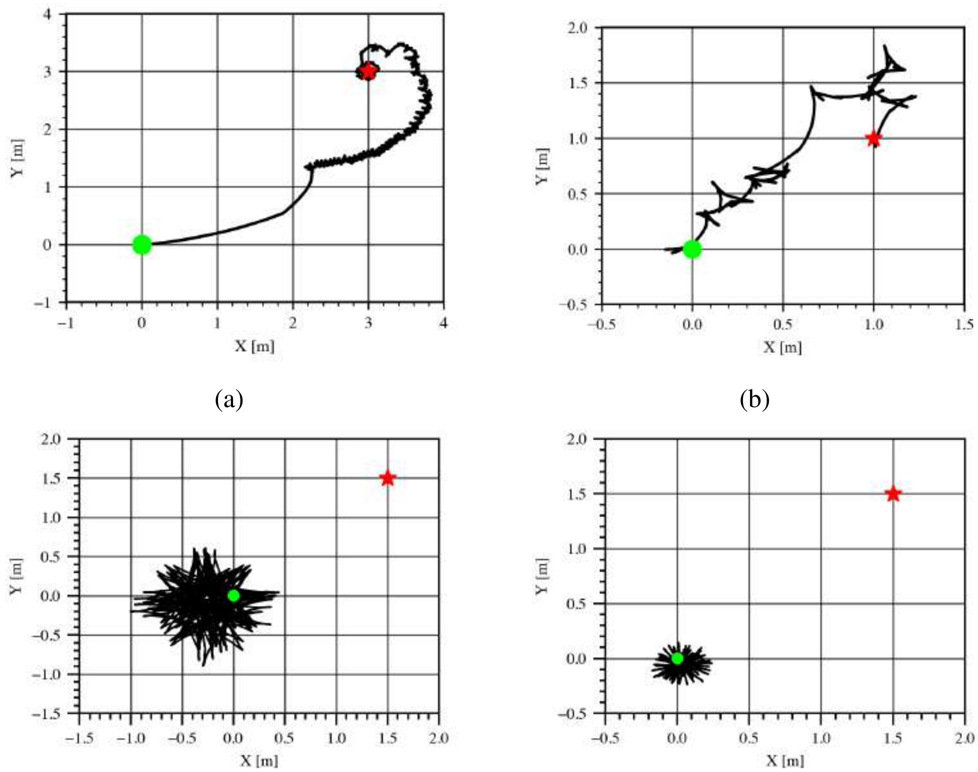
Figure 43: Direct Averaging for the unicycle model with forward tuning

In the laboratory experiments, we investigated various dither frequencies to allow the system to expand the local exploration area and obtain more accurate gradient estimates. However, this adjustment did not significantly impact the results, as shown in Fig. 44d, noting that only the linear chirp signal was used as the source. In the Lie Bracket ESC experiment, previously shown in Fig. 41, the seeker took approximately 23 minutes to converge to the source. Based on those results, it was assumed that it would take a while to converge; however, with the forward tuning, the robot shtill had not moved any cloer to the source after 45 minutes. The effectiveness of the Lie Bracket system can be attributed to its continuous forward motion, which enhances the accuracy of cost estimation as it approaches the source, as depicted in Fig. 41. In this figure, the seeker progresses toward the source and alters its path once it gets close, indicating that the seeker must be in close proximity to the source for optimal performance. Based on our preliminary tests, we predict that unless the seeker is in close proximity to the source, it will struggle to obtain a suitable variation in the cost function needed to estimate the gradient and reach convergence. This assumption is made regarding the forward tuning method, given that the Lie Bracket method has proven capable of overcoming local extrema. Additionally, we recognize that the presence of sound reflections in the testing area poses challenges, particularly near the corners of the room. In those positions, the directionality of the sensor may lead to inconsistent gradient estimations, resulting in ambiguity regarding the correct direction of motion toward the optimal position.

Currently, the cost function is calculated as the Root Mean Squared (RMS) of the acoustic signal, with the cost function being the sound pressure level (SPL) as measured in dBs. However, it may be more appropriate to utilize the Fast Fourier Transform (FFT) instead. Using the FFT would allow us to identify an optimal frequency window, with frequencies outside this range being disregarded. This approach provides a more accurate method for locating the source compared to simply measuring sound intensity using RMS. Lastly, the system may require an extended testing duration to ensure convergence, rendering this approach less practical for ongoing testing. Given that the lab experiment could not converge to the source, we will shift our focus to the angular tuning method. The forward tuning approach will need further investigation to confirm the issues and assumptions discussed in this section and address them with future adjustments. Later, in Section 5.5.4, we will discuss the application of a light source which is simpler to use for benchmarking against existing methods in the lab and makes it easier to manage effects like background light and reflections in indoor spaces.

(B-2) Angular Tuning

Following forward tuning tests using the acoustic source, we shifted our focus to direct aver-



(a) (b)

(c) (d)

Figure 44: Direct averaging with forward tuning results: (a) Numerical simulation using a virtual acoustic source, and $\omega = 2.5$, and $k_p = 0.08$, (b) Gazebo simulation using a virtual acoustic source and $\omega = 10$, (c) Laboratory experiment using a real acoustic source with $\omega = 3.0$, and (d) Laboratory experiment using a real acoustic source and $\omega = 1.0$. Common parameters used for Gazebo and laboratory tests: $a = 0.05$, $k_p = 1.0$. Tests (c) and (d) are adjusting the dither frequencies (ω) where the seeker failed to converge to the acoustic source.
KEY: Seeker trajectory = ▬ , Seeker initial position = ● , Source = ★

aging with angular tuning design, shown in Fig. 45, based on the method presented in [53]. In this design, v_{com} remains constant and ω_{com} is tuned, as seen in (5.20). The new gain factor k_d introduced in this approach acts as a smoothing variable during the convergence process.

$$\dot{\theta}_v = a\omega \cos \omega t + k_p \xi \sin \omega t - k_d \xi^2 \sin \omega t \tag{5.20}$$

The vehicle is able to reach the virtual acoustic source with the numerical simulation shown in Fig. 46. The impact of parameter k_d is evident in Figs. 46a and 46b, where the numerical simulations demonstrate the difference in the robot's trajectory. Figure 46a displays how with a larger k_d, the system undergoes a more "roundabout" transient phase but eventually settles into a residual annular trajectory around the source. Figure 46b with $k_d = 0$ show trajectories with a more direct path to the source, althought the trajectories can overshoot the source. This highlights the importance of tuning the parameter k_d based on system requirements. If time is a constraint, then k_d should be

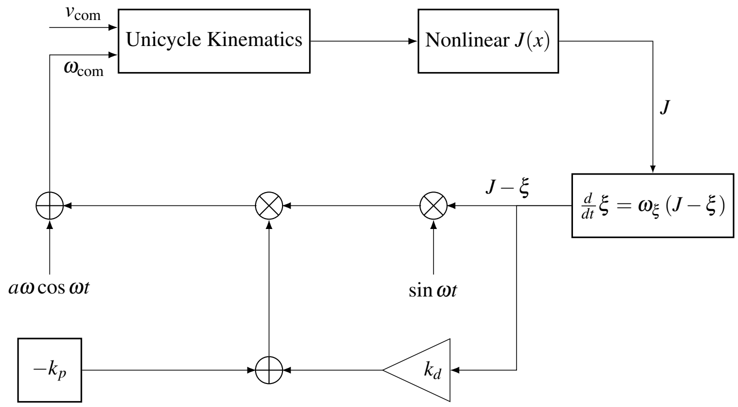
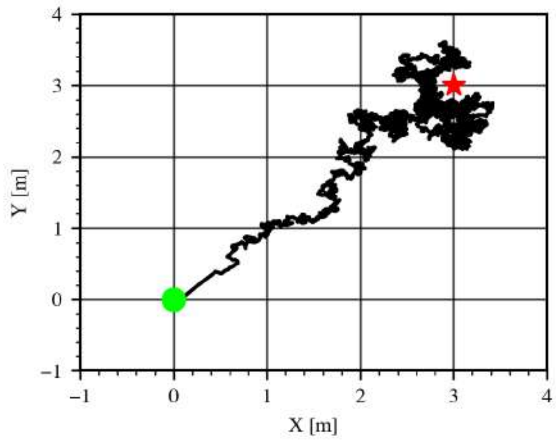


Figure 45: Direct averaging with angular tuning for the unicycle model

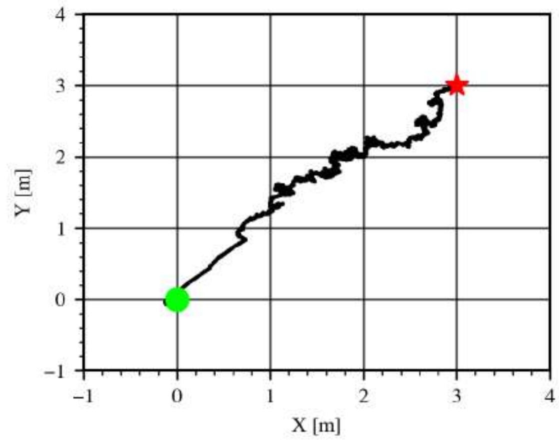
equal to 0 so that the system converges to the source rapidly. Otherwise, then it is acceptable for the system to hover around the source. The Gazebo simulation depicted in Fig. 47 illustrates two experiments conducted using the Gazebo simulation environment. The results shown in Fig. 47b utilized parameters analogous to those employed in the numerical simulation, while the parameters in Fig. 47a were modified to comply with the speed constraints of TurtleBot3. It is important to note that the second Fig. 47b portrays a scenario that is not feasible, as the angular speed achieves 3 rad/s, surpassing the TurtleBot3's maximum limit for angular velocity. Nevertheless, this demonstrates the potential for convergence with those parameters. The optimal simulation for the TurtleBot3, which adheres to the specified limits is presented in Fig. 47a. This configuration gradually converges to a neighborhood around the source (1m, 1m), in contrast to the second configuration that targeted (3m, 3m). As indicated by the figure and the associated parameters, the TurtleBot3 operates at a very slow speed, resulting in extended convergence times of approximately 500 seconds. Due to the constant forward velocity, the robot has the potential to overshoot the source unless system is sufficiently decelerated, the figures do not show this since the simulation was stopped just after the robot converged to a neighborhood of the source. To enhance the system's sensitivity in reaching the sound source without overshooting, substantial reductions in forward velocity are necessary. In numerical simulations (with results not shown here), we decreased the forward velocity to $v_{\text{com}} = 0.001$, yielding a slower convergence time of $1,500$ seconds to a source positioned at (3m, 3m). The results from the laboratory experiment, which utilized only the linear chirp signal as the source with the acoustic source, were not successful due to an insufficient cost fluctuations for the ESC. We conducted a test with the quadratic source which worked but we continued to run into issues with the estimations of the real acoustic source. These findings suggest that the angular tuning approach may not be practical for the nonholomic kinematics, given the considerable reduction in speed required for effective convergence without overshoot. By implementing a new method which uses a rotating sensor, we aim to address some of these challenges, as it will facilitate gradient estimation from the rotating sensor and allow for the adjustment of both angular and forward velocities.

5.5.3 Rotating Microphone Frame Hardware Design

The idea behind the rotating microphone frame is to use a sensor which rotates and samples the environment. Unlike the case with statically mounted microphone (Section 5.5.2), where the seeker's



(a)



(b)

Figure 46: Effect of the parameter k_d on direct averaging with angular tuning using numerical simulation with a virtual acoustic source: a) $k_d = 30$, b) $k_d = 0.0$. Other parameters were $a = 0.5$, $k_p = 100$, $\omega = 30$, $\omega_\xi = 2$, and $v_{\text{com}} = 0.15$.

KEY: Seeker trajectory = ▬, Seeker initial position = ●, Source = ★.

position is perturbed to estimate the field derivatives, here the sensor's position is perturbed to provide the estimations. This requires additional components from the experiments with statically mounted microphone. The rotating microphone frame will be mounted on an additional platform atop of the TurtleBot3, allowing the frame to spin around freely.

The following experiments detail our investigation to find suitable hardware components for the rotating microphone concept. The first experiment was motivated by the desire to find a quiet motor to spin the microphone frame while not interfering with the microphone's readings. The second experiment was motivated by the need to find a suitable microphone frame size so the sensor can scan the vehicle's surroundings and detect enough of a difference in sound pressure level in its environment.

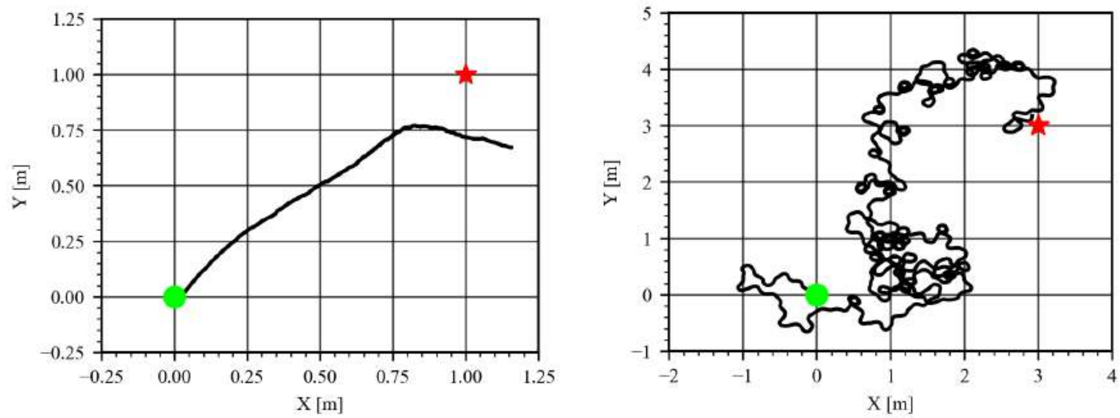
5.5.3.1 Motor Selection

The goal of the following experiment is to find the appropriate motor hardware for the rotating microphone frame application.

Experiment Objectives

The objective of this experiment is to determine an appropriate motor to drive the rotation of the microphone frame. A suitable motor should meet the following criteria:

- Generates minimal noise so it does not affect the readings from the microphone at the end of the frame.
- Capable of a full 360 degree rotation both clockwise and counterclockwise.
- Compact size so that it can be implemented with the TurtleBot3 and other vehicles in the lab.



(a) (b)

Figure 47: Direct averaging with angular tuning using Gazebo simulation with a virtual acoustic source. Part b) employs parameters that are close to what used in the numerical simulation of Fig. 46. Parameters used for the tests are

$$\begin{array}{r|c} \text{a)} & a = 0.005,\, k_p = 0.1,\, k_d = 0,\, \omega = 20,\ \omega_\xi = 1.0, v_{\text{com}} = 0.003 \\ \text{b)} & a = 0.5,\, k_p = 50,\, k_d = 0,\, \omega = 5,\ \omega_\xi = 2.0, v_{\text{com}} = 0.15 \end{array}$$

KEY: Seeker trajectory = ▬, Seeker initial position = ●, Source = ★.

- Able to operate using 5V supplied by an onboard Raspberry Pi. Minimal current draw within the range 100 to 200 mA.
- Needs to be implemented with an absolute encoder to track the angular position of the microphone frame during rotation. The motor may come with an encoder preinstalled, or a separate add-on encoder device can be used in combination to track the rotation.

Experiment Setup

Using parts available in the lab, several different motors were selected for analysis. As shown in Fig. 48, a Stepper Online Nema 17 stepper motor, a Hi-TEC HS-311 servo motor, and a Dynamixel XL430 servo motor were tested against each other.

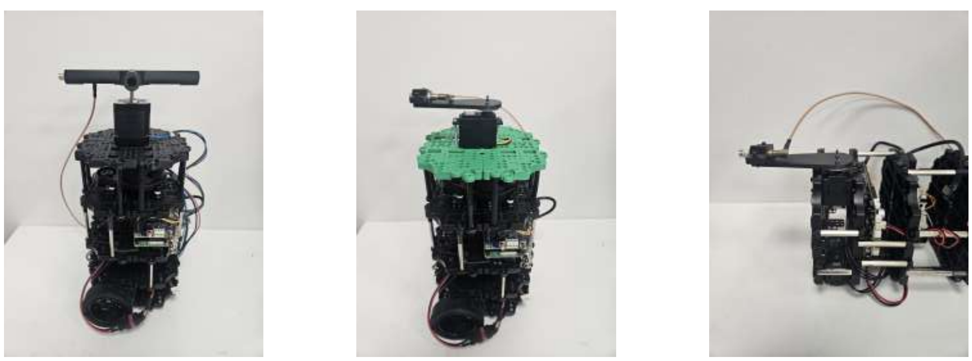
With these parts, several experiments were conducted where the microphone frame was spun back and forth by an active motor while the microphone sensor was recording audio simultaneously. Both the Nema 17 stepper and the Hi-TEC HS-311 servo were mounted atop TurtleBot3 (see Fig. 49). The microphone frame had to be mounted differently for the Dynamixel XL430 servo motor as this motor is originally used to drive the TurtleBot3's wheels. To use this motor, the robot was laid on its side with the microphone frame attached where the tire would normally be mounted (see Fig. 49c).

In all experiments a speaker was placed one foot away from the robot, while turned to face the vehicle. All microphone frames held the sensor at a radius of 3 inches from the motor's axle to be consistent. Tests were conducted with no sound playing, to capture the noise of just the active motor. Afterwards, we played linear chirp test tones from the speaker, to see how the active motor would affect the signal the microphone is trying to capture. The linear chirp test tone used in this experiment varies the instantaneous frequency of the signal linearly with time. This sound sweeps frequencies from 0 up to 7,500 Hz. This allows us to test how the magnitude of the signal recorded by the TurtleBot3's microphone varies over a wide range of frequencies.

In all tests, the motors would spin exactly the same way. Starting from rest, it would spin 180 degrees clockwise, pause, spin 180 degrees counterclockwise, pause again, and restart the process. This path allows the microphone sensor to scan the surrounding environment in front of the robot.

(a) (b) (c)

Figure 48: Motors tested: (a) Nema 17 stepper motor; (b) Hi-TEC HS-311 servo motor; and (c) Dynamixel XL430 servo motor.



(a) (b) (c)

Figure 49: Rotating microphone frames with: (a) Nema 17 stepper motor, (b) Hi-TEC HS-311 servo, and (c) Dynamixel XL430 servo.

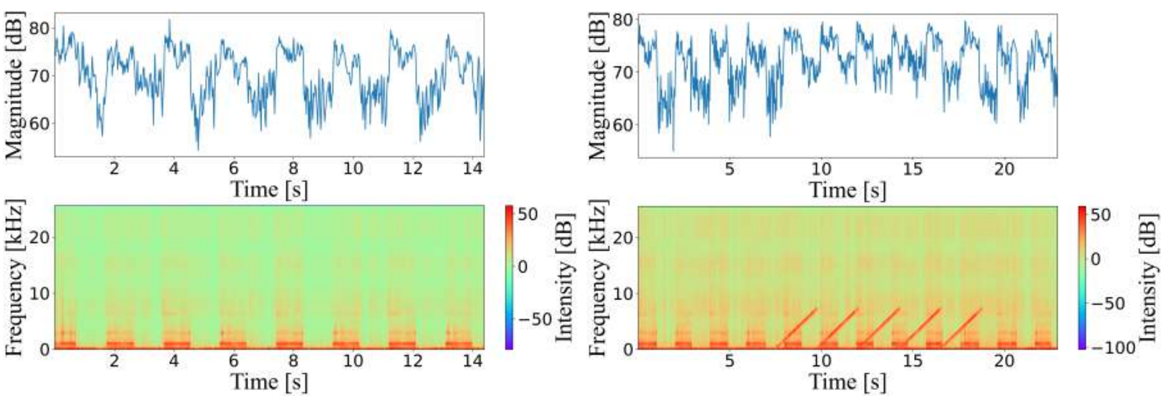
Results and Discussion

The Nema 17 stepper motor spins quite slowly. Subjectively, this motor creates noticeable humming sound during operation at frequencies under 2500 Hz, as shown in Fig. 50. When the motor pauses briefly before turning back the other way, the spectrogram clears up and the Root Mean Squared (RMS) sound pressure level (SPL) drops by about 5 to 10 dB. This level of motor noise can potentially mask the sound from the signal of interest.

The HS-311 servo motor was the next motor tested. Subjectively, the motor creates a high pitched squeal when it starts its spin. Compared to the Nema 17 stepper motor, this servo spins much faster. Observing the signal's spectrogram in Fig. 51, one can see the noisy areas where the servo is spinning, and the clearer areas where it is paused. Since this motor is fast, the noisy areas appear thinner, however the RMS magnitudes reveal that the noise generated is louder than before. Note that the noise generated by the HS-311 servo motor appears primarily at higher frequencies (around 7,000 Hz and 12,500 Hz), however there is still an effect on the measurements of the signal of interest.

Unlike the first two motors, Dynamixel servo motor is able to alter its rotation speed. With this motor, several tests were conducted to study the effect of the motor's RPM on the noise generated and picked up by the microphone. The results pictured in Fig. 52 were captured when the servo was spinning slowly at about 10 RPM. The brief spikes seen in the spectrograms and the RMS plots are from the servo accelerating up to speed from rest. This creates a very brief period of loud noise, but once the servo reaches a constant rotation speed, the noise seen in the spectrogram is minimal. In Fig. 52b, the linear chirp sounds are very clearly captured, and the effect of the background noise from the Dynamixel motor appears minimal.

Our experiments indicate that the best way to reduce the noise created by the motor is to spin it rather slowly. Further tests (not shown here) indicate a range of 10 to 20 RPM is acceptable for the noise generated by an active Dynamixel servo motor.

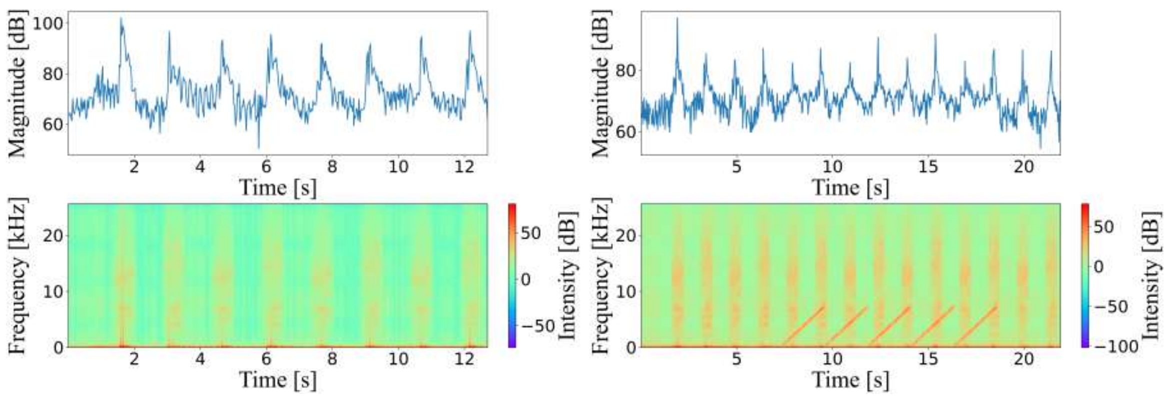


(a) (b)

Figure 50: Testing the Nema 17 stepper motor with: (a) background noise with an active motor and (b) a linear chirp test tone played by the speaker. Top: RMS magnitudes of the signal, Bottom: Signal Spectrogram.

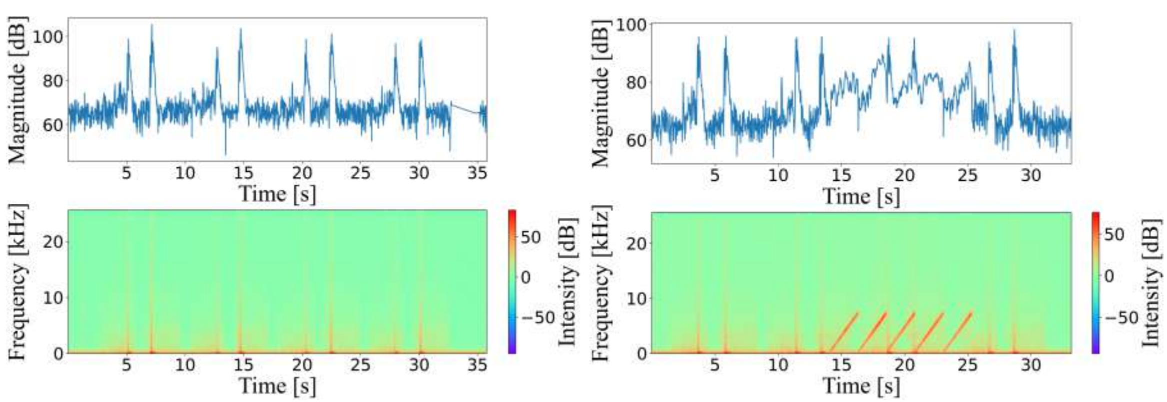
Overall these results indicate a need for a servo motor similar to the Dynamixel XL430 model. This motor meets almost all of the requirements outlined earlier, and it even allows a user to alter the motor's speed. For our purposes, we can set it to spin at slower speeds to decrease noise. The only drawback to the Dynamixel XL430 is that it requires 11.1V to power it, thus it cannot be run using the Raspberry Pi's 5V pins.

These findings motivated the search for a similar servo motor that meets our power requirement, and has adjustable speeds. The Parallax feedback servo motor in Fig. 53a was chosen since it meets all requirements. This servo motor offers bidirectional, continuous rotation, with built in speed controls allowing speeds of up to 120 RPM. The device also offers an internal absolute encoder which allows us to calculate the angular position of the microphone frame attached to the servo motor shaft. Additionally this servo motor is compact enough for the ground vehicle and can be easily implemented on other types of vehicles. Tests shown in Fig. 54 show an active motor spinning slowly back and forth at a speed of 20 RPM. The motor does not create a lot of noise, and signals captured



(a) (b)

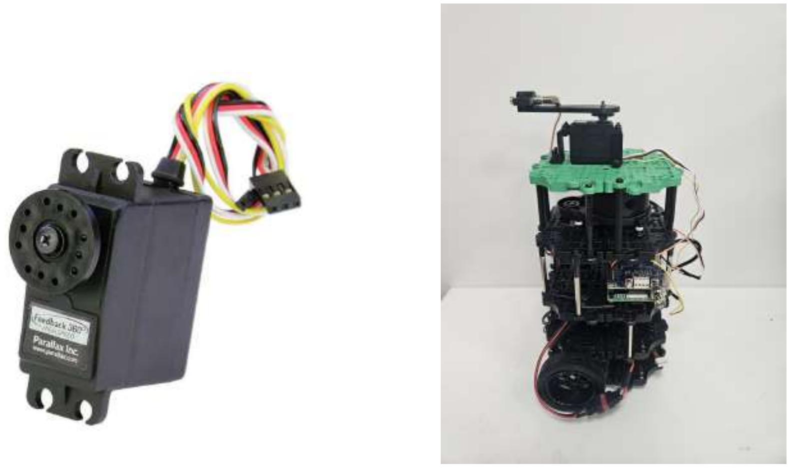
Figure 51: Testing the Hi-TEC HS-311 servo motor with: (a) background noise with an active motor, and (b) a linear chirp test tone played by the speaker. Top: RMS magnitudes of the signal, Bottom: Signal Spectrogram.



(a) (b)

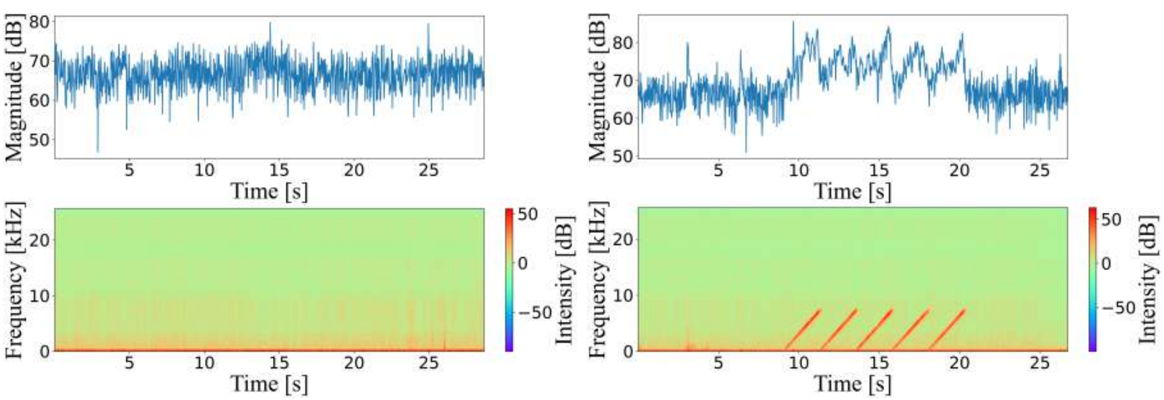
Figure 52: Testing the Dynamixel XL430 servo motor with: (a) background noise with an active motor, and (b) a linear chirp test tone played by the speaker.Top: RMS magnitudes of the signal, Bottom: Signal Spectrogram.

by the microphone appear very clearly. This Parallax servo motor will be used in experimental testing moving forward. This motor can be seen mounted on the vehicle shown in Fig. 55.



(a) (b)

Figure 53: The Parallax Feedback 360 servo motor: (a) individual part, and (b) mounted atop the TurtleBot3.



(a) (b)

Figure 54: Testing the Parallax servo motor with: (a) background noise with an active motor, and (b) a linear chirp test tone played by the speaker. Top: RMS magnitudes of the signal, Bottom: Signal Spectrogram

5.5.3.2 Microphone Characterization

The goal of this experiment is to characterize the microphone's sensitivity to sounds coming from different directions and distances to the source. Microphone frames of various sizes will be compared in order to find the optimal frame arm length, i.e. distance from the microphone sensor to the rotation axis of the frame's motor. In addition to these, TurtleBot3 will be placed at various

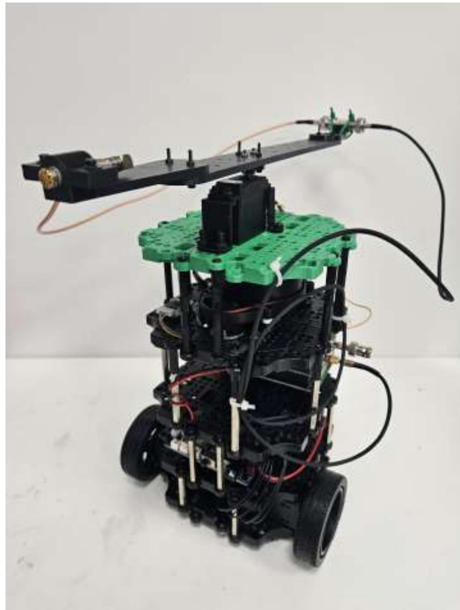


Figure 55: TurtleBot3 with a rotating sensor frame mounted atop the vehicle.

distances from the speaker in order to study the performance of the rotating microphone frame as the vehicle moves further away from the sound source.

Experiment Overview

The rotating microphone frame atop the TurtleBot3 is used to move the sensor to different angular positions (with respect to the vehicle's body) and perform measurements at each position to sample the cost function J. These acoustic samples (in decibels) are used by the ESC algorithms to estimate the gradient of the acoustic field in the environment. The ESC will then use the gradient estimations to update the estimated optimal parameter and determine the direction for the vehicle's motion to improve the cost. In order to get a sufficiently accurate estimate of the gradient however, there needs to be an appreciable difference in the measured sound levels as the microphone sensor changes its angular position.

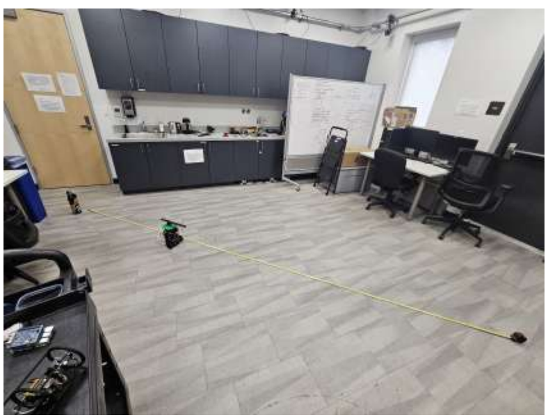
As the microphone frame rotates, the sensor traces out a circular path above the TurtleBot3, collecting samples within a 2D plane. Changing the size of the microphone frame changes the distance from motor's axle to the sensor at the end of the frame's arm. This changes the radius of the microphone's circular path. The goal of this experiment is to find a microphone frame arm size that is large enough to see an appreciable difference in dB as the frame is rotated but not so large where it would be impractical to implement.

There are several factors to consider when selecting a microphone frame size. Smaller rotating frames are compact and easier to implement on the TurtleBot3 and other vehicles. However, microphone data will be collected in a smaller radius, thus the sensor can only scan a small part of its environment. A bigger rotating frame can sample data in a larger radius, meaning it can scan more of its environment and potentially detect larger changes in dB values. However, bigger frames mean the TurtleBot3 needs to be given a wide berth, which can make it difficult for the vehicle to avoid nearby obstacles. An appropriate rotating microphone frame will be large enough so that the microphone can scan enough of its environment to detect a change in dB value, without being impractically large to implement.

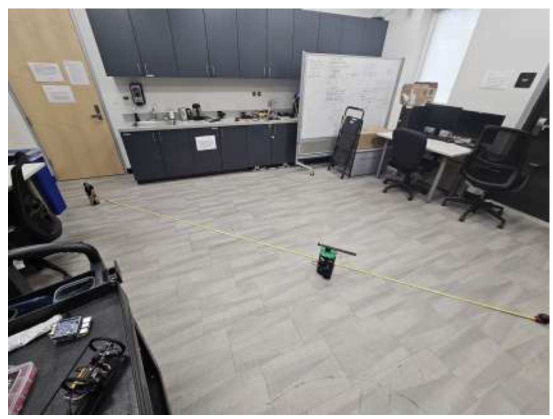
Experiment Setup

In this test, a speaker is placed atop a water bottle, and used to play a looping linear chirp test tone at a constant volume. Arm lengths of 1", 2", 3", 4", 5", 6", 8", and 10" were used to hold the microphone sensor atop the TurtleBot3, several of which are seen in Fig. 57. The vehicle itself was placed at distances of 1ft, 5ft, 10ft, and 15ft away from the source of the sound seen in Fig. 56. The microphone frame atop the TurtleBot3 was rotated in 15 degree increments. The test would start with the microphone frame at 0 degrees (i.e. pointing directly at the speaker), then turn the arm in 15 degree increments clockwise, until it would finish at an angle of 180 degrees (i.e. pointing directly away from the speaker). Each time the microphone frame turns, it would pause for 30-40 seconds, using the microphone to record audio at this specific angular position.

The data gathered from this test will be used to create microphone directionality plots. These are polar plots which demonstrate the microphone's sensitivity to sounds coming at the sensor from different directions. These plots show how the decibel reading from the microphone changes with respect to an angular position, and over various frequency bands of interest. The microphone directionality plots can then be used to analyze the performance of different frame arm lengths.



(a)



(b)

Figure 56: The TurtleBot3 was placed at various distances to the sound source. Shown are (a) 5 ft, and (b) 10 ft.

Results and Discussion

In this experiment, the same linear chirp test tone described in experiment I is used. This tone allows us to visualize how the magnitude changes with respect to angular position over several frequencies of interest. The results of our testing reveal the following trends shown in Tables 3 to 5. Figs. 58 to 60 display the performance of a microphone frame arm length of 6".

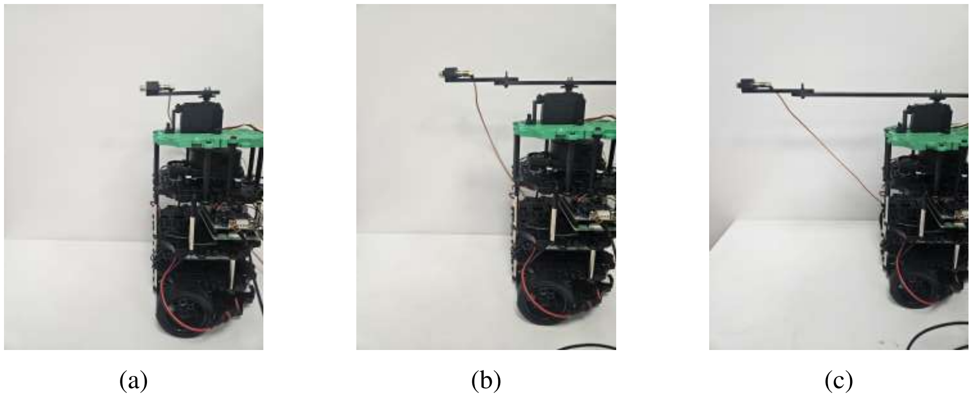


Figure 57: A comparison of some of the different microphone frame arm length sizes tested: (a) 3 inches, (b) 6 inches, and (c) 10 inches.

Distance From Source	Remarks
1ft	Drop of about 6-8 dB as the arm sweeps from 0 to 180 degrees over most frequencies.
5ft	Drops of about 4-8 dB for frequencies over 5000 Hz. Difficult to capture a change in decibel data at lower frequencies at this distance.
10ft	Difficult to capture any difference in dB level as the arm sweeps from 0 to 180 degrees over most frequencies.

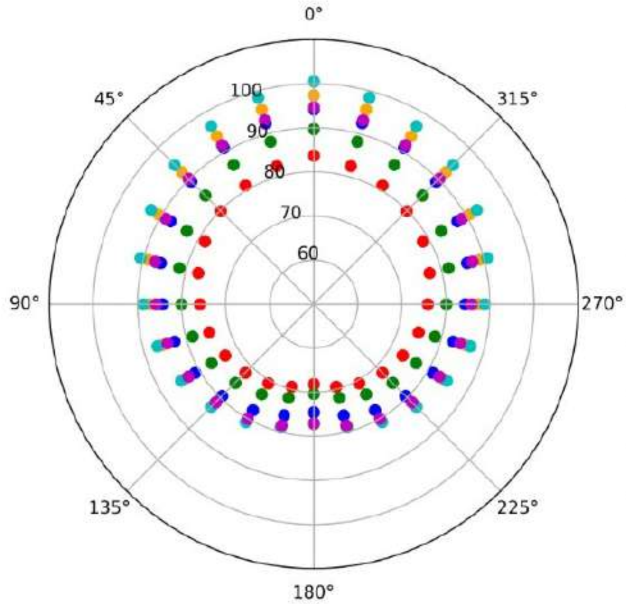
Table 3: Summary of results for microphone frames with short arm lengths of 1", 2", and 3".

Distance From Source	Remarks
1ft	Drop of about 10-20 dB as the arm sweeps from 0 to 180 degrees over most frequencies.
5ft	Drops of about 5-10 dB as the arm sweeps from 0 to 180 degrees for frequencies over 5000 Hz. Drops of about 3-5 dB seen at lower frequencies.
10ft	Drop of about 3-6 dB as the arm sweeps from 0 to 180 degrees for frequencies over 5000 Hz. Lower frequencies see drops of around 2-3 dB.

Table 4: Summary of results for microphone frames with moderate arm lengths of 4", 5", and 6".

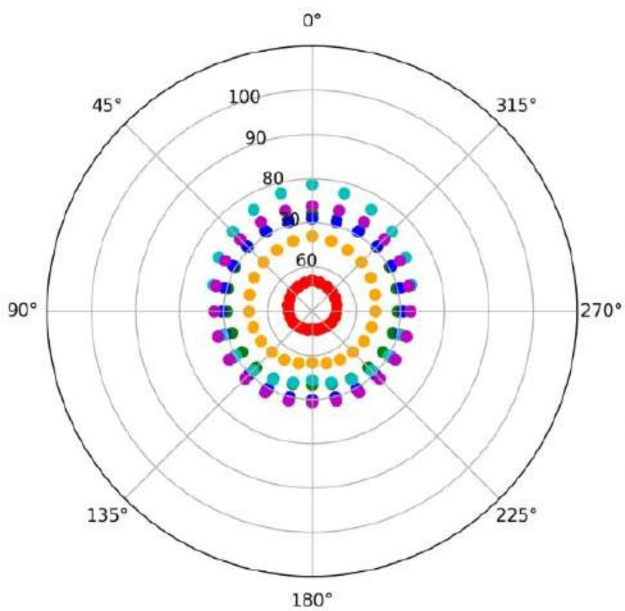
Distance From Source	Remarks
1ft	Drop of about 10-20 dB as the arm sweeps from 0 to 180 degrees over most frequencies.
5ft	Drop of 10-15 dB for frequencies over 5000 Hz. Lower frequencies see drops of around 4-5 dB.
10ft	Drop of about 5-10 dB for frequencies over 5000 Hz. Lower frequencies see drops of around 2-5 dB.

Table 5: Summary of results for microphone frames with long arm lengths of 8", and 10".



Frequency of Interest	Total dB Change from 0° to 180°
2000 Hz	15.6
3000 Hz	19.4
4000 Hz	19.7
5000 Hz	20.2
6000 Hz	23.5
7000 Hz	17.7

Figure 58: Servo arm length 6 inches, distance to source 1 ft.

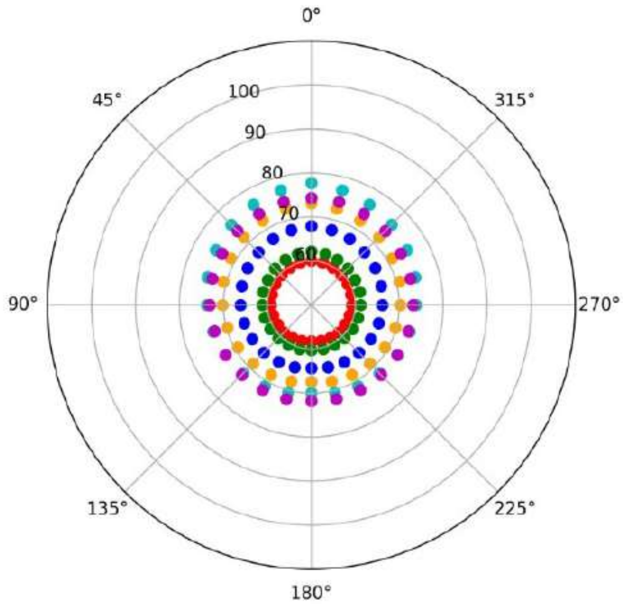


Frequency of Interest	Total dB Change from 0° to 180°
2000 Hz	3.0
3000 Hz	5.4
4000 Hz	1.0
5000 Hz	5.3
6000 Hz	13.1
7000 Hz	3.3

Figure 59: Servo arm length 6 inches, distance to source 5 ft.

Conclusions

The experimental results show that in general, shorter frame arms (1-3 inches) are not practical at long distance. The portion of the surrounding environment they can sample is just not large enough to really detect a noticeable difference in the sound pressure level. It will be difficult to estimate the gradient of the surrounding acoustic field without detecting any difference. Moderate



Frequency of Interest	Total dB Change from 0° to 180°
2000 Hz	1.9
3000 Hz	2.0
4000 Hz	3.6
5000 Hz	5.7
6000 Hz	7.7
7000 Hz	2.6

Figure 60: Servo arm length 6 inches, distance to source 10 ft.

arm lengths (4-6 inches) work better, they have satisfactory performance at the distances we were able to test in the lab. There is a noticeable change in the decibel level as we spin the microphone around atop the TurtleBot3. At long range, we were able to capture enough of a difference in dB at high frequency ranges that is adequate for our purpose. The longer arm lengths have the best performance. Since they are able to sample a larger portion of the surrounding environment, they see the the biggest change in sound pressure level with a change in the sensor's angular position. Longer arms (8-10 inches) can yield the best estimate of the gradient of the surrounding acoustic field, however these long arms cause implementation problems with a small vehicle like the TurtleBot3.

We decided to move forward with a microphone frame with a 6 inch arm length. Our experiments show that this frame size is large enough to sample enough of the environment, while being small enough to not have issues in practical settings.

5.5.4 GESC with a Rotating Sensor

Having designed the rotating sensor platform for the TurtleBot3, we now need to incorporate the rotating sensor into the ESC. Since there are wires connecting the microphone or photoresistor to the TurtleBot3, we seek to limit the servo arm angle ϕ such that $\phi(t) \in [-\pi/2, \pi/2]$ for all t to avoid entangling the wires. This constraint prevents using sinusoidal perturbations, but we want to find perturbation signals p which have m's for estimating the derivatives in the relative frame. Thus, we use the baseline sensor arm angle profile of

$$\phi(t) = \begin{cases} -\pi/2 + \pi \cdot (t \bmod 2), & \text{if } t \bmod 2 < 1, \\ \pi/2 - \pi \cdot ((t-1) \bmod 2), & \text{otherwise.} \end{cases} \tag{5.21}$$

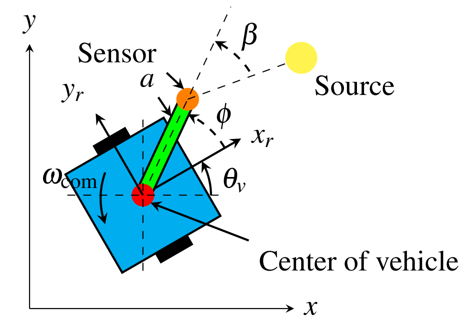


Figure 61: UGV with a rotating sensor.

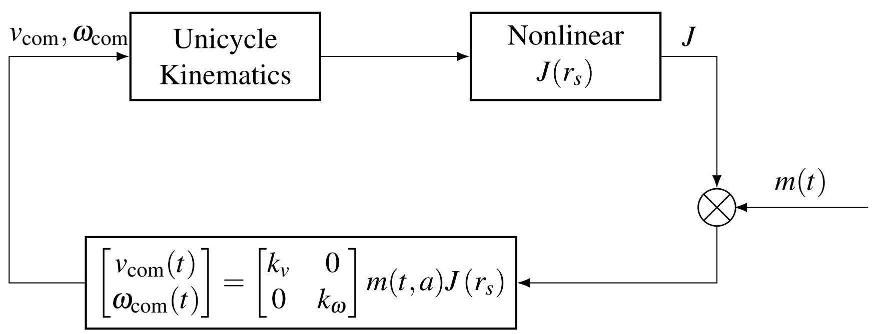
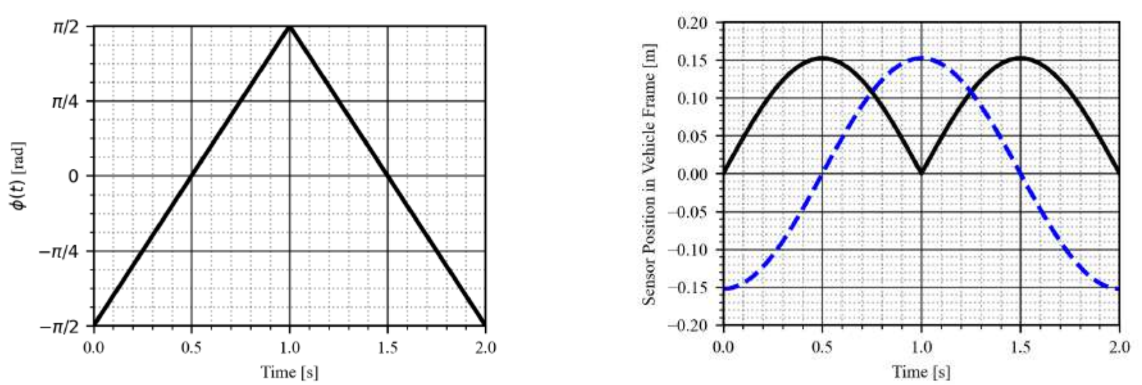


Figure 62: Block diagram for the nonholonomic unicycle gradient-based ESC



(a) (b)

Figure 63: (a) The $\phi(t)$ function over time and (b) The sensor position in the x-y relative frame over time.
KEY for b): $p_{x_r} =$ ▬ , $p_{y_r} =$ ▪▪▪.

which is a periodic function with a period of 2. Perturbations of the sensor in the relative frame is then

$$ap(t) = a \begin{bmatrix} p_{x_r}(t) \\ p_{y_r}(t) \end{bmatrix} = a \begin{bmatrix} \cos(\phi(t)) \\ \sin(\phi(t)) \end{bmatrix} \tag{5.22}$$

where p_{x_r} and p_{y_r} are the perturbations in the relative frame of the vehicle. From the theory in Section 4.1, we know that

$$m(t,a) = \begin{bmatrix} 2\cos(\phi(t))/a\left(1-\frac{8}{\pi^2}\right) \\ 2\sin(\phi(t))/a \end{bmatrix} \tag{5.23}$$

can be used to estimate the gradient with $\hat{g}(\omega t, r_c, a) = m(\omega t, a)J(r_s)$. With the gradient estimate, we can construct a feedback law with the commanded linear and angular velocities as

$$\begin{bmatrix} v_{\text{com}} \\ \omega_{\text{com}} \end{bmatrix} = -\begin{bmatrix} k_v & 0 \\ 0 & k_\omega \end{bmatrix} \hat{g}(\omega t, r_c, a) \tag{5.24}$$

The logic behind this controller is simple, we simply need to move the robot forward when the robot's forward axis is somewhat aligned in the direction of steepest descent and rotate the robot so that its forward axis is more aligned with direction of steepest descent. With the controller designed, we can now investigate the performance of a GESC with a rotating sensor against the traditional statically mounted sensors.

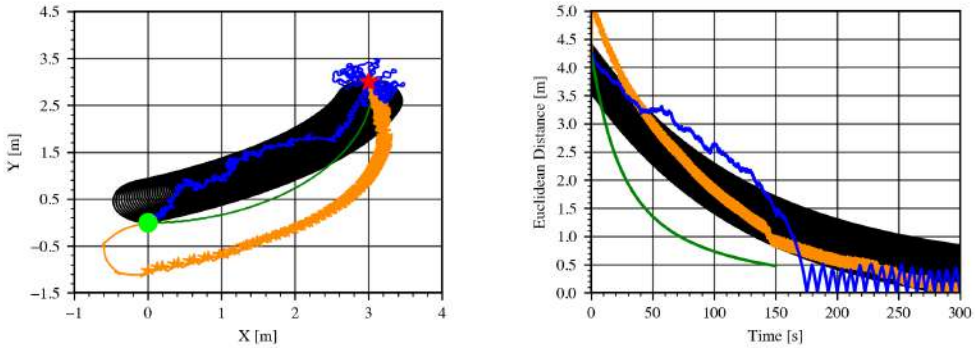
Experiments Setup

The experimental setup for the rotating sensor, with an arm length of 0.152m (or 6") as seen in Fig. 23, mirrors the test setup outlined in the preceding section on testing types (Section 5.4). Along with analyzing the acoustic cost function (described in Section 5.3.4.1) as our primary source, we discuss some results obtained with a light source (described in Section 5.3.4.1) due to unresolved issues with the acoustic cost function in the context of indoor laboratory experiments. The types of sensors required to analyze these acoustic and light sources include a microphone to measure sound pressure levels and a photoresistor diode to monitor changes in light intensity through resistance variations. Additionally, we evaluate various distances (discussed in Section 5.4) and orientations to assess the system's robustness; however, not all results from these tests are included, as some details do not significantly impact the overall analysis. We will show a selection of successful results and discuss any challenges encountered during testing, as well as potential tests for future experiments.

Results and Discussions

We will be comparing results on Lie Bracket and direct averaging methods with our proposed design of a rotating sensor. This will be shown with the three different testing methods discussed in Section 5.4, including the numerical simulation, Gazebo simulation, and real-world laboratory experiments. Figs. 64, 65, and 66 compare the performance of the GESC methods with static sensor mount (Lie Bracket method [42], the angular tuning method [53], and forward tuning method [24]), and the GESC with a rotating sensor mount, in response to different types of signal sources. The results for the GESC rotating sensor design demonstrate enhanced robustness in robot's velocity control while achieving a faster convergence to the source for both the virtual quadratic and virtual acoustic sources. When we refer to convergence in the following results, we define the seeker as converging to the source when it reaches a vicinity of approximately 0.5 meters from the source. This classification will be applied to the results of all three different testing methods described.

The Lie Bracket, angular tuning, and forward tuning approaches do not converge as quickly as the rotating sensor method when applied with the virtual quadratic source, virtual acoustic sources,



(a) (b)

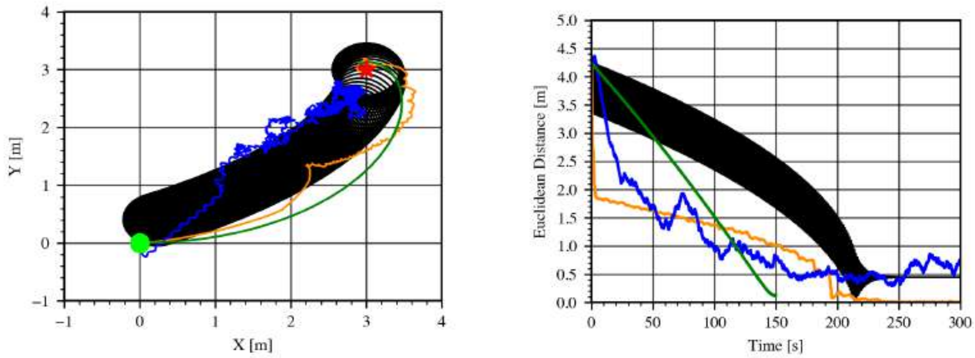
Figure 64: Comparison of different methods in numerical simulations using a virtual quadratic source with (a) the trajectory and (b) Euclidean distance vs. time. Parameters for each method are:

GESC Lie bracket [7]	$\omega_\xi = 0.1$, $\omega = 0.5$, and $k = 0.15$
GESC angular-tuning [53]	$a = 0.5$, $k_p = 70$, $k_d = 0$, and $\omega = \omega_\xi = 5$
GESC forward-tuning[24]	$\omega = 10$, $a = 0.1$, $k_p = 0.5$, $\omega_\xi = 2$
GESC rotating-sensor	$k_v = 0.00015$ and $k_\omega = -0.003$

KEY: GESC Lie bracket = , GESC angular-tuning = , GESC forward-tuning = GESC w/rotating-sensor = , Seeker initial position = , and Source =

and virtual light source. The GESC rotating sensor follows a smoother path and achieves faster convergence. Notably, the angular tuning, forward tuning, and Lie Bracket methods, as shown in Fig. 64, respectively converge at approximately 175 seconds and 200 seconds for both forward tuning and Lie Bracket, whereas the GESC rotating sensor converges in about 140 seconds for the virtual quadratic source. For the virtual acoustic source shown in Fig. 65, the angular tuning, forward tuning, and Lie Bracket converge in approximately 170 seconds, 190 seconds, and 210 seconds, respectively, while the rotating sensor converges in 135 seconds. As indicated in Section 5.3.2, the virtual acoustic cost function presents a more realistic scenario, which accounts for the extended convergence time for all methods. The parameters for our k gains are deliberately small float values for the rotating sensor. This choice is aimed to slow the system down to comply with the TurtleBot3 speed constraints, thereby replicating real-world testing conditions as closely as possible. Before testing the algorithm on the real robot, we confirm that it works in the Gazebo simulation. From the simulation shown in Fig. 67, the rotating sensor methods is able to converge to the source with a more direct path and with the fastest convergence time. The other methods take more than 200 seconds to reach the light source while the rotating sensor takes roughly 100 seconds to converge. In addition to the virtual acoustic source, the virtual light source was also tested, and all four methods were evaluated. The Gazebo simulation demonstrated that, when the initial robot orientation was set to 45°, all methods successfully converged to the source. The Lie bracket method required 420 seconds, the forward tuning method took 975 seconds, the angular tuning took 180 seconds, and the rotating sensor method completed in 75 seconds (Table 7), indicating that this approach achieves faster convergence compared to the others.

In the laboratory experiments for the real acoustic source we ran into similar issues as seen in



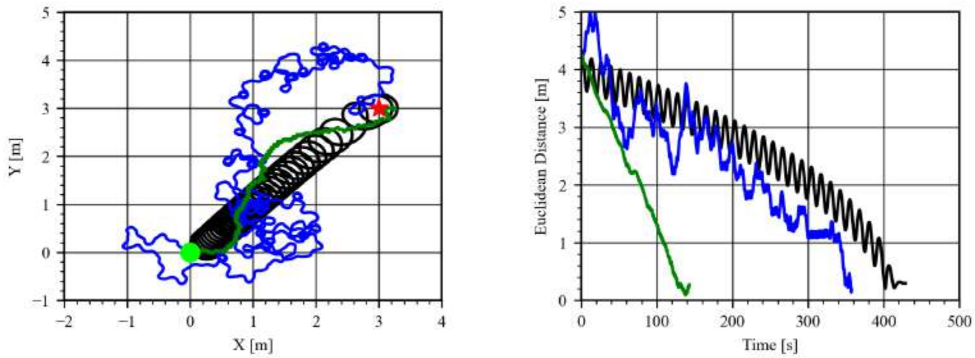
(a) (b)

Figure 65: Numerically simulated trajectories for various methods on a virtual acoustic source with (a) the trajectory in $x-y$ position and (b) The Euclidean distance vs. time. Parameters for each method used are

GESC Lie bracket [7]	$\omega_\xi = 0.01$, $\omega = 5$, $k = 0.75$
GESC angular-tuning [53]	$a = 0.5$, $k_p = 50$, $k_d = 0$, $\omega = 5 = \omega_\xi$
GESC forward-tuning [24]	$\omega_\xi = 1$, $\omega = 2.5$, $a = 0.005$, $k_p = 0.08$
GESC rotating-sensor	$k_v = 0.000004$, $k_\omega = 0.0008$

KEY: GESC Lie bracket = ▬ , GESC angular-tuning = ▬ , GESC forawrd-tuning = ▬ , GESC w/rotating-sensor = ▬ , Seeker initial position = ● , Source = ★

Section 5.5.2 with the cost function, where the seeker was only able to converge to the source with the initial distance of (1m, 1m). The seeker performed effectively when it started within 1 meter of the source, but challenges emerged for initial distances beyond 1 meter. This issue may stem from flat regions in the cost function, where gradients are minimal, and/or from saddle points or equilibrium-like regions, which can similarly stall optimization due to insufficient fluctuation of the cost value for accurate estimation. Another potential explanation for this phenomenon is the strong presence of sound ray reflections near the corners of the room. In such positions, the sensor directionality may lead to inconsistent gradient estimations, resulting in ambiguity regarding the correct direction of motion. Due to issues with the acoustic source, we pivot to using a light source for the testing which was previously discussed in Section 5.3.4.1. In the testing with the photo-resistor source, as illustrated in Fig. 68, the design of the rotating sensor demonstrates an improved convergence rate when compared to alternative methods. When examining the results in Table 7, with the robot's initial heading angle oriented towards the source (45°), the Lie bracket method took approximately 330 seconds to converge. The angular tuning process required 190 seconds, the forward tuning took 360 seconds, and the rotating frame method achieved convergence to the light source in 54 seconds. We can conclude that the rotating sensor design converges more quickly and efficiently compared to the other methods. The Lie bracket method follows a relatively inefficient path toward the source, which results in higher power consumption due to extensive area exploration. The angular tuning method reaches the source in a relatively reasonable timeframe; however, the forward velocity must be limited to approximately 0.02 m/s to allow the system sufficient time to estimate



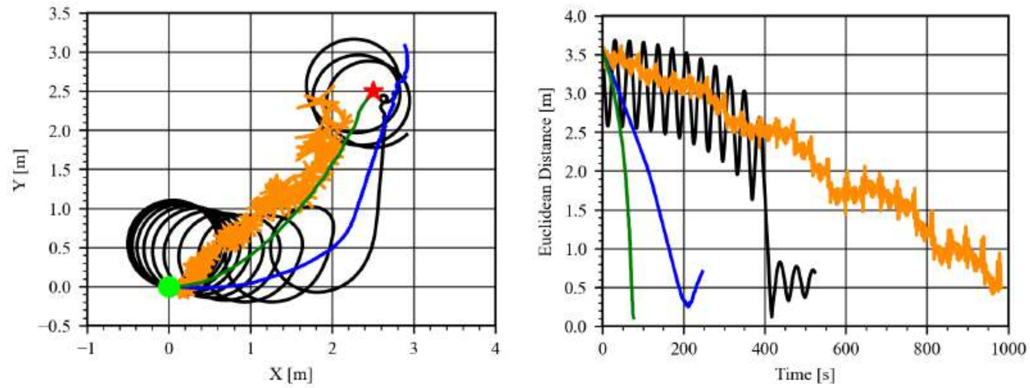
(a) (b)

Figure 66: Gazebo simulation trajectories for the various methods acting on a virtual acoustic source. (a) The trajectories in the $x-y$ plane, (b) The Euclidean distance over time. Parameters for each method used are

GESC Lie bracket	$\omega_\xi = 0.1$, $\omega = 0.5$, $k = 0.15$
GESC angular-tuning	$a = 0.5$, $k_p = 50$, $k_d = 0$, $\omega = 5$, $\omega_\xi = 2$
GESC rotating-sensor	$k_v = 0.000015$, $k_\omega = 0.003$

KEY: GESC Lie bracket = ▬, GESC angular-tuning = ▬, GESC w/rotating-sensor = ▬, Seeker initial position = ● Source = ★.

the source location and appropriately adjust the angular velocity. If the forward velocity is too high, the robot risks passing by or overshooting the source and then slowly corrects its position, as discussed in Section 5.5.2.2. Additionally, our testing environment is limited in space, a slower forward speed helps to prevent the robot from hitting surrounding walls before convergence, ensuring proper alignment with the source. The forward tuning method is capable of locating the source; however, its local exploration process is significantly slower than other methods and requires considerable energy, making it less practical for frequent use. The rotating sensor approach reaches the source via a more direct path, which is more energy-efficient since it relies on estimations derived from the rotating sensor rather than the robot's velocities. It then guides the robot in that direction. The key limitations of the rotating sensor design include the rotation speed of the sensor and the overall speed of the robot. Examining the other initial orientations up to $-45°$ (Table 7), all four methods demonstrate varying convergence times. The forward tuning and lie bracket methods successfully converge across all initial orientations, whereas the angular tuning and rotating sensor methods encounter difficulties in converging for the initial orientation of $-90°$ and $-135°$. Both of these methods struggle at these starting orientations since they are pointed away from the source. The angular tuning method requires the sensor to face toward the source in order to have the best chance of converging within a reasonable timeframe. The rotating sensor is unable to reach the source in these orientations for a couple reasons, one has to do with light source intensity or the cost function. In the simulation at $-90°$ the robot is able to converge, as our cost function models the source as an ideal point, not accounting for the fact that real-world light intensity may be lower. For example, in the laboratory, if the light source batteries are older, the light will appear dimmer compared to



(a) (b)

Figure 67: Gazebo simulation trajectories for the various methods acting on a virtual light source with: (a) the trajectory in $x-y$ plane and (b) The Euclidean distance vs. time. Parameters for each method used can be found in Table. 6. All seekers start at the origin with an initial heading angle of $\theta_{v,0} = 0^o$ and the source is located at $r^* = (2.5, 2.5)$ meters.

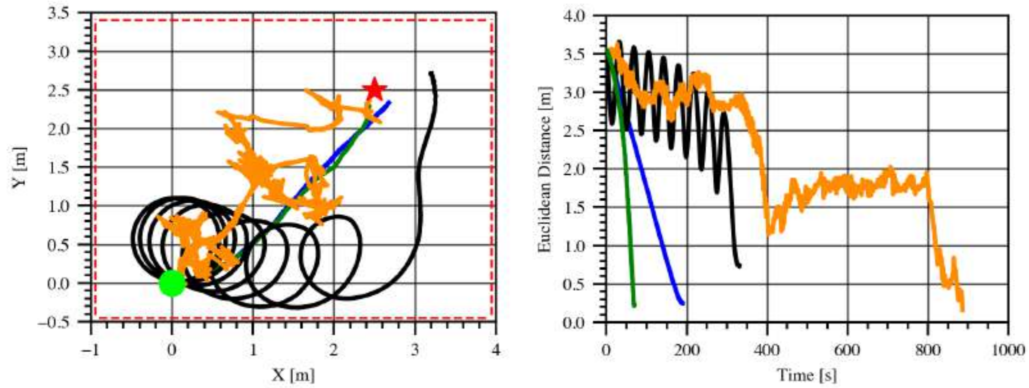
KEY: GESC Lie Bracket= , GESC angular-tuning= , GESC forward-tuning= , GESC w/rotating-sensor = , Room Walls = , Seeker initial position = , Source = .

new batteries. We tested this and observed improved convergence rates after replacing the batteries. Another significant factor affecting convergence at $-90°$ and $-135°$ relates to hardware limitations in the rotation mechanism. The arm needs to be able to rotate beyond $-90°$ to $90°$, so that the system can obtain an estimate of the source's position. Currently, the rotation range does not allow the robot to estimate what is behind it, but with minor modifications to the arm's rotation capabilities, the vehicle could successfully converge, making this approach more robust than the angular tuning method. This was verified through simulations involving increased arm rotation, which resulted in successful convergence and was faster than both the lie bracket and forward tuning methods.

The testing with a real acoustic source will be re-examined with future testing in order to get some usable results and compare the methods. As stated earlier, the calculation of the sound data needs modifications to account for a range of frequencies instead of measuring Root Mean Squared (RMS) value or sound pressure level (SPL) of the data. This will enable the system to be more

Table 6: General parameters for the methods used for both Gazebo simulation and laboratory testing with the light source

Lie bracket[7]		**Angular tuning** [53]		**Forward tuning**[24]		**Rotating Sensor**	
symbol	value	symbol	value	symbol	value	symbol	value
ω	0.18	ω	3.0	ω	1.6	ω	20.0
v_{com}	0.1	v_{com}	0.02	ω_{com}	0.15	kv	–0.05
h	1.0	h	0.05	h	0.5	$k\omega$	–0.5
k	6.0	α	0.3	α	0.1		
μ	0.05	c	80.0	c	35.0		
		d	0.0				



(a) (b)

Figure 68: Laboratory experiment comparing GESC methods which include Lie bracket, Direct averaging angular tuning and forward tuning, as well as the rotating sensor design using a photo-resistor. (a) the trajectory in $x-y$ plane, (b) The Euclidean distance vs. time. Parameters for the methods used can be found in Table 6. All seekers start at the origin with an initial heading angle of $\theta_{v,0}$ and the source is located at $r^* = (2.5, 2.5)$ meters.

KEY: GESC Lie bracket with static sensor = ▬, GESC angular-tuning with static sensor = ▬, GESC forward-tuning with static sensor = ▬, GESC with rotating-sensor = ▬, Room Walls = ▪▪▪▪, Seeker initial position = ●, Source =★.

precise in estimating a signal to converge as we are processing the data for optimal frequencies rather than calculating the intensity of the sound. Another potential way to increase the estimation accuracy is to increase the range of rotation of the sensor to 360° by additional hardware changes; this may potentially mitigate the effect of sound reflections near the corners. From some initial testing in the Gazebo simulation (not shown here), we anticipate there to be an improvement in the convergence time. Therefore, once implemented, we will also retest the robot with a full rotating sensor to determine the improved convergence rates. Additionally, we had successful results with the Gazebo simulation for the virtual acoustic source, therefore, we can make an assumption that with the real sound source we should see improved results. For access to videos of the tested methods, please refer to the hyperlinks provided in the footnote. [1]

5.5.5 Adaptive Methods with a Rotating Sensor

In this section, we analyze the performance of two different adaptive methods: RMSProp ESC and AdaGrad ESC. We will make use of the TurtleBot3 equipped with a rotating sensor seen in Fig. 69. With this vehicle, we utilize two different choices of spin profile denoted as "full" and "back-and-forth" rotation to investigate the ESC's performance for different ways of spinning the sensor. In this section, we discuss the derivative estimation signals used and present the block diagrams for these ESC schemes, then perform extremum seeking experiments for both light and acoustic-based cost functions. We utilize the results from light extremum seeking experiments to highlight the benefits of these adaptive methods over a Gradient ESC. In acoustic extremum seeking experiments, we

[1] Playlist of GESC videos for light source.

Table 7: Method Performance at Various Initial Heading Angles (Gazebo vs. Lab)(seconds)[s] with a Light Source

Angle ($\theta_{v,0}$) / **Method**	**45°**		**0°**		**–45°**	
	Gazebo	Lab	Gazebo	Lab	Gazebo	Lab
Forward Tuning	975	363	1150	893	975	828
Angular Tuning	180	189	182	190	185	218
Lie Bracket	420	332	520	343	620	300
Rotating Sensor	**75**	**54**	**80**	**70**	**82**	**110**

Angle ($\theta_{v,0}$) / **Method**	**–90°**		**–135°**	
	Gazebo	Lab	Gazebo	Lab
Forward Tuning	1500	883	1200	425
Angular Tuning	250	N/A	N/A	N/A
Lie Bracket	690	302	650	487
Rotating Sensor	**128**	N/A	N/A	N/A

demonstrate the ability of each of the ESC schemes to converge on the source's position as it is moved to different locations in the environment.

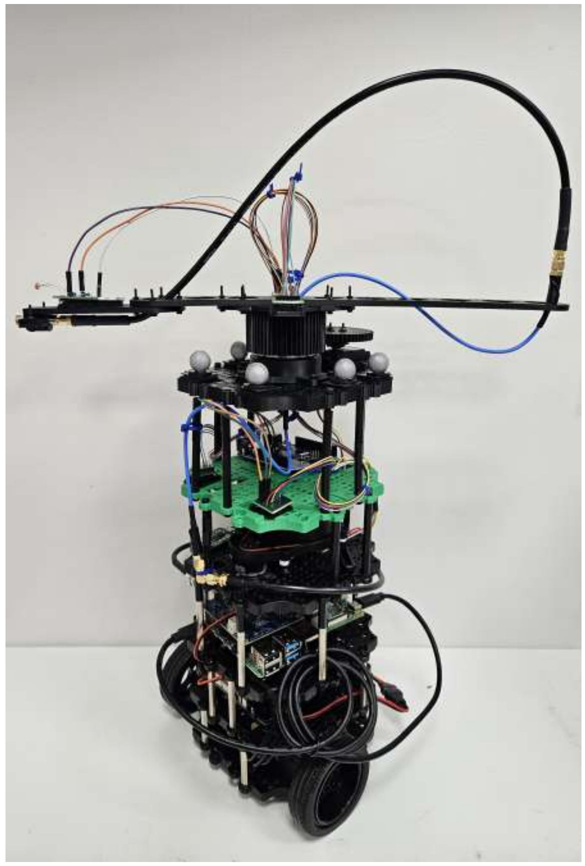
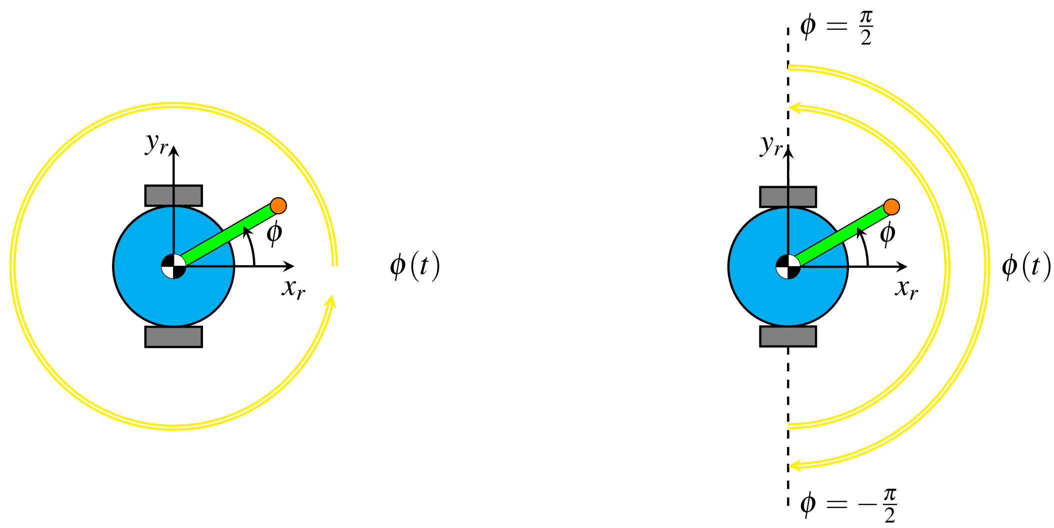


Figure 69: TurtleBot3 vehicle equipped with a slip ring which allows the rotating sensor frame to spin freely without wrapping up any of the wires on top.

5.5.5.1 Derivative Estimation

Both the RMSProp and AdaGrad ESC methods require estimation of both gradient terms and the gradient outer product to operate. We take the perturbation signals from Eq. 5.22 where these signals merely represent the position of the sensor relative to the vehicle's center. The perturbation signals are a function of the rotating frame's angular position $\phi(t)$. The two different spin profiles in Fig. 70 give two different angular position functions for experimentation. In this section, we utilize perturbation-based estimation techniques from Section 4.1.2 to develop gradient and gradient outer product estimation signals ($m(t)$ and $M(t)$) to accommodate these two different choices of spin profiles as discussed below.



(a) (b)

Figure 70: Spin profiles tested with adaptive ESC schemes: (a) full rotation, and (b) back-and-forth rotation in an envelope of $-\frac{\pi}{2} \leq \phi \leq \frac{\pi}{2}$ radians.

Full Rotation Spin Profile

In our implementation, we typically set the rotation speed of the sensor frame to be a constant, we denote this speed with the variable ω_s. For the full rotation spin profile, the frame angular position function is given by the following

$$\phi(t) = \omega_s \cdot t \tag{5.25}$$

One can estimate the gradient with the following demodulation signals

$$m(t) = \frac{2}{a} \cdot \begin{bmatrix} \cos(\phi(t)) \\ \sin(\phi(t)) \end{bmatrix} \tag{5.26}$$

The gradient squared terms for RMSprop-based ESC (RMSpESC) are estimated using

$$M(t) = \frac{1}{a^2} \cdot \begin{bmatrix} 3 - 4\sin^2(\phi(t)) \\ 4\sin^2(\phi(t)) - 1 \end{bmatrix} \tag{5.27}$$

and the following demodulation signals estimate the half vectorized gradient outer product terms for AdaGrad ESC (AGESC)

$$M(t) = \frac{1}{a^2} \cdot \begin{bmatrix} 3 - 4\sin^2(\phi(t)) \\ 2\sin(\phi(t)) \\ 4\sin^2(\phi(t)) - 1 \end{bmatrix} \tag{5.28}$$

Back-and-Forth Rotation Spin Profile

For a rotating sensor frame spinning back-and-forth (BnF) in front of the vehicle within an envelope of $-\frac{\pi}{2} \leq \phi(t) \leq \frac{\pi}{2}$, the frame angular position function is given by the following

$$\phi(t) = \begin{cases} -\frac{\pi}{2} + \omega_s t & t \leq \frac{\pi}{\omega_s} \\ \frac{\pi}{2} - \omega_s(t - \frac{\pi}{\omega_s}) & \frac{\pi}{\omega_s} < t \leq \frac{2\pi}{\omega_s} \end{cases} \tag{5.29}$$

One can estimate the gradient with the following demodulation signals

$$m(t) = \frac{2}{a} \cdot \begin{bmatrix} \frac{\pi^2}{\pi^2-8}\cos(\phi(t)) - \frac{2}{\pi^2-8} \\ \sin(\phi(t)) \end{bmatrix} \tag{5.30}$$

The gradient squared terms for RMSpESC are estimated using

$$M(t) = \frac{1}{a^2} \cdot \begin{bmatrix} 117.293 & -17.970 \\ -17.970 & 5.420 \end{bmatrix} \cdot \begin{bmatrix} \cos^2(\phi(t)) - \frac{4}{\pi}\cos(\phi(t)) + \frac{4}{\pi^2} \\ \sin^2(\phi(t)) \end{bmatrix} \tag{5.31}$$

and the following demodulation signals estimate the half vectorized gradient outer product terms for AGESC

$$M(t) = \frac{1}{a^2} \cdot \begin{bmatrix} 117.293 & 0.0 & -17.970 \\ 0.0 & 4.351 & 0.0 \\ -17.970 & 0.0 & 5.420 \end{bmatrix} \cdot \begin{bmatrix} \cos^2(\phi(t)) - \frac{4}{\pi}\cos(\phi(t)) + \frac{4}{\pi^2} \\ 2(\sin(\phi(t)))(\cos(\phi(t)) - \frac{2}{\pi}) \\ \sin^2(\phi(t)) \end{bmatrix} \tag{5.32}$$

5.5.5.2 Implementation of Adaptive Methods

We implement the RMSpESC and AGESC adaptive methods utilizing the control schemes shown in the following block diagrams. Note that the gradient demodulation signals $(m(t))$ and gradient outer product demodulation signals $(M(t))$ are interchangeable based on the user's choice of spin profile.

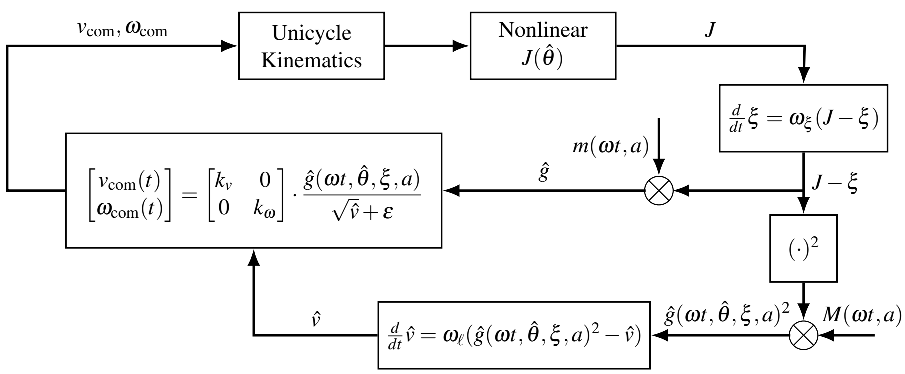


Figure 71: Block diagram for the RMSpESC scheme utilized with the nonholonomic unicycle for light source seeking experiments.

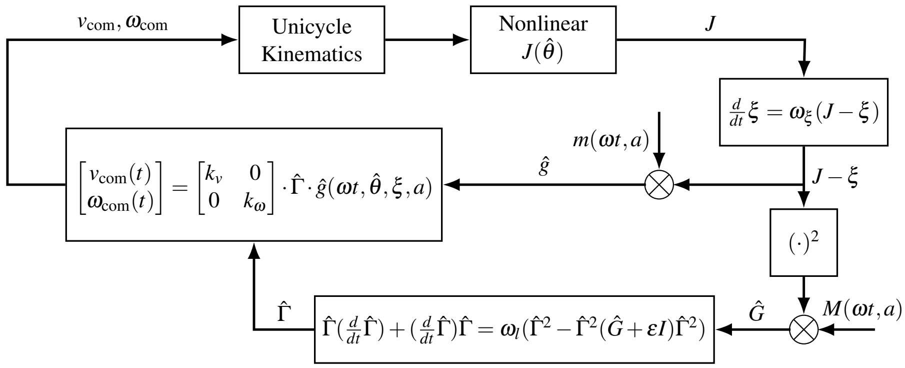


Figure 72: Block diagram for the AGESC scheme utilized with the nonholonomic unicycle for light source seeking experiments.

For acoustic extremum seeking experiments, we introduce an additional low-pass filter to filter the cost function values before inputting them further into the controller. Since microphone measurements are affected by noise, the noise issue persists when the sensor data is converted to a cost function value. The low-pass filter helps attenuate this noise, improving the performance of the ESC. We introduce the low-pass filter parameter ω_j into all acoustic extremum seeking schemes.

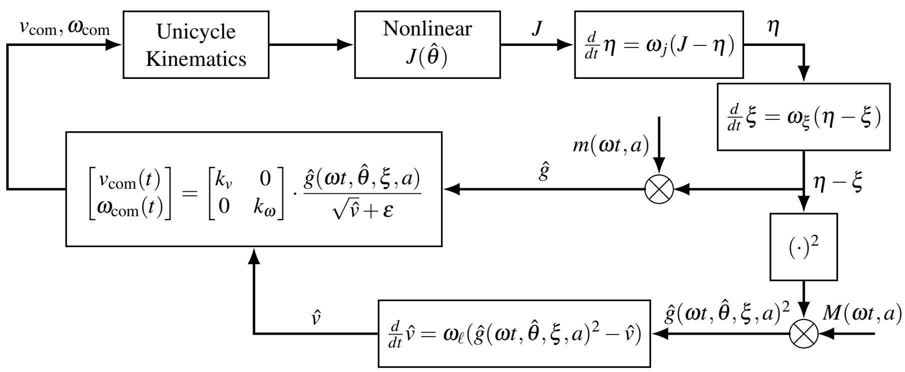


Figure 73: Block diagram for the RMSpESC scheme utilized with the nonholonomic unicycle for acoustic source seeking experiments.

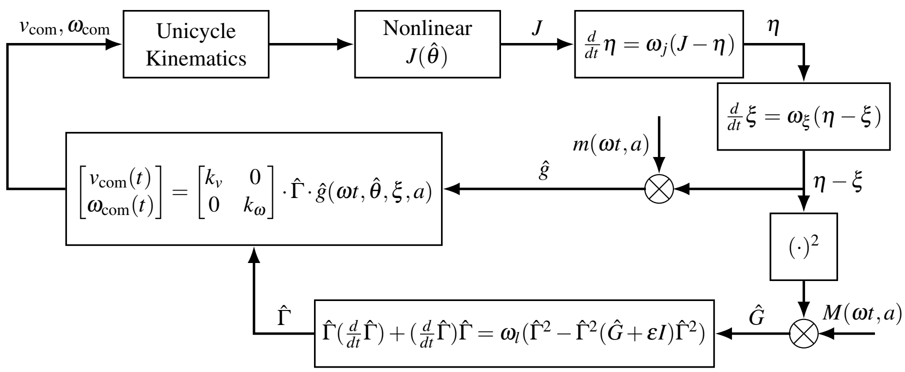


Figure 74: Block diagram for the AGESC scheme utilized with the nonholonomic unicycle for acoustic source seeking experiments.

5.5.5.3 Light Extremum Seeking Experiments

RMSProp ESC Performance

We evaluate the performance of the RMSpESC scheme shown in Fig. 71 with light-based cost functions in the laboratory environment. We utilize the photoresistor sensor to obtain both resistance-based and voltage-based cost values to evaluate the performance of the ESC for the different cost function. In the following experiments, different colors are used to document a sweep of different initial heading angles $\theta_{v,0}$ seen in Fig. 32c. In all the following experiments, the initial sensor angular position is set to be $\phi_0 = -\frac{\pi}{2}$ with respect to the relative vehicle frame regardless of the spin profile used. Figures 75-78 show the path traveled by the seeker and a time history of the distance to the source for the parameter sweep of $\theta_{v,0} = 45°$ to $\theta_{v,0} = -135°$. The gray dashed line in the time history plot describes an absolute distance of 0.5 meters from the source. When the vehicle crosses this threshold, it has completed its objective of navigating to the source, and we take this as the convergence time of the experiment. For the RMSpESC scheme, we used the same parameters (seen in Tables 16 and 17) regardless of the cost function or spin profile choice, with no retuning required.

The results for both the voltage and resistance cost values tabulated in Tables 8 and 9 show that the full rotation spin profile consistently yields slower convergence than the back-and-forth profile. The convergence time with the back-and-forth spin profile was around 10 to 20 seconds faster than the convergence time of full rotation spin profile for initial heading angles $\theta_{v,0} = 45°$, $0°$ where the seeker starts with its heading somewhat facing towards the source. However, as the seeker's initial heading turns away from the source (i.e., $\theta_{v,0} = -45°$, $-90°$, $-135°$), the difference in convergence time between the two choices of spin profile grows to about 40 to 50 seconds. Although the full rotation spin profile appears to be slower, it does converge more consistently than the choice of a back-and-forth rotation spin profile. For initial headings like $\theta_{v,0} = -135°$ where the seeker is facing directly opposite of the source, the seeker did not converge to the source with the back-and-forth rotation spin profile. In this case, the photoresistor sensor's incidence angle β (shown in Fig. 32c) is very large as the arm sweeps out only an arc in front of the vehicle. Since the sensor is turned away from the light for most of the spin profile, it is unable to really detect a change in light intensity, i.e., from its perspective the surroundings appear "dark". Experiments comparing the full-rotation to the back-and-forth rotation spin profiles at an $\theta_{v,0} = -135°$ can be seen in this YouTube playlist.

In the experiments using resistance-based cost values shown in Figs. 77 and Fig. 78, one can start to see slight variations with the path the seeker takes towards the source. There is a noticeable deviation between the paths corresponding to different $\theta_{v,0}$; however, this does not significantly impact the convergence time of the experiment. For the most part, the path taken by the RMSpESC seeker is rather direct, confirming the efficiency of the method.

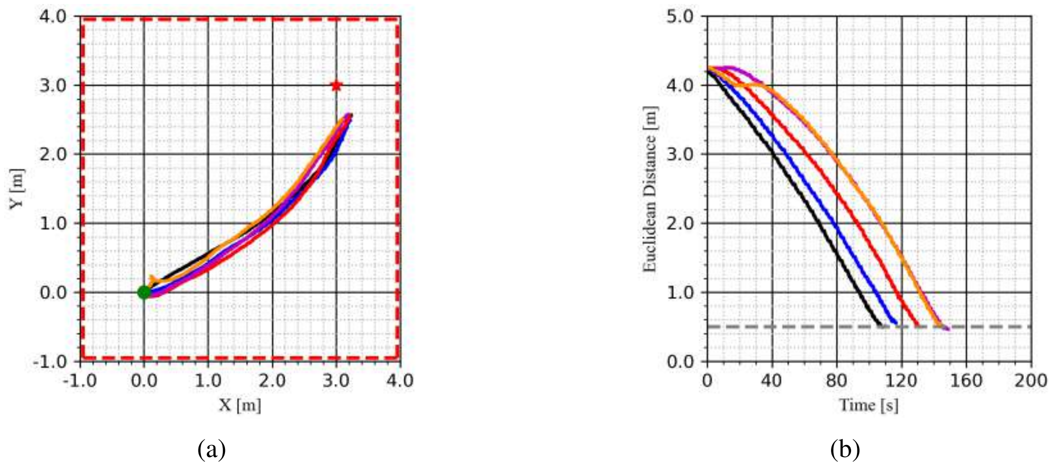
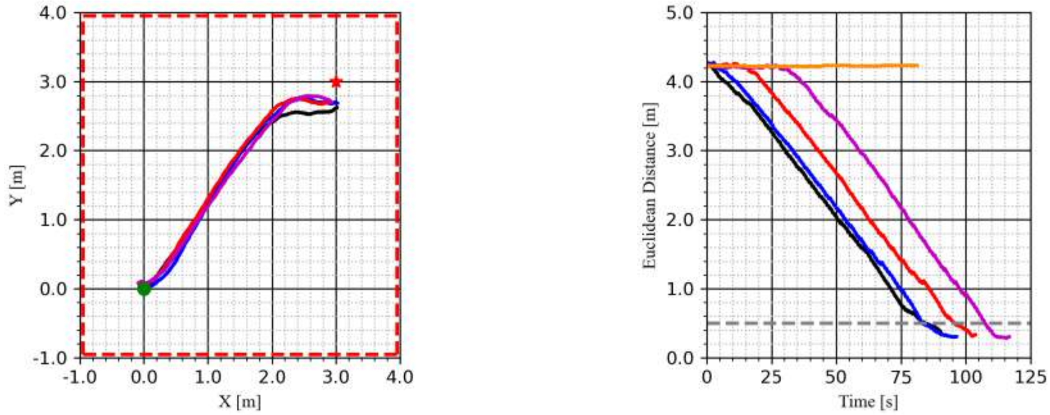


Figure 75: RMSpESC light source seeking laboratory experiment with voltage-based cost values and utilizing a full rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 16.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Angle $\theta_{v,0}$ of 45° = ▬, Angle $\theta_{v,0}$ of 0° = ▬, Angle $\theta_{v,0}$ of $-45°$ = ▬, Angle $\theta_{v,0}$ of $-90°$ = ▬, Angle $\theta_{v,0}$ of $-135°$ = ▬.



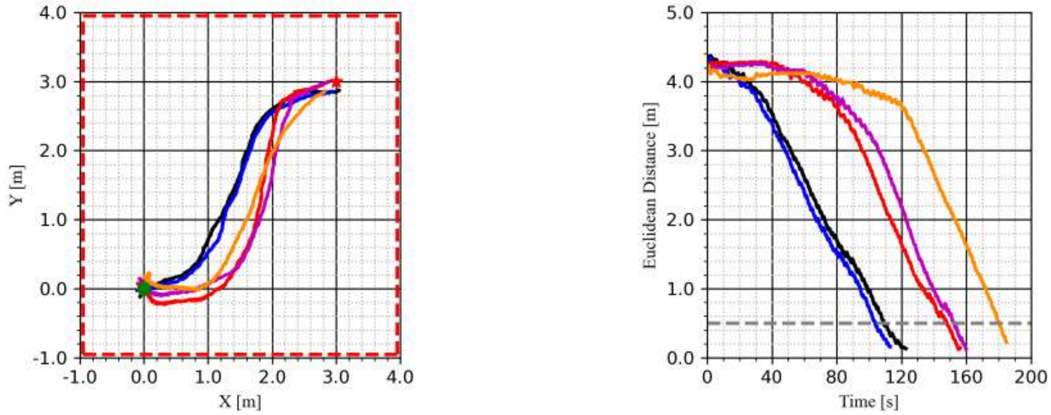
(a) (b)

Figure 76: RMSpESC light source seeking laboratory experiment with voltage-based cost values and utilizing a back-and-forth rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 16.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Angle $\theta_{v,0}$ of 45° = ▬, Angle $\theta_{v,0}$ of 0° = ▬, Angle $\theta_{v,0}$ of $-45°$ = ▬, Angle $\theta_{v,0}$ of $-90°$ = ▬, Angle $\theta_{v,0}$ of $-135°$ = ▬.

Table 8: RMSpESC seeker convergence time results (in seconds) for voltage-based cost values at various initial heading angles.

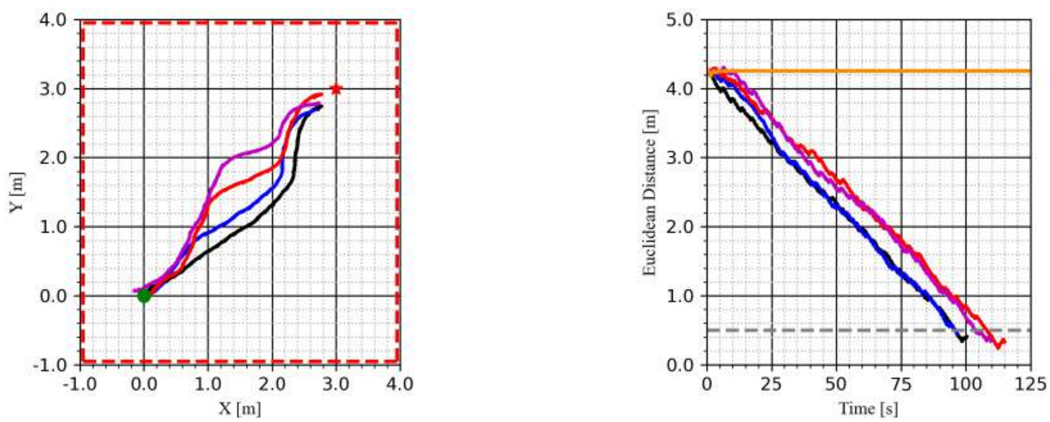
Angle $\theta_{v,0}$	**Full Rotation**	**Back-and-Forth Rotation**
45°	108	84
0°	120	83
$-45°$	131	96
$-90°$	146	107
$-135°$	145	N/A



(a) (b)

Figure 77: RMSpESC light source seeking laboratory experiment with resistance-based cost values and utilizing a full rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 17.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Angle $\theta_{v,0}$ of 45° = ▬▬, Angle $\theta_{v,0}$ of 0° = ▬▬, Angle $\theta_{v,0}$ of −45° = ▬▬, Angle $\theta_{v,0}$ of −90° = ▬▬, Angle $\theta_{v,0}$ of −135° = ▬▬.



(a) (b)

Figure 78: RMSpESC light source seeking laboratory experiment with resistance-based cost values and utilizing a back-and-forth rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 17.

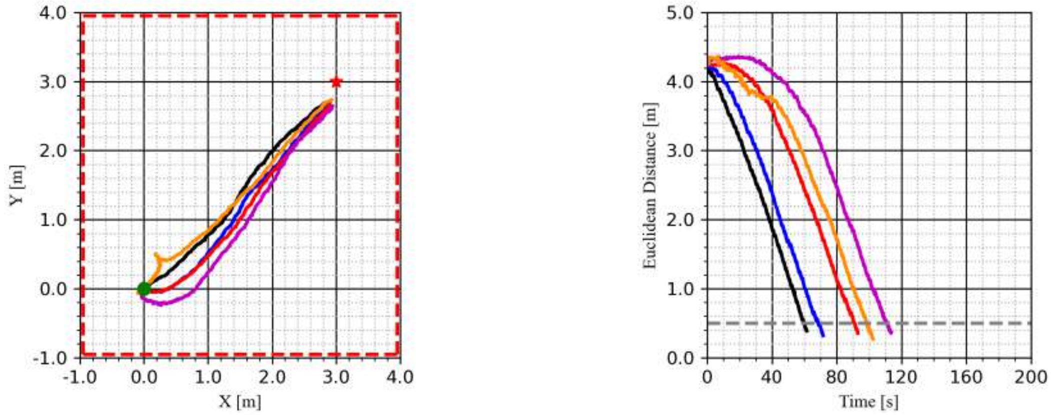
KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Angle $\theta_{v,0}$ of 45° = ▬▬, Angle $\theta_{v,0}$ of 0° = ▬▬, Angle $\theta_{v,0}$ of −45° = ▬▬, Angle $\theta_{v,0}$ of −90° = ▬▬, Angle $\theta_{v,0}$ of −135° = ▬▬.

Table 9: RMSpESC seeker convergence time results (in seconds) for resistance-based cost values at various initial heading angles.

Initial Angle $\theta_{v,0}$	**Full Rotation**	**Back-and-Forth Rotation**
$45°$	109	96
$0°$	104	96
$-45°$	148	108
$-90°$	153	104
$-135°$	181	N/A

AdaGrad ESC Performance

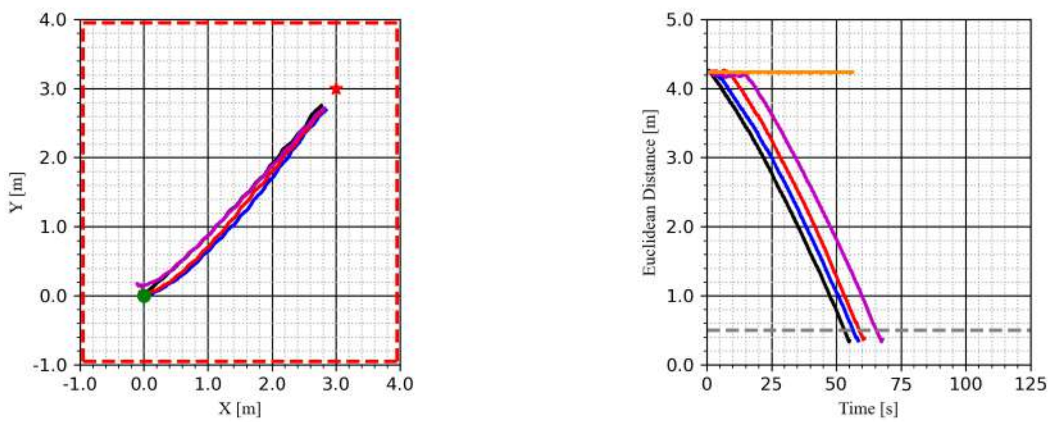
This section describes the performance of the AGESC method with a light source. Like before, we evaluate the performance as the vehicle's initial heading is varied and the initial sensor angular position is set to be $\phi_0 = -\frac{\pi}{2}$. Figures 79 and 80 show the performance of the AGESC with the voltage-based cost function, while Figs. 81 and 82 show the performance of the AGESC with the resistance-based cost function. While the RMSpESC method required no parameter adjustments for the different spin profiles, AGESC requires re-tuning for ω_l, k_v, and k_ω when the spin profile is changed as seen in Tables 16 and 17. This is necessary because changing the spin profile changes the gradient and gradient outer product demodulation signals $m(t)$ and $M(t)$, respectively, so the aforementioned parameters must be adjusted to accommodate this. In addition, AGESC parameters are sensitive to the choice of cost function. This is because voltage cost values are on the order of 10^{-2} while resistance cost values are on the order of 10^5, this difference in magnitude necessitates retuning of the AGESC. We can observe from the results in Tables 10 and 11 that the full rotation spin profile converges slightly slower than the back-and-forth profile which is consistent with the results previously obtained for the RMSpESC method.



(a) (b)

Figure 79: AGESC light source seeking laboratory experiment with voltage-based cost values and utilizing a full rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 16.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Angle $\theta_{v,0}$ of 45° = ▬▬, Angle $\theta_{v,0}$ of 0° = ▬▬, Angle $\theta_{v,0}$ of −45° = ▬▬, Angle $\theta_{v,0}$ of −90° = ▬▬, Angle $\theta_{v,0}$ of −135° = ▬▬.



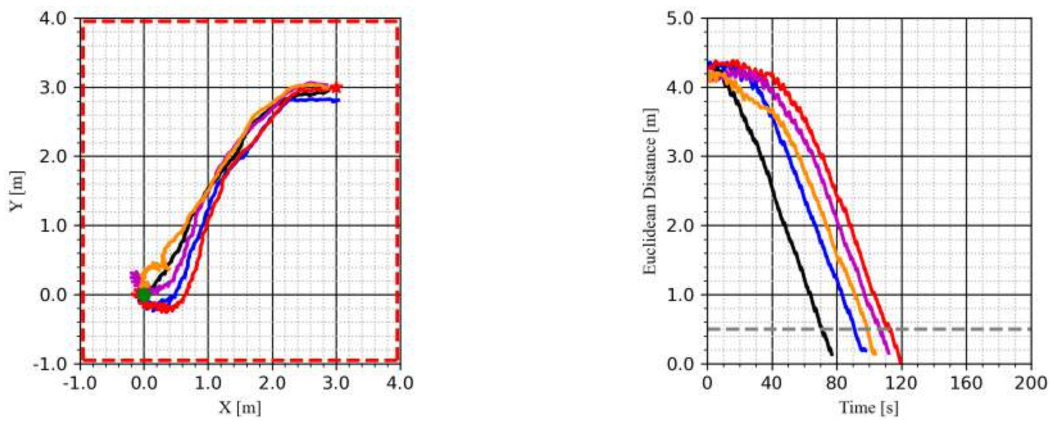
(a) (b)

Figure 80: AGESC light source seeking laboratory experiment with voltage-based cost values and utilizing a back-and-forth rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 16.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Angle $\theta_{v,0}$ of 45° = ▬▬, Angle $\theta_{v,0}$ of 0° = ▬▬, Angle $\theta_{v,0}$ of −45° = ▬▬, Angle $\theta_{v,0}$ of −90° = ▬▬, Angle $\theta_{v,0}$ of −135° = ▬▬.

Table 10: AGESC seeker convergence time results (in seconds) for voltage-based cost values at various initial heading angles.

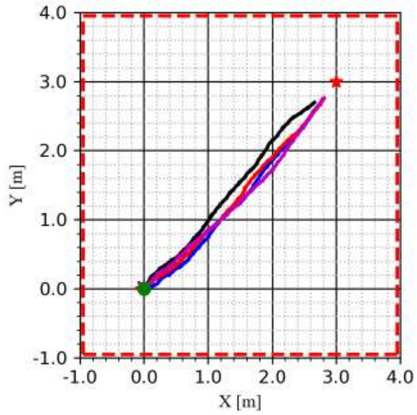
Initial Angle $\theta_{v,0}$	**Full Rotation**	**Back-and-Forth Rotation**
45°	60	53
0°	69	57
-45°	91	59
-90°	110	65
-135°	99	N/A



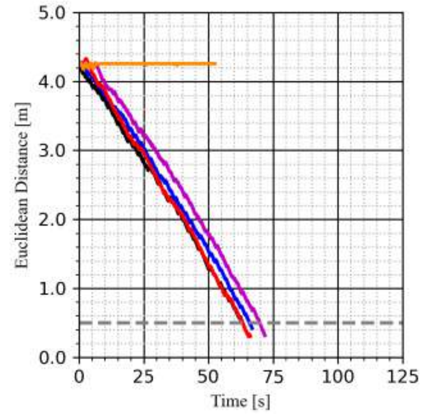
(a) (b)

Figure 81: AGESC light source seeking laboratory experiment with resistance-based cost values and utilizing a full rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 17.

KEY: Seeker Initial Position = , Source Position = , Room Boundaries = , Convergence Threshold = , Angle $\theta_{v,0}$ of 45° = , Angle $\theta_{v,0}$ of 0° = , Angle $\theta_{v,0}$ of -45° = , Angle $\theta_{v,0}$ of -90° = , Angle $\theta_{v,0}$ of -135° = .



(a)



(b)

Figure 82: AGESC light source seeking laboratory experiment with resistance-based cost values and utilizing a back-and-forth rotation spin profile: (a) trajectory comparison, and (b) convergence rate comparison. Parameters for this experiment are given in Table 17.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Angle $\theta_{v,0}$ of 45° = ▬, Angle $\theta_{v,0}$ of 0° = ▬, Angle $\theta_{v,0}$ of $-45°$ = ▬, Angle $\theta_{v,0}$ of $-90°$ = ▬, Angle $\theta_{v,0}$ of $-135°$ = ▬.

Table 11: AGESC seeker convergence time results (in seconds) for voltage-based cost values at various initial heading angles.

Initial Angle $\theta_{v,0}$	**Full Rotation**	**Back-and-Forth Rotation**
45°	71	63
0°	91	66
$-45°$	112	63
$-90°$	106	70
$-135°$	99	N/A

Adjusting Rotating Sensor Frame Length

In this section, we are interested in adjusting the rotating frame's arm length a to see if this affects the convergence rate in both simulation and laboratory experiments. This rotating frame arm length a represents the amplitude of our perturbation signal in our extremum seeking scheme. According to Theorem 4.1, in the limit as $a \to 0$, the gradient estimates converge to the true gradient of the objective function. If decreasing the rotating frame arm length improves the gradient estimate as theory suggests, perhaps this boosts the convergence rate in our source seeking experiments.

To remain consistent with the work in Section 5.5.3.2, we utilize the arm lengths of 3", 6" and 10" (or 0.0762, 0.1524, 0.254 meters, respectively), which can be visualized in Fig. 57. We test this effect in numerical and Gazebo simulations, as well as in the laboratory. For all simulation and laboratory experiments documented in this section, we utilize the RMSpESC method, the back-and-forth rotation spin profile, and the parameters shown in Table 12.

The results for several simulations and laboratory experiments are shown in Fig. 84. In these

Table 12: Parameters tuned for the RMSpESC light source seeking vehicle for resistance-based cost values for experiments in Figs. 84, 85, and 86.

Symbol	Value
a	0.1524 m
ω_s	2.094 rad/s
ω_ξ	1.0
ω_ℓ	1.0
ε	0.1
k_v	0.1
k_ω	0.5
$v_{\max}$	0.1 m/s
$\omega_{\max}$	0.5 rad/s

figures, numerical simulation, gazebo simulation, and laboratory experiments are plotted against each other for ease of comparison. Observing the vehicle trajectories in the left column, we see that the seeker takes a rather direct route towards the source. There is no significant difference between the predicted trajectory in simulation, and the one achieved experimentally. The time history of the seeker distance from the source is shown to give a sense of the convergence rates for these experiments. In these experiments, we do not see any significant effect on convergence when varying the rotating sensor frame's arm length. All simulations converged to the source in roughly 70 to 80 seconds, while the laboratory experiments converged in approximately 90 to 100 seconds. While theory suggests that the gradient estimate improves with a decreasing a, this does not seem to have an effect on the performance of the source seeker.

While the arm length did not affect the speed of convergence, it did help with another issue we were facing in the laboratory experiments. For this light based cost function, we had some issues getting our vehicle to converge to the source's position consistently when it started at longer distances (not shown here) facing away from the light. Referring to Fig. 83, this is an approximate visualization of the cost data we would expect to get from these lab experiments. Notice the presence of these large shelves at larger radii and larger values of β. These shelves are caused by the way we are capturing this light intensity in laboratory experiments. We are using an Arduino board which forces all photoresistor sensor data through an analog to digital conversion, meaning a value has to be fit to one of 1024 distinct resistances levels. The larger shelves seen at higher cost values correspond to the robot at further distances from source and when the sensor is not facing the light. If the robot is in a scenario like this, we found that the longer arm length of 10 inches allowed us to more easily escape these cost value shelves since it can sweep out and explore a larger amount of the vehicle's surroundings. However note the presence of the dark purple shelf at the top, this corresponds the the maximum resistance value that can be represented by our setup. If the vehicle is deep within the purple region where the sensor's reading is effectively maxed out, we see no spatial variation of the cost function and the ESC algorithm will fail. This limitation means there is an effective range of approximately 4 to 5 meters from the light source where our ESC algorithm can operate. To reiterate, these limitations come from our choice of Arduino and photoresistor setup used to capture light intensity. To see how this vehicle drives toward the source in the laboratory, please see the following videos for the lab experiments with a three inch arm [2], a six inch arm [3], and a ten inch arm [4].

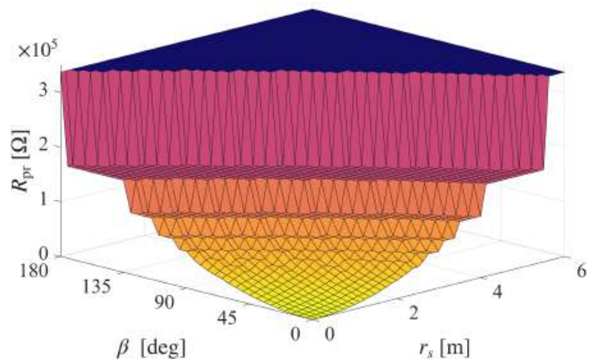


Figure 83: This figure is restated from Fig. 28, a longer rotating frame arm length helps the seeker better operate with this choice of cost function.

Moving Light Source

These tests validated the capability of our adaptive methods to find the static light source in our laboratory environment. We have performed additional tests with the RMSpESC (with the parameters in Table 12 to assess our algorithm's capability to respond to time varying cost functions. In the experiment shown in Fig. 85, a student moves the light source in an hourglass pattern once the vehicle has driven close to the source's position. Observing Fig. 85b, one can clearly see how the cost value decreases when the vehicle becomes close to the light, and then spikes as the light source is moved to a different position. Nevertheless, the seeker is able to turn about and converge to the light source's new position in its environment. In the experiment shown in Fig. 86, the light source is placed atop of another vehicle as it drives around in a circle. One can see in Fig. 86a that the seeker is moving in circles as it is tracking the source orbiting around it. The seeker was set to slow speeds so that it would not collide with the other vehicle, however even with this constraint it tracked the moving light source quite well. Figure 86b shows the cost value time history of the experiment. The cost value remains low as the seeker tracks the source.

5.5.5.4 Acoustic Extremum Seeking Experiments

We conduct the following set of acoustic extremum seeking experiments at the San Diego State University Backdoor Recording Studio. This choice of testing environment was made as an alternative to the DSIM lab, whose acoustic profile was discussed in Section 5.3.3. This recording studio seen in Fig. 34 provides a space to test our ESC schemes in a more acoustically controlled environment than the DSIM lab. Figure 35 shows a diagram of the recording studio environment, the borders of the test area are marked with ArUco markers, and special coordinates are denoted on the floor by masking tape. Although the recording studio is quite limited in terms of floor space, the heavy curtains around the borders of the room lessen the sound reflections in the room by dampening sound.

[2] Three inch arm test from long distance
[3] Six inch arm test from long distance
[4] Ten inch arm test from long distance

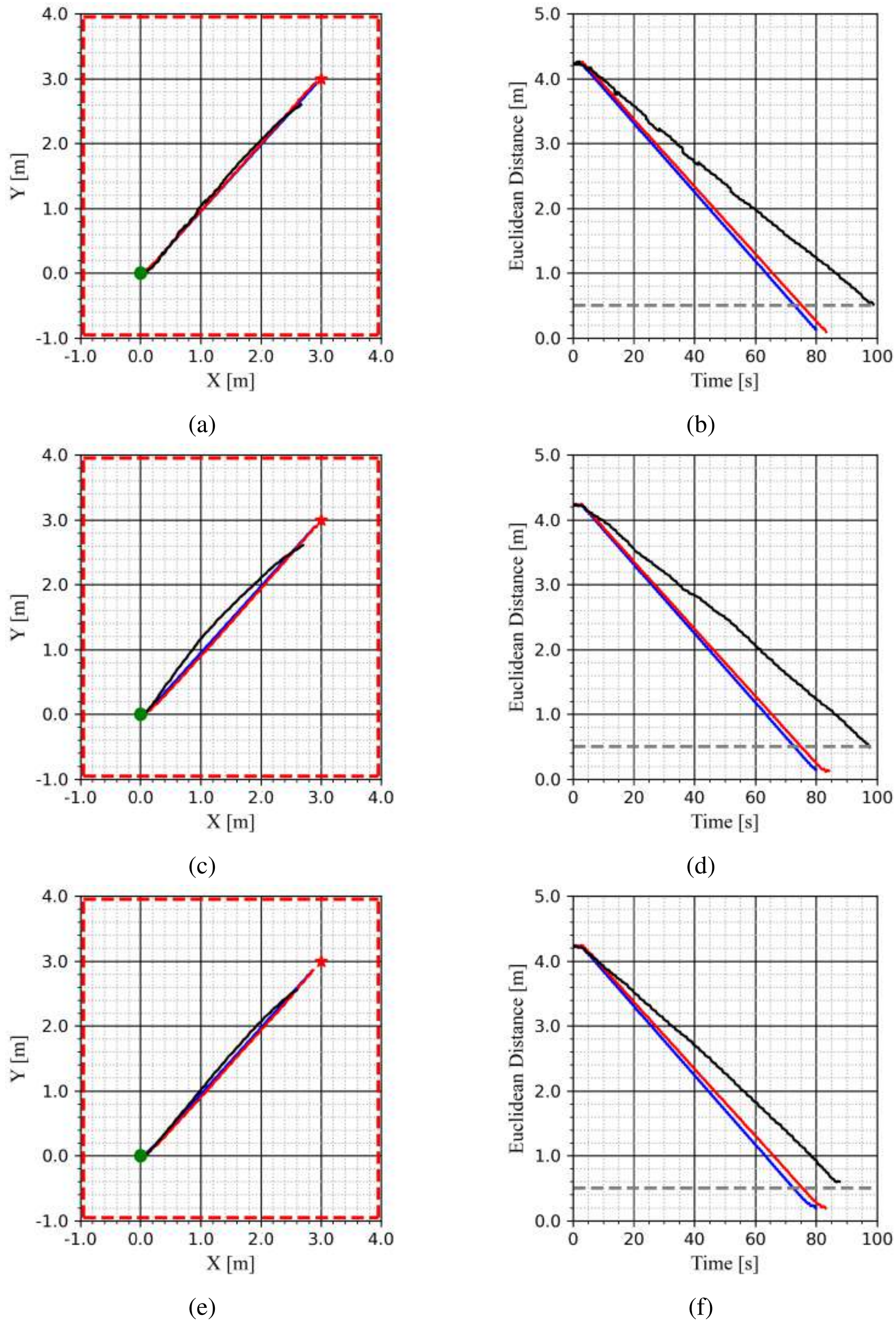
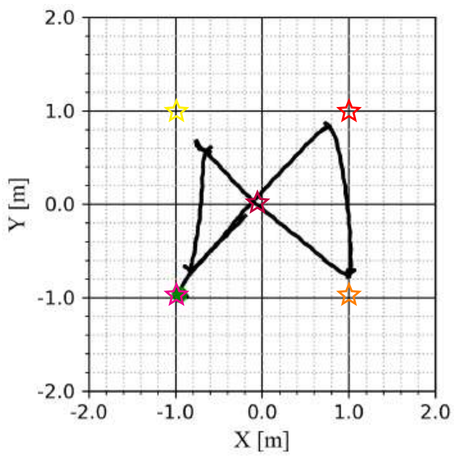
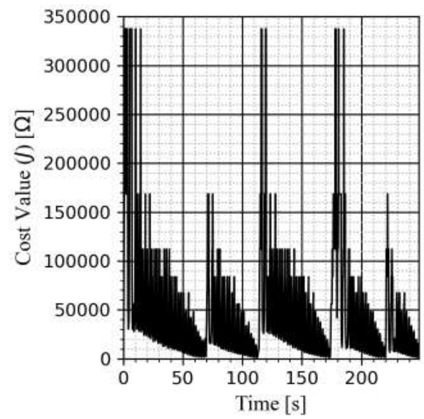


Figure 84: RMSpESC light source seeking with resistance-based cost values. The rotating sensor frame arm length: (a, b) 3 inches (0.0762m), (c, d) 6 inches (0.1524m), and (e, f) 10 inches (0.254m). Figures (a, c, e) show the seeker trajectory while figures (b, d, f) show the Euclidean distance to the source. Parameters for these experiments are given in Table 12.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Numerical Simulation = ▬▬, Gazebo Simulation = ▬▬, Laboratory Experiment = ▬▬.



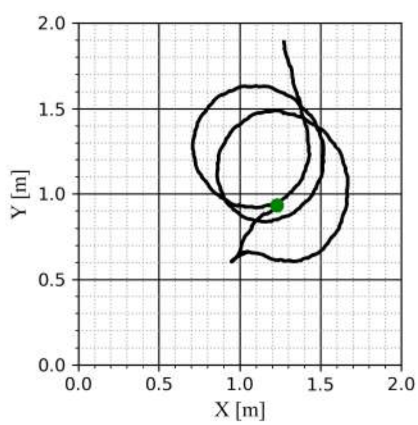
(a)



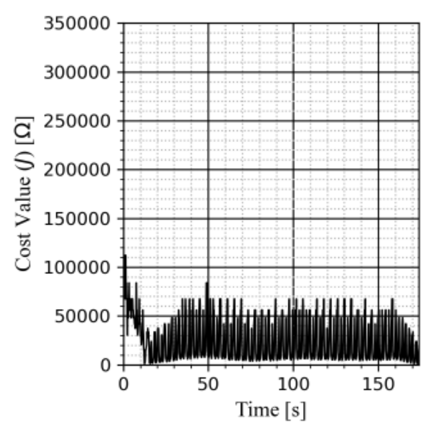
(b)

Figure 85: Laboratory experiment with RMSpESC seeking a moving light source in an hourglass pattern with resistance-based cost values and utilizing a back-and-forth rotation spin profile: (a) the vehicle's trajectory, and (b) a time history of the R_{pr} cost value. Parameters for this experiment are given in Table 12. The video of this experiment is available here.

KEY: Seeker Initial Position = ●, Source Position 1 = ☆, Source Position 2 = ☆, Source Position 3 = ☆, Source Position 4 = ☆, Source Position 5 = ☆.



(a)



(b)

Figure 86: Laboratory experiment with RMSpESC tracking a time-varying light source atop another TurtleBot3 with resistance-based cost values and utilizing a back-and-forth rotation spin profile: (a) the vehicle's trajectory, and (b) a time history of the R_{pr} cost value. Parameters for this experiment are given in Table 12. The video of this experiment is available here.

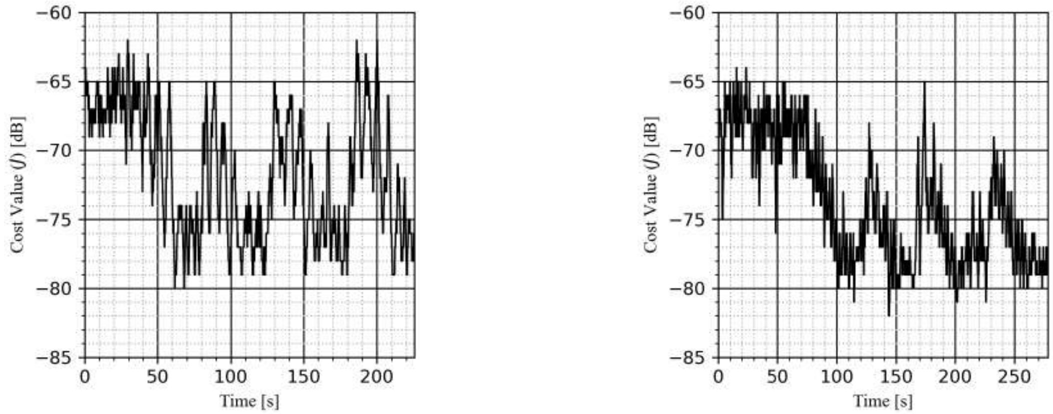
KEY: Seeker Initial Position = ●.

The sound dampening of the acoustic studio worked quite well, and we were able to enable the acoustic extremum seeker to converge to the source's position quite reliably. With this, we began experimenting with all the various ESC schemes we tested before with light sources. Like before, we are still interested in testing two different spin profiles with these ESC methods, and one can view the acoustic experiments in the playlist linked here. In these videos, we are testing the capability of the ESC method to navigate towards a sound source in its environment. Similar to experiments conducted before with the light sensor (specifically see the experiment video corresponding to Fig. 85), the acoustic source is moved to four different locations in the studio. In our experiments, we wait until seeker converges to a small neighborhood of about 1 meter away from the sound source, once this is achieved the sound source is moved to a different position in the room.

Figures 87 through 91 all show the time history of the cost value during the experiment. Figures 87, 88, and 90 feature experiments with the GESC, RMSpESC, and AGESC methods, respectively. These experiments serve as a proof of concept for acoustic source seeking, we confirm that the seeker is able to converge to the location of the sound source for all of these ESC schemes and for both choices of spin profile. Figure 89 features the experiments with RMSpESC with another choice of acoustic cost value based off the sound pressure level in the 2000 Hz octave band. Finally, Fig. 91 features two failed experiments performed with the AGESC method, we use these to highlight the limitations of our ESC scheme in this indoor studio environment.

In all of these experimental results, we can clearly see the four distinct times the seeker minimizes its cost value as it approaches the sound source. Interestingly, the seeker vehicle never completely closes the distance to the sound source, rather it consistently gets within 1 meter of the source and then seems to oscillate in place, as seen in an experiment with the GESC method presented in Fig. 87. We speculate that this is due to destructive interference in the acoustic signal being so close to the sound source. Destructive interference affects the sound pressure level reading we measure with our sensor, making the source appear "quieter" by one to two dBs as the sensor gets close. However, we can see in our successful experiments that when the sound source changes position, the seeker vehicle is able to find the new position of the source, and settle into a neighborhood close to it. Although the extremum seeking vehicle is able to converge to the sound source's location quite consistently in this testing environment, we still do have some issues if the seeker drives too close to the walls of the test area. Figure 91 shows the results of two failed experiments we had with the AdaGrad ESC method. In this experiment captured in this video, the seeker drove too close to the walls of the room and was unable to correctly drive to the source of the sound elsewhere. Even with the improved sound dampening in this environment, it appears that sound reflections still cause issues with local extrema along the walls of the test area. These sound reflections seem to be an issue for any acoustic ESC test in an indoor environment, however we hope this becomes less of a problem as we work on conducting acoustic ESC testing in an outdoor environment. Moreover, we are developing other methods to enable escape from local extrema as denoted in Section 2.

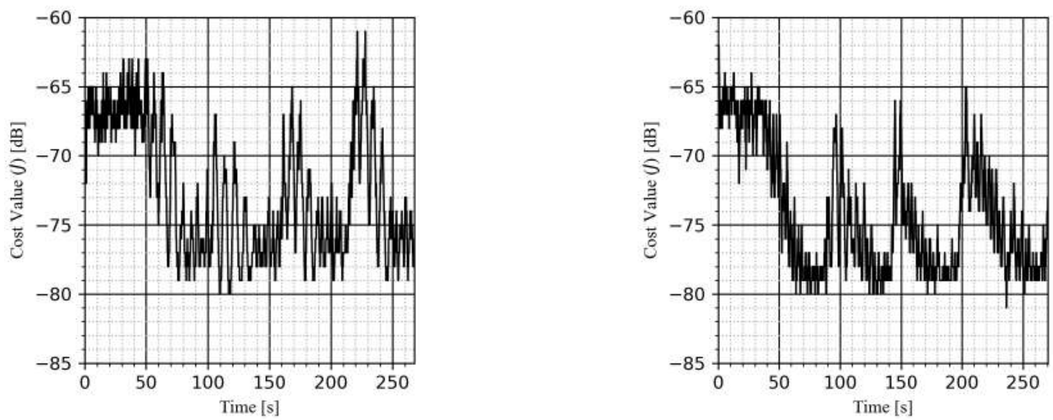
In addition to taking the sound pressure level as our acoustic cost value, we conducted some experiments where we performed a spectral decomposition, and took the reading from the 2000 Hz octave band as the objective function value [51, Appendix B.4]. Figure 89 shows the results of the experiments with this choice of alternative acoustic cost value. With our speaker playing a 2000 Hz test tone, and now utilizing a cost value based specifically on the 2000 Hz octave band, we were able to get results similar to the experiments performed with the RMS-SPL cost value before these results are shown in Fig. 89a.



(a)

(b)

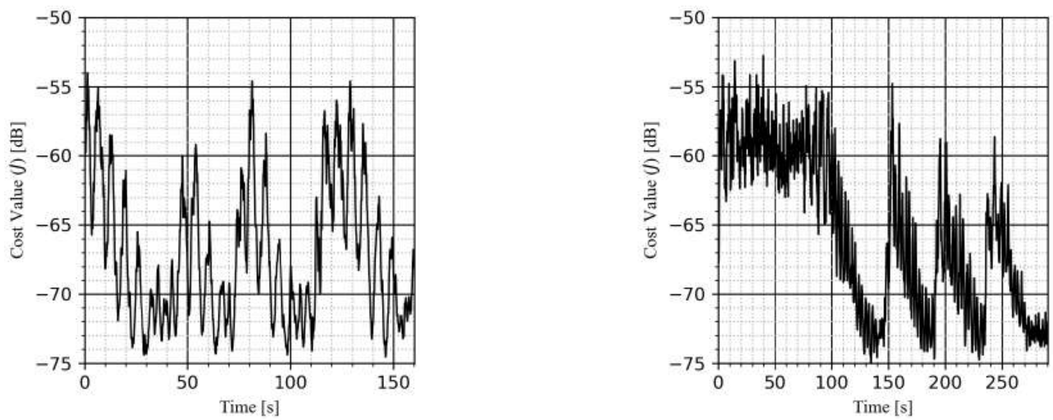
Figure 87: Studio experiment with GESC seeking a moving acoustic source with RMS sound pressure level cost values and a sensor frame with: (a) a full rotation spin profile (seen in this video), and (b) a back-and-forth rotation spin profile (seen in this video). Parameters for this experiment are given in Table 13.



(a)

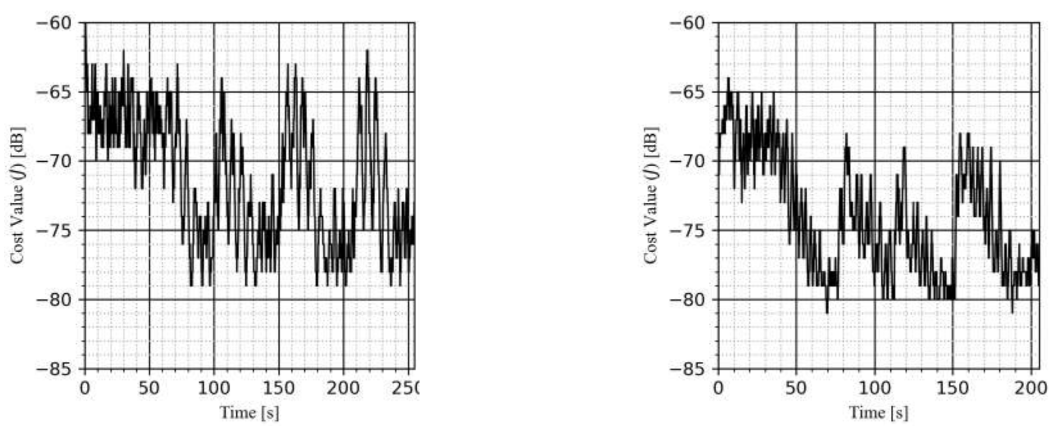
(b)

Figure 88: Studio experiment with RMSpESC seeking a moving acoustic source with RMS sound pressure level cost values and a sensor frame with: (a) a full rotation spin profile (seen in this video), and (b) a back-and-forth rotation spin profile (seen in this video). Parameters for this experiment are given in Table 13.



(a) (b)

Figure 89: Studio experiment with RMSpESC seeking a moving acoustic source with octave band sound pressure level cost values and a sensor frame with: (a) a full rotation spin profile (seen in this video), and (b) a back-and-forth rotation spin profile (seen in this video). Both experiments here took the reading in the 2000 Hz octave band as the cost value, we can see how these cost values are lower magnitude than the previous experiment in Fig. 88 utilizing the RMS sound pressure level as the cost value. Parameters tuned for this experiment are the same ones as shown in Table 13.



(a) (b)

Figure 90: Studio experiment with AGESC seeking a moving acoustic source with RMS sound pressure level cost values and a sensor frame with: (a) a full rotation spin profile (seen in this video), and (b) utilizing a back-and-forth rotation spin profile (seen in this video). Parameters for this experiment are given in Table 13.

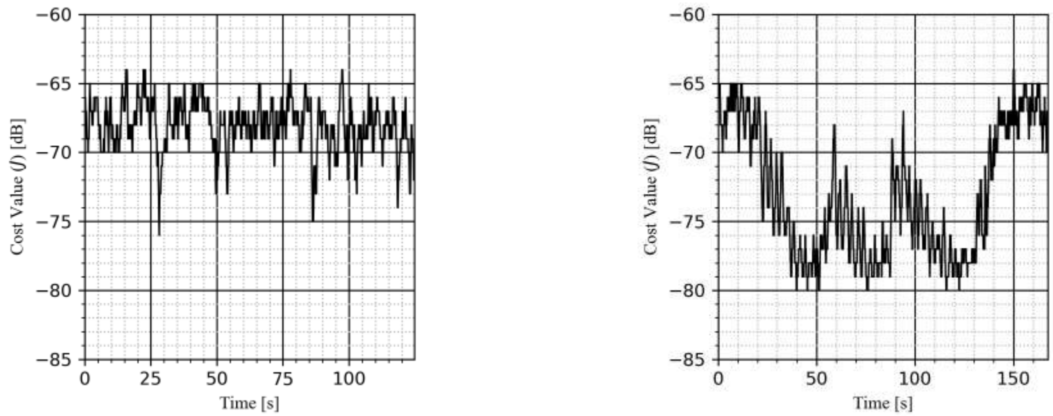


Figure 91: Failed studio experiment with AGESC seeking a moving acoustic source with RMS sound pressure level cost values and a sensor frame with: (a) a back-and-forth rotation spin profile (seen in this video), and (b) a back-and-forth rotation spin profile (seen in this video). Parameters tuned for this experiment are the same as seen in Fig. 90 and Table 13.

Table 13: Parameter table for acoustic source seeking studio experiments with RMS sound pressure level cost values. Note for conciseness, the Back-and-Forth rotation spin profile is denoted as "BnF" below.

	GESC		RMSpESC		AGESC	
Symbol	**Full**	**BnF**	**Full**	**BnF**	**Full**	**BnF**
ϕ_0	$-90°$	$-90°$	$-90°$	$-90°$	$-90°$	$-90°$
a	0.33 m	0.33 m	0.33 m	0.33 m	0.33 m	0.33 m
ω_s	1.047 rad/s	1.571 rad/s	1.047 rad/s	1.571 rad/s	1.047 rad/s	1.571 rad/s
ω_j	2.0	2.0	2.0	2.0	2.0	2.0
ω_ξ	1.0	1.0	1.0	1.0	1.0	1.0
ω_ℓ	-	-	1.0	1.0	$1e^{-5}$	$1e^{-7}$
ε	-	-	0.1	0.1	0.1	0.1
k_v	1.0	1.0	0.1	0.1	0.1	0.1
k_ω	5.0	5.0	0.5	0.5	0.5	0.5
$v_{\max}$	0.05 m/s	0.05 m/s	0.05 m/s	0.05 m/s	0.05 m/s	0.05 m/s
$\omega_{\max}$	0.25 rad/s	0.25 rad/s	0.25 rad/s	0.25 rad/s	0.25 rad/s	0.25 rad/s

5.5.5.5 Comparison with the GESC Methods

Light Source Experiments

We performed a similar set of light source seeking experiments for a GESC seeker with a rotating sensor frame (discussed in 5.5.4) to compare with our adaptive ESC methods. These results are compiled into Tables 14 and 15 for easy comparison. In general, the back-and-forth rotation profile was faster than the full rotation profile, however there are some scenarios where this back-and-forth profile falters, as seen when $\theta_{v,0} = -135°$. In this scenario, the sensor sweeps through an arc segment and only obtains cost value information from the front and sides of the robot. The full rotation spin profile is able to obtain cost value information from all directions, and with this it is able to converge to sources positioned directly behind the vehicle quite reliably. The GESC and AGESC methods achieved a bit faster convergence times than RMSpESC, this conclusion holds for both cost values (voltage and resistance) and both spin profiles tested. In all experiments regardless of ESC method used, the seeker took a rather direct path towards the source.

Table 14: Convergence time results (in seconds) for light source seeking laboratory experiments with voltage-based cost values. The Back-and-Forth rotation spin profile is denoted as "BnF".

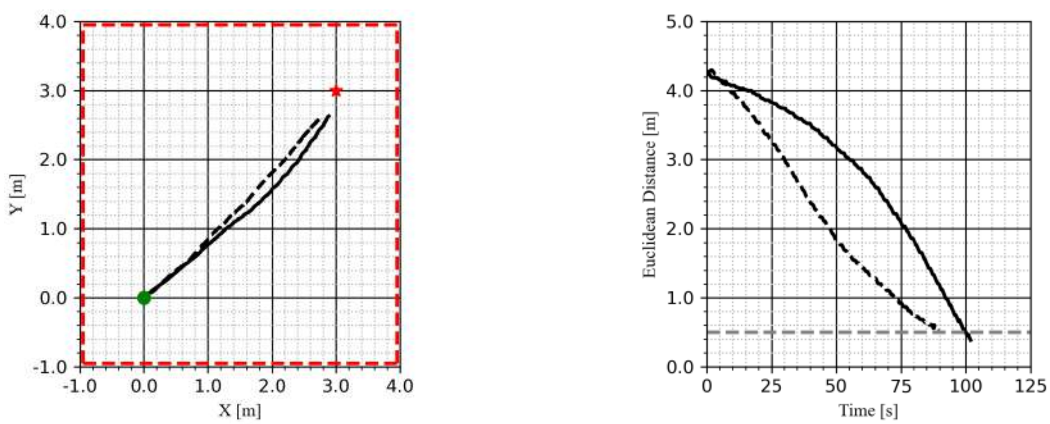
	GESC		**RMSpESC**		**AGESC**	
Angle $\theta_{v,0}$	**Full**	**BnF**	**Full**	**BnF**	**Full**	**BnF**
$45°$	100	55	108	84	60	55
$0°$	114	57	120	83	69	57
$-45°$	129	69	131	96	91	59
$-90°$	154	84	146	107	110	65
$-135°$	120	N/A	145	N/A	99	N/A

Table 15: Convergence time results table for light source seeking laboratory experiments with resistance-based cost values. Note for conciseness, the Back-and-Forth rotation spin profile is denoted as "BnF" below.

	GESC		**RMSpESC**		**AGESC**	
Angle $\theta_{v,0}$	**Full**	**BnF**	**Full**	**BnF**	**Full**	**BnF**
$45°$	87	82	109	96	71	63
$0°$	87	71	104	96	91	66
$-45°$	88	76	148	108	112	63
$-90°$	87	82	153	104	106	70
$-135°$	91	N/A	181	N/A	99	N/A

The GESC parameters we used to obtain those results in Tables 14 and 15 are documented in Tables 16 and 17 for both voltage and resistance-based cost values respectively. Across all experiments, the forward velocity constraint $v_{\max} = 0.1$ m/s and angular velocity constraint $\omega_{\max} = 0.5$ rad/s were used. In addition, regardless of the spin profile choice, the rotating sensor was moving at a speed of $\omega_s = 2.094$ rad/s. These constraints force the seeker to drive slow, this is done so that the seeker does not drive through its environment faster than it can explore it with the rotating sensor. One could potentially improve the convergence rate of the extremum seeker by increasing the speed of local exploration. To do this, one would need to increase the speed of the servo with a higher value of ω_s, and would need more processing power to ensure that the sensor adequately samples the distribution of the cost function over the period of rotation. In our implementation, we kept this servo rotation speed relatively slow because we are concerned with the structural integrity of the rotating frame seen in Fig. 69 at high speeds. Future iterations of the rotating sensor frame should ensure the assembly is robust at high speeds.

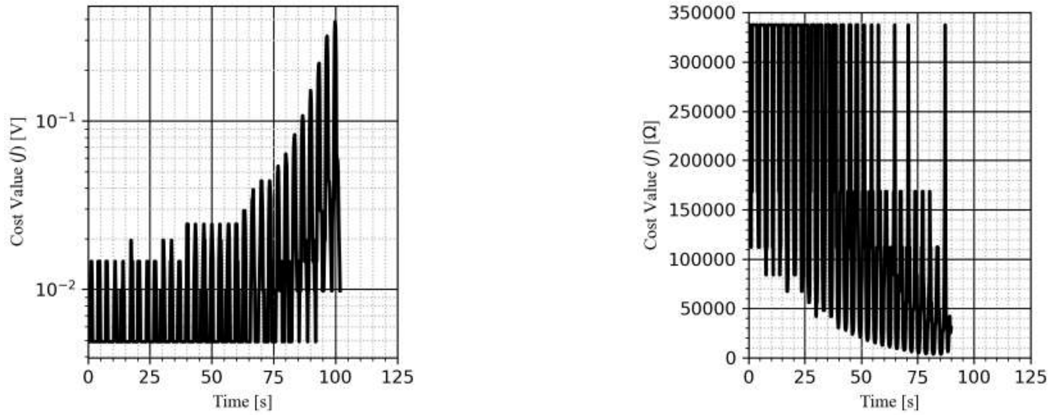
The main difference between adaptive and GESC methods in the light experiments was the convergence rate observed with the two different cost functions. Figures 92, 94, and 95 show a convergence comparison on a voltage versus resistance-based cost function for the GESC, RMSpESC, and AGESC methods, respectively. In all of these figures, we plot the results of the seeker utilizing a full rotation spin profile, at an initial heading of $\theta_{v,0} = 45°$. For the GESC method, we can see in Fig. 92b a distinct difference in the convergence rate between the two experiments. The cost value time history for both of the experiments is seen in Fig. 93, note in Fig. 93a, the voltage reading of the sensor is maximized at the source's position, while in Fig. 93b the resistance reading is minimized at the source's position. The GESC parameters for both the voltage and resistance cost values are tuned to achieve good performance when the seeker starts far from the light source. For voltage-based experiments, both the magnitude of the cost values and the strength of the gradient increase as the seeker converges to the location of the source, as seen in Fig. 93a. This can potentially increase the convergence rate of the GESC method, leading to higher speeds in close range of the source. For resistance-based experiments, the magnitude of the cost values and the strength of the gradient decrease as the seeker converges to the source's location as seen in Fig. 93b. In this scenario, we see much slower convergence at close range when the cost values have dropped considerably. These results highlight the problem a GESC method has with a change in cost function; not only will parameters need to be re-tuned, the cost function itself may affect the rate of convergence. One can see that the adaptive methods shown in Fig. 94b and Fig. 95b are more invariant to our change in the cost function. Both RMSpESC and AGESC show no noticeable change in convergence rate compared to the GESC, this is a quite useful property since it boosts our ESC performance even when there are deficiencies in the gradient of the cost function.



(a) (b)

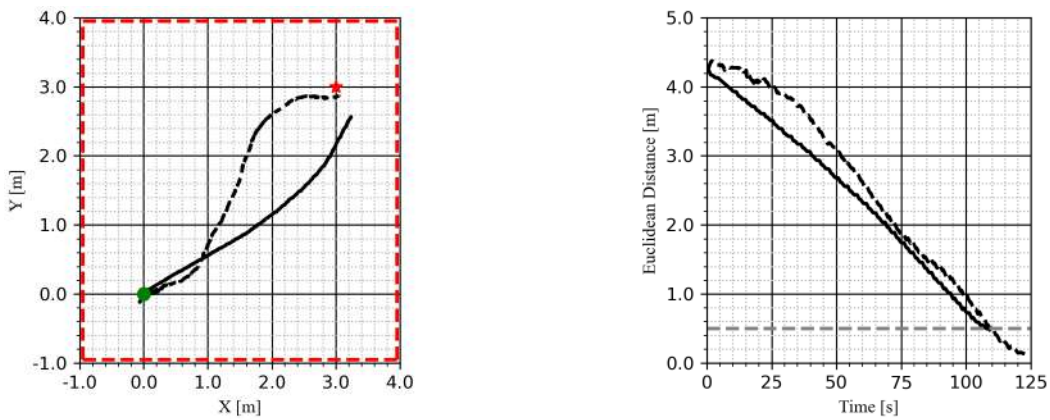
Figure 92: Laboratory experiment with GESC seeking a light source with voltage or resistance-based cost functions with a full rotation spin profile: (a) the vehicle's trajectory, and (b) a time history of the Euclidean distance to the source. Parameters for this experiment are given in Tables 16 and 17.

KEY: Seeker initial position = ●, Source position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Laboratory experiment with voltage-based cost values = ▬▬, Laboratory experiment with resistance-based cost values = ▪▪▪▪.



(a) (b)

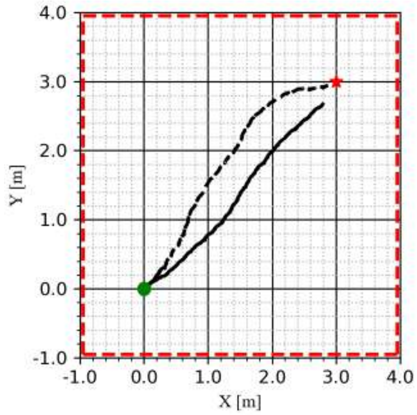
Figure 93: GESC light source seeking voltage-based versus resistance-based cost value visualization corresponding to the experiments in Fig. 92: (a) the time history of voltage-based cost values, and (b) the time history of the resistance-based cost values. In (a), the voltage reading of the sensor is maximized at the source's position, while in (b) the resistance reading is minimized at the source's position.



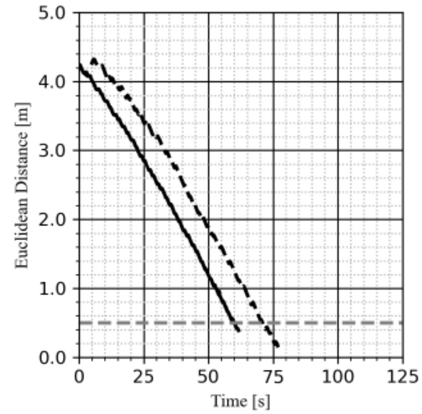
(a) (b)

Figure 94: Laboratory experiment with RMSpESC seeking a light source with voltage or resistance-based cost functions and a full rotation spin profile: (a) the vehicle's trajectory, and (b) a time history of the Euclidean distance to the source. Parameters for this experiment are given in Tables 16 and 17.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Laboratory Experiment with voltage-based cost values = ▬▬, Laboratory experiment with resistance-based cost values = ▪▪▪▪.



(a)



(b)

Figure 95: Laboratory experiment with AGESC seeking a light source with voltage or resistance-based cost functions and a full rotation spin profile: (a) the vehicle's trajectory, and (b) a time history of the Euclidean distance to the source. Parameters for this experiment are given in Tables 16 and 17.

KEY: Seeker Initial Position = ●, Source Position = ★, Room Boundaries = ▪▪▪▪, Convergence Threshold = ▪▪▪▪, Laboratory Experiment with voltage-based cost values = ▬▬, Laboratory experiment with resistance-based cost values = ▪▪▪▪.

Table 16: Parameter table for light source seeking laboratory experiments with voltage-based cost values. Note for conciseness, the Back-and-Forth rotation spin profile is denoted as "BnF" below.

	GESC		**RMSpESC**		**AGESC**	
Symbol	**Full**	**BnF**	**Full**	**BnF**	**Full**	**BnF**
ϕ_0	$-90°$	$-90°$	$-90°$	$-90°$	$-90°$	$-90°$
a	0.18 m	0.18 m	0.18 m	0.18 m	0.18 m	0.18 m
ω_s	2.094 rad/s	2.094 rad/s	2.094 rad/s	2.094 rad/s	2.094 rad/s	2.094 rad/s
ω_ξ	1.0	1.0	1.0	1.0	1.0	1.0
ω_ℓ	-	-	1.0	1.0	$1e^{-2}$	$1e^{-5}$
ε	-	-	0.1	0.1	0.1	0.1
k_v	1.0	1.0	0.1	0.1	5.0	0.5
k_ω	5.0	5.0	0.5	0.5	25.0	2.5
$v_{\max}$	0.1 m/s	0.1 m/s	0.1 m/s	0.1 m/s	0.1 m/s	0.1 m/s
$\omega_{\max}$	0.5 rad/s	0.5 rad/s	0.5 rad/s	0.5 rad/s	0.5 rad/s	0.5 rad/s

Table 17: Parameter table for light source seeking laboratory experiments with resistance-based cost values. Note for conciseness, the Back-and-Forth rotation spin profile is denoted as "BnF" below.

	GESC		RMSpESC		AGESC	
Symbol	**Full**	**BnF**	**Full**	**BnF**	**Full**	**BnF**
ϕ_0	$-90°$	$-90°$	$-90°$	$-90°$	$-90°$	$-90°$
a	0.18 m	0.18 m	0.18 m	0.18 m	0.18 m	0.18 m
ω_s	2.094 rad/s	2.094 rad/s	2.094 rad/s	2.094 rad/s	2.094 rad/s	2.094 rad/s
ω_ξ	1.0	1.0	1.0	1.0	1.0	1.0
ω_ℓ	-	-	1.0	1.0	$1e^{-13}$	$1e^{-14}$
ε	-	-	0.1	0.1	0.1	0.1
k_v	$1e^{-7}$	$1e^{-7}$	0.1	0.1	$1e^{-5}$	$1e^{-5}$
k_ω	$5e^{-7}$	$5e^{-7}$	0.5	0.5	$5e^{-5}$	$5e^{-5}$
$v_{\max}$	0.1 m/s	0.1 m/s	0.1 m/s	0.1 m/s	0.1 m/s	0.1 m/s
$\omega_{\max}$	0.5 rad/s	0.5 rad/s	0.5 rad/s	0.5 rad/s	0.5 rad/s	0.5 rad/s

Acoustic Source Experiments

In our acoustic extremum seeking experiments in Section 5.5.5.4, we were able to validate the capability of GESC, RMSpESC, and AGESC to consistently converge to the acoustic source's position. The three methods were all capable of tracking the sound source as it moved to different locations in the environment, and all methods completed the experiment in a similar amount of time. These experiments serve as a proof-of-concept for acoustic extremum seeking using a vehicle with a rotating sensor frame.

Conclusion

With regards to all the experiments conducted in Section 5.5.5, the GESC and AGESC methods needed constant retuning when the cost function type was changed because of the magnitude of the cost values in an experiment. Note that the RMS-SPL acoustic based function has cost values $J = \mathscr{O}\left(10^1\right)$, the voltage-based function has values $J = \mathscr{O}\left(10^{-2}\right)$, and the resistance-based function has values $J = \mathscr{O}\left(10^5\right)$. In addition, the different choices of spin profile necessitated retuning since the back-and-forth rotation estimation signals in Eq. (5.30) to (5.32) are two orders of magnitude larger than the full rotation estimation signals in Eq. (5.26) to (5.28). In particular, the forward velocity gain k_v, angular velocity gain k_ω, and filter parameter ω_l needed to be re-tuned for these various source seeking experiments. The RMSpESC method requires very little retuning to achieve comparable convergence results as the other methods. As seen in Tables 16 and 17 the RMSpESC parameters are identical regardless of the spin profile choice for these light source seeking experiments. The only major change to the parameters for acoustic source seeking experiments in Table 13 was with the servo rotation speed ω_s and the velocity constraints ($v_{\max}$, and $\omega_{\max}$) but note this change was applied to all methods, not just RMSpESC. In practice the RMSpESC proves to be quite flexible and easy to work with; this makes it our adaptive ESC method of choice in future work.

5.6 Unmanned Surface Vessel

Unmanned underwater vessels (USV) are another platform that can be used to implement the ESC algorithms. The implementation on this platform will provide knowledge on source seeking algorithms performance in outdoor environments where the seeker vehicle is navigating on the highly reflective water surface. Moreover, compared to indoor or underwater environments, the sound waves are influenced by factors such as refraction and atmospheric turbulence, which introduce unique complexities to the signal field. We consider two USV platforms for this implementation: a custom-built small platform suitable for pool testing with multiple vehicles, and a larger platform intended for open water testing.

Two undergraduate students designed and built the small USV platform (see Fig. 96a). The larger platform was acquired from Blue Robotics for open-water testing (see Fig. 96b).

(a)

(b)

Figure 96: (a) DISM custom-built USV and (b) BlueBoat.

5.6.1 Design and manufacturing of a small USV platform for ESC testing

5.6.1.1 Initial Design

The team was given the task of building a USV with the following requirements:

- total manufacturing cost of the vehicle $< \$2000$
- ease of reproduction and and manufacturing, allowing for efficient scaling to support multi-vehicle testing
- small enough to support multi-vehicle testing in a pool environment
- capable of supporting up to 10 lbs of payload, with dedicated space allocated for additional sensors and equipment
- lightweight for easy portability
- powerful onboard processing capabilities
- uses differential thrust for locomotion

Design criteria

The design of the USV was divided into three sections: the body, the electronics, and the microphone mount. Each section was made to meet a set of design specifications, which aimed to balance the durability and functionality of the USV's design while also meeting the project's requirements discussed previously.

Body

The vessel's hulls and their associated connectors were fully 3D printed. Marine epoxy and a rubberized coating was used to treat the surface of the 3D printed parts. The 3D printed design makes the vehicle lightweight and facilitates replication for additional units.

Electronics and control

The electronics were integrated in a waterproof electronic box mounted on top of the connecting arms. Control is managed by an onboard Pixhawk 6C Mini, with PX4 flight control firmware that is used for telemetry and motor actuation commands. PX4 was chosen over Ardupilot due to more robust ROS integration and pre-made Gazebo software-in-the-loop simulation environments. Computational loads are handled by a companion Raspberry Pi 4B, running Ubuntu for ROS communication. It communicates with the flight controller via a serial connection. Another Raspberry Pi 4B is used for microphone data collection, using a Diligent MCC172 DAQ-Hat. It sends microphone data over an Ethernet connection to the companion Raspberry Pi.

Testing

After connection, the hulls were placed in water for 8 hours to ensure they could withstand submersion with no leaks. Additional tests showed as well that it can withstand a maximum extra payload of 10 lbs. Lastly, radio control of the vehicle was tested in a pool. This test showed smooth motion and adequate maneuverability for ESC implementation (video of pool test available here). The USV remained under budget, with a final manufacturing cost of $1,500.

5.6.1.2 Additional Design Changes

Design of Microphone Mount

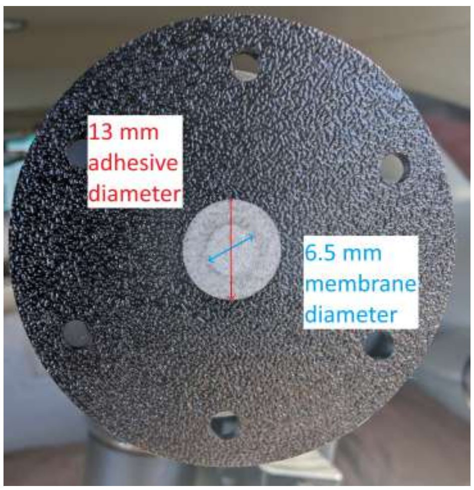
To facilitate ESC algorithm testing on the surface vessel, it needs a sensor capable of continuous rotation. The design requirements are as follows:

- Fit in the free 6"x8" space on top of the E-box.
- Needs to support continuous 360^o rotation of the rotor
- Have a water-resistant enclosure for the B&K microphone, with minimal acoustic interference
- The rotating arm should not interfere with the Holybro GPS unit
- Create a water-resistant seal around any penetrations through the E-box

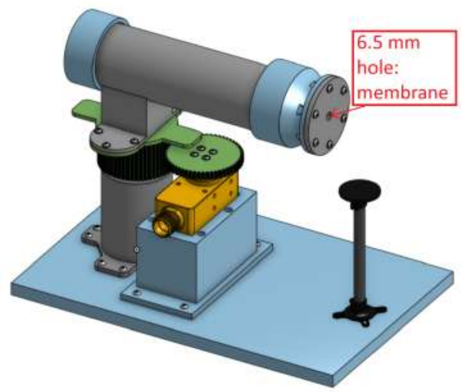
The solution developed for TurtleBot3, illustrated in Fig. 69 in Section 5.5.5 fills many of these requirements. It has a small enough footprint to fit in the allowed space on the E-box, it uses a slip-ring to achieve continuous rotation. The only modifications that need to be made are to make the penetrations through the E-box water resistant, and to change the dimensions to allow the rotor to be driven by a servo in a waterproof enclosure, such as the one used on the BlueROV2, illustrated in Fig. 108 of Section 5.7.2.

The largest challenge is to properly protect the expensive B&K microphone. Water-resistant acoustic membranes are a common component used in consumer electronics, for example, the membranes used in smartphones to allow submersion in water. An ePTFE membrane from Gore was chosen as an acoustic membrane to allow an audio-permeable barrier to the elements.

Figures 97b and 97c show the 3D assembly and the final production of the servo and microphone mounts, respectively. A critical design constraint highlighted in Fig.97a was the 6.5 mm diameter hole to accommodate the 6.5 mm diameter of the acoustic membrane, and the 13 mm diameter of the attached adhesive.



(a)



(b)

(c)

Figure 97: USV rotating microphone mount: (a) Gore acoustic membrane with membrane and adhesive diameters, (b) CAD assembly model for the rotating microphone mount , and (c) final 3D printed design of the rotating microphone mount.

5.6.2 Future plans

The physical constuction of the USV is complete, the first algorithm to be tested in the pool is RMSpESC. The ROS-ESC framework expanded on in Section 5.2 outputs a continuous stream of twist velocity setpoints published to a ROS2 topic. Currently, the USV's twist to motor actuation controller is not able reach the velocity setpoints, preventing a true test of the ESC method. The motor speed controller needs to be calibrated such that the USV reaches the velocity setpoints given by the ROS-ESC framework.

5.7 Unmanned Underwater Vehicle

This section discusses the transition from two-dimensional (2D) ESC (using UGVs and USVs) to the application of ESC in three-dimensional (3D) underwater environments. To achieve this, two submersibles were selected for testing the ESC algorithms. The first was an unmanned underwater vehicle (UUV), shown in Fig.98, built by a senior design capstone project. The capstone design project's key objective was to design a lightweight and low-cost UUV capable of processing data and running ESC algorithms using an onboard computer, a novelty in the submersible industry. The project also included the design of a hydrophone mount that could rotate for acoustic data collection and survive harsh ocean conditions. The UUV was finished and tested in a pool, passing its design requirements. While the UUV has not yet been used for ESC testing due to necessary hardware changes identified during development, these are planned to be resolved in the future.

Alongside the senior design UUV project, an additional submersible for ESC testing was developed. The BlueROV2 by Blue Robotics, shown in Fig.99, is offered as a modifiable test platform for scientific research. While it lacks the powerful onboard processing capabilities of our custom-built UUV, the BlueROV2 offers greater reliability and ease of use, making it an ideal platform for initial ESC testing. Modifications to integrate the hydrophone mount with the BlueROV2 are discussed later in Section 5.7.2.

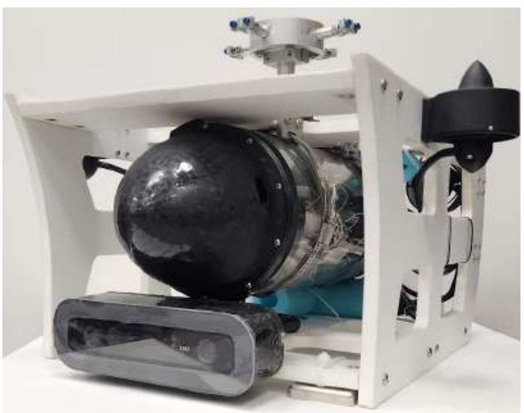


Figure 98: DISM custom-built UUV

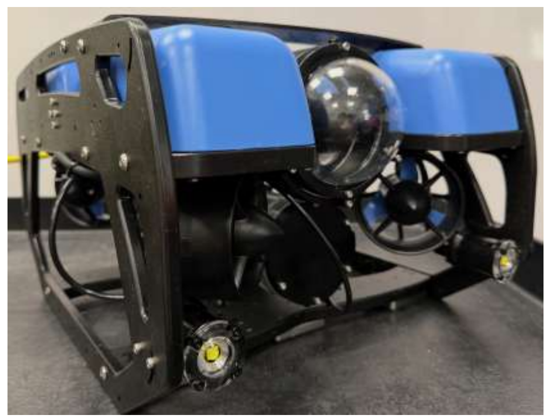


Figure 99: BlueROV2

5.7.1 UUV Hardware Design

Project Requirements

To create an effective test bench for ESC testing, a capstone senior design team (made of six undergraduate students) was given the task of building a UUV with the following requirements:

- total manufacturing cost of the vehicle $<$ \$3500 (excluding sensors cost)
- a rotating mount designed to hold four hydrophones atop the vehicle
- powerful onboard processing capabilities
- lightweight for easy portability in ocean testing

With these goals in mind, the UUV would satisfy the needs for testing ESC in 3D space. Using its high-powered internal computer, it would be able to handle complex algorithm processing directly onboard, enabling full autonomy without the need for external computing resources.

Design criteria

The design of the UUV was divided into three sections: the frame, the hydrophone mount, and the electronics. Each section was made to meet a set of design specifications, which aimed to balance the durability and functionality of the UUV's design while also meeting the project's needs.

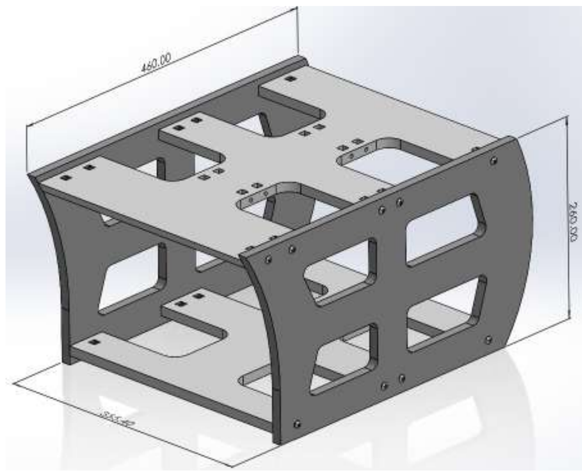


Figure 100: UUV frame model.

The frame design of the UUV, shown in Fig.100, emphasized material selection, modularity, and operational efficiency in challenging underwater environments. By utilizing marine-grade HDPE plastic for the structural panels, the design ensures low water absorption and high corrosion resistance in saltwater conditions. The boxy shape of the frame enhances durability and minimizes deformation in turbulent waters, while also simplifying production and assembly. Finally, the open-frame facilitates ease of maintenance and modification, while also contributing to the UUV's overall lightweight structure.

Hydrophone mount

The hydrophone mount in Fig.101 was engineered to support up to four RESON TC4013 Hydrophones for underwater acoustic signal detection. It was made from marine-safe PETG plastic and aluminum, which made the mount resistant to saltwater and robust. Clamps to hold the hydrophones are positioned at the end of three-inch arms extending from a central hub. The clamps feature an adjustable attachment point that allows the hydrophone's angle to be changed for testing. The central hub holds the hydrophone arms while being clamped to a vertical aluminum shaft. At the end of this shaft, a stepper motor generates the torque necessary to spin the hydrophones degrees back and forth, creating the sweeping motion required for the ESC controller. The stepper motor is enclosed in a plastic PETG housing that offers protection from the elements and includes holes for mounting to the top of the UUV.

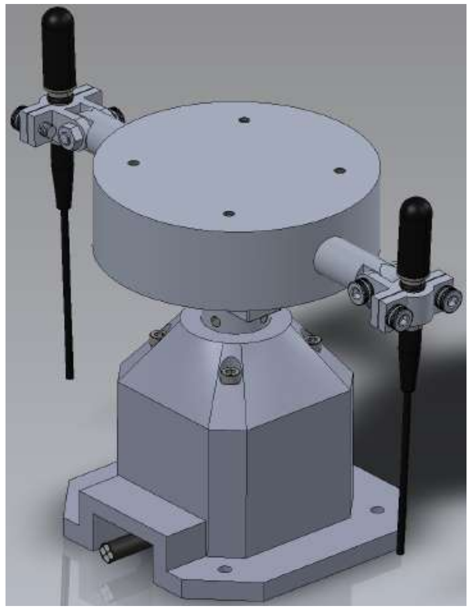


Figure 101: Hydrophone mount model.

Electronics

The electronics board of the UUV shown in Fig.102 integrates multiple devices and components to manage ESC calculations, acoustic data collection, and vehicle movement. The Nvidia Jetson Orin Nano serves as the primary processing unit, using data collected from the hydrophones to perform algorithm computations. A Raspberry Pi Model 4B, paired with a DAQ-HAT data acquisition system, captures the acoustic data from the hydrophones and sends it to the Jetson for processing. The hydrophone mount's stepper motor is driven by an Arduino Uno R3 connected to a stepper motor driver. Power from the onboard battery is managed by the distribution board, supplying energy to the board's components. Finally, the thrusters are operated by the Jetson based on the ESC calculations.

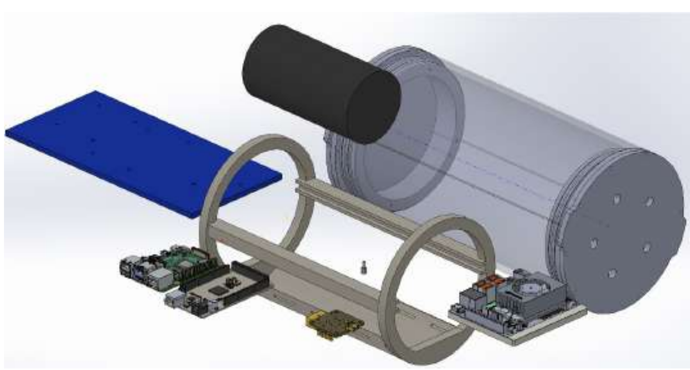


Figure 102: Electronics board model.

Final design and testing

The UUV's final design testing was conducted at a local YMCA pool, where it was evaluated against its project requirements. The UUV remained under budget, with a final project cost of $3,368. The hydrophone mount functioned underwater, using its stepper motor to rotate a hydrophone back and forth. The onboard Nvidia Jetson Orin Nano provided robust ESC algorithm processing capabilities, enabling autonomous operation. The UUV had a final weight of 12.2 kg, light enough to be carried by a single person. Although the UUV met all design criteria, new hardware requirements found late in the UUV's development str need for further ESC testing. Since the arm length of the hydrophone mount was found to impact ESC algorithm performance, a redesign of the mount is necessary to account for this development. Additionally, smaller design details still need refinement for full functionality. To speed up the ESC implementation, the BlueROV2 was selected as a secondary test vehicle, functioning out of the box with only minor modifications. The UUV is planned to be updated in parallel with the BlueROV2's development and testing.

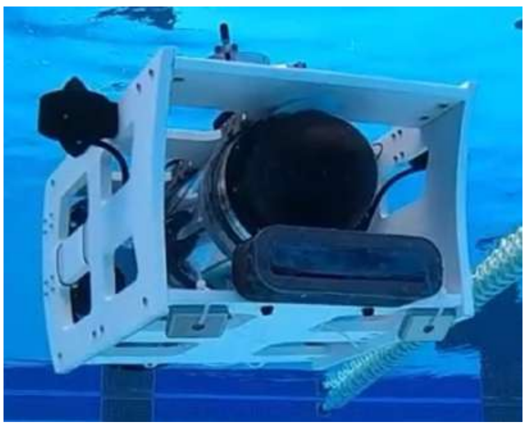


Figure 103: DISM custom-built UUV.

Figure 104: UUV team photo.

5.7.2 BlueROV2 Customization

As discussed in the underwater vehicle overview, the BlueROV2 was selected as a secondary platform for testing extremum seeking control algorithms in 3D. While it lacks the onboard computational capability of the custom-built UUV, its ability to connect to external high-powered computers combined with superior ease of use, makes it an effective platform for initial ESC experiments. This section details the hardware modifications and experiments that were required to prepare the BlueROV2 for ESC research.

Initial Hardware Modifications

To enable acoustic ESC testing, the BlueROV2 required several modifications. The primary additions include the integration of hydrophone mounts and supporting electronics for power, data acquisition, and processing. An additional electronics enclosure, shown in Fig. 105, was mounted to the underside of the BlueROV2 using a payload extension. This compartment housed the Raspberry Pi, DAQ-HAT, and other supplementary components responsible for hydrophone control and signal acquisition.

To power the new electronics, waterproof connections were added to transfer data and power between the BlueROV2's main and lower enclosures. As illustrated in Fig.106, the wiring was routed through rubber tubing sealed at both ends using Blue Robotics Penetrators. A 3D-printed electronics board was then designed to securely mount all electrical components inside the lower enclosure, completing the installation.

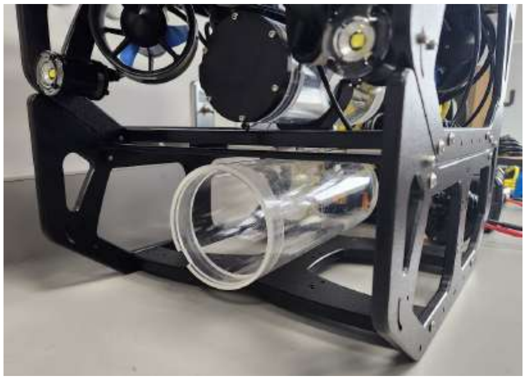


Figure 105: Added electronics enclosure to BlueROV2.

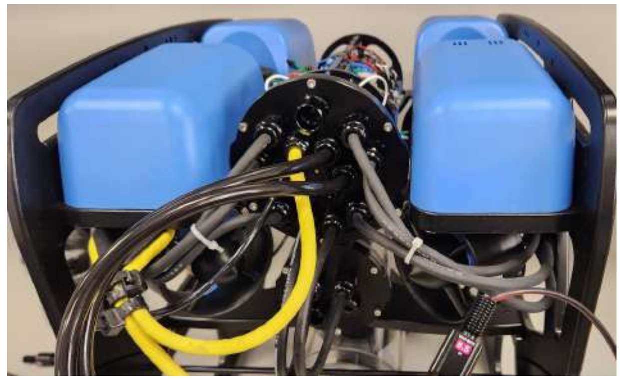


Figure 106: BlueROV2 wiring.

Experiments to Improve Hydrophone Mount

One of the primary changes from the senior design UUV project was that the hydrophone mount required further design optimization. Based on the experiments in Section 5.5.3.2, arm length was shown to significantly affect ESC performance. To improve the design, a new hydrophone mount was developed for the BlueROV2 with interchangeable arm lengths of 3", 6", and 10". The 6" arm can be seen in Fig. 107 below.

Figure 107: Hydrophone frame with 6" arm length.

Using interchangeable arm lengths, pool experiments were planned to determine the optimal length for ESC performance. Additional experiments were also scheduled to identify the influence of noise generated by the mount's servo motor and BlueROV2's thrusters. The servo chosen for the tests was the Parallax Feedback 360. This servo motor was selected to drive the hydrophone frame due to its closed-loop feedback capability and prior success in TurtleBot3 experiments in section 5.5.3.2. Table 18 summarizes the test plans to optimize the hydrophone mount design.

Table 18: Planned experiments to improve BlueROV2 hydrophone mount.

Experiments	Details
Hydrophone Orientation Test	The hydrophone will record sound in a pool at varying distances and arm lengths to identify an optimal hydrophone mount design.
Rotating Mount Motor Noise Test	After finding an optimal arm length, the next test will gather data while the hydrophone is rotating. This will be used to identify noise generated by the mount's servo motor.
BlueROV2 Thruster Noise Test	Since the thrusters of the BlueROV2 generate noise while running, this test will run the thrusters at various speeds and directions to identify how they affect data collection.

Hardware Modifications For Experiments

To prepare BlueROV2 for the experiments, additional hardware modifications were required. During the process of installing the hydrophone mount to the ROV2, one of the primary challenges of the modification was waterproofing the servo motor while joining the servo's spline to hydrophone frame. Early attempts tried to waterproof a 3D-printed enclosure using marine-grade epoxy and watertight bearings. This proved unsuccessful after multiple iterations due to small gaps in the material. The issue was resolved with the use of the Underwater Servo SER-20XX, a metal enclosure manufactured by Blue Trails Engineering. This enclosure, compatible with Blue Robotics penetrators, safely housed the Parallax servo underwater. It also allowed the wiring of the servo to be passed through waterproof tubing into the lower electronics enclosure. To complete the servo installation, a custom mounting plate was 3D-printed to attach the servo to the hydrophone mount, which was then affixed to the top frame of the BlueROV2.

The next needed addition to the BlueROV2 customization process was installing the RESON TC4013 hydrophone. This process presented a challenge similar to the servo enclosure addition, needing a method to waterproof a connection between the hydrophone and electronics enclosure. An initial prototype was designed using a similar method to the UUV, where a plug was fitted between the hydrophone's wire and its penetrator. This prototype proved to be faulty, leaking during testing. After this prototype, a new design was created by utilizing the included O-ring near the sensor's tip, suitable for creating a waterproof seal. A flexible rubber tube was fitted over this O-ring with its other end sealed by a Blue Robotics penetrator passing into the electronics enclosure. The excess tubing was then coiled around a 3D-printed wire wrap, providing sufficient slack to allow the hydrophone assembly to rotate. The result of the installation can be seen in Fig. 108.

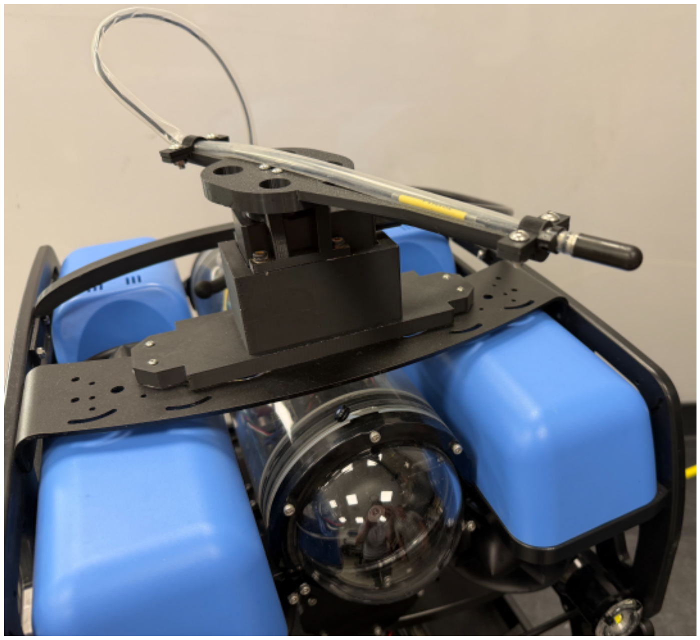


Figure 108: Hydrophone frame installed on top of BlueROV2. A plastic mounting plate connects the hydrophone mount to the BlueROV2. The mount includes a 3D printed connector holding the metal servo enclosure and the 6" arm mounted to the servo's output shaft. The hydrophone is mounted to the arm and is connected via a flexible rubber tube wrapped behind the mount.

At the time of this report, several preliminary tests of the modified BlueROV2 had been completed. However, due to ongoing data collection and analysis, the results of these experiments are not yet ready for inclusion in this report. The next steps involve completing the remaining experiments, followed by finalization the BlueROV2's customization.

5.7.3 BlueROV2 Numerical Simulations

In parallel with the BlueROV2 customization and sensor integration experiments, one of the key objectives of this project was to develop a method for simulating and testing the BlueROV2. As discussed in Section 5.4, three testing environments are required to evaluate the vehicle's dynamic behavior under ESC control. Therefore, through the use of numerical simulations, semi-realistic Gazebo simulations, and real-world pool experiments, the BlueROV2 will demonstrate the capability of extremum seeking control with rotating sensors in 3D.

5.7.3.1 ROV2 Particle Dynamics

The first of these environments is a numerical simulation developed in Python 3 using the existing ESC software package of the DSIM laboratory. The dynamic model of the BlueROV2, shown in Fig. 109, along with the mathematical formulation presented below, was used within this package to demonstrate the control functionality.

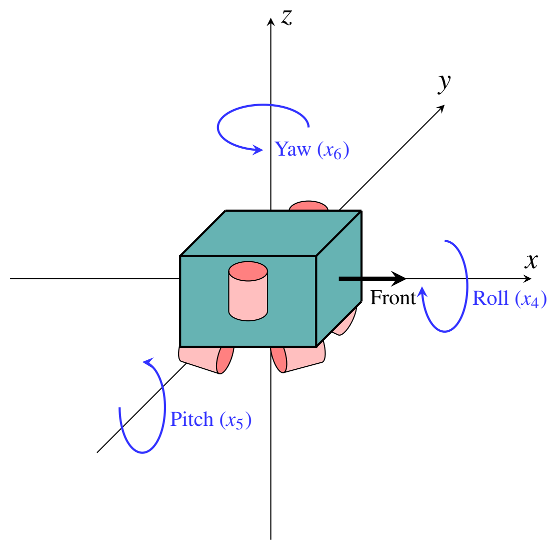


Figure 109: ROV2 particle dynamics model.

The ROV2 operates with five degrees of freedom, allowing it to maneuver freely within three-dimensional space. One of the key advantages of the BlueROV2 is that its software automatically manages vehicle balance and thruster forces. As a result, only basic velocity and orientation commands are required to achieve motion, simplifying the simulation process. For this simplified control model, the ROV2 can move along each translational axis and is capable of rotation in roll and yaw. However, due to its thruster configuration, the vehicle cannot pitch. This is accounted for in simulations such as Fig. 110 where the pitch input $\dot{x_4}$ is always set to zero.

The BlueROV2's simulation inputs are mapped to the vector $u = [\dot{x_1}, \dot{x_2}, \dot{x_3}, \dot{x_4}, \dot{x_5}, \dot{x_6}]$, where $\dot{x_1}, \dot{x_2}, \dot{x_3}$ are the BlueROV2's translational velocity inputs and $\dot{x_4}, \dot{x_5}, \dot{x_6}$ correspond to an angular velocity input representing roll, pitch, and yaw. Using these inputs, the translational equation of motion of the BlueROV2 in the vehicle frame are

$$\dot{x}_{\text{trans}} = \begin{bmatrix} \dot{x_1} \\ \dot{x_2} \\ \dot{x_3} \end{bmatrix} \tag{5.33}$$

The variable $\dot{x}_{\text{trans}}$ is a vector that represents the translational velocity components of the submersible as a function of the input vector u. For clarity, an example use case for the equation would be if the BlueROV2 is facing a certain direction, an input $\dot{x_1}$ would move the BlueROV2 forward in the direction it is facing, regardless of its orientation globally.

To represent the rotational dynamics of the ROV2, rotation matrices are defined for each angular velocity component. These matrices describe the vehicle's rotation about the roll, pitch, and yaw axes, shown by

$$R_{x_4} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(x_4) & -\sin(x_4) \\ 0 & \sin(x_4) & \cos(x_4) \end{bmatrix} \tag{5.34}$$

$$R_{x_5} = \begin{bmatrix} \cos(x_5) & 0 & \sin(x_5) \\ 0 & 1 & 0 \\ -\sin(x_5) & 0 & \cos(x_5) \end{bmatrix} \tag{5.35}$$

$$R_{x_6} = \begin{bmatrix} \cos(x_6) & -\sin(x_6) & 0 \\ \sin(x_6) & \cos(x_6) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{5.36}$$

Combining the individual angular components into a single rotation matrix becomes

$$R = R_{x_6} R_{x_5} R_{x_4} \tag{5.37}$$

The composite rotation matrix R is then converted to a 6×6 block form rotation matrix B to allow the 6-element input u to be applied. This manipulation is given by

$$B = \begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix} \tag{5.38}$$

The block form rotation matrix B is then multiplied by the input u, resulting in the vehicle's velocity in the global frame, vector $\dot{x}$. The vector is the combination of the translational and the rotational velocities components representing the BlueROV2's motion.

$$\dot{x} = Bu \tag{5.39}$$

The resulting dynamics equations were then implemented as a numerical simulation to verify functionality. In Fig. 110, the BlueROV2 dynamics model was given an input u for an initial 100 seconds to produce forward translation along the x-axis. Then after the movement, the input was changed by introducing a translational input along the y-axis and a rotational yaw input.

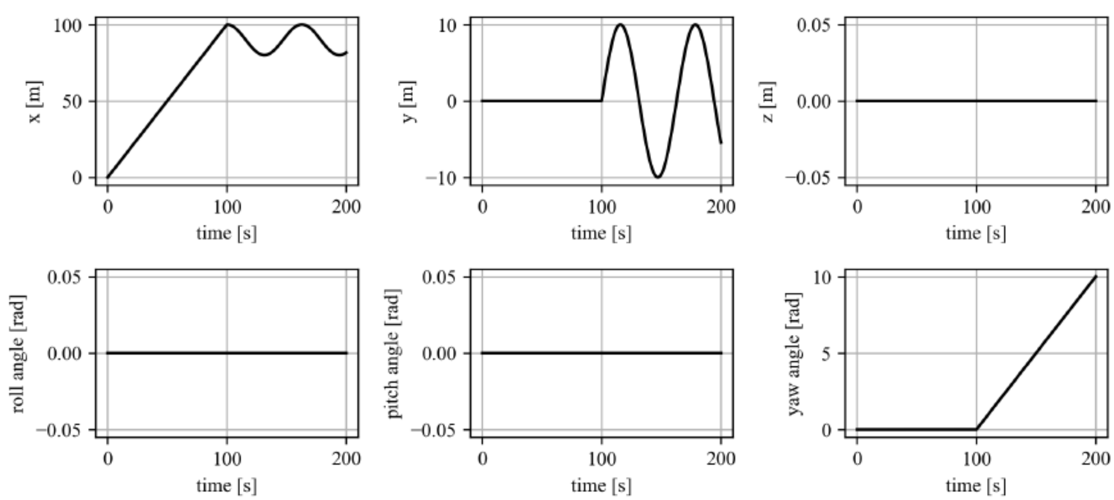


Figure 110: ROV2 particle dynamics numerical simulation. Position versus time graphs are shown for each axis. The parameters include the initial input $u = [1.0, 0.0, 0.0, 0.0, 0.0, 0.0]$ for $t < 100$ and final input $u = [0.0, 1.0, 0.0, 0.0, 0.0, 0.1]$.

The resulting plots confirm that the simulated BlueROV2 followed the commanded inputs accurately. As seen in the graph, the input u in the local reference frame was correctly mapped in the figure's global frame throughout the movement. These results verify the implementation of the BlueROV2's dynamic equations and demonstrate the model's ability to replicate the submersible's motion. Using this simulation as a foundation for future work, Section 5.7.3.2 will utilize the newly developed BlueROV2 particle dynamics to test future ESC algorithms

5.7.3.2 GESC Particle Simulations

Building upon the ROV2 dynamic model developed in Section 5.7.3.1, the next stage of the numerical simulation involved implementing an extremum seeking controller. To accomplish this, the first step was to apply a proven, pre-existing controller to the model as a proof of concept before developing the rotating-sensor ESC technique. For this initial implementation, the article by J. Cochran [1] provided a method for 3D ESC on an underactuated vehicle. The vehicle moves by maintaining a constant forward velocity while tuning pitch and yaw to achieve convergence. The vehicle model used in Cochran's work is described in Fig.111.

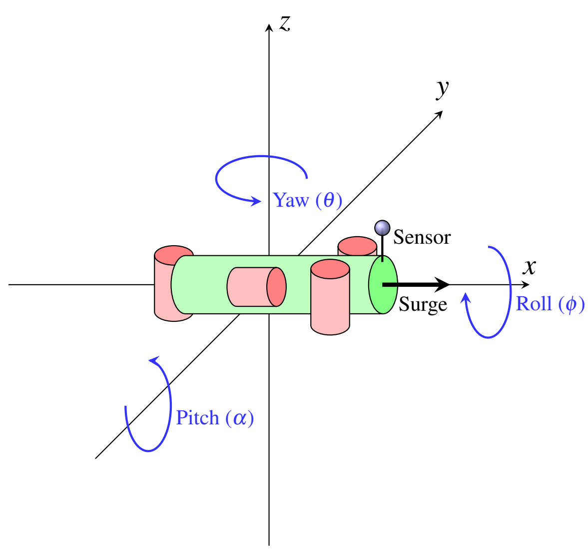


Figure 111: Underactuated vehicle re-creation based on [1].

In Fig. 111, the vehicle's pitch, roll, and yaw were represented by α, ϕ, and θ, respectively. To maintain consistency with the ROV2 dynamic model, these variables were redefined as x_4, x_5, x_6. The translational components were also redefined becoming x_1, x_2, x_3. A constant surge velocity was applied in x_1's direction, while pitch x_5 and yaw x_6 were tuned by the controller to drive the vehicle toward the extremum of the cost function J. The block diagram in Fig. 112 outlines the controller's design used in Cochran's method.

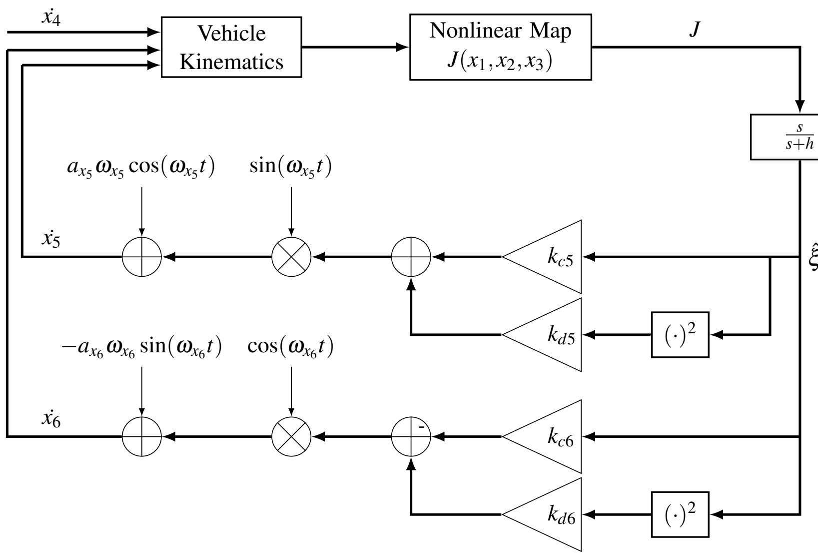


Figure 112: 3D static sensor GESC control block diagram re-creation from [1].

Deriving the control inputs from the diagram, the states become equal to

$$\dot{x}_4 = 0 \tag{5.40}$$

$$\dot{x}_5 = a_{x_5}\omega_{x_5}\cos(\omega_{x_5}t) + \sin(\omega_{x_5}t)(k_{c_{x_5}}\xi + k_{d_{x_5}}\xi^2) \tag{5.41}$$

$$\dot{x}_6 = -a_{x_6}\omega_{x_6}\sin(\omega_{x_6}t) + \cos(\omega_{x_6}t)(k_{c_{x_6}}\xi + k_{d_{x_6}}\xi^2) \tag{5.42}$$

where $a_{x_5}\omega_{x_5}\cos(\omega_{x_5}t)$ and $-a_{x_6}\omega_{x_6}\sin(\omega_{x_6}t)$ represent the perturbations and $\sin(\omega_{x_5}t)(k_{c_{x5}}\xi + k_{d_{x5}}\xi^2)$ and $\cos(\omega_{x_6}t)(k_{c_{x6}}\xi + k_{d_{x6}}\xi^2)$ are the demodulation signal scaled by their gains and bias terms. Using these inputs, the GESC controller effectively moves the underactuated vehicle to converge the source.

3D GESC Implementation With A Static Sensor For ROV2

To demonstrate that the BlueROV2's particle dynamics model can be used for ESC numerical simulation, the 3D static sensor ESC method used by Cochran can be applied. However, because the BlueROV2 does not support pitch control, an adjustment to the controller is necessary. One possible modification would be to tune roll instead of pitch; however, in practice, the ROV2's buoyancy distribution causes it to remain naturally upright. This physical limitation makes roll tuning impractical. To account for this, the pitch tuning was instead replaced with vertical position tuning. This method was internally referred to as Z-bobbing. The block diagram shown in Fig. 113 illustrates this modification.

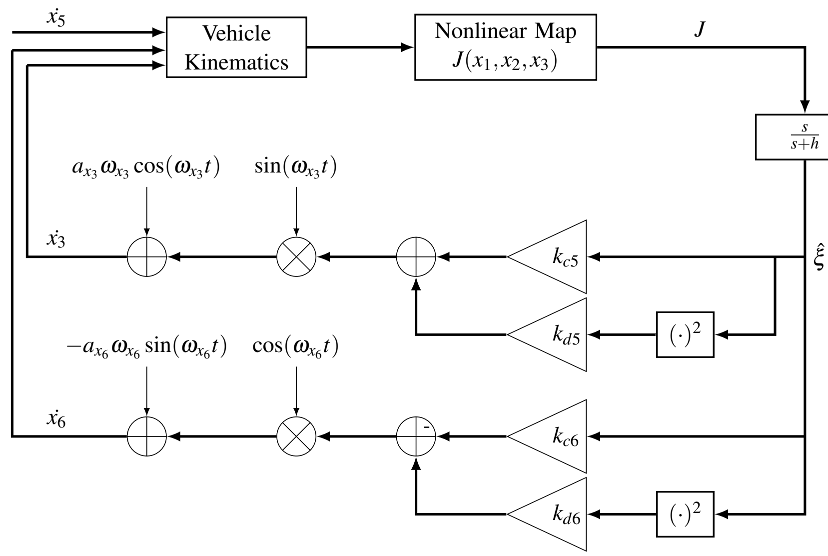


Figure 113: Z-Bobbing block diagram.

In this updated block diagram, the pitch perturbation signal $a_{x_5}\omega_{x_5}\cos(\omega_{x_5}t)$ used in Cochran's original design was replaced with a vertical perturbation $a_{x_3}\omega_{x_3}\cos(\omega_{x_3}t)$. The new signal modulates the BlueROV2's height along z-axis represented by x_6. The pitch angle x_5 is fixed to zero throughout the simulation to account for the dynamic limitations and the vehicles roll x_4 also remains at zero for stability. The remaining control parameters in the diagram are also redefined to match the nomenclature of the submersible's dynamics, completing the modification. This results in the controller producing oscillatory vertical motion while converging towards the source. Fig.114 describes the motion of the resulting numerical simulation.

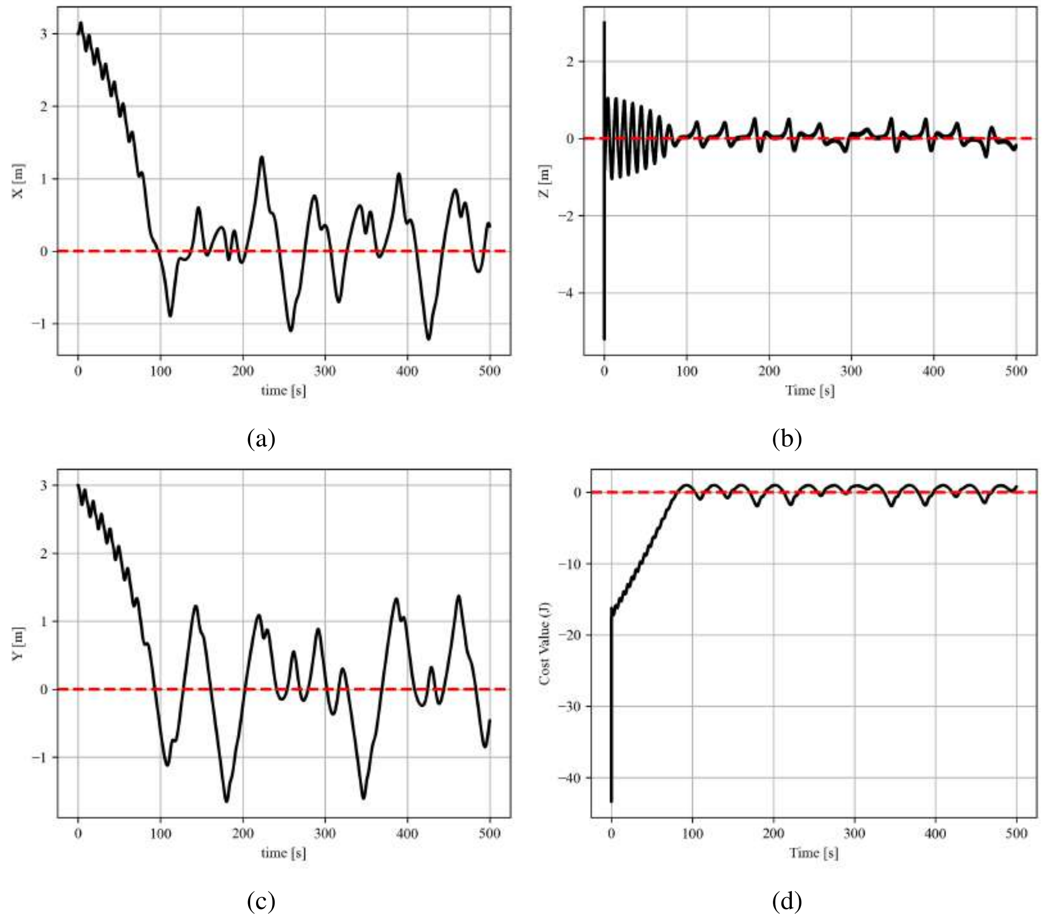


Figure 114: Z-Bobbing numerical simulations using a quadratic cost function to demonstrate the functionality of the BlueROV2 dynamics for GESC: (a) x-position versus time, (c) y-position versus time, (b) z-position versus time, (d) cost value based versus time. Parameters for these experiments are: $\dot{x_1} = 0.1\text{m/s}$, $c_{x_3} = 100, c_{x_6} = 100, d_z = 100, d_{x_6} = 100$, $\omega_\xi = 10$.
KEY: ▪▪▪ = Source position.

2D GESC Implementation With A Rotating Sensor For ROV2

After validating the particle dynamics of the BlueROV2 for use in ESC, the next project step was to integrate a rotating sensor. To simplify the initial implementation, a 2D version of the controller was first developed based on previous work using TurtleBot3 in Section 5.5.4. Differing from the Turtlebot3 implementation, the BlueROV2 has the capability to move along each axis without turning. This simplifies the controller, allowing it to rely only on translational movements $\dot{x_1}, \dot{x_2}, \dot{x_3}$ while keeping angular inputs $\dot{x_4}, \dot{x_5}, \dot{x_6}$ at zero. The resulting block diagram utilizing the simplified controller is shown in Fig. 115.

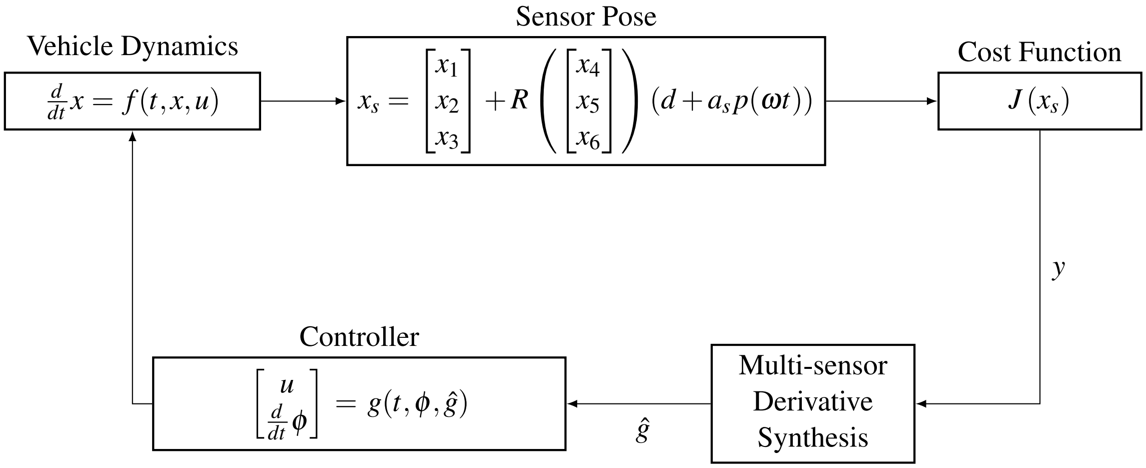
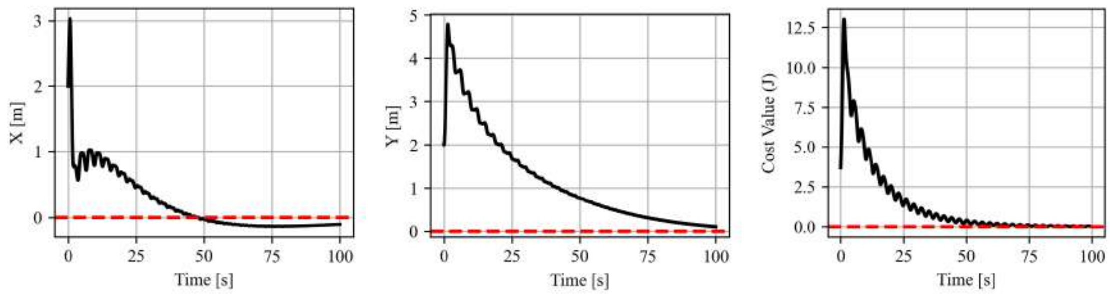


Figure 115: ROV2 2D rotating sensor simulation.

The block diagram begins with the Sensor Pose block, where the sensor's position x_s is calculated and transformed from its local reference frame to the global frame. This process begins in Eq. (5.43), where the perturbation signal $p(\omega t)$ extracts the components of the sensor's angle, ϕ_s in its local frame as a function of ω and t. Then the arm length of the sensor's frame a_s is multiplied by the perturbation signal and added onto the displacement from the vehicle to the sensor's frame d to obtain the local coordinates of the sensor. The rotation matrix R as a function of x_4, x_5, x_6 subsequently rotates the local sensor coordinates to the vehicle's frame of reference. Finally, the global position of the vehicle is added to the transformed sensor coordinates to compute the global position of the sensor.

$$p(\omega t) = \begin{bmatrix} \cos(\phi_s(\omega t)) \\ \sin(\phi_s(\omega t)) \\ 0 \end{bmatrix} \tag{5.43}$$

Following the Sensor Pose block, x_s is passed into the Nonlinear Map containing the quadratic cost function $J = 0.5{x_{s_1}}^2 + 0.5{x_{s_2}}^2$. Utilizing the cost function and its derivatives up to a desired order, the cost values are generated and processed to calculate a single estimated gradient $\hat{g}$ in the Multi-Sensor Derivative Synthesis block. The controller then receives the estimated gradient and generates the submersible's control input u and any auxiliary parameters ϕ. Following the controller, vehicle dynamics receives the command and drives the BlueROV2 towards the minimum of the cost function at θ_*. The resulting simulation, shown in Fig.116, illustrates the vehicle's motion and convergence.



(a) (b) (c)

Figure 116: 2D GESC simulation using a quadratic cost function to demonstrate the use of a rotating sensor on the BlueROV2: (a) x-position versus time, (b) y-position versus time, (c) cost value versus time. Parameters for this experiment are: $a = 0.1524$ m, $\omega_{\xi} = 2.094$ rad/s, $k_v = 0.04$, $k_{\omega} = 0.04$. **KEY**: ▪▪▪ = Source position.

5.7.4 Future Plans

Following the completion of the ROV2 test experiments in Section 5.7.2, the next step is to finalize the submersible's hardware. For full 3D sensing capability, at least three hydrophone mounts are required to obtain derivative information along all axes. Therefore, two additional hydrophone mounts will be added to the sides of the vehicle with appropriate arm lengths to support 3D ESC using rotating sensors.

In parallel with the updated hardware, the finalization of the multi-sensor derivative estimation theory will be completed by adding rotations. This addition will enable the development of a 3D ESC numerical simulation using rotating sensors. Once complete, more realistic simulations will be made using Gazebo. A Gazebo simulation will allow for a testing environment that includes effects such as buoyancy and hydrodynamics. Using a simulation that more closely replicates real life testing will then be used to implement the technique on the physical ROV2.

Once the hardware modifications and Gazebo simulations are complete, the BlueROV2 will be prepared and ready for pool testing. The result of this test will validate the multi-sensor derivative estimation theory and the application of extremum seeking control using rotating sensors in 3D.

6. Accomplishments and Future Work

6.1 Accomplishments

During this reporting period, we have worked on the theoretical framework in parallel with the operational framework to leverage the knowledge gained from real-life implementations to inform the theoretical design. We have made the following advances for each of these frameworks:

Theoretical framework — We completed the following works: (1) obtained conditions for perturbation-based derivative estimates using a single rotating sensor (T1-1); (2) developed a multi-sensor derivative estimation approach integrating information from multiple static sensors (under T1-2); (3) proved practical stability of model-free ESC from their associated model-based ESC (T2-1); (4)

proved global asymptotic stability for interconnected adaptive gradient systems using various techniques (T2-2); (5) refined the practical stability proof for RMSpESC (T3-1); and (6) designed AGESC control (under T3-2).

Operational framework — We completed considerable portions of Tasks 10-2 and 10-4. We implemented the AGESC method onboard a UGV to successfully seek static and moving light and acoustic sources. We conducted extensive experiments and validated the effectiveness of the rotating sensor frame in seeking static and moving light and acoustic sources indoors while comparing GESC, RMSpESC, and AGESC, and conventional ESCs (with static sensors). We developed and tested an additional design for the rotating sensor frame of the UGV to extend the rotation envelope from 180° to full 360°. This new full rotation design makes the convergence quite independent from the seeker's initial heading and enables reliable convergence to sources positioned directly behind the seeker. A rotating sensor frame similar to the UGV's was adopted for the USV, but it required three additional, now-completed modifications: making the electronics box penetrations water resistant, changing the dimensions to fit a servo-driven rotor in a water-proof enclosure, and protecting the sensor. The customization of the UUV, BlueROV2, presented significant challenges in two primary areas, which have since been successfully resolved: (1) hardware integration and space management for the hydrophone mount and its supporting electronics (power, data acquisition, and processing); and (2) waterproofing the connections (for added wires) and servo enclosure to withstand extended periods of submersion. In parallel with BlueROV2 customization, we have developed the BlueROV2 particle dynamics and used that to complete the numerical simulations for GESC with a static sensor in a 3D map and GESC with a rotating sensor in a 2D environment.

6.2 Future Work

We will continue to work concurrently on the theoretical and operational frameworks, as detailed below.

Theoretical framework

Task 1-2- Multiple Sensor Estimation: We demonstrated through the averaging technique that derivative estimates $\hat{g}$ of the nonlinear field J can be obtained from a single rotating sensor in 2D maps. Additionally, we developed a new multi-sensor derivative estimation approach that integrates information from multiple static sensors to generate a single fitted gradient estimate of the cost function J. The next step is to extend this approach to accommodate rotating sensor data in 3D maps.

Tasks 3, 4, and 5- AdaGM and AccGM optimizers: We studied the stability of general model-free ESC and interconnected systems and demonstrated the practical stability of RMSpESC on multivariable strictly convex maps. We have to yet show the semiglobal practical uniform asymptotic stability properties for the following systems: (1) ESC with the heavy ball optimizer (HBESC system), and (2) ESC with the Adagrad optimizer (AGESC system). The proofs will utilize the results of our works on interconnected systems and/or stability of model-free systems. Next, we design a new blend ESC (BESC) which merges the properties of AdaGM and AccGM and leverage their respective strengths to regularize the convergence rate and improve convergence along flat regions and non-convex regions.

Task 6- Escaping Local Minima in Highly Non-Convex Maps: Accelerated gradient-based ESC can escape some extrema, but not all extrema, and not always. We plan to build upon the BESC by creating additional strategies designed to ensure escape from undesired extrema particularly in cases where HBESC fails.

Task 7- Safety with Log Barrier Functions: We plan to prove safety and stability of log barrier functions for the GESC before implementing them onboard the unmanned vehicles and implementing them for the other ESC systems of interest.

Task 8- Moving Source: Most sources of interest are not stationary but rather move throughout the environment. We plan to first complete the proof of semiglobal Input-to-State practical stability (sGISpS) for the GESC before attempting to prove a sGISpS for the AdaGM and AccGM.

Operational framework

We plan to conduct additional UGV, USV, and UUV demonstrations, as detailed below.

UGV demonstrations: The plan for UGV involves implementing: (1) HBESC and BESC onboard the vehicle to seek light and acoustic sources using a rotating microphone frame (Task 10-2), (2) the escape strategies in highly non-convex to converge to the desired acoustic source (Task 10-3), and (3) the ESC with Log Barrier Functions for validation in cluttered environments (Task 10-4).

USV demonstrations: The plan for USV includes implementing RMSpESC, HBESC, and BESC onboard the vehicle to seek acoustic sources using a rotating microphone frame (Task 10-2).

UUV demonstrations (Task 10-2): The current plan for UUV involves: (1) completing pool tests to guide the design of the hydrophone mount for the BlueRov2; (2) finalizing the design and integration of the three rotating hydrophone mounts; (3) integrating a sonar with the BlueROV2 for mapping and localization; (4) adapting existing UUV simulation packages, which requires addressing challenges related to poor documentation and environment dependency; and (5) implementing GESC and RMSpESC on the vehicle to seek acoustic sources using the three rotating hydrophone frames in a pool.

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List of Symbols, Abbreviations, and Acronyms

Latin Symbols

Symbol	Description	Symbol	Description
a	Dither amplitude	q_x, q_y, q_z	Positive constants
d	Sensor displacement	Q	Covariance matrix
D	Derivative operator	r	Position of the vehicle in the global frame
g	Gradient	$r(t)$	Vector of continuous, mean-zero, bounded, linear independent, (almost) periodic signals
G	Gradient outer product	R	Cross-variance between the signals
m	Demodulation signal	R	Resistance
H	Hessian	R	Rotation matrix
I	Electrical current or Intensity	S	Max decibel reading
J	Cost function	t	Time
k	Gain or learning rate	T	Time period
L_p	Sound pressure level	v	Forward velocity
$m(t)$	Time-varying gradient estimation filter	v	Gradient vector element-wise squared
p	Pressure	V	Lyapunov Function or Voltage
P	Total power radiated from the light source	x, y, z	State vector
P_m	Polynomial of order m	(x,y)	Position of the vehicle in the global frame
$p(t)$	Sinusoidal dither signal		

Greek Symbols

Symbol	Description	Symbol	Description
α	Class $\mathscr{K}_\infty$ function or multi-index or pitch	λ	Lagrange multiplier
β	Class $\mathscr{K}\mathscr{L}$ function	ξ	Derivative value or Washout filter state
β	Angle between the vector normal to the sensor's surface and the vector pointing from the sensor to the signal source	ρ	Continuous (almost) periodic function
γ	Logarithm of inverse Hessian	$\rho(t)$	Extended dither signal
γ	Gamma condition for the integral Lyapunov function	ρ_q	Gain of the ISS system
Γ	Inverse Hessian	τ	Fast time scale
δ	Small parameter	ϕ	Sensor angle in the relative frame or roll
η	Auxiliary state	ω	Dither frequency or angular velocity
ε	Washout filter parameter	ω_l	Low-pass Ricatti filter gain
θ	State or robot heading angle or yaw	ω_ξ	Washout filter gain

Mathematical Symbols

Symbol	Description	Symbol	Description
D^+	Upper right Dini derivative	$\widetilde{\langle\cdot\rangle}$	Error coordinate value
D_-	Lower left Dini derivative	∇	Gradient
$\dot{\langle\cdot\rangle}$	Derivative	∇^2	Hessian
$\hat{\langle\cdot\rangle}$	Estimate	$\langle\cdot\rangle^T$	General matrix transpose
$\overline{\langle\cdot\rangle}$	Averaged	$\langle\cdot\rangle_*$	Optimal
$\overline{\hat{\langle\cdot\rangle}}$	Averaged estimate	$\langle\cdot\rangle^*$	Source

Subscripts

Symbol	Description	Symbol	Description
0	Initial condition	pr	Photoresistor
c	Center	r	Relative frame
com	Commanded	ref	Reference
d	Derivative	s	Sensor
i, j	Element in the ith row and jth column	source	Source
ℓ	Light	trans	Translation
meas	Measured	*	Optimal
p	Proportional		

Acronyms

AccGM	Accelerated Gradient Methods
AdaGM	Adaptive Gradient Methods
AGESC	AdaGrad ESC
DAQ	Data Acquisition
ESC	Extremum Seeking Control
GAS	Globally Asymptotically Stable
GESC	Gradient Extremum Seeking Control
GUAS	Globally Uniformly Asymptotically Stable
HAT	Hardware Attached on Top
HBESC	Heavy Ball ESC
HBK	Hottinger, Brüel & Kjær
RMS	Root Mean Squared
RMSprop	Root Mean Squared Propagation
RMSpESC	RMSprop-based ESC
sGPUAS	Semiglobally Uniformly Practically Asymptotically Stable
SPL	Sound Pressure Level
UGV	Unmanned Ground Vehicle
USV	Unmanned Surface Vehicle
UUV	Unmanned Underwater Vehicle